MASTER NUMERICS - BASE44 BUILD PACKAGE
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PART 1 - BUILD PROMPT (BASE44_PROMPT.md)

Master Numerics -- full Base44 build package

You have TWO files:
1. **This prompt** (the full app spec).
2. **‘curriculum-export.json‘** -- the COMPLETE real curriculum: 21 physics worlds,
   14 math districts, 134 lessons, 1,176 sections, **1,051 questions**, and 74 embedded
   YouTube videos, with all lesson prose, worked examples, answers, and explanations.

Use the JSON as the source of truth for content -- import it verbatim and render it.
Generate code/UI from the prompt; do NOT rewrite or shorten the curriculum.

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curriculum-export.json -- schema

```
{
  "app": "Master Numerics",
  "physics": [ World, ... ],   // 21 cosmic worlds, K -> graduate
  "math":    [ World, ... ]    // 14 city/nature districts, K -> graduate
}

World = {
  slug, name, subtitle, description, gradeRange, scaleLabel,
  palette: { accent, glow },
  lessons: [ Lesson, ... ]
}

Lesson = {
  slug, title, tagline, estMinutes, xpReward, difficulty (1-5),
  requiresMath?  (slug of a math lesson this physics lesson depends on),
  sections: [ Section, ... ],
  questions: [ Question, ... ]
}

Section = { kind, title?, content }  // kind ?
  HERO            content: { scene, headline, sub }
  CONTEXT         content: { markdown }
  CONCEPT         content: { markdown }
  SIMULATION      content: { simId, intro }            // interactive mini-sim by id
  WORKED_EXAMPLES content: { examples: [{ title, problem, steps[], answer }] }
  VIDEOS          content: { videos: [{ youtubeId, title }] }   // embed these
  CONCEPT_CHECK   content: { intro }                   // shows scope=CONCEPT_CHECK Qs
  PRACTICE        content: { intro }                   // shows scope=PRACTICE Qs
  DEEPER_DIVE     content: { markdown }
  SUMMARY         content: { takeaways: [string], formulas: [{ label, tex }] }

Question = {
  scope: "CONCEPT_CHECK" | "PRACTICE",
  kind: "MCQ"|"TRUE_FALSE"|"NUMERIC"|"MATCHING"|"FILL_BLANK"|"ORDER"|"PROOF"|"SYMBOLIC"|"GRAPH",
  prompt, options?, answer, difficulty? (1-5), hint?, explanation
}
// answer by kind:
//   MCQ -> correct index (number)        TRUE_FALSE -> boolean
//   NUMERIC -> { value, tolerance }      MATCHING -> right[] in left order
//   FILL_BLANK -> string[][] (accepted strings per blank)
//   ORDER/PROOF -> [] (options already in correct order; learner reorders)
//   SYMBOLIC -> { expr, vars?, tolerance? }
//   GRAPH -> { expr, domain?, variable?, tolerance?, mode?, samples? }
//           mode:"draw" => render a sketch-the-curve canvas instead of a text field
```
All markdown/prompts use **KaTeX** math in ‘$...$‘ / ‘$$...$$‘ (escape literal money as ‘\$‘).

APP SPEC

Build a gamified "digital university" called **Master Numerics** -- an immersive journey
through mathematics and physics from kindergarten through graduate level. Curriculum is
DATA (imported from ‘curriculum-export.json‘), not hand-written pages. Two themed,
Duolingo-style scrollable lesson-path worlds, a local in-browser answer-checking engine,
embedded videos, downloadable worksheets, formula flashcards, and a restrained
black-and-white + one electric-blue-accent design. Free-demo mode: ALL content is open,
the store sells ONLY cosmetics, and nothing about learning is paywalled.

Design system (strict)
- Monochrome base (near-black bg ‘#06070B‘, off-white text ‘#F4F6FB‘) + ONE accent:

  electric blue ‘#2D7DFF‘. Color is a REWARD SIGNAL only -- semantic tokens: accent (blue),
  success (green), alert (red), star (gold ‘#F5B83C‘), plus neutral surface/line/fg.
- **NO EMOJI** anywhere -- use clean line icons and SVG shapes.
- Real math notation everywhere via **KaTeX**.
- Honor ‘prefers-reduced-motion‘ (animations cross-fade/skip).
- Fully responsive: no overlapping text at mobile or desktop widths.
- Subtle ever-present starfield behind every screen. Physics = cosmic theme;
  Math = city/nature theme.

Two worlds + zoom transition
- Physics "Journey" = a COSMIC scroll; Math "City" = a CITY/NATURE scroll. Both render as
  a Duolingo-style path of unit nodes: a mastery ring around each node, 0-3 stars above,
  unit name below; done / current / locked / coming-soon states. "Locked" nodes are still
  tappable to preview -- never a dead end.
- A "warp"/zoom hyperspace transition animates travel between the two worlds and between
  units (reduced-motion safe -> instant navigate when reduced motion is on).
- Per-LESSON and per-QUESTION difficulty indicators (1-5) as small pips/stars.
- Cross-subject prerequisite: when a physics lesson has ‘requiresMath‘, show a
  "need a refresher?" card linking straight to that math lesson (working jump link).

Lesson player
Render every section in order: HERO -> CONTEXT -> CONCEPT -> SIMULATION -> WORKED_EXAMPLES ->
VIDEOS (embed the YouTube ids) -> CONCEPT_CHECK -> PRACTICE -> DEEPER_DIVE -> SUMMARY.
Quizzes are stepped (one question at a time) with instant per-question explained feedback,
then a summary; PRACTICE sets record a scored attempt.

Local grading engine (no third-party services)
Grade IN THE BROWSER and ALWAYS return an explained result (why right/wrong):
- NUMERIC: tolerance compare. MCQ/TRUE_FALSE/FILL_BLANK/MATCHING/ORDER/PROOF: direct.
- SYMBOLIC: algebraic equivalence by RANDOM SAMPLING (1/2 = 0.5, (x+1)^2 = x^2+2x+1).
- GRAPH: sample the entered function vs the target over a domain; show a live preview plot
  (target hidden until graded).
- GRAPH ‘mode:"draw"‘: a sketch-the-curve canvas (mouse + touch) -- drag a row of nodes to
  trace a graph; grade node heights vs the target with a generous tolerance; reveal the
  target curve after grading.
- Expression evaluator: use RADIANS; treat adjacent identifiers as one token (explicit ‘*‘
  for multiply); single-letter or underscore-suffixed variables (‘x‘, ‘x_f‘).

Gamification
XP per lesson/challenge; daily XP goal; day streaks; mastery per lesson (0-1 ring) + 0-3
stars and a "Lesson mastered!" state; ranks with progress; daily challenges with claimable
rewards; level-up / reward toasts (icon-based, no emoji).

Store -- COSMETIC ONLY (no learning paywall)
Earn coins; spend ONLY on cosmetics: mascot/avatar outfits, hats, pets, profile
backgrounds, auras, titles, badges, path skins, cosmetic loot crates. A friendly mascot
with moods (idle/happy/thinking/celebrate) drawn with SVG/icons. Free-demo flag: when the
paywall is OFF (default), EVERY lesson/world is open and NO subscribe prompts, trial
countdowns, or upgrade banners appear anywhere.

Pages / features
Onboarding (mascot-guided: pick subject + grade -> recommended start); Auth; Dashboard
(welcome, streak, daily goal, daily challenges, rank progress, quick links, continue);
Physics Journey map + World + Lesson; Math City map + District + Lesson; downloadable
WORKSHEETS per lesson (printable / Save-as-PDF, optional answer key, regenerate reshuffle);
FLASHCARDS (spaced repetition, SM-2: Again/Hard/Good/Easy); FORMULAS reference sheet
(KaTeX, favorites, printable); graphing CALCULATOR; NOTEBOOK scratchpad; social
(leaderboard, friends, real-time head-to-head quiz BATTLE); achievements; profile +
customization; in-app NOTIFICATIONS (bell + feed + toasts); settings (theme, reduced
motion, high contrast); admin view.

Acceptance criteria
[x] Lesson player renders all section kinds incl. embedded YouTube videos.
[x] All question kinds graded locally with explained feedback, incl. the drawn/sketch canvas.
[x] Downloadable worksheets; formula flashcards.
[x] Two themed scroll worlds (cosmic physics, city/nature math) as Duolingo paths, with a
  zoom/warp transition between them.
[x] Per-lesson AND per-question difficulty indicators (1-5).
[x] Cross-subject math-prerequisite surfacing with a working jump-to-math link.
[x] Streaks, XP, mastery; cosmetic-only store with NO lesson paywall; in-app notifications.
[x] Restrained B&W + electric-blue UI; color only in reward states; NO emoji; no overlapping
  text at mobile/desktop; real KaTeX math; reduced-motion-safe animations.
[x] ALL imported content rendered: 21 physics worlds + 14 math districts, 134 lessons, 1,051
  questions, 74 videos -- K -> graduate, both subjects, verbatim from curriculum-export.json.

PART 2 - FULL CURRICULUM DATA (curriculum-export.json)

```
{
  "app": "Master Numerics",
  "generatedAt": "2026-06-15T22:37:27.899Z",
  "physics": [
    {
      "slug": "the-moon",
      "name": "The Moon",
      "subtitle": "Where the journey begins",
      "description": "A silent, silver world a quarter of a million miles away. Here you will meet gravity, learn why the Moon shines without burning, and watch its face change night after night.",
      "gradeRange": "K-2",
      "scaleLabel": "3x10^6 m",
      "palette": {
        "accent": "#C9CED6",
        "glow": "#8E97A8"
      },
      "lessons": [
        {
          "slug": "what-is-gravity",
          "title": "What Is Gravity?",
          "tagline": "The invisible pull that holds the universe together",
          "estMinutes": 12,
          "xpReward": 100,
          "sections": [
            {
              "kind": "HERO",
              "content": {
                "scene": "moon",
                "headline": "The Invisible Pull",
                "sub": "You just landed on the Moon. Take one hop... and you float six times higher than you ever could on Earth. Why? Let’s find out."
              }
            },
            {
              "kind": "CONTEXT",
              "title": "Why does this matter?",
              "content": {
                "markdown": "Drop your pencil. Where did it go? Down. It always goes down. It never floats up to the ceiling, and it never drifts sideways out the window.\n\nThat is not an accident. Something is pulling on that pencil -- and on you, your dog, the ocean, and even the Moon itself. We call that pull **gravity**. Gravity is why rain falls, why slides are fun, why basketballs come back down after a shot, and why you don’t float out of bed at night.\n\nAstronauts who walked on the Moon felt gravity too -- just much less of it. They could leap like superheroes and carry heavy packs like they were pillows. Understanding why is your first step on this journey through the cosmos."
              }
            },
            {
              "kind": "CONCEPT",
              "title": "The big idea",
              "content": {
                "markdown": "**Gravity is a pull between things.** Everything that is made of stuff -- scientists say everything with *mass* -- pulls on everything else. You pull on your chair. Your chair pulls on you. But those pulls are far too tiny to feel.\n\nFor gravity to feel strong, something needs to be **huge**. A planet, for example.\n\n**Earth is enormous**, so its pull is strong. Earth’s gravity grabs everything near it and pulls it toward the ground -- toward the center of the Earth. That pull on *you* has a name you already know: your **weight**. When you stand on a scale, you are measuring how hard Earth is pulling you down.\n\nNow here is the Moon trick. **The Moon is much smaller than Earth** -- about 80 Moons would fit inside one Earth. Smaller means a weaker pull. The Moon’s gravity is only about **one sixth** as strong as Earth’s. So on the Moon:\n\n- You would weigh about 6 times less.\n- You could jump about 6 times higher.\n- A dropped feather would still fall -- just slowly and gently.\n\nNotice that last one. The Moon **does** have gravity. Things don’t float away on the Moon; they fall, softly. (Astronauts float on the space station for a different reason -- they are actually *falling around* the Earth, which we will explore in a later world.)\n\nOne more job gravity does, and it is a big one: **gravity is the invisible string that keeps the Moon circling the Earth**, and keeps the Earth circling the Sun. Without gravity, the Moon would fly off in a straight line into the dark, and so would we.\n\nSo remember the three rules of gravity:\n\n1. Gravity **pulls**, it never pushes.\n2. **Bigger things pull harder.**\n3. Gravity pulls everything **toward the center** of the big thing -- that’s what \"down\" means."
              }
            },
            {
              "kind": "SIMULATION",
              "title": "Try it: drop lab",
              "content": {
                "simId": "gravity-drop",
                "intro": "Drop a ball on Earth, then on the Moon. Watch the timer -- the Moon’s weaker pull means a slower, floatier fall. Then invent your own planet with the gravity slider."
              }
            },
            {
              "kind": "WORKED_EXAMPLES",
              "title": "Worked examples",
              "content": {
                "examples": [
```

```
                  {
                    "title": "Which way will it fall?",
                    "problem": "Maya holds an apple over her head in Australia and lets go. People say Australia is on the ’bottom’ of the Earth. Does the apple fall toward the sky or toward the ground?",
                    "steps": [
                      "Rule 3 says gravity pulls toward the **center of the Earth**.",
                      "In Australia, the ground is the direction toward Earth’s center -- just like everywhere else.",
                      "’Down’ simply means ’toward the center,’ no matter where you stand on the planet."
                    ],
                    "answer": "The apple falls to the ground. Down is toward Earth’s center everywhere -- nobody falls off the bottom of the world."
                  },
                  {
                    "title": "The Moon backpack",
                    "problem": "Leo’s space backpack feels as heavy as 6 bricks on Earth. About how heavy would it feel on the Moon?",
                    "steps": [
                      "The Moon’s pull is about one sixth of Earth’s pull.",
                      "So everything feels about 6 times lighter there.",
                      "6 bricks of heaviness / 6 = 1 brick of heaviness."
                    ],
                    "answer": "It would feel like about 1 brick. Same backpack, same stuff inside -- just a gentler pull."
                  },
                  {
                    "title": "The invisible string",
                    "problem": "The Moon zooms through space, yet it never flies away from Earth. What holds it?",
                    "steps": [
                      "Gravity is a pull between *any* two things with mass.",
                      "Earth is huge, so its pull reaches all the way out to the Moon.",
                      "That pull constantly bends the Moon’s path into a circle around us, like a ball swung on a string."
                    ],
                    "answer": "Earth’s gravity. It acts like an invisible string keeping the Moon in orbit."
                  }
                ]
              }
            },
            {
              "kind": "VIDEOS",
              "title": "Watch and wonder",
              "content": {
                "videos": [
                  {
                    "youtubeId": "suQDwZcnJdg",
                    "title": "Gravity | The Dr. Binocs Show"
                  },
                  {
                    "youtubeId": "ljRlB6TuMOU",
                    "title": "Defining Gravity | Crash Course Kids"
                  },
                  {
                    "youtubeId": "dh5wGSLw1p4",
                    "title": "Why Do Things Float in Space? | SciShow Kids"
                  }
                ]
              }
            },
            {
              "kind": "CONCEPT_CHECK",
              "title": "Quick check",
              "content": {
                "intro": "Three quick questions -- answer right away and see how you did."
              }
            },
            {
              "kind": "PRACTICE",
              "title": "Practice problems",
              "content": {
                "intro": "Show what you know! Use the hints if you get stuck -- then submit to see your score."
              }
            },
            {
              "kind": "DEEPER_DIVE",
              "title": "Deeper dive: the feather and the hammer",
              "content": {
                "markdown": "About 400 years ago, the scientist **Galileo** claimed something strange: without air pushing back, a heavy thing and a light thing should fall **together** -- gravity speeds them up equally.\n\nOn Earth, air ruins the test. A feather flutters because air catches it like a tiny parachute.\n\nBut the Moon has **no air at all**. In 1971, astronaut **David Scott** stood on the Moon holding a hammer in one hand and a falcon feather in the other. He let go of both at the same time... and they hit the dust **at exactly the same moment**. Galileo was right -- it just took 400 years and a trip to the Moon to film the proof."
```

```
          }
        },
        {
          "kind": "SUMMARY",
          "title": "Mission summary",
          "content": {
            "takeaways": [
              "Gravity is a pull between all things that have mass.",
              "Bigger objects pull harder -- Earth’s pull is strong, the Moon’s is about 1/6 as strong.",
              "Your weight is how hard a planet pulls on you, so you’d weigh less on the Moon.",
              "Gravity pulls toward the center of a planet -- that is what ’down’ means.",
              "Gravity keeps the Moon orbiting Earth like an invisible string."
            ],
            "formulas": [
              {
                "label": "Moon weight",
                "tex": "\\text{weight on Moon} \\approx \\tfrac{1}{6} \\times \\text{weight on Earth}"
              }
            ]
          }
        }
      ],
      "questions": [
        {
          "scope": "CONCEPT_CHECK",
          "kind": "TRUE_FALSE",
          "prompt": "Gravity can push things away from a planet.",
          "answer": false,
          "explanation": "Gravity only pulls. It always draws things together, never pushes them apart."
        },
        {
          "scope": "CONCEPT_CHECK",
          "kind": "MCQ",
          "prompt": "What makes a ball fall to the ground when you drop it?",
          "options": [
            "Wind",
            "Earth’s gravity pulling it",
            "The ball pushing itself",
            "Magnets"
          ],
          "answer": 1,
          "explanation": "Earth’s gravity pulls the ball toward the center of the Earth -- that’s why dropped things
 go down."
        },
        {
          "scope": "CONCEPT_CHECK",
          "kind": "TRUE_FALSE",
          "prompt": "The Moon has no gravity at all.",
          "answer": false,
          "explanation": "The Moon does have gravity -- about one sixth of Earth’s. Things still fall there, just
more gently."
        },
        {
          "scope": "PRACTICE",
          "kind": "MCQ",
          "prompt": "Why is the Moon’s gravity weaker than Earth’s?",
          "options": [
            "The Moon is colder",
            "The Moon is much smaller than Earth",
            "The Moon is farther from the Sun",
            "The Moon spins slower"
          ],
          "answer": 1,
          "hint": "Think about rule 2: what makes a pull strong?",
          "explanation": "Smaller mass means a weaker pull. The Moon is much smaller than Earth, so its gravity is
weaker."
        },
        {
          "scope": "PRACTICE",
          "kind": "MCQ",
          "prompt": "A robot weighs as much as 12 bricks on Earth. About how heavy would it feel on the Moon?",
          "options": [
            "12 bricks",
            "6 bricks",
            "2 bricks",
            "0 bricks -- it would float"
          ],
          "answer": 2,
          "hint": "Divide by 6.",
          "explanation": "Moon gravity is about 1/6 of Earth’s: 12 / 6 = 2 bricks of heaviness."
        },
        {
```

```
            "scope": "PRACTICE",
            "kind": "TRUE_FALSE",
            "prompt": "If you dropped a ball on the Moon, it would float away into space.",
            "answer": false,
            "hint": "Remember the feather and hammer.",
            "explanation": "It would fall to the ground -- slowly and softly, because the Moon’s gravity is gentle but
 real."
          },
          {
            "scope": "PRACTICE",
            "kind": "MCQ",
            "prompt": "What keeps the Moon traveling around the Earth instead of flying off into space?",
            "options": [
              "A giant invisible wall",
              "Earth’s gravity",
              "The Sun’s heat",
              "Wind from Earth"
            ],
            "answer": 1,
            "hint": "It works like an invisible string.",
            "explanation": "Earth’s gravity constantly pulls the Moon, bending its path into an orbit around us."
          },
          {
            "scope": "PRACTICE",
            "kind": "MCQ",
            "prompt": "Kiri stands at the South Pole and drops a snowball. Which way does it fall?",
            "options": [
              "Up into the sky",
              "Sideways",
              "Toward the ground",
              "It floats"
            ],
            "answer": 2,
            "hint": "Where does gravity always pull?",
            "explanation": "Gravity pulls toward Earth’s center, so the snowball falls to the ground -- ’down’ works
the same everywhere on Earth."
          }
        ]
      },
      {
        "slug": "why-does-the-moon-glow",
        "title": "Why Does the Moon Glow?",
        "tagline": "The biggest mirror in the night sky",
        "estMinutes": 10,
        "xpReward": 100,
        "sections": [
          {
            "kind": "HERO",
            "content": {
              "scene": "moon",
              "headline": "Borrowed Light",
              "sub": "The Moon shines bright enough to read by -- yet it makes no light of its own. So where does
moonlight really come from?"
            }
          },
          {
            "kind": "CONTEXT",
            "title": "Why does this matter?",
            "content": {
              "markdown": "For thousands of years, people told stories about the glowing Moon. Sailors steered ships
by it. Farmers harvested under it. You have probably seen it shine through your window.\n\nBut here’s the mystery: the
Moon is just a giant ball of **rock and dust**. Rocks don’t glow. Pick one up tonight -- does it shine? No. So why does
that rock in the sky blaze silver-bright?\n\nThe answer teaches us something powerful: some things in the sky **make**
light, and other things only **bounce** it. Telling those apart is one of an astronomer’s most important skills."
            }
          },
          {
            "kind": "CONCEPT",
            "title": "The big idea",
            "content": {
              "markdown": "There are two kinds of bright things:\n\n1. **Light makers**, like the Sun, fires, and
lamps. They create their own light.\n2. **Light bouncers**, like mirrors, the white pages of a book, and... the Moon.
They only reflect light that hits them.\n\n**The Moon is a light bouncer.** Moonlight is really **sunlight** that
traveled to the Moon, bounced off its dusty gray surface, and then traveled on to your eyes. Every moonbeam is
secondhand sunshine.\n\nYou can prove the idea with a flashlight and a gray ball in a dark room. The ball is invisible
in the dark -- until your flashlight hits it. Then it ’glows.’ Turn the flashlight off and the glow is gone instantly.
The Moon works exactly the same way, with the Sun as the flashlight.\n\nHere’s something surprising: the Moon is
actually a **bad** mirror. Its dusty surface is dark gray, like an old road. It bounces back only a little of the
sunlight that hits it. So why does it look dazzling? Two reasons:\n\n- The Sun is **incredibly** bright, so even a small
 bounce is a lot of light.\n- We usually see the Moon against the **black night sky**, where even a little light looks
brilliant -- the way one candle looks bright in a dark room but invisible at noon.\n\nThat also solves another puzzle:
you can sometimes see the **Moon in the daytime**! It’s up there bouncing sunlight just the same, but against a bright
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blue sky it looks pale and ghostly instead of brilliant.\n\nAnd because moonlight is bounced sunlight, only the half of
the Moon **facing the Sun** is ever lit. The other half is in darkness. Hold that thought -- it is the secret behind the
 Moon’s changing shapes, which is your very next lesson."
              }
            },
            {
              "kind": "SIMULATION",
              "title": "Try it: light the Moon",
              "content": {
                "simId": "moon-phases",
                "intro": "Drag the Moon around its orbit. Notice the bright half of the Moon always faces the Sun -- the
 Moon never makes light, it only catches it."
              }
            },
            {
              "kind": "WORKED_EXAMPLES",
              "title": "Worked examples",
              "content": {
                "examples": [
                  {
                    "title": "Flashlight test",
                    "problem": "Sam shines a flashlight on a gray ball in a dark room, then turns the flashlight off.
What happens to the ball’s glow, and what does this tell us about the Moon?",
                    "steps": [
                      "With the flashlight on, light bounces off the ball into Sam’s eyes -- the ball looks bright.",
                      "With the flashlight off, there is no light to bounce. The ball goes dark instantly.",
                      "The ball never made light; it only reflected it -- exactly like the Moon and the Sun."
                    ],
                    "answer": "The glow disappears the instant the light is off. The Moon, like the ball, only shines
while sunlight is hitting it."
                  },
                  {
                    "title": "Moon vs. stars",
                    "problem": "Stars make their own light, but the Moon only bounces light. So why does the Moon look
so much brighter than the stars?",
                    "steps": [
                      "Brightness depends on how much light reaches your eyes -- and distance matters enormously.",
                      "Stars are unimaginably far away; their light spreads out across trillions of kilometers.",
                      "The Moon is our next-door neighbor, so its bounced sunlight arrives strong."
                    ],
                    "answer": "Because the Moon is so much closer. A nearby lamp outshines a distant bonfire."
                  },
                  {
                    "title": "The daytime Moon",
                    "problem": "Ana sees the Moon at 3 in the afternoon and says, ’The Moon broke! It’s only supposed to
 come out at night.’ Is the Moon broken?",
                    "steps": [
                      "The Moon is in the sky day and night -- it orbits Earth on its own schedule, not ours.",
                      "It bounces sunlight in the daytime just as it does at night.",
                      "Against the bright blue sky, the Moon simply looks faint instead of brilliant."
                    ],
                    "answer": "Nothing is broken. The daytime Moon is normal -- it just looks pale next to the bright
sky."
                  }
                ]
              }
            },
            {
              "kind": "VIDEOS",
              "title": "Watch and wonder",
              "content": {
                "videos": [
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                    "youtubeId": "kFR7HLRMx_o",
                    "title": "Phases of the Moon | Learn Bright"
                  },
                  {
                    "youtubeId": "vwfbdPyzgDo",
                    "title": "Over (to) The Moon | Crash Course Kids"
                  }
                ]
              }
            },
            {
              "kind": "CONCEPT_CHECK",
              "title": "Quick check",
              "content": {
                "intro": "Let’s see if the idea stuck."
              }
            },
            {
              "kind": "PRACTICE",
```

```
              "title": "Practice problems",
              "content": {
                "intro": "Time to shine -- bounce these questions back correctly!"
              }
            },
            {
              "kind": "DEEPER_DIVE",
              "title": "Deeper dive: how good a mirror is the Moon?",
              "content": {
                "markdown": "Scientists measure how well something bounces light with a number called **albedo**. A perfect mirror has an albedo of 1 (it bounces everything). Fresh snow is about 0.8.\n\nThe Moon? Just **0.12** -- it reflects only about 12% of the sunlight that hits it, about the same as worn asphalt on a road. If you could hold a piece of the Moon next to a piece of charcoal, they wouldn’t look wildly different!\n\nThe full Moon only *seems* dazzling because the Sun delivers a staggering amount of light, and because our eyes compare it to the blackness of space around it. Astronomy is full of these tricks of comparison -- keep your eyes sharp."
              }
            },
            {
              "kind": "SUMMARY",
              "title": "Mission summary",
              "content": {
                "takeaways": [
                  "The Moon makes no light of its own -- it is rock and dust.",
                  "Moonlight is sunlight bouncing off the Moon’s surface to your eyes.",
                  "Only the half of the Moon facing the Sun is ever lit.",
                  "The Moon looks bright because the Sun is powerful and the night sky is dark.",
                  "The daytime Moon is normal -- it just looks pale against a bright sky."
                ],
                "formulas": []
              }
            }
          ],
          "questions": [
            {
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              "kind": "MCQ",
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                "The Moon makes it",
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                "City lights from Earth",
                "Stars shining on the Moon"
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              "answer": 1,
              "explanation": "Moonlight is reflected sunlight -- the Moon is a light bouncer, not a light maker."
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              "explanation": "The Moon is a ball of rock and dust. Only the Sun in our solar system makes its own light by burning (fusing) fuel."
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              "prompt": "You can sometimes see the Moon during the day.",
              "answer": true,
              "explanation": "The Moon is often up in the daytime -- it just looks pale against the bright sky."
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            {
              "scope": "PRACTICE",
              "kind": "MCQ",
              "prompt": "Which of these makes its own light?",
              "options": [
                "A mirror",
                "The Moon",
                "The Sun",
                "A white wall"
              ],
              "answer": 2,
              "hint": "Light makers vs. light bouncers.",
              "explanation": "The Sun is a light maker. Mirrors, walls, and the Moon only bounce light that hits them."
            },
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              "scope": "PRACTICE",
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              "prompt": "If the Sun suddenly stopped shining, what would happen to moonlight?",
              "options": [
                "Nothing -- the Moon glows on its own",
                "Moonlight would disappear",
```

```
                "The Moon would get brighter",
                "The Moon would turn blue"
              ],
              "answer": 1,
              "hint": "What is moonlight made of?",
              "explanation": "No sunlight means nothing to bounce -- the Moon would go dark, like the gray ball when the
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              "prompt": "The half of the Moon facing away from the Sun is dark.",
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              "hint": "Think about the flashlight and the ball.",
              "explanation": "Only the side facing the Sun catches light. The far side at that moment is in shadow."
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night?",
              "options": [
                "It is covered in tiny mirrors",
                "The Sun is very bright and the night sky is very dark",
                "It is hot",
                "Earth lights it up"
              ],
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              "hint": "Think of one candle in a dark room.",
              "explanation": "Even a weak bounce of the Sun’s enormous light looks brilliant against the black sky."
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              "options": [
                "The Moon is much closer to us",
                "The Moon is bigger than stars",
                "Stars are turned off at night",
                "The Moon is hotter"
              ],
              "answer": 0,
              "hint": "A nearby lamp vs. a distant bonfire.",
              "explanation": "Distance! Stars are unimaginably far away, while the Moon is our close neighbor, so its
bounced light reaches us strong."
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                "sub": "Crescent, half, full, gone -- the Moon seems to change shape every night. But the Moon never
changes at all. So what’s really going on?"
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thin smile, then a half circle, then a glowing full disc, then back to a sliver -- and then it starts over.\n\nLong
before clocks and calendars, people used these shapes to **count time**. The word *month* even comes from *Moon*!
Farmers planted by it, sailors planned voyages by it, and many holidays around the world are still set by the Moon’s
shape today.\n\nBut the Moon is a solid ball of rock. Rock doesn’t melt away and regrow each month. The changing shapes
-- called **phases** -- are one of the sky’s greatest illusions, and you’re about to see through it."
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              "kind": "CONCEPT",
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                "markdown": "Three facts you already know combine into the answer:\n\n1. The Moon makes no light -- it
only **bounces sunlight** (last lesson!).\n2. So **one half** of the Moon is always lit -- the half facing the Sun.\n3.
```

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The Moon **travels around Earth** about once a month.\n\nNow put it together. As the Moon circles us, we see its sunlit half **from different angles**. Sometimes the bright half faces us straight on. Sometimes it faces away. Usually we see part bright and part dark.\n\n**The phases, in order:**\n\n- **New Moon** -- the Moon sits between Earth and the Sun. Its lit side faces *away* from us, so we see almost nothing. The Moon seems to vanish.\n- **Waxing Crescent** -- a thin sliver of the lit side peeks at us. *Waxing* means **growing**.\n- **First Quarter** -- we see half of the lit side: a bright half circle. (It’s called a quarter because the Moon is one quarter of the way around its trip.)\n- **Waxing Gibbous** -- more than half bright, still growing. *Gibbous* means bulging.\n- **Full Moon** -- Earth is between the Sun and Moon, and the entire sunlit face beams at us. The whole disc glows.\n- **Waning Gibbous** -- now shrinking. *Waning* means **shrinking**.\n- **Last Quarter** -- half lit again, the *other* half this time.\n- **Waning Crescent** -- a final thin sliver... and then New Moon again.\n\nThe full cycle takes about **29.5 days** -- roughly a month.\n\nThe most important thing to remember: **the Moon itself never changes shape.** It is always a complete ball, always half lit by the Sun. The only thing that changes is **how much of that lit half we can see from Earth**. The phases are a matter of *viewpoint*, not of the Moon growing and shrinking."
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                      "The lit half of the Moon always faces the Sun.",
                      "With the Moon between us and the Sun, its lit half points *away* from Earth.",
                      "We are looking at the dark half -- so we see (almost) nothing."
                    ],
                    "answer": "New Moon. The Moon is there, but its bright side is facing away from us."
                  },
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                    "problem": "Tonight is a Full Moon. About how long until the next Full Moon?",
                    "steps": [
                      "The full cycle of phases takes about 29.5 days.",
                      "Each phase returns once per cycle.",
                      "So Full Moon to Full Moon is one whole cycle."
                    ],
                    "answer": "About 29 and a half days -- roughly one month."
                  },
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                    "problem": "Min watches the Moon for three nights. The bright part gets bigger each night. Is the Moon waxing or waning, and is a Full Moon coming soon or just passed?",
                    "steps": [
                      "Waxing means the lit part we see is growing; waning means shrinking.",
                      "Min sees it growing, so the Moon is waxing.",
                      "Waxing phases lead up to the Full Moon."
                    ],
                    "answer": "The Moon is waxing, and a Full Moon is on its way."
                  }
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                    "title": "Moon Phases | Crash Course Astronomy"
                  },
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                    "title": "Moon Phases for Kids | Twinkl"
                  }
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              "content": {
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              "content": {
                "markdown": "A common mix-up: many people think the dark part of the Moon is **Earth’s shadow**. It isn’t! During phases, Earth’s shadow is nowhere near the Moon. The dark part is simply the Moon’s own **night side** -- the half facing away from the Sun.\n\nEarth’s shadow does occasionally sweep across the Moon, but that is a special event called a **lunar eclipse**. It can only happen at a Full Moon, when Sun, Earth, and Moon line up almost perfectly -- and the Moon often turns an eerie copper-red as Earth’s atmosphere bends red sunlight into the shadow. Eclipses are rare; phases happen every single night. Different machinery entirely!"
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                  "The Moon orbits Earth about every 29.5 days, giving one full cycle of phases.",
                  "Order: New -> Crescent -> First Quarter -> Gibbous -> Full -> Gibbous -> Last Quarter -> Crescent -> New.",
                  "Waxing = growing toward Full. Waning = shrinking toward New.",
                  "The dark part of the Moon is its own night side -- not Earth’s shadow."
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                  }
                ]
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              "options": [
                "New Moon",
                "First Quarter",
                "Full Moon",
                "Crescent"
              ],
              "answer": 2,
              "explanation": "At Full Moon, Earth is between the Sun and Moon, so the whole lit face points at us."
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              "options": [
                "The bright part is shrinking",
                "The bright part is growing",
                "The Moon is melting",
                "The Moon is full"
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              "answer": 1,
              "explanation": "Waxing means growing -- the lit part we see gets bigger each night until Full Moon."
            },
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```
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              "options": [
                "It moves behind the Sun and hides",
                "Its sunlit side faces away from Earth",
                "Earth’s shadow covers it",
                "It turns invisible"
              ],
              "answer": 1,
              "hint": "Where is the lit half pointing?",
              "explanation": "The Moon is between us and the Sun, so its bright half faces the Sun -- and its dark half
faces us."
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              "prompt": "Put these in order, starting from New Moon: Full, Crescent, Quarter.",
              "options": [
                "Full -> Quarter -> Crescent",
                "Crescent -> Quarter -> Full",
                "Quarter -> Full -> Crescent",
                "Crescent -> Full -> Quarter"
              ],
              "answer": 1,
              "hint": "The light grows a little at a time.",
              "explanation": "After New Moon the light grows: a thin Crescent first, then a Quarter (half lit), then the
 Full Moon."
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              "prompt": "About how many days does one full cycle of Moon phases take? (Give your answer in days.)",
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              "hint": "It’s close to one month.",
              "explanation": "The cycle from New Moon to New Moon takes about 29.5 days -- the original ’month.’"
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              "answer": false,
              "hint": "Remember the deeper dive.",
              "explanation": "That dark part is the Moon’s own night side. Earth’s shadow only touches the Moon during a
 rare lunar eclipse."
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              "options": [
                "It will grow",
                "It will shrink",
                "It will stay full",
                "It will turn red"
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              "answer": 1,
              "hint": "What comes after Full -- waxing or waning?",
              "explanation": "After Full Moon the Moon wanes: the visible lit portion shrinks night by night toward the
Last Quarter."
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beside footprints that will outlive every castle on Earth."
              }
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completely different rules. The sky above them was **black at noon**. The silence was total -- clap your hands and you’d
 hear nothing. Their boot prints pressed into dust that hadn’t moved in a billion years.\n\nNASA is planning to send
astronauts back to the Moon, and someday people may live there. Anyone who does will need to answer very practical
questions: How long is a day? How hot does it get? Why must you always wear a suit? This lesson is your astronaut
briefing."
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            },
            {
              "kind": "CONCEPT",
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                "markdown": "**A ’day’ means one full spin.** Earth spins around once every 24 hours -- that’s why we
get a sunrise every morning. The Moon spins too, but **very slowly**: one full day-night cycle on the Moon takes about
**29.5 Earth days**.\n\nThat means if you stood on the Moon:\n\n- **Daytime would last about two weeks.** The Sun would
crawl across the sky so slowly you could barely see it move.\n- **Night would also last about two weeks** -- fourteen
Earth-days of darkness.\n\n**The sky is black even at noon.** Earth’s daytime sky is blue because our **air** scatters
sunlight everywhere above us (a whole lesson awaits you in the Atmosphere world!). The Moon has **no air at all**. No
air means nothing to scatter light -- so the Sun blazes in a perfectly black, star-less-looking sky. Sunlit ground,
black sky, all at once.\n\nNo air also means:\n\n- **No sound.** Sound travels by shaking air. No air, no sound --
astronauts talk by radio.\n- **No wind and no rain.** Nothing ever blows the dust around. That’s why astronaut
**footprints from 1969 are still there today**, sharp as the day they were made, and could last millions of years.\n-
**Wild temperatures.** On Earth, air acts like a blanket, spreading warmth around. With no blanket, the Moon’s daytime
ground roasts to about **120 degC** (hotter than boiling water) and its night plunges to about **-130 degC** (far colder
 than Antarctica). Space suits are really wearable air-conditioned blankets.\n\nOne last wonder: the Moon spins
**exactly** once for every trip it makes around Earth. Because of that perfect match, **the same side of the Moon always
 faces us**. People on Earth had never seen the far side until a spacecraft flew around and photographed it in 1959!"
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you went to school for one whole Moon daytime, about how many Earth days would you be there?",
                    "steps": [
                      "One full Moon day-night cycle is about 29.5 Earth days.",
                      "Daytime is about half of that cycle.",
                      "29.5 / 2 ~= 14.75."
                    ],
                    "answer": "About 14 to 15 Earth days of nonstop daylight. Pack a very big lunch."
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                  {
                    "title": "The silent hammer",
                    "problem": "An astronaut hammers a flag pole into the Moon’s surface. Her friend stands ten steps
away, watching. What does the friend hear?",
                    "steps": [
                      "Sound needs something to travel through -- usually air.",
                      "The Moon has no air between the two astronauts.",
                      "Vibrations from the hammer can’t cross the gap (though the hammering astronaut might feel buzzing
 through her own suit and bones)."
                    ],
                    "answer": "Nothing at all. The friend sees the hammer strike in total silence -- that’s why suits
have radios."
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                    "title": "Footprints forever",
                    "problem": "On Earth, a footprint on the beach disappears within hours. Why are 50-year-old
footprints on the Moon still perfect?",
                    "steps": [
                      "On Earth, wind blows sand and rain washes prints away.",
                      "Wind and rain are made of moving air and water.",
                      "The Moon has no air and no liquid water -- nothing ever disturbs the dust."
                    ],
                    "answer": "With no wind, rain, or weather of any kind, nothing erases them. They may last millions
of years."
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```

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            "title": "Over (to) The Moon | Crash Course Kids"
          },
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          "The Moon has no air: the daytime sky is black, and there is no sound, wind, or rain.",
          "With no air blanket, temperatures swing from about +120 degC to -130 degC.",
          "No weather means footprints and tracks can last for millions of years.",
          "The Moon is tidally locked -- the same side always faces Earth."
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        "About 29.5 Earth days",
        "One year"
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            "answer": false,
            "explanation": "Sound needs air to travel. With no air on the Moon, astronauts must use radios."
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            "prompt": "Why do footprints on the Moon last so long?",
            "options": [
              "The dust is sticky like glue",
              "There is no wind or rain to erase them",
              "Robots protect them",
              "The Moon is frozen solid"
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            "answer": 1,
            "hint": "What erases footprints on a beach?",
            "explanation": "No air means no wind; no water means no rain. Nothing ever disturbs the dust."
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            "prompt": "Why does the Moon get both much hotter AND much colder than Earth?",
            "options": [
              "It is closer to the Sun",
              "It has no blanket of air to hold and spread warmth",
              "It is made of metal",
              "Its rock burns easily"
            ],
            "answer": 1,
            "hint": "What does Earth’s air act like at night?",
            "explanation": "Air works like a blanket, trapping and spreading heat. Without it, the sunlit side roasts
and the night side freezes."
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that?",
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            "explanation": "29.5 / 2 ~= 14.75 -- about two weeks of continuous daylight."
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              "Her gloves are too soft",
              "Sound cannot travel without air",
              "It is too cold for sound",
              "The Moon absorbs all noise"
            ],
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            "hint": "What does sound travel through?",
            "explanation": "Sound is a vibration carried by air (or water, or solids). In the Moon’s airless space
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              }
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                    "steps": [
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```
                    "Down means toward Earth’s center for everyone.",
                    "Ben’s toast falls toward the center -- which is toward his kitchen floor.",
                    "Ruby’s toast also falls toward the center -- which is toward *her* kitchen floor."
                  ],
                  "answer": "Both pieces land on their own kitchen floors (butter-side down, probably). Nobody’s toast
 flies into space."
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                  "title": "The paper trick",
                  "problem": "Lena drops a flat sheet of paper and a crumpled ball of the same kind of paper. The ball
 lands first. Did crumpling make the paper heavier?",
                  "steps": [
                    "Crumpling changes the shape, not the amount of paper -- the weight is identical.",
                    "The flat sheet has a big surface, so air pushes up on it like a parachute.",
                    "The crumpled ball slips through the air with little resistance."
                  ],
                  "answer": "No -- same weight! Air resistance slowed the flat sheet. Shape, not weight, made the
difference."
                },
                {
                  "title": "Why higher falls hurt more",
                  "problem": "A ball dropped from your hand hits gently. The same ball dropped from a balcony hits
hard. The ball didn’t get heavier -- what changed?",
                  "steps": [
                    "Falling things speed up the longer they fall.",
                    "From the balcony, the ball falls for more time.",
                    "More time falling means a much faster landing speed."
                  ],
                  "answer": "Its speed. A longer fall gives gravity more time to speed the ball up, so it lands much
faster and hits harder."
                }
              ]
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about **9.8 meters per second of speed, every second** (when air isn’t interfering). We call this number **g**.\n\nAfter
 1 second of falling: about 10 m/s. After 2 seconds: about 20 m/s. After 3 seconds: about 30 m/s -- already highway
speed!\n\nOn the Moon, g is only about **1.6** -- which is exactly why everything there fell in dreamy slow motion. When
 you’re older you’ll write this as the famous formula **v = g*t** (speed = gravity x time). You’ve already discovered
what it means."
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                "’Down’ means toward the center of the Earth -- everywhere on the planet.",
```

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                "There is no bottom of the Earth; nobody can fall off.",
                "Falling objects speed up the longer they fall.",
                "Heavy and light objects fall together -- air resistance, not weight, makes feathers drift.",
                "Earth's gravity (g ~= 9.8) is about six times stronger than the Moon's."
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                }
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            }
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              "Toward the center of the Earth",
              "Toward the floor of your house only",
              "Away from the Sun"
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            "answer": 1,
            "explanation": "Down points toward Earth's center -- which is why it works the same in Canada, Australia,
and everywhere else."
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            "prompt": "People in Australia are upside-down and could fall off the Earth.",
            "answer": false,
            "explanation": "Gravity pulls everyone toward Earth's center, so everyone feels right-side-up and firmly
grounded."
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            "prompt": "A falling ball moves faster after falling for 3 seconds than after 1 second.",
            "answer": true,
            "explanation": "Falling objects keep speeding up -- gravity adds more speed every second."
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happens?",
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              "The small rock lands first",
              "They land at the same time",
              "Neither falls"
            ],
            "answer": 2,
            "hint": "Remember the hammer and feather on the Moon.",
            "explanation": "Without air resistance mattering, heavy and light objects fall together -- the extra pull
on the heavy rock is canceled by its extra sluggishness."
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              "The flat sheet catches more air",
              "The ball is rounder so gravity likes it",
              "Crumpling removes gravity"
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            "hint": "Think parachute.",
            "explanation": "Air pushes up against the flat sheet's large surface like a parachute. The crumpled ball
meets little air resistance. The weight never changed."
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          "explanation": "Dropped objects everywhere fall toward Earth’s center -- which is always toward the ground
 beneath your feet."
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            "A longer fall gives you more landing speed",
            "The air is thinner up high",
            "Gravity is stronger at the top"
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          "hint": "What happens to speed during a fall?",
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landing."
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forces: the pushes and pulls behind every movement on Earth."
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wagon is pulled. A ball is kicked (a push from a foot), caught (a pull-to-a-stop by hands), and thrown again.\n\nNothing
 -- not a ball, not a bike, not a planet -- starts moving, stops moving, or changes direction **by itself**. Something
must push or pull it. Scientists call every push and every pull a **force**, and forces are the steering wheel of the
entire universe. Even gravity, your old friend from the Moon, is just one famous example of a pull."
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*do*. A force can:\n\n- **Start** something moving (kick a resting ball),\n- **Stop** something moving (catch the
ball),\n- **Speed it up or slow it down** (pedal harder; squeeze the brakes),\n- **Change its direction** (bat hits ball
 -- the ball was flying one way and leaves another),\n- **Squash or stretch it** (squeeze clay, stretch a rubber
band).\n\n**Bigger force, bigger change.** Tap a toy car and it rolls a little. Shove it and it races. The harder the
push, the faster things speed up.\n\n**Heavier things are harder to change.** Push an empty shopping cart: easy. Push a
full one with the *same strength*: it barely gets going. More stuff (more **mass**) means more force is needed for the
same change. This is also why a rolling bowling ball is harder to stop than a rolling tennis ball.\n\n**The hidden
force: friction.** Roll a ball across grass and it slows down and stops -- but you never see anything touch it.
Something did: **friction**, a force made wherever two surfaces rub. Friction always pushes *against* sliding. Rough
surfaces (grass, carpet, sandpaper) make lots of friction; smooth surfaces (ice, polished floors) make little. That’s
why the same hockey puck glides forever on ice but dies quickly on pavement.\n\nFriction isn’t the enemy, though.
**Without friction you couldn’t walk** -- your shoes would slip like on perfect ice. Brakes, knots, and even holding a
pencil all depend on it.\n\nSo whenever anything changes how it’s moving, ask the physicist’s question: **what pushed or
 pulled it?** There is always an answer."
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                    "steps": [
                      "Pulling the sled uphill -- a pull from Ava (while gravity pulls the sled downhill).",
                      "Pushing off -- a push from Ava’s hands/feet to start moving.",
                      "Dragging boots -- friction between boots and snow, a force that slows the sled to a stop."
                    ],
                    "answer": "Three star players: Ava’s pull (up), Ava’s push (start), and friction’s push-back (stop) -- with gravity pulling downhill the whole time."
                  },
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                    "title": "The two carts",
                    "problem": "Jo pushes an empty cart and a full cart with exactly the same strength. Which speeds up more, and why?",
                    "steps": [
                      "Both carts get the same push (same force).",
                      "The full cart has much more mass.",
                      "More mass means the same force produces a smaller change in motion."
                    ],
                    "answer": "The empty cart speeds up more. Same push + less mass = bigger change."
                  },
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                    "problem": "The same hockey puck is slid with the same push across ice and across carpet. On which surface does it travel farther, and what force makes the difference?",
                    "steps": [
                      "After the push ends, only friction is acting to slow the puck.",
                      "Ice is smooth: very little friction, so the puck keeps most of its speed.",
                      "Carpet is rough: lots of friction, which quickly steals the puck’s motion."
                    ],
                    "answer": "Farther on ice. Friction -- strong on carpet, weak on ice -- is the force that decides."
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                    "title": "What Are Push and Pull Forces?"
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                  "Bigger forces make bigger changes in motion.",
                  "Heavier (more massive) objects need more force for the same change.",
                  "Friction is a hidden force between rubbing surfaces that resists sliding -- and makes walking possible.",
                  "Forces come in pairs: push something and it pushes back on you."
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                  }
                ]
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                "A kind of energy drink",
                "A push or a pull",
                "Something only machines have",
                "A type of magnet"
              ],
              "answer": 1,
              "explanation": "Every force is a push or a pull -- from a kick to gravity itself."
            },
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              "prompt": "A ball sitting still in the grass will eventually start rolling all by itself.",
              "answer": false,
              "explanation": "Nothing starts moving without a force. The ball waits for a push or pull -- forever, if needed."
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              "options": [
                "starts motion",
                "stops motion",
                "has no effect",
                "creates gravity"
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              "explanation": "Your hands apply a force that stops the ball’s motion. Stopping is just as much a force’s job as starting."
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              "options": [
                "Nothing different",
                "Makes the truck speed up more",
                "Makes the truck heavier",
                "Reverses gravity"
              ],
              "answer": 1,
              "hint": "Bigger force, bigger...?",
              "explanation": "A bigger force creates a bigger change in motion -- the truck speeds up more."
            },
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                "The gym floor pushes the ball forward",
                "Balls dislike carpet",
                "Gravity is weaker in the gym"
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              "hint": "Which surface is rougher?",
              "explanation": "Rough carpet rubs hard against the ball -- that friction force quickly slows it. The
smooth floor rubs much less."
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              "explanation": "Without friction your shoes couldn’t grip the ground, brakes couldn’t stop bikes, and
knots wouldn’t hold. Friction is essential."
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moving?",
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                "The empty wagon",
                "The full wagon",
                "They’re the same",
                "Neither will move"
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              "explanation": "More mass resists changes in motion more. The brick-filled wagon needs a bigger force for
the same get-up-and-go."
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                "The water pushing back on her",
                "The pool walls",
                "Wind"
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              "hint": "Forces come in pairs.",
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Same trick as a rocket!"
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empty air. No strings. No touching. Real invisible force."
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 hold cabinet doors closed, and spin inside every electric motor -- fans, blenders, electric toothbrushes, electric
cars.\n\nAnd magnets do something most forces can’t: they **work without touching**. Every push or pull from the last
lesson needed contact -- foot on ball, hand on wagon. Magnets reach across empty space, just like gravity does.
(Spoiler: planet Earth itself is a gigantic magnet, and that fact has guided sailors home for a thousand years.)"
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(steel is mostly iron). Bring a magnet near a paperclip and the clip leaps to it.\n\nBut magnets are picky! They ignore
plastic, wood, glass, paper, rubber -- and even most metals. A soda can (aluminum), a gold ring, a copper coin: no pull
at all. **Magnetic != metal.** Only iron-family metals join the club. The test is simple: if a magnet grabs it, there’s
iron or steel inside.\n\n**Every magnet has two ends, called poles:** a **north pole (N)** and a **south pole (S)**. You
 can’t have one without the other -- snap a magnet in half and each piece instantly has its own N and S!\n\nThe poles
follow one golden rule:\n\n- **Opposite poles attract.** N pulls on S. Click!\n- **Same poles repel.** N pushes away N;
S pushes away S. Try to force them together and you’ll feel an invisible springy cushion fighting your fingers.\n\nThat
repelling push is special -- it’s one of the only ways to feel an **invisible push** with your bare hands.\n\n**Magnetic
 force passes through things.** A magnet can hold paper to the fridge because its pull reaches *through* the paper. It
works through cardboard, plastic, glass, even your hand. But the force fades fast with distance -- a magnet that snaps
up a paperclip from one centimeter away may be helpless from ten.\n\n**The biggest magnet you know is under your feet.**
 Deep inside Earth, churning liquid iron turns our whole planet into a giant (gentle) magnet, with magnetic poles near
the North and South Poles. A **compass** is just a tiny magnet balanced on a pin -- Earth’s magnetism pulls its N end to
 point north. With that one trick, explorers crossed oceans long before satellites existed."
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opposites snap together, sames shove apart."
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nail. Which stick?",
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                      "Magnets attract iron and steel only (steel contains iron).",
                      "Steel spoon: yes. Iron nail: yes.",
                      "Plastic isn’t metal at all; aluminum is metal but not the iron family."
                    ],
                    "answer": "The steel spoon and the iron nail stick. The ruler and the soda can are ignored -- being
metal isn’t enough; it must be the iron family."
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                    "title": "The stubborn magnets",
                    "problem": "Omar pushes the N end of one magnet toward the N end of another. They refuse to touch,
sliding apart like ghosts. What’s happening?",
                    "steps": [
                      "Same poles repel -- N pushes N away.",
                      "The closer Omar forces them, the stronger the invisible push grows.",
                      "The magnets slide aside to escape, like two wrong ends of a spring."
                    ],
                    "answer": "Same-pole repulsion. Flip one magnet around (N to S) and they’ll snap together
instantly."
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it work through paper but fail when she adds a thick book in between?",
                    "steps": [
                      "Magnetic force passes through non-magnetic stuff like paper.",
                      "But the force weakens quickly with distance.",
                      "A thick book pushes the magnet too far from the steel door."
                    ],
                    "answer": "Paper is thin, so the magnet is still close to the steel and holds on. The book adds
distance, the force fades, and the drawing falls."
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                "title": "Magnetism | The Dr. Binocs Show"
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hatch on a beach, swim thousands of kilometers across the ocean for years -- and then find their way back to *the very 
same beach* to nest. Migrating birds fly continent to continent through cloudy nights with no landmarks. Salmon, 
lobsters, and even some bacteria navigate the same way: scientists have shown they sense Earth’s magnetic field, 
carrying a kind of built-in compass.\n\nExactly *how* their bodies detect the field is still being figured out -- tiny 
magnetic crystals in their cells and special light-sensing proteins in their eyes are both suspects. It is a genuine 
open mystery of science. Maybe you’ll be the one to solve it."
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              "Magnets pull on iron and steel -- not on plastic, wood, or most other metals.",
              "Every magnet has a north pole and a south pole; you can never have just one.",
              "Opposite poles attract; same poles repel.",
              "Magnetic force works without touching and passes through thin materials, but fades with distance.",
              "Earth is a giant magnet -- that’s why a compass needle points north."
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\\leftrightarrow S\\ \\text{repel}"
              }
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            "A steel paperclip",
            "A rubber band",
            "A wooden bead"
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          "answer": 1,
          "explanation": "Magnets attract iron and steel. Plastic, rubber, and wood are completely ignored."
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          "prompt": "What happens when two NORTH poles are pushed toward each other?",
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            "They snap together",
            "They push each other away",
            "They melt",
            "Nothing"
          ],
```

```
            "answer": 1,
            "explanation": "Same poles repel -- you can feel the invisible push fighting your fingers."
          },
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            "prompt": "A magnet must touch a paperclip to pull on it.",
            "answer": false,
            "explanation": "Magnetic force reaches across space -- the clip jumps to the magnet before they touch."
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            "scope": "PRACTICE",
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            "prompt": "An aluminum soda can is metal, but a magnet won’t stick to it. Why?",
            "options": [
              "The can is too smooth",
              "Magnets only attract iron-family metals",
              "The can is too cold",
              "Soda blocks magnetism"
            ],
            "answer": 1,
            "hint": "Magnetic != metal.",
            "explanation": "Only iron and steel (and a few cousins like nickel) respond to magnets. Aluminum, copper, and gold do not."
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            "kind": "MCQ",
            "prompt": "The N pole of magnet A snaps onto one end of magnet B. Which pole did it grab?",
            "options": [
              "B’s north pole",
              "B’s south pole",
              "Either one",
              "Magnets have no poles"
            ],
            "answer": 1,
            "hint": "Opposites...",
            "explanation": "Opposites attract: a north pole clicks onto a south pole."
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            "prompt": "If you cut a magnet in half, one piece will be all-north and the other all-south.",
            "answer": false,
            "hint": "Poles always come in pairs.",
            "explanation": "Each half instantly becomes a complete magnet with its own N and S. Poles can never be separated."
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            "options": [
              "the needle is heavier on one end",
              "Earth itself is a giant magnet",
              "the North Pole is shiny",
              "wind pushes it"
            ],
            "answer": 1,
            "hint": "What’s churning deep inside Earth?",
            "explanation": "Liquid iron moving in Earth’s core makes the whole planet magnetic, and the tiny needle-magnet lines up with it."
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            "prompt": "A magnet holds a note to the fridge through a sheet of paper. What would make it drop the note?",
            "options": [
              "Using two sheets of gift-wrap",
              "Adding a thick book between magnet and fridge",
              "Turning the magnet sideways",
              "Drawing on the note"
            ],
            "answer": 1,
            "hint": "Magnetic force fades with...?",
            "explanation": "Magnetic force weakens quickly with distance. A thick book pushes the magnet too far from the steel door to hold on."
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            "content": {
              "markdown": "Right now, while you read this, it is the middle of the night for kids on the other side of the world. When you eat breakfast, they’re sound asleep. When you go to bed, their school day is starting.\n\nFor most of human history, people believed the obvious: that the Sun circles around the Earth, rising in the east and setting in the west. It *looks* exactly like that! It took bold thinkers like **Copernicus** and **Galileo** to prove the opposite -- and the real answer explains time zones, sunrises, why your shadow moves, and why ’sunset’ is a beautiful illusion."
            }
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              "markdown": "**The Earth spins.** Our planet rotates around an invisible line through the North and South Poles -- its **axis** -- like a giant carousel, making one complete turn every **24 hours**. You, your house, and your whole country are riders on that carousel, moving right now at hundreds of kilometers per hour. (You don’t feel it because the spin is perfectly smooth -- just as you don’t feel speed in a smooth airplane.)\n\n**The Sun, meanwhile, shines steadily** from one direction. And here’s the key picture: at any moment, the Sun can only light up **half the ball**. Shine a flashlight on a globe and you’ll see it instantly -- one half bright, one half dark. Always.\n\n- The half facing the Sun is having **daytime**.\n- The half facing away is having **night**.\n\nAs Earth turns, you ride from the dark half into the bright half. From where you stand, it looks like the Sun is climbing up over the horizon -- we call it **sunrise**, but really it’s *you-rise*: your part of Earth is turning to face the Sun. Twelve hours later you ride back into the shadow, and the Sun seems to sink -- **sunset**, or really *you-set*.\n\nBecause Earth spins toward the **east**, the Sun always appears to rise in the **east** and set in the **west**, everywhere on the planet.\n\nThis also explains **shadows**. In the morning, with the Sun low in the east, your shadow stretches long toward the west. At midday, the Sun is high and your shadow shrinks to a puddle at your feet. By evening it stretches long again -- toward the east this time. Your shadow is a sundial, silently tracking the Earth’s turn.\n\nAnd remember the Moon world: the Moon makes this same kind of turn, but takes almost a month. Earth’s brisk 24-hour spin is why our days feel just right -- and as you’ll learn someday, it’s no accident life likes it that way."
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                  "problem": "Pia shines a flashlight at a globe in a dark room. How much of the globe is lit, and what does the lit part represent?",
                  "steps": [
                    "Light travels in straight lines, so it can only strike the side facing it.",
                    "Exactly half the ball is bright; the other half is in shadow.",
                    "The bright half is ’daytime’; the shadowed half is ’night.’"
                  ],
                  "answer": "Half the globe is lit. That bright half is everywhere on Earth currently having daytime; the dark half is having night."
                },
                {
                  "title": "Breakfast and bedtime",
                  "problem": "It’s 8 a.m. in Vancouver. Kenji in Japan, on roughly the opposite side of the Earth, is doing what?",
                  "steps": [
                    "Opposite sides of the spinning Earth face opposite ways.",
                    "If Vancouver faces the Sun (morning), Japan faces away (night).",
                    "Their clock reads roughly 12 hours different."
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                  ],
                  "answer": "It’s around midnight for Kenji -- he’s asleep. One planet, two opposite times of day."
                },
                {
                  "title": "The shrinking shadow",
                  "problem": "At 9 a.m. Theo’s shadow is longer than he is. By noon it has shrunk tiny. What
happened?",
                  "steps": [
                    "Shadow length depends on how high the Sun appears in the sky.",
                    "As Earth turns through the morning, Theo’s spot rotates to face the Sun more directly -- the Sun
appears higher.",
                    "A higher Sun shines more from above, casting a shorter shadow."
                  ],
                  "answer": "Earth’s rotation ’raised’ the Sun in Theo’s sky, so his shadow shrank. By evening it will
 stretch long again, pointing the other way."
                }
              ]
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                  "title": "Earth’s Rotation & Revolution | Crash Course Kids"
                },
                {
                  "youtubeId": "yo53v51bIi4",
                  "title": "What Causes Day and Night?"
                }
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            "content": {
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            }
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            "content": {
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            "title": "Deeper dive: proving the spin",
            "content": {
              "markdown": "How do we *know* Earth spins, rather than the Sun circling us? For centuries no one could
prove it directly. Then in 1851, French physicist **L?on Foucault** hung a heavy ball on a 67-meter wire from the
ceiling of the Panth?on in Paris and set it swinging.\n\nA swinging pendulum keeps its swing-direction fixed in space.
Yet hour by hour, the crowd watched the pendulum’s path slowly turn relative to the floor... because the **floor -- the
Earth -- was turning underneath it.** People could literally watch the planet rotate.\n\nFoucault pendulums still swing
in science museums all over the world today. Find one, watch it for an hour, and you’ll see the Earth turn with your own
 eyes."
            }
          },
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                "Earth rotates on its axis once every 24 hours.",
                "The Sun lights only half the Earth at a time: the lit half has day, the dark half has night.",
                "Sunrise and sunset are caused by Earth’s turning -- the Sun isn’t moving around us.",
                "Earth spins toward the east, so the Sun appears to rise in the east and set in the west.",
                "Shadows are long in morning and evening and shortest near midday, tracking the spin."
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                {
                  "label": "One day",
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                }
              ]
            }
          }
        ],
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      "Earth spinning on its axis",
      "Clouds covering the Sun",
      "The Moon blocking sunlight"
    ],
    "answer": 1,
    "explanation": "Earth’s rotation carries each place into sunlight (day) and back into shadow (night) every 24 hours."
  },
  {
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    "prompt": "When it is daytime where you live, it is daytime everywhere on Earth.",
    "answer": false,
    "explanation": "Only the half of Earth facing the Sun has daytime. The other half is in night at that very moment."
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    "prompt": "How many hours does Earth take to spin around once?",
    "answer": {
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      "tolerance": 0.5
    },
    "explanation": "One full rotation takes 24 hours -- that’s exactly what we call a day."
  },
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    "kind": "MCQ",
    "prompt": "At sunset, what is really happening?",
    "options": [
      "The Sun flies below the ocean",
      "Your part of Earth is rotating away from the Sun",
      "The Sun shrinks for the night",
      "Earth stops spinning"
    ],
    "answer": 1,
    "hint": "Who is actually moving?",
    "explanation": "The Sun stays put. Your spot on the carousel is turning away from it, so the Sun appears to sink behind the horizon."
  },
  {
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    "kind": "MCQ",
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    "options": [
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      "In the west",
      "In the east",
      "Straight overhead"
    ],
    "answer": 2,
    "hint": "You turn toward the thing that seems to ’come up.’",
    "explanation": "Spinning eastward means the eastern horizon turns to face the Sun first -- sunrise is always in the east."
  },
  {
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    "answer": true,
    "hint": "Where is the Sun at midday?",
    "explanation": "Near midday the Sun is highest, shining down from above, so your shadow shrinks to its smallest."
  },
  {
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    "kind": "MCQ",
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    "options": [
      "Noon",
      "6 in the morning",
      "Midnight",
      "Sunset"
    ],

"answer": 2,
"hint": "Opposite side = facing the opposite way.",
"explanation": "The opposite side of the planet faces away from the Sun -- it’s around midnight there."
},
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"scope": "PRACTICE",
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"prompt": "Why don’t you feel the Earth spinning, even though it carries you at hundreds of km/h?",
"options": [
"Earth spins too slowly to matter",
"The motion is perfectly smooth and steady",
"Gravity cancels all motion",
"We actually do feel dizzy all day"
],
"answer": 1,
"hint": "Do you feel speed in a smooth, steady airplane?",
"explanation": "We only feel changes in motion -- bumps, speeding up, slowing down. Earth’s spin is utterly smooth, so our bodies notice nothing."
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"sub": "Above your head right now: a column of air 100 kilometers tall, pressing down with the weight of a small car per square meter. Why aren’t you crushed?"
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}
},
{
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"markdown": "**Air is matter.** It’s made of molecules -- far too small to see, but real. A bedroom’s worth of air weighs about as much as a house cat. An empty school gym holds roughly a few hundred kilograms of air. ’Empty’ is never really empty.\n\n**Air pressure is the weight of all the air above you.** Stack books on your hand: each added book presses harder. Now realize the atmosphere is a stack about **100 kilometers tall**, and you’re at the bottom. All that air pushes on you with a pressure scientists measure as about **101 kilopascals (kPa)** at sea level -- roughly **1 kilogram pressing on every square centimeter** of your body, or ten tonnes on a desk!\n\n**So why aren’t you crushed?** Two reasons. First, air pushes in **all directions** -- down, up, and sideways -- because its molecules fly every which way. The pushes on each side of your hand balance out. Second, the air *inside* you (lungs, ears, blood) pushes **outward** with the same pressure. Inside push equals outside push: you feel nothing. You only notice pressure when it gets **unbalanced** -- like when an elevator ride leaves old-pressure air trapped in your ears. The ’pop’ is your inner ear equalizing. Equilibrium restored!\n\n**Pressure drops as you climb.** Go up a mountain and there’s less air left above you, so the stack weighs less. The rule of thumb: pressure roughly **halves every 5.5 km** you climb.\n\n- Sea level: ~100 kPa\n- 5.5 km (higher than most mountains): ~50 kPa\n- 11 km (airliner cruising height): ~25

```
 kPa\n- 100 km: nearly zero -- the official edge of space (the **K?rm?n line**).\n\nThat’s why mountaineers on Everest
carry oxygen tanks, why airplane cabins are pressurized, and why a sealed chip bag from the seaside puffs up like a
balloon in the mountains: the air **inside** the bag keeps its push, while the air outside pushes back less. Unbalanced
pressure always wins arguments."
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thin out, and the sealed balloon swell as the outside push weakens."
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                    "problem": "Air pressure is about 100 kPa at sea level and roughly halves every 5.5 km. Estimate the
 pressure outside an airliner at 11 km.",
                    "steps": [
                      "11 km is two ’halvings’: 5.5 km + 5.5 km.",
                      "First halving: 100 kPa -> 50 kPa at 5.5 km.",
                      "Second halving: 50 kPa -> 25 kPa at 11 km."
                    ],
                    "answer": "About 25 kPa -- a quarter of sea-level pressure. This is why cabins must be pressurized
for passengers."
                  },
                  {
                    "title": "The crushed bottle",
                    "problem": "Aisha seals an empty plastic bottle on a mountaintop, then drives down to the beach. At
the bottom, the bottle is crumpled and dented. Who crushed it?",
                    "steps": [
                      "Sealed at altitude, the bottle traps thin, low-pressure mountain air inside.",
                      "At sea level, the outside air pushes with full ~100 kPa.",
                      "Inside push (weak) < outside push (strong) -- the imbalance squeezes the bottle."
                    ],
                    "answer": "The atmosphere did. The stronger sea-level air outside out-pushed the weak mountain air
trapped inside and crumpled the bottle."
                  },
                  {
                    "title": "How a straw really works",
                    "problem": "People say you ’suck’ juice up a straw. What actually pushes the juice into your
mouth?",
                    "steps": [
                      "Sucking removes air from inside the straw, lowering the pressure there.",
                      "The atmosphere still presses with full force down on the juice in the cup.",
                      "That outside push shoves juice up the straw toward the low-pressure zone."
                    ],
                    "answer": "Atmospheric pressure pushes the juice up. A straw is really a pressure-imbalance machine
-- in true vacuum-of-space, straws wouldn’t work at all."
                  }
                ]
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                    "title": "Weather School 4 Kids: Air Pressure"
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              "content": {
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history. He fitted two copper half-spheres together, pumped the air **out** of the inside... and invited two teams of
horses to pull them apart.\n\nSixteen horses heaved. The spheres held.\n\nNothing was gluing them -- nothing was *inside
 at all*. The atmosphere outside was simply pressing the halves together with tonnes of force, with no inside air to
push back. When von Guericke opened a valve and let air hiss back in, the spheres fell apart in his hands.\n\nThe
’Magdeburg hemispheres’ proved the invisible ocean is mighty. Every vacuum cleaner, suction cup, and drinking straw is a
 tiny tribute to those sixteen confused horses."
              }
            },
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                  "Air is matter -- it has mass and weight.",
                  "Air pressure is the weight of the air above pressing on everything, in all directions: ~101 kPa at
sea level.",
                  "You aren’t crushed because pressure pushes equally from all sides, and from inside you too.",
                  "Pressure drops with altitude -- roughly halving every 5.5 km -- fading to ~zero at the 100 km K?rm?n
line.",
                  "Unbalanced pressure does work: it pops ears, crushes bottles, and pushes juice up straws."
                ],
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                    "label": "Halving rule",
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                  }
                ]
              }
            }
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              "answer": true,
              "explanation": "Air is made of molecules -- real matter. The whole atmosphere weighs about 5.5 quadrillion
 tonnes."
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              "prompt": "What is air pressure?",
              "options": [
                "The temperature of air",
                "The push from the weight of air above and around us",
                "Wind speed",
                "How clean the air is"
              ],
              "answer": 1,
              "explanation": "Air pressure is the push of air’s weight -- about 1 kg per square centimeter at sea level,
 in every direction."
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              "prompt": "As you climb a tall mountain, air pressure...",
              "options": [
                "increases",
                "stays exactly the same",
                "decreases",
                "reverses direction"
              ],
              "answer": 2,
              "explanation": "Higher up, there’s less air stacked above you, so the pressure drops -- roughly halving
every 5.5 km."
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            {
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```

```
              "prompt": "Why doesn’t the atmosphere’s huge pressure crush your body?",
              "options": [
                "Skin is pressure-proof armor",
                "Air pushes equally from all sides, and your insides push back out",
                "The pressure only pushes downward",
                "Gravity cancels it"
              ],
              "answer": 1,
              "hint": "Think balance: inside vs. outside.",
              "explanation": "Pressure acts in all directions, and the air and fluids inside you push outward just as
hard -- balanced pushes feel like nothing."
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(in kPa) at 16.5 km -- three halvings up.",
              "answer": {
                "value": 12.5,
                "tolerance": 2
              },
              "hint": "Halve 100 three times.",
              "explanation": "100 -> 50 -> 25 -> 12.5 kPa. At 16.5 km, only about an eighth of the atmosphere remains
above you."
            },
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              "options": [
                "the chips release gas when excited",
                "the inside air keeps its push while the outside push weakens",
                "mountain cold expands plastic",
                "gravity is weaker up high"
              ],
              "answer": 1,
              "hint": "Which side of the bag changed?",
              "explanation": "The trapped sea-level air inside still pushes hard; the thin mountain air outside pushes
back less. The imbalance inflates the bag."
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              "options": [
                "Your tongue’s lifting power",
                "The juice’s desire to be drunk",
                "Atmospheric pressure pushing on the juice in the cup",
                "Static electricity"
              ],
              "answer": 2,
              "hint": "You only removed air from the straw...",
              "explanation": "Lowering the pressure inside the straw lets the full atmospheric push on the cup’s surface
 shove juice up the tube."
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              "prompt": "Your ears pop in a fast elevator because the air pressure outside your eardrum changed before
the air inside caught up.",
              "answer": true,
              "hint": "Pop = equalize.",
              "explanation": "Exactly -- the pop is trapped inner-ear air equalizing with the new outside pressure,
rebalancing the pushes on your eardrum."
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                "scene": "atmosphere",
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                "sub": "Wind has knocked over trees, sailed ships around the planet, and is probably ruffling a flag
near you right now. But what *is* wind -- and who keeps pushing it?"
              }
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            },
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                "markdown": "Weather decides whether your soccer game happens, what you wear, and what farmers can grow.
 Severe weather -- storms, hurricanes, heat waves -- shapes lives around the world. Forecasting it accurately saves
thousands of lives every year.\n\nAnd remarkably, almost all of it -- every breeze, cloud, and thunderstorm -- is
powered by a single engine: **the Sun heating the Earth unevenly**. Master that one idea, and the daily weather report
transforms from mysterious jargon ('a low-pressure system is moving in...') into a story you can actually follow."
              }
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                "markdown": "Three gears make the weather engine turn:\n\n**Gear 1: The Sun heats Earth unevenly.**
Sunlight warms dark pavement faster than grass, land faster than the sea, and the equator far more than the poles. The
air sitting above warm ground gets warmed too -- Earth's surface acts like a stove burner under the ocean of
air.\n\n**Gear 2: Warm air rises, cool air sinks.** When air warms, its molecules zip around faster and spread out --
the air becomes **less dense** (lighter for its size), so it floats upward through the cooler air around it, exactly
like a bubble rising in water or a hot-air balloon lifting off. Cool, denser air sinks to take its place. This endless
vertical loop is called **convection**.\n\n**Gear 3: Wind is air rushing from high pressure to low pressure.** Where air
 rises, less air is left pressing on the ground -- a **low-pressure** zone. Where air sinks and piles up, pressure runs
**high**. And air, like a crowd leaving a stadium, always flows from the packed place to the empty place: **from high
pressure toward low pressure. That horizontal flow is wind.** Bigger pressure difference = faster wind. A hurricane is
simply this engine running at terrifying full throttle around an extreme low.\n\n**See all three gears at the beach.**
On a sunny afternoon, land heats up faster than the sea. Air over the land warms and rises (low pressure on land);
cooler, heavier sea air slides in to replace it. That's the famous afternoon **sea breeze**, blowing from water to land.
 At night the engine reverses: land cools quickly, the sea stays warm, and a gentle **land breeze** drifts out to sea.
One beach, a complete weather system in miniature.\n\n**Bonus gear: rising air builds clouds.** Rising warm air carries
invisible water vapor upward. High up, it cools, and the vapor condenses into countless droplets -- a **cloud**. Pile up
 enough rising, moist air and you get rain. So next time you see a towering summer cloud, you're watching Gear 2 in
action, kilometers tall."
              }
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                "simId": "sea-breeze",
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over land, cool sea air sliding in underneath. Then switch to night and watch the whole engine run backwards."
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land, or land to sea?",
                    "steps": [
                      "Sun heats land faster than water; air over land warms and rises.",
                      "Rising air leaves lower pressure over the land.",
                      "Cooler, denser sea air flows in toward that low pressure."
                    ],
                    "answer": "From sea to land -- the classic afternoon sea breeze. Flags on the beach flap inland."
                  },
                  {
                    "title": "The radiator loop",
                    "problem": "A heater sits under a window on one side of a room, yet the whole room warms up. Trace
the air's journey.",
                    "steps": [
                      "Air touching the heater warms, becomes less dense, and rises to the ceiling.",
                      "It drifts across the ceiling, cooling as it goes, then sinks on the far side.",
                      "Cool floor-level air flows back toward the heater to be warmed in turn -- a convection loop."
                    ],
                    "answer": "A circulating loop: up at the heater, across the ceiling, down the far wall, back along
the floor. The room is a tiny atmosphere."
                  },
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                    "title": "Reading the forecast",
                    "problem": "The forecaster says: 'A strong low-pressure system arrives tomorrow.' Should you expect
calm sunshine or wind and possible rain -- and why?",
                    "steps": [
                      "A low means air is rising there, and surrounding air rushes inward -- that's wind.",
                      "Rising air cools, and its moisture condenses into clouds.",
                      "Lots of rising, moist air often means rain."
                    ],
```

```
                    "answer": "Wind and likely rain. Lows = rising air = wind, clouds, and wet weather. Highs, with their gently sinking air, usually bring calm, clear skies."
                  }
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                  },
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                    "title": "Air Pressure Lesson for Kids"
                  }
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              "content": {
                "markdown": "Scale the beach breeze up to the whole planet. The equator is Earth’s permanent ’hot land,’ the poles its permanent ’cool sea.’ Giant convection loops form, with warm equatorial air rising and flowing poleward high above, while cooler air returns along the surface.\n\nBut there’s a twist -- literally. Because the **Earth is spinning** (remember the great carousel?), these planet-sized winds get deflected sideways as they travel, an effect called the **Coriolis effect**. The result: steady wind highways like the **trade winds**, which sailors rode west across the Atlantic for centuries, and the **jet streams** -- rivers of air 10 km up, screaming along at 200+ km/h, which is why your flight east is faster than your flight west.\n\nEvery local breeze belongs to this single, planet-wide, Sun-powered, spin-twisted machine. Meteorologists model it with supercomputers; you now understand its gears."
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                  "The Sun heats Earth’s surface unevenly -- that’s the energy source for all weather.",
                  "Warm air is less dense and rises; cool air is denser and sinks (convection).",
                  "Rising air creates low pressure; sinking, piling air creates high pressure.",
                  "Wind is air flowing from high pressure to low pressure -- bigger difference, stronger wind.",
                  "Sea breeze by day, land breeze by night; rising moist air builds clouds and rain."
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                    "tex": "\\text{wind: } P_{high} \\longrightarrow P_{low}"
                  }
                ]
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              "prompt": "What is wind?",
              "options": [
                "Air made of special wind molecules",
                "Air moving from high pressure toward low pressure",
                "The Earth exhaling",
                "Sound waves you can feel"
              ],
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            "answer": 1,
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            "explanation": "It’s the other way around: warm air is less dense and rises; cool, denser air sinks."
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              "The Moon",
              "Earth’s hot core",
              "The Sun",
              "Ocean salt"
            ],
            "answer": 2,
            "explanation": "The Sun’s uneven heating of Earth’s surface drives convection, pressure differences, and
therefore all wind and weather."
          },
          {
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            "kind": "MCQ",
            "prompt": "On a sunny afternoon, why does the sea breeze blow from the water toward the land?",
            "options": [
              "Fish push the air",
              "Waves splash air ashore",
              "Warm air rises over the hot land and cool sea air flows in to replace it",
              "The sea is higher than the land"
            ],
            "answer": 2,
            "hint": "Which heats faster, sand or sea?",
            "explanation": "Land heats faster, its air rises (low pressure), and the cooler, denser air over the sea
slides inland to fill the gap."
          },
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            "scope": "PRACTICE",
            "kind": "MCQ",
            "prompt": "A hot-air balloon rises because the heated air inside it is...",
            "options": [
              "heavier than outside air",
              "less dense than outside air",
              "magnetic",
              "full of helium"
            ],
            "answer": 1,
            "hint": "Same reason warm air rises anywhere.",
            "explanation": "Heating spreads the air’s molecules out, making it less dense than the surrounding cool
air -- so it floats upward, balloon and all."
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            "scope": "PRACTICE",
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            "prompt": "A bigger difference in pressure between two places makes a stronger wind between them.",
            "answer": true,
            "hint": "Think of a crowded room with one open door vs. a whole open wall.",
            "explanation": "The pressure difference is the push. Small difference, gentle breeze; huge difference,
storm-force winds."
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            "options": [
              "catches fire",
              "cools down and its water vapor condenses into droplets",
              "freezes the sky",
              "collides with the Moon"
            ],
            "answer": 1,
            "hint": "What happens to the invisible moisture as the air climbs and cools?",
            "explanation": "As rising air cools high above the ground, the invisible water vapor it carries condenses
into billions of tiny droplets -- a cloud."
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          {
            "scope": "PRACTICE",
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            "prompt": "The forecast shows a HIGH pressure system parked over your town this weekend. Most likely
```

weather?",
              "options": [
                "Calm and clear",
                "Hurricane",
                "Constant thunderstorms",
                "Snow, regardless of season"
              ],
              "answer": 0,
              "hint": "What is the air doing in a high -- rising or gently sinking?",
              "explanation": "In high-pressure zones air slowly sinks, which suppresses clouds -- typically calm, fair weather. Lows bring the wind and rain."
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          "title": "Why Is the Sky Blue?",
          "tagline": "Sunlight, scattered -- and why sunsets burn orange",
          "estMinutes": 14,
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                "sub": "The most-asked science question in history has a real answer -- and the same answer explains blue noons, orange sunsets, and the black sky of the Moon."
              }
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              "content": {
                "markdown": "Every kid on Earth has asked it. Most adults still can’t answer it. Yet ’why is the sky blue?’ is a real physics question with a beautiful, complete answer -- one that was only nailed down about 150 years ago by the physicist **Lord Rayleigh**.\n\nThe answer is bigger than one color. It explains why sunsets glow orange and red, why the Moon’s daytime sky is dead black, why distant mountains look hazy and bluish, and even helps explain why Mars has a butterscotch sky. One mechanism, half the beauty in the sky."
              }
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                "markdown": "**Step 1: Sunlight is secretly every color at once.** The Sun’s white light is a mixture of all the rainbow’s colors -- red, orange, yellow, green, blue, violet -- blended together. A prism (or raindrops making a rainbow) can unmix them. Each color is a light wave with a different **wavelength**: red waves are long and lazy; blue and violet waves are short and choppy.\n\n**Step 2: Air bumps light around.** The atmosphere is full of molecules (mostly nitrogen). When sunlight streams through, the light waves jostle these molecules, and the molecules fling some of the light off in random new directions. This is called **scattering**.\n\n**Step 3: Here’s the secret -- blue scatters far more than red.** Short, choppy waves get bounced around much more easily than long, lazy ones. Blue light scatters roughly **ten times more** than red light. (Violet scatters even more, but the Sun sends less violet and our eyes are weak at seeing it -- so blue wins.)\n\n**Put it together:** as sunlight crosses the sky, its blue part gets knocked around the atmosphere again and again, bouncing molecule to molecule, until blue light is raining down on you **from every direction of the sky**. Look anywhere away from the Sun and you see that scattered blue. **The sky is blue because you’re seeing sunlight’s blue, bounced down to your eyes by the air itself.**\n\n**Now the sunset trick.** At noon, sunlight punches almost straight down through the atmosphere -- a short trip. But at sunrise and sunset, the Sun sits on the horizon, and its light must **slice sideways through hundreds of kilometers of air** to reach you -- a journey dozens of times longer. Along that marathon, almost *all* the blue (and even the green) gets scattered away to other places, leaving what survives the trip: **orange and red**. The Sun blushes, and clouds catch the leftover fire.\n\n**Final proof: remove the air.** No air, no scattering, no color. On the Moon -- as you saw in World 1 -- the Sun blazes in a pitch-black daytime sky. Astronauts orbiting Earth watch our thin glowing blue rim from above: the atmosphere itself, doing its scattering trick, visible from space."
              }
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```
                "problem": "Explain, using path length, why the same Sun looks white-yellow at noon but deep orange at sunset.",
                "steps": [
                  "At noon, light crosses the atmosphere almost vertically -- the shortest possible path.",
                  "Only some blue is scattered out; the remaining mix still looks white-yellow.",
                  "At sunset the path through air is dozens of times longer, so nearly all the short-wavelength light (blue, then green) is scattered away en route."
                ],
                "answer": "The long sunset path strips out the blues and greens, leaving the long-wavelength survivors -- orange and red -- to reach your eyes."
              },
              {
                "title": "The Moon’s black noon",
                "problem": "The Sun shines on the Moon just as it shines on Earth. Why is the Moon’s daytime sky black instead of blue?",
                "steps": [
                  "A blue sky requires something to scatter sunlight across the dome of the sky.",
                  "The scatterers on Earth are air molecules.",
                  "The Moon has no atmosphere -- nothing to bounce the light around."
                ],
                "answer": "No air, no scattering: sunlight travels in straight lines, the Sun glares fiercely, and the rest of the sky stays black."
              },
              {
                "title": "Blue mountains",
                "problem": "From far away, distant mountain ranges often look hazy and bluish. Connect this to sky physics.",
                "steps": [
                  "Between you and a far mountain lie many kilometers of air.",
                  "That air scatters sunlight -- preferentially blue -- into your line of sight.",
                  "The farther the mountain, the more scattered blue light is layered over its image."
                ],
                "answer": "You’re seeing a ’mini-sky’ of scattered blue light added between you and the mountains. The Blue Ridge Mountains are literally named for Rayleigh scattering."
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                "title": "Why Is the Sky Blue? | SciShow Kids"
              },
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                "youtubeId": "ehUIlhKhzDA",
                "title": "Why Is the Sky Blue? | NASA Space Place"
              }
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          "content": {
            "markdown": "Change the atmosphere, change the sky.\n\n**Mars** has an ultra-thin atmosphere of carbon dioxide laced with fine rusty dust. The dust scatters light differently than gas molecules do: the Martian daytime sky glows a **butterscotch tan** -- and at sunset, the area around the Sun turns **blue**. Mars has blue sunsets! NASA’s rovers have photographed them.\n\n**Titan**, Saturn’s giant moon, is wrapped in a thick orange smog of nitrogen and hydrocarbons -- its sky is a deep orange murk where the Sun is just a brighter patch.\n\nAstronomers now study the colors of skies on **exoplanets** -- worlds orbiting other stars -- by analyzing starlight that filters through their atmospheres. The same Rayleigh physics you just learned is one of the tools being used, right now, to search distant skies for signs of life."
          }
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                  "White sunlight is a mix of all colors, each with a different wavelength.",
                  "Air molecules scatter sunlight -- short blue wavelengths about ten times more than long red ones
(Rayleigh scattering).",
                  "The sky is blue because scattered blue sunlight reaches your eyes from every direction.",
                  "Sunsets are orange-red because the long sideways path scatters the blues and greens away first.",
                  "No atmosphere means no scattering: the Moon’s daytime sky is black."
                ],
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                    "label": "Rayleigh’s law",
                    "tex": "\\text{scattering} \\propto \\frac{1}{\\lambda^4}\\quad(\\text{shorter } \\lambda
\\Rightarrow \\text{much more scattering})"
                  }
                ]
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              "explanation": "White light is all colors blended; prisms and raindrops can unmix it into a rainbow."
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              "options": [
                "are painted blue",
                "scatter blue light much more than red light",
                "reflect the ocean",
                "absorb every color except blue"
              ],
              "answer": 1,
              "explanation": "Short blue wavelengths are scattered all over the sky by air molecules, so blue arrives at
 your eyes from every direction. (The ocean reflection story is a myth!)"
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              "options": [
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                "Violet",
                "Green",
                "Red"
              ],
              "answer": 3,
              "explanation": "Long, lazy red wavelengths slip through air with the least scattering -- which is exactly
why they survive the long sunset journey."
            },
            {
              "scope": "PRACTICE",
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              "prompt": "Why does the setting Sun look orange-red?",
              "options": [
                "The Sun cools down in the evening",
                "Its light crosses a much longer stretch of air, which scatters the blue away",
                "Dust from the day settles on it",
                "Our eyes get tired of blue"
              ],
              "answer": 1,
              "hint": "Compare the light’s path at noon vs. at the horizon.",
              "explanation": "Near the horizon, sunlight slices sideways through far more atmosphere. The blues and
greens scatter away along the route, leaving orange and red to arrive."
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            {
              "scope": "PRACTICE",
              "kind": "MCQ",
              "prompt": "An astronaut on the Moon looks up at noon. The sky is...",
              "options": [
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                "black",
                "orange",
                "green"
```

```
              ],
              "answer": 1,
              "hint": "What does the Moon lack?",
              "explanation": "No atmosphere means nothing to scatter sunlight across the sky. The Sun blazes against
blackness."
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              "hint": "Would landlocked deserts have blue skies?",
              "explanation": "A popular myth! Skies are blue over deserts and oceans alike, because the cause is
scattering by air molecules -- not reflection from water."
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              "options": [
                "mountains are made of blue rock",
                "the many kilometers of air in between scatter blue light into your view",
                "snow turns blue at a distance",
                "your eyes lose focus"
              ],
              "answer": 1,
              "hint": "There’s a ’mini-sky’ between you and the peaks.",
              "explanation": "All that intervening air does its usual trick, layering scattered blue light over the
distant view -- atmospheric perspective."
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              "kind": "MCQ",
              "prompt": "Violet light scatters even more than blue, yet the sky doesn’t look violet. One big reason
is...",
              "options": [
                "violet light is illegal",
                "the Sun emits less violet and our eyes detect it poorly",
                "violet is too heavy to stay up",
                "clouds eat violet"
              ],
              "answer": 1,
              "hint": "Think about both the source (Sun) and the detector (your eyes).",
              "explanation": "The Sun’s light contains less violet than blue, and human eyes are far more sensitive to
blue -- so the sky’s mixture looks blue to us."
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hours straight. The sky is stronger than it looks."
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 -- wax wings melting in the Sun. Then, on a cold December morning in 1903, the **Wright brothers** flew a homemade
machine for 12 seconds over a North Carolina beach, and the world changed forever.\n\nToday over 100,000 flights take
off every day. None of it is magic, and none of it works without the previous three lessons: planes fly by making a
**deal with the air** -- the very air whose weight, pressure, and motion you’ve just mastered. Time to cash in
everything you’ve learned."
              }
            },
            {
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              "title": "The big idea",
              "content": {
                "markdown": "**Air is substantial -- so it can push hard.** You learned air has mass and presses with
real force. Stick your hand out a car window (carefully!): at highway speed, the air shoves your hand like a living
```

thing. A wing is a machine for organizing that shove.\n\n**Four forces wrestle over every plane:**\n\n- **Weight** -- gravity pulling the plane down (old friend!).\n- **Lift** -- the upward push from air on the wings.\n- **Thrust** -- engines pushing the plane forward.\n- **Drag** -- air resistance pulling backward (the same force that slowed your flat sheet of paper).\n\nTo fly level, **lift must equal weight**. To hold speed, **thrust must equal drag**. Flying is a balancing act between these four, every second.\n\n**So where does lift come from?** Two views of the same deal:\n\n**View 1 -- the push-back (Newton).** A wing is tilted and shaped so that, as it slices forward, it **deflects huge amounts of air downward** -- both off its bottom surface and by pulling air down along its curved top. And remember the rule from Pushes and Pulls: *every push gets a push back.* The wing throws air **down**; the air throws the wing **up**. A big airliner’s wings hurl tonnes of air downward every second -- that’s the upward shove holding 500 tonnes aloft.\n\n**View 2 -- the pressure difference (Bernoulli).** Air sweeping over the wing’s curved top moves faster than the air below, and faster-moving air presses more weakly. Result: **strong pressure below, weak pressure above** -- a net upward squeeze. Try it yourself: hold a strip of paper to your lips, drooping down, and blow hard across its **top**. The strip rises! You weakened the pressure above it, and the still air below pushed it up.\n\nBoth views are correct -- they’re two descriptions of one event: the wing organizing the air.\n\n**The two levers of lift:** go **faster** (more air handled per second -- this is why planes need long runways to build speed before lifting off) or meet **more air** (bigger wings, or thicker air). Which explains two famous facts: planes climbing into the thin air at 11 km must fly far faster to grip enough air... and on the airless Moon, no wing, propeller, or helicopter can work *at all*. Rockets only fly there because they bring their own push -- throwing fuel down instead of air."
          }
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                "title": "The four-force checkup",
                "problem": "A plane is cruising in a perfectly straight line at constant speed and constant height. Compare its four forces.",
                "steps": [
                  "Constant height means the vertical forces balance: lift = weight.",
                  "Constant speed means the horizontal forces balance: thrust = drag.",
                  "Any imbalance would mean climbing/sinking or speeding up/slowing down."
                ],
                "answer": "Lift equals weight, and thrust equals drag. Steady flight is a perfect four-way tie."
              },
              {
                "title": "Why runways are long",
                "problem": "A jet sits at the start of a 3 km runway. Why can’t it simply lift straight up the moment its engines roar?",
                "steps": [
                  "Lift comes from wings moving through air -- no forward speed, no air deflected, no lift.",
                  "Lift grows as speed grows: the wings handle more air each second.",
                  "The runway is where the plane builds speed until lift finally exceeds weight."
                ],
                "answer": "Wings only work with airflow. The runway lets the plane accelerate until its wings deflect enough air downward for lift to beat its weight -- rotation, and up it goes."
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              {
                "title": "Helicopter on the Moon?",
                "problem": "Could a powerful helicopter fly on the Moon? Could a rocket? Explain the difference.",
                "steps": [
                  "A helicopter’s spinning blades are wings: they fly by throwing AIR downward.",
                  "The Moon has no air to throw -- the blades would spin uselessly in vacuum.",
                  "A rocket carries its own material to throw: it hurls burning fuel downward and gets pushed up in return."
                ],
                "answer": "Helicopter: impossible -- nothing to push against. Rocket: works perfectly, because it brings its own ’air’ in the form of fuel. (Mars, with its thin air, needed NASA’s Ingenuity helicopter to have huge, fast blades -- and it flew!)"
              }
            ]
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                "title": "How Airplanes Fly! | SciShow Kids"
              },

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              "title": "How Does an Airplane Fly? | AumSum"
            },
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              "title": "How Do Airplanes Work? | Brain Candy TV"
            }
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          "markdown": "On December 17, 1903, at Kitty Hawk, North Carolina, Orville Wright lay flat on the wing of
 the *Flyer* while his brother Wilbur steadied the wingtip. The engine -- built in their bicycle shop -- clattered to
life. The first flight lasted **12 seconds** and covered **37 meters**: shorter than the economy cabin of a modern
747.\n\nWhat made the brothers succeed where famous, well-funded scientists failed? **They tested everything.**
Distrusting published data, they built their own **wind tunnel** out of a wooden box and a fan, and measured lift on
over 200 miniature wing shapes through a long winter. They flew over a thousand glider flights before ever adding an
engine. Their real invention wasn’t just the plane -- it was the modern method of aerospace engineering: *measure, test,
 revise.*\n\nSixty-six years separate Kitty Hawk’s 12 seconds from boots on the Moon. The same air you breathed today
taught humanity to fly."
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            "Four forces govern flight: lift up, weight down, thrust forward, drag back.",
            "Level flight at steady speed means lift = weight and thrust = drag.",
            "Wings make lift by deflecting air downward -- the air pushes back up (and pressure above the wing is
lower than below).",
            "More speed or more (denser) air means more lift -- hence long runways, and faster flight in thin
high-altitude air.",
            "No air, no lift: wings are useless on the Moon, which is why we go there on rockets."
          ],
          "formulas": [
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            }
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        "options": [
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          "Lift, magnetism, heat, sound",
          "Weight, friction, light, wind",
          "Thrust, gravity, bounce, glow"
        ],
        "answer": 0,
        "explanation": "Lift (up), weight (down), thrust (forward), and drag (backward) -- flight is the contest
between these four."
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      {
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        "prompt": "For a plane to fly level, its lift must equal its weight.",
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"explanation": "Balanced vertical forces mean no climbing or sinking -- lift exactly cancels weight in level flight."
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"options": [
"being lighter than air, like a balloon",
"deflecting air downward so the air pushes the wing up",
"flapping",
"magnetically repelling the ground"
],
"answer": 1,
"explanation": "The wing throws air down; by the push-back rule, the air throws the wing up. (Equivalently: lower pressure above, higher below.)"
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"scope": "PRACTICE",
"kind": "MCQ",
"prompt": "Why does a jet need a long runway before takeoff?",
"options": [
"To warm up its tires",
"To build enough speed for its wings to generate lift greater than its weight",
"Pilots enjoy the view",
"To use up extra fuel"
],
"answer": 1,
"hint": "No airflow over the wings means no...?",
"explanation": "Wings only lift when air flows over them. The runway run builds the speed at which deflected air finally out-pushes the plane’s weight."
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"scope": "PRACTICE",
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"prompt": "You blow hard across the TOP of a drooping paper strip and it rises. Why?",
"options": [
"Your breath is hot and heat lifts paper",
"Fast air on top means lower pressure there, so the still air below pushes the strip up",
"Sound waves lift it",
"Paper is magnetic"
],
"answer": 1,
"hint": "Fast-moving air presses more...?",
"explanation": "Speeding up the air above lowers its pressure; the unchanged, stronger pressure beneath pushes the strip upward -- a wing in miniature."
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"prompt": "A helicopter with a powerful enough engine could fly on the Moon.",
"answer": false,
"hint": "What do spinning blades push against?",
"explanation": "Rotor blades fly by throwing air downward. The Moon has no air to throw -- only rockets, which hurl their own fuel, can ’push’ in a vacuum."
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"prompt": "At 11 km altitude the air is about four times thinner than at sea level. To keep enough lift, high-flying jets must...",
"options": [
"fly much faster",
"fly much slower",
"turn off their engines",
"open the windows"
],
"answer": 0,
"hint": "Two levers of lift: speed and amount of air.",
"explanation": "Thin air means less air met per second -- the jet compensates with speed, handling more of the thin air every second to keep lift equal to weight."
},
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"prompt": "A plane’s thrust becomes greater than its drag while lift still equals weight. What happens?",
"options": [
"It speeds up",
"It slows down",
"It sinks",
"Nothing"
],

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              "answer": 0,
              "hint": "Which pair of forces went out of balance -- vertical or horizontal?",
              "explanation": "The horizontal tug-of-war is now won by thrust, so the plane accelerates forward. The
balanced vertical pair keeps it level while it does."
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      "description": "Four hundred kilometers up, the Space Station circles the Earth every 90 minutes. In
near-weightlessness -- where friction nearly vanishes -- the true rules of force show themselves: pushes and pulls,
simple machines, work and power, and the energy that ties it all together.",
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                "sub": "300 km up, astronauts float as if weightless. A single fingertip push sends one drifting across
the cabin. Up here, with friction nearly gone, the real rules of force finally show themselves."
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              "kind": "CONTEXT",
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              "content": {
                "markdown": "Why does a kicked ball eventually stop, but a spacecraft coasts for years? Why is it hard
to start a heavy cart moving -- and just as hard to stop it once it’s rolling?\n\nThe answer to every one of these is
**force**: a push or a pull. Forces start motion, stop motion, speed things up, slow them down, and turn them. Learn how
 forces work and you can explain a soccer kick, a seatbelt, a rocket launch, and the drift of an astronaut all with the
same handful of ideas."
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                "markdown": "**A force is a push or a pull.** We measure force in **newtons (N)**. Holding a small apple
 takes about 1 N. Forces have a **size** and a **direction** -- a 10 N push north is different from a 10 N push
south.\n\n**Forces add up.** Two people pushing a box the same way combine their forces. Pushing opposite ways, they
subtract. The single force left over after all the pushes and pulls are combined is the **net force** $F_{net}$.\n\n-
**Balanced forces** ($F_{net} = 0$): the pushes cancel. The object keeps doing exactly what it was doing -- staying
still, or moving at steady speed in a straight line.\n- **Unbalanced forces** ($F_{net} \\ne 0$): there’s a leftover
push. Now -- and only now -- the motion **changes**: it speeds up, slows down, or turns.\n\n**This is the great idea of
inertia.** Objects resist changes to their motion. A still object stays still; a moving object keeps moving -- *unless
an unbalanced force acts on it*. So a ball doesn’t ’naturally’ stop. Something stops it: the unbalanced forces of
**friction** and air resistance, quietly pushing backward. Remove them, and motion never quits -- which is exactly why a
 spacecraft coasts across the solar system and an astronaut drifts until a wall pushes back.\n\n**More force, or less
mass, means a bigger change.** Push the same cart harder and it speeds up faster. Load the cart with bricks (more
**mass**) and the same push barely budges it. Heavy things have more inertia -- they resist changes more
stubbornly.\n\n**Every push comes with a push back.** Push off a wall while floating in space and *you* shoot backward
-- the wall pushed you exactly as hard as you pushed it. Forces always come in pairs, in opposite directions. A rocket
throws gas downward; the gas throws the rocket up. A swimmer pushes water back; the water pushes the swimmer forward."
              }
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                    "problem": "The red team pulls a rope left with 400 N. The blue team pulls right with 350 N. What is
 the net force, and which way does the rope move?",
```

```
                    "steps": [
                      "The pulls are in opposite directions, so subtract them.",
                      "$F_{net} = 400 - 350 = 50$ N.",
                      "The leftover 50 N points toward the stronger (red) side."
                    ],
                    "answer": "A net force of 50 N to the left -- the rope (and blue team) accelerates leftward. If both
 teams pulled with 400 N, the net force would be zero and nothing would move."
                  },
                  {
                    "title": "Why the seatbelt saves you",
                    "problem": "A car stops suddenly in a crash. Use inertia to explain why an unbelted passenger keeps
moving forward.",
                    "steps": [
                      "Before the crash, car and passenger move forward together at speed.",
                      "The crash applies a huge backward force to the car, stopping it.",
                      "No such force acts on the passenger -- by inertia they keep moving forward at the old speed."
                    ],
                    "answer": "Inertia keeps the passenger moving until something exerts a force on them. The seatbelt
is that force, slowing them with the car instead of the windshield doing it."
                  },
                  {
                    "title": "Astronaut push-off",
                    "problem": "A floating astronaut gently pushes a heavy supply crate. Both drift apart. Why does the
astronaut move faster than the crate?",
                    "steps": [
                      "The push pairs are equal: the astronaut pushes the crate, the crate pushes back equally.",
                      "Equal forces, but different masses.",
                      "The smaller-mass astronaut has less inertia, so the same force changes their motion more."
                    ],
                    "answer": "Equal and opposite forces act, but the lighter astronaut accelerates more than the
massive crate -- so the astronaut zips off faster."
                  }
                ]
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                "intro": "Three quick questions -- no resistance, please."
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                "intro": "Push through these five."
              }
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              "content": {
                "markdown": "For 2,000 years people believed Aristotle: that things stop moving because stopping is
their natural state, and motion needs a constant push to continue.\n\n**Galileo** doubted it. He rolled balls down one
ramp and up another, and noticed the ball always tried to climb back to its starting height. Make the second ramp
gentler, and the ball rolled farther to reach that height. So what if the second ramp were perfectly flat and
frictionless? Galileo reasoned the ball would roll **forever**, never finding its height.\n\nHe couldn’t build a
frictionless surface -- but he could imagine one. That leap, later sharpened by Newton, overturned two millennia of
physics: motion doesn’t need a cause; *changes* in motion do. Centuries later, spacecraft coasting silently between the
planets are the proof Galileo never got to see."
              }
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                  "A force is a push or pull, measured in newtons (N), with size and direction.",
                  "Net force is what’s left after all forces combine. Balanced (net = 0) means no change in motion.",
                  "Inertia: objects resist changes to their motion. Things stop because of friction, not ’naturally’.",
                  "The same force changes a light object’s motion more than a heavy one’s (less mass, less inertia).",
                  "Forces come in equal, opposite pairs -- push the wall and it pushes you back."
                ],
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                    "tex": "F_{net} = \\sum F \\quad (F_{net}=0 \\Rightarrow \\text{motion unchanged})"
                  }
                ]
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              "a push or a pull",
              "the speed of an object",
              "the weight of air"
            ],
            "answer": 1,
            "difficulty": 1,
            "explanation": "A force is simply a push or a pull, measured in newtons."
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            "prompt": "If the forces on an object are balanced (net force zero), its motion cannot change.",
            "answer": true,
            "difficulty": 2,
            "explanation": "Balanced forces leave nothing to change the motion -- the object stays still or keeps
moving steadily in a straight line."
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            "prompt": "A hockey puck slides across the ice and slowly stops. What stopped it?",
            "options": [
              "It ran out of motion",
              "Friction -- an unbalanced backward force",
              "Gravity pulling it down",
              "Nothing; stopping is natural"
            ],
            "answer": 1,
            "difficulty": 2,
            "explanation": "Without friction the puck would glide on forever. The small backward force of friction is
what slows and stops it."
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            "prompt": "Two students push a cart the same direction, one with 30 N and one with 45 N. What is the net
force on the cart, in newtons?",
            "answer": {
              "value": 75,
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            "difficulty": 2,
            "hint": "Same direction means add.",
            "explanation": "Forces in the same direction add: $30 + 45 = 75$ N."
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          {
            "scope": "PRACTICE",
            "kind": "NUMERIC",
            "prompt": "In a tug-of-war, the left team pulls with 520 N and the right team pulls with 480 N. What is
the size of the net force, in newtons?",
            "answer": {
              "value": 40,
              "tolerance": 0
            },
            "difficulty": 2,
            "hint": "Opposite directions means subtract.",
            "explanation": "Opposite pulls subtract: $520 - 480 = 40$ N, toward the stronger (left) team."
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            "scope": "PRACTICE",
            "kind": "MCQ",
            "prompt": "A spacecraft far from any planet shuts off its engines. What happens to its motion?",
            "options": [
              "It quickly slows and stops",
              "It keeps coasting at the same speed and direction",
              "It speeds up forever",
              "It falls straight down"
            ],
            "answer": 1,
            "difficulty": 3,
            "hint": "What forces act on it out there?",
            "explanation": "With essentially no friction or other unbalanced force, inertia carries it onward at
constant velocity -- possibly for centuries."
          },
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              "answer": true,
              "difficulty": 2,
              "hint": "Forces come in pairs.",
              "explanation": "Every force has an equal and opposite partner. The wall’s push back is what sends you drifting away."
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              "kind": "MCQ",
              "prompt": "Two identical pushes are applied: one to a loaded cart, one to an empty cart. Which speeds up more?",
              "options": [
                "The loaded cart",
                "The empty cart",
                "They speed up equally",
                "Neither moves"
              ],
              "answer": 1,
              "difficulty": 3,
              "hint": "Less mass means less inertia.",
              "explanation": "With less mass (less inertia), the empty cart’s motion changes more for the same force, so it accelerates more."
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              }
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                "markdown": "How did ancient builders raise stone blocks heavier than houses? How does a small child lift a grown adult on a seesaw? How can one arm crank a car up off the ground?\n\nNone of it breaks the rules of force -- it **reroutes** them. A **simple machine** lets a small force do a big job, by spreading that force over a longer distance. Ramps, levers, pulleys, and wheels are the building blocks of every complex machine ever made, from a wheelbarrow to a robotic arm in orbit."
              }
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                  "problem": "A loading ramp lets workers push a 600 N barrel up with only 200 N of force. What is the ramp’s mechanical advantage?",
                  "steps": [
                    "MA compares output (the load) to input (your effort).",
                    "$MA = \\dfrac{\\text{load}}{\\text{effort}} = \\dfrac{600}{200}$.",
                    "$= 3$."
                  ],
                  "answer": "MA = 3. The ramp tripled the worker’s force -- but the barrel must travel about three times farther along the slope than its height gain."
                },
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                  "title": "Seesaw balance",
                  "problem": "A small child (200 N) sits 3 m from a seesaw’s pivot. Where must a 600 N adult sit to balance it?",
                  "steps": [
                    "A lever balances when force x distance is equal on both sides.",
                    "Child’s side: $200 \\times 3 = 600$.",
                    "Adult’s side must also equal 600: $600 \\times d = 600$, so $d = 1$."
                  ],
                  "answer": "The adult sits 1 m from the pivot. Sitting closer to the fulcrum, the heavy adult balances the light child far out -- the lever in action."
                },
                {
                  "title": "Counting pulley sections",
                  "problem": "A pulley system supports a load with 4 sections of rope sharing the weight. Roughly what effort lifts a 400 N load?",
                  "steps": [
                    "Each supporting rope section carries an equal share of the load.",
                    "Four sections share 400 N: $400 \\div 4 = 100$ N each.",
                    "You pull with about that one share."
                  ],
                  "answer": "About 100 N -- one quarter of the load (MA ~= 4). The catch: you must pull about 4 m of rope to raise the load 1 m."
                }
              ]
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              "markdown": "The ancient Greek genius **Archimedes** understood levers so deeply he reportedly boasted, *’Give me a place to stand and a lever long enough, and I will move the Earth.’*\n\nHe wasn’t entirely joking. With a long enough lever and a fulcrum, the *force* needed to lift even the Earth would be small -- though you’d have to move your end of the lever an absurd, cosmic distance, for an unimaginable length of time. The force shrinks; the distance explodes. The deal always holds.\n\nArchimedes turned this insight into war machines, water screws still used for irrigation today, and the foundations of mechanics. Every robotic arm, crane, and excavator is a descendant of his lever."
            }
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                "Simple machines change a force’s size or direction to make work easier.",
                "They never give free energy: less force always costs more distance.",
                "Mechanical advantage MA = load / effort tells you how many times the force is multiplied.",
                "Lever, inclined plane, pulley, wheel-and-axle, wedge, and screw are the six classics.",
                "Friction means real machines help a bit less than the ideal -- lubrication recovers some."
              ],
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                {
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              "options": [
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                "Use a smaller force, spread over a longer distance",
                "Remove friction completely",
                "Make objects weightless"
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              "explanation": "Machines trade force for distance -- a smaller push over a longer path does the same job."
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conserved."
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                  "Crowbar",
                  "Doorknob",
                  "Axe"
                ],
                "right": [
                  "Inclined plane",
                  "Lever",
                  "Wheel and axle",
                  "Wedge"
                ]
              },
              "answer": [
                "Inclined plane",
                "Lever",
                "Wheel and axle",
                "Wedge"
              ],
              "difficulty": 3,
              "explanation": "A ramp is an inclined plane, a crowbar is a lever, a doorknob is a wheel and axle, and an
axe is a wedge."
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advantage?",
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              "explanation": "$MA = 800 \\div 160 = 5$. The lever multiplied your force five times."
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              "prompt": "A ramp has a mechanical advantage of 4. If you push with 75 N, what load (in newtons) can you
move up it?",
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"hint": "load = MA x effort.",
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"multiplies your force by 10",
"changes the direction of your pull, so you pull down to raise the flag",
"removes the flag’s weight",
"creates energy"
],
"answer": 1,
"difficulty": 2,
"hint": "One fixed pulley doesn’t add rope sections.",
"explanation": "A single fixed pulley redirects your effort (pull down to lift up) but doesn’t multiply force; its MA is about 1."
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"answer": true,
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"explanation": "A lever balances when force x distance matches on both sides, so a small force far out can balance a big force close in."
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"options": [
"The math is wrong",
"Some effort is lost to friction as heat",
"Gravity gets stronger",
"Levers shrink over time"
],
"answer": 1,
"difficulty": 3,
"hint": "What turns into heat in moving parts?",
"explanation": "Friction in moving parts wastes a little of your effort as heat, so the real output force is a bit less than the ideal."
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"sub": "In physics, straining to hold a heavy box that doesn’t move counts as no work at all. The definition is strict -- and once you know it, energy starts to make sense."
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direction of the force**. One joule is one newton pushing through one meter.\n\n**The strict part:** if nothing moves, **no work is done**, no matter how hard you strain. Holding a heavy box still: $d = 0$, so $W = 0$. Pushing a wall that won’t budge: zero work. Carrying a box across a flat room: the lifting force is *upward* but the motion is *sideways*, so that force does no work either. Work needs motion **along the force**.\n\n**Work is the transfer of energy.** When you do work on something, you give it energy -- you speed it up, or lift it higher. That’s why work and energy share the same unit, the joule. (This is the secret behind simple machines: they can’t change the work needed, only split it into a smaller force over a longer distance -- same $F \\times d$.)\n\n**Power is how fast you do work:**\n\n$$P = \\frac{W}{t}$$\n\nPower P (in **watts, W**) is work divided by the time t taken. One watt is one joule per second. Run up the stairs and you do the same work as walking up -- but in less time, so more power. A 100 W bulb uses 100 joules every second. A car engine might deliver tens of thousands of watts; a rocket, billions."
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                    "problem": "You lift a 50 N box straight up onto a shelf 2 m high. How much work do you do?",
                    "steps": [
                      "Use $W = F \\times d$ with the force and the distance along it.",
                      "$W = 50 \\times 2$.",
                      "$= 100$ J."
                    ],
                    "answer": "100 joules -- and that energy is now stored in the box as height (potential energy), ready to be released if it falls."
                  },
                  {
                    "title": "The strict definition",
                    "problem": "A waiter carries a 20 N tray steadily across a level room, 8 m. How much work does the carrying force do on the tray?",
                    "steps": [
                      "The supporting force points up; the tray moves sideways.",
                      "Work counts only motion in the direction of the force.",
                      "There is no motion along the upward force, so $d = 0$ for that force."
                    ],
                    "answer": "Zero work (on the tray, by the lifting force). Tiring, yes -- but physics work needs motion along the force, and here there is none."
                  },
                  {
                    "title": "Same work, more power",
                    "problem": "Two students each do 600 J of work climbing a staircase. Ana takes 10 s; Ben takes 5 s. Who is more powerful, and by how much?",
                    "steps": [
                      "Power is work over time: $P = W / t$.",
                      "Ana: $600 / 10 = 60$ W.",
                      "Ben: $600 / 5 = 120$ W."
                    ],
                    "answer": "Ben, at 120 W versus Ana’s 60 W -- double the power for the same work, because he did it in half the time."
                  }
                ]
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            },
            {

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                "Work = force x distance moved along that force: W = F x d, measured in joules.",
                "No motion (or motion across the force) means zero work, no matter the effort.",
                "Doing work transfers energy -- that’s why both use the joule.",
                "Power = work / time: P = W / t, measured in watts (joules per second).",
                "Same work done faster means more power."
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                  "tex": "W = F \\times d"
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                  "label": "Power",
                  "tex": "P = \\frac{W}{t}"
                }
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            "options": [
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              "you think hard",
              "an object is heavy"
            ],
            "answer": 1,
            "difficulty": 2,
            "explanation": "Work needs a force AND motion along that force: W = F x d."
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            "answer": false,
            "difficulty": 2,
            "explanation": "No motion means distance is zero, so the work is zero -- however tiring it feels."
          },
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            "kind": "MCQ",
            "prompt": "Power measures...",
            "options": [
              "how much force you use",
              "how fast work is done",
              "how heavy something is",
              "total distance travelled"
            ],
            "answer": 1,
            "difficulty": 2,
            "explanation": "Power is the rate of doing work: P = W / t, in watts."
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            "difficulty": 2,
            "hint": "W = F x d.",
            "explanation": "$W = 40 \\times 6 = 240$ J."
          },
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            "scope": "PRACTICE",
            "kind": "NUMERIC",
            "prompt": "A motor does 1500 J of work in 5 s. What is its power output, in watts?",
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              "value": 300,
              "tolerance": 0
            },
            "difficulty": 2,
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              "hint": "P = W / t.",
              "explanation": "$P = 1500 \\div 5 = 300$ W."
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              "answer": {
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                "tolerance": 0
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              "difficulty": 2,
              "hint": "The distance is the height lifted.",
              "explanation": "$W = F \\times d = 25 \\times 4 = 100$ J, now stored as the object’s height."
            },
            {
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              "prompt": "Two elevators do the same work lifting a load. Elevator A takes 4 s; elevator B takes 8 s.
Which has greater power?",
              "options": [
                "Elevator A",
                "Elevator B",
                "They are equal",
                "Neither uses power"
              ],
              "answer": 0,
              "difficulty": 3,
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              "explanation": "Same work in less time means more power, so the faster Elevator A is more powerful."
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              "prompt": "A simple machine lets you do a job with less force, but the total work (force x distance) stays
 about the same.",
              "answer": true,
              "difficulty": 3,
              "hint": "Force down, distance up -- same product.",
              "explanation": "Machines split the same work into a smaller force over a longer distance; ignoring
friction, F x d is unchanged."
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                "sub": "Energy can hide as motion, height, heat, light, or chemical fuel -- and flip between them
endlessly. But the grand total never changes. Not once, anywhere, ever."
              }
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thrown ball becomes heat when it lands. Energy is the single thread running through every event in the universe.\n\nThe
deepest law in all of science is that this thread is **never cut**: energy is only ever **transformed** from one form to
 another, never created or destroyed. Understand the forms of energy and the law that ties them together, and you hold
the master key to physics, chemistry, biology, and engineering alike."
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 it’s measured in **joules**. It comes in many forms:\n\n- **Kinetic energy** -- the energy of **motion**. Anything
moving has it: a thrown ball, flowing water, a zooming spacecraft. More speed or more mass means more kinetic energy.\n-
 **Potential energy** -- **stored** energy, waiting to be released. A raised hammer has **gravitational** potential
energy (height); a stretched spring or drawn bow has **elastic** potential energy; food and fuel hold **chemical**
potential energy in their bonds.\n- **Thermal energy** -- the energy of jiggling particles; we feel it as heat.\n-
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**Light (radiant) energy** -- carried by light, like the sunlight crossing space.\n- **Electrical energy** -- carried by
 moving charges through a wire.\n- **Sound energy** -- carried by vibrations through a material.\n\n**Energy
transforms.** Almost everything interesting is one form of energy turning into another:\n\n- A roller coaster trades
**gravitational potential** (at the top) for **kinetic** (at the bottom), then back again on the next hill.\n- A
flashlight turns **chemical** (battery) -> **electrical** -> **light** and **thermal**.\n- A plant turns **light** ->
**chemical** (sugar). You eat it and turn **chemical** -> **kinetic** (movement) and **thermal** (body heat).\n\n**The
law of conservation of energy:** in any change, **the total energy stays the same**. Energy is never created from
nothing and never truly destroyed -- only moved and reshaped. When a bouncing ball finally stops, its energy didn’t
vanish; it spread out as heat and sound. That’s also why a true ’perpetual motion machine’ is impossible: there’s no way
 to get out more energy than you put in.\n\n**Friction is energy’s escape hatch.** In real machines and motions, some
energy always ’leaks’ into thermal energy (heat) and sound, spreading out where it’s hard to use again. The total is
still conserved -- but the useful part shrinks. That’s why nothing is 100% efficient."
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                    "problem": "A coaster car has 50,000 J of gravitational potential energy at the top of a hill.
Ignoring friction, what is its kinetic energy at the bottom?",
                    "steps": [
                      "Energy is conserved: the total stays 50,000 J.",
                      "At the bottom, the height (potential) is gone -- converted to motion.",
                      "So all 50,000 J has become kinetic energy."
                    ],
                    "answer": "50,000 J of kinetic energy. The potential energy of height turned entirely into the
energy of motion. (Real coasters lose a little to friction, so it’s slightly less.)"
                  },
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                    "title": "Trace the flashlight",
                    "problem": "List the chain of energy transformations when you switch on a battery flashlight.",
                    "steps": [
                      "The battery stores chemical potential energy.",
                      "Switching on lets it flow as electrical energy through the bulb.",
                      "The bulb converts that to light energy -- and some thermal energy (it gets warm)."
                    ],
                    "answer": "Chemical -> electrical -> light + thermal. The warmth you feel is the ’leaked’ part --
proof no transformation is perfectly efficient."
                  },
                  {
                    "title": "Where did the bounce go?",
                    "problem": "A dropped ball bounces a little lower each time, then stops. Energy is conserved, so
where did its energy go?",
                    "steps": [
                      "Each bounce, some kinetic energy converts to thermal energy and sound.",
                      "The ball and floor warm up slightly; you hear the ’thud’.",
                      "After many bounces, all the original energy has spread out as heat and sound."
                    ],
                    "answer": "It became thermal energy (heat) and sound, spread into the ball, floor, and air. Nothing
was destroyed -- just scattered into hard-to-use forms."
                  }
                ]
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failed. Whenever it seemed to break, physicists discovered a hidden form of energy rather than abandon the law.\n\nIn
the 1930s, certain radioactive decays seemed to lose a little energy. Rather than give up conservation, **Wolfgang
Pauli** boldly predicted an invisible, nearly massless particle carrying the missing energy away. Decades later the
**neutrino** was found -- exactly as conservation demanded. Trillions are streaming through your body right
now.\n\nEinstein deepened the law with $E = mc^2$: mass itself is a super-concentrated form of energy. The Sun shines by
 converting a tiny bit of its mass into the light that, eight minutes later, warms your face. Energy bookkeeping, on a
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cosmic scale, always balances."
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                  "Energy is the ability to do work, measured in joules.",
                  "Key forms: kinetic (motion), potential (stored -- gravitational, elastic, chemical), thermal, light, 
electrical, sound.",
                  "Energy constantly transforms from one form to another.",
                  "Conservation of energy: the total is never created or destroyed, only reshaped.",
                  "Friction leaks energy into heat and sound, so no real process is 100% efficient."
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                "Potential energy",
                "Kinetic energy",
                "Chemical energy",
                "No energy"
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              "difficulty": 1,
              "explanation": "Kinetic energy is the energy of motion -- anything moving has it."
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              "options": [
                "kinetic energy",
                "gravitational potential energy",
                "sound energy",
                "electrical energy"
              ],
              "answer": 1,
              "difficulty": 2,
              "explanation": "Height stores gravitational potential energy, ready to convert to motion if the book 
falls."
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              "explanation": "Conservation of energy forbids it -- energy is only transformed, never created or 
destroyed. Perpetual motion machines are impossible."
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kinetic energy at the bottom is about...",
              "options": [
                "0 J",
                "40,000 J",
                "80,000 J",
                "160,000 J"
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              "answer": 2,
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              "hint": "Total energy is conserved.",
              "explanation": "All the potential energy converts to kinetic, so about 80,000 J -- energy is conserved."
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            "prompt": "Order the energy transformations when sunlight ends up powering your bike ride.",
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              "Light energy from the Sun",
              "Chemical energy stored in plants (food)",
              "Chemical energy in your muscles",
              "Kinetic energy of the moving bike"
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            "answer": [],
            "difficulty": 3,
            "hint": "Follow the energy from the Sun to the road.",
            "explanation": "Sunlight -> food (chemical) -> muscle (chemical) -> motion (kinetic): a chain of
transformations, all conserving total energy."
          },
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            "scope": "PRACTICE",
            "kind": "MCQ",
            "prompt": "A bouncing ball gradually stops. Conservation of energy says its energy...",
            "options": [
              "was destroyed",
              "turned mostly into heat and sound",
              "turned into more height",
              "became new matter"
            ],
            "answer": 1,
            "difficulty": 3,
            "hint": "What warms up, and what do you hear?",
            "explanation": "The energy spreads out as thermal energy (heat) and sound -- conserved, but scattered into
 hard-to-use forms."
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            "options": [
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              "chemical -> electrical -> kinetic",
              "kinetic -> chemical -> light",
              "sound -> electrical -> chemical"
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            "answer": 1,
            "difficulty": 3,
            "hint": "Start with what a battery stores.",
            "explanation": "A battery’s chemical energy becomes electrical energy in the motor, which becomes kinetic
energy of the moving car."
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            "prompt": "Because of friction, real machines always lose some energy as heat, so none are 100%
efficient.",
            "answer": true,
            "difficulty": 2,
            "hint": "Friction is energy’s escape hatch.",
            "explanation": "Friction converts some useful energy into spread-out thermal energy and sound, so
efficiency is always below 100%."
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the physics that telescopes are built on: how waves carry energy, how sound and light travel, and how the full
electromagnetic spectrum reveals an invisible universe.",
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                "markdown": "Drop a pebble in a pond and rings spread outward. Shout, and your voice reaches a friend. Flip on a lamp and light fills the room. Earthquakes shake whole cities from kilometers away.\n\nAll four are **waves** -- and they share the same rules. A wave is how **energy travels from place to place without matter making the whole journey**. Learn the anatomy of a wave once, and you’ve started to understand sound, light, music, radio, and the telescopes that read the universe."
              }
            },
            {
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                "markdown": "**A wave carries energy, not matter.** Watch a duck on rippling water: the ripples race past, but the duck just bobs **up and down** in place. The water doesn’t travel to shore -- the *wave* does, passing its energy along.\n\n**Two types of wave:**\n\n- **Transverse** -- the material wiggles *across* the direction the wave travels. Water ripples, a shaken rope, and light are transverse.\n- **Longitudinal** -- the material squashes and stretches *along* the direction of travel. Sound is longitudinal: air gets pushed into compressions and pulled into gaps.\n\n**The anatomy of a wave:**\n\n- **Amplitude** -- how big the wiggle is (the height of a crest). Bigger amplitude means **more energy** -- a louder sound, a brighter light, a taller ocean wave.\n- **Wavelength** ($\\lambda$) -- the distance from one crest to the next.\n- **Frequency** ($f$) -- how many waves pass each second, measured in **hertz (Hz)**. High frequency means crests arrive quickly.\n- **Wave speed** ($v$) -- how fast the wave travels.\n\n**The wave equation ties them together:**\n\n$$v = f \\times \\lambda$$\n\nSpeed equals frequency times wavelength. For a fixed speed, squeezing the wavelength shorter forces the frequency higher, and vice-versa. This one relationship runs through everything that follows: a high musical note is a high frequency; a short-wavelength light wave is high frequency too.\n\n**Waves bounce, bend, and blend.** They **reflect** off barriers (an echo), **refract** (bend) when they pass into a new material at an angle, and **interfere** when they overlap -- crests adding to crests (louder/brighter) or crests filling troughs (silence/darkness). These behaviors are the fingerprints that tell scientists they’re looking at a wave at all."
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                      "Use $v = f \\times \\lambda$.",
                      "$v = 4 \\times 0.5$.",
                      "$= 2$ m/s."
                    ],
                    "answer": "2 m/s. Four waves per second, each half a meter long, march past at two meters every second."
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                    "problem": "Two ocean waves have the same wavelength, but wave A is twice as tall as wave B. Which carries more energy, and what changed?",
                    "steps": [
                      "Amplitude is the height of the wave.",
                      "Energy increases with amplitude.",
                      "Wave A’s greater height means greater amplitude."
                    ],
                    "answer": "Wave A carries more energy. Taller waves (greater amplitude) carry more energy -- which is why a big wave hits harder than a ripple of the same length."
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                    "title": "The bobbing duck",
                    "problem": "A wave travels across a lake toward the shore. Does the water itself travel to the shore? What does?",
                    "steps": [
                      "A floating duck shows the water’s true motion: up and down in place.",
                      "The crests move forward, but the water returns to where it started.",
                      "What moves forward is the wave’s energy."
                    ],
                    "answer": "The water mostly stays put, bobbing up and down; the wave (energy) is what travels to shore. Waves move energy, not matter."
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        "markdown": "When an earthquake strikes, it sends out **seismic waves** -- both transverse and
longitudinal -- that race through the planet. Networks of sensors around the world catch them.\n\nHere’s the clever
part: different waves travel at different speeds and bend as they pass through materials of different density, exactly
like light refracting. By timing when each wave arrives and how it bent, geologists have **mapped the inside of the
Earth** -- a place no drill will ever reach. That’s how we know Earth has a solid inner core, a liquid outer core, and a
 thick rocky mantle: we read it in the waves.\n\nThe same trick, aimed at the sky instead of the ground, lets
astronomers read the insides of stars from the way they ring and pulse -- a field called *asteroseismology*."
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          "Transverse waves wiggle across the travel direction (light); longitudinal waves squash along it
(sound).",
          "Amplitude = wave height = energy; wavelength = crest-to-crest; frequency = waves per second (Hz).",
          "Wave speed: v = f x lambda ties frequency and wavelength together.",
          "Waves reflect, refract, and interfere -- their tell-tale behaviors."
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        "frequency",
        "speed"
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              "explanation": "$f = v \\div \\lambda = 20 \\div 4 = 5$ Hz."
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              "options": [
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                "A larger amplitude",
                "A slower speed",
                "A lower frequency"
              ],
              "answer": 1,
              "difficulty": 2,
              "hint": "Think wave height.",
              "explanation": "Greater amplitude (a taller wave) means more energy -- louder sound, brighter light,
bigger ocean wave."
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higher.",
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              "explanation": "If v is constant and lambda shrinks, then f must rise so that f x lambda still equals v."
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                "disappearing"
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                "markdown": "Music, speech, a warning siren, a doctor’s ultrasound, a bat hunting in the dark -- all are **sound**. Sound is how we communicate, enjoy music, and even peer inside the human body and the ocean floor.\n\nAnd the rule that sound needs a **medium** explains one of the strangest facts about space: it’s utterly silent. Master sound and you’ll understand pitch, loudness, echoes, sonic booms, and why the movies get space battles completely wrong."
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                    "steps": [
                      "Light arrives almost instantly, so the 6 s is sound’s travel time.",
                      "Distance = speed x time = $343 \\times 6$.",
                      "$\\approx 2058$ m, about 2 km."
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                    "answer": "About 2 kilometers away. A handy rule: every 3 seconds of delay is roughly 1 km."
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                    "title": "High note or low note?",
                    "problem": "A singer raises the frequency of the note she’s holding. Does the pitch go up or down, and does the speed of the sound in the room change?",
                    "steps": [
                      "Pitch is set by frequency: higher frequency, higher pitch.",
                      "Raising the frequency raises the pitch.",
                      "Sound speed depends on the medium (the air), not the note -- so it stays about 343 m/s."
                    ],
                    "answer": "The pitch goes up; the speed of sound stays the same. In a fixed medium, only frequency and wavelength trade off (v = f lambda)."
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                  {
                    "title": "Silence in space",
                    "problem": "A movie shows a spaceship exploding with a huge BOOM. What’s wrong with this, physically?",
                    "steps": [
                      "Sound is a vibration that needs particles to travel through.",
                      "Space is a near-perfect vacuum -- almost no particles.",
                      "With no medium, the vibration can’t propagate."
                    ],
                    "answer": "There would be no sound. In the vacuum of space, explosions are silent -- the movie boom is artistic license, not physics."
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        "content": {
          "markdown": "Bats and dolphins hunt in darkness using **echolocation**: they emit high-frequency squeaks and listen for the echoes bouncing back, building a sound-picture of their surroundings precise enough to snag a single insect mid-flight.\n\nHumans copied the trick. **Sonar** maps the ocean floor and finds submarines by timing sound echoes through water. **Ultrasound** scanners send sound far above human hearing into the body and reconstruct images of a beating heart -- or a baby -- from the echoes, with no radiation at all.\n\nThe same idea even helps the blind: some people learn to click their tongues and read the returning echoes to navigate. Sound, it turns out, is a way of seeing."
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            "Pitch comes from frequency; loudness comes from amplitude (measured in decibels).",
            "Humans hear roughly 20 Hz-20,000 Hz; below is infrasound, above is ultrasound.",
            "Sound travels about 343 m/s in air -- faster in water and steel -- much slower than light.",
            "The flash-to-thunder delay measures distance: about 1 km per 3 seconds."
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            }
          ]
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          "amplitude",
          "speed",
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            "wavelength"
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          "difficulty": 2,
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the nearest meter.",
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            "tolerance": 1
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          "difficulty": 2,
          "hint": "distance = speed x time.",
          "explanation": "$d = 343 \\times 5 = 1715$ m."
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3 s)?",
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            "About 3 km",
            "About 9 km",
            "About 27 km"
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          "difficulty": 3,
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          "explanation": "9 s / 3 s per km ~= 3 km away."
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          "answer": true,
          "difficulty": 3,
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          "explanation": "Sound moves faster in stiffer, denser media -- much faster in steel than in air."
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            "Light travels far faster than sound",
            "Sound is created later",
            "The bang is quiet"
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          "answer": 1,
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flash."
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            "a rainbow",
            "no drag"
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Bend it with a curved piece of glass and you can capture a galaxy."
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lets you see, plants eat, cameras capture, and telescopes reach across the universe to look billions of years into the
past.\n\nUnderstanding how light **travels in straight lines, reflects, and refracts** explains mirrors, shadows,
rainbows, lenses, eyeglasses, cameras, microscopes, and the great space telescopes. It’s the physics of seeing itself."
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 why a blocked light makes a sharp **shadow**, and why you can’t see around corners.\n\n**Light is incredibly fast:**
about **300,000,000 m/s** in space -- fast enough to circle the Earth 7 times in one second. Nothing travels faster.
Because the speed is finite, the light from distant stars is old: we see them as they were long ago.\n\n**Three things
light does when it meets a surface:**\n\n- **Reflection** -- it bounces. A mirror reflects so cleanly you see an image.
The rule: the angle coming in equals the angle going out. Rough surfaces scatter light every which way, which is how you
 see ordinary objects.\n- **Refraction** -- it **bends** when it passes from one material into another (air into water,
say) because it changes speed. This is why a straw in a glass looks broken at the surface, and why a pool looks
shallower than it is.\n- **Absorption** -- it’s soaked up and turned to heat. A black shirt absorbs most light (and
warms up); a white shirt reflects most.\n\n**Lenses bend light to form images.** A lens is shaped glass that refracts
light in a controlled way:\n\n- A **convex** (bulging) lens bends parallel light rays together to a **focal point**. It
can magnify and is the heart of magnifying glasses, cameras, the human eye, and telescopes.\n- A **concave** (hollow)
lens spreads rays apart.\n\n**Your eye is an optical instrument.** Its lens focuses light onto the retina at the back.
If the focus lands a bit short or long, the world looks blurry -- and **eyeglasses add a lens** to fix exactly where the
 light converges. **Telescopes and microscopes** use the same lens physics, just scaled to gather faint, distant light
or magnify the tiny.\n\n**Curved mirrors do it too.** The biggest telescopes don’t use lenses at all -- they use giant
**curved mirrors** to collect and focus starlight, because a huge mirror is far easier to build than a huge lens."
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of light explains this?",
                    "steps": [
                      "Light from the underwater part of the straw passes from water into air.",
                      "Crossing that boundary, the light changes speed and bends -- refraction.",
                      "Your brain assumes light travelled straight, so the straw appears displaced."
                    ],
                    "answer": "Refraction. Light bends as it leaves the water, shifting the straw’s apparent position so
 it looks broken at the surface."
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                    "title": "Why telescopes are big",
                    "problem": "Astronomers always want bigger telescope mirrors. What is the main advantage of a larger
 mirror (or lens)?",
                    "steps": [
                      "A telescope’s job is to gather light from faint, distant objects.",
                      "A bigger mirror has more area to catch light.",
                      "More light gathered means fainter, farther objects become visible."
                    ],
                    "answer": "A larger mirror collects more light, revealing fainter and more distant objects -- which
is why observatories build ever-bigger mirrors."
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                    "problem": "A star is 100 light-years away. What does it mean to say we ’see it as it was 100 years
ago’?",
                    "steps": [
                      "Light travels fast but not instantly.",
                      "Light from that star took 100 years to reach us.",
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```
                      "So the image arriving now left the star 100 years ago."
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                    "answer": "We’re seeing 100-year-old light. Looking far into space is literally looking back in time -- telescopes are time machines."
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                  "Light travels in straight-line rays at about 300,000,000 m/s -- the fastest speed there is.",
                  "At a surface, light can reflect (bounce), refract (bend when changing speed), or be absorbed (become heat).",
                  "Refraction explains the ’broken’ straw and why pools look shallow.",
                  "Convex lenses focus light to a point and magnify; the eye, cameras, and telescopes all rely on this.",
                  "Big telescopes use large curved mirrors to gather more light and reveal fainter, farther objects."
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                "circles"
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cooler."
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        "Light refracts (bends) as it leaves the water",
        "The straw really bends",
        "Reflection from the glass"
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position."
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        "spreads them apart",
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        "absorbs them",
        "reflects them backward"
      ],
      "answer": 1,
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      "explanation": "A convex lens converges parallel rays to a focal point, which lets it focus and magnify
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        "make them heavier",
        "gather more light to see fainter objects",
        "reflect sound",
        "use less glass"
      ],
      "answer": 1,
      "difficulty": 2,
      "hint": "More area, more light.",
      "explanation": "A bigger mirror collects more light, revealing fainter and more distant objects."
    },
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        "removing light",
        "reflecting images"
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              "hint": "It’s all about the focal point landing on the retina.",
              "explanation": "Corrective lenses bend incoming light so it focuses exactly on the retina, sharpening the image."
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              "content": {
                "markdown": "Your phone, the microwave oven, the TV remote, a hospital X-ray, the warmth of a fire -- every one of these uses **light your eyes cannot see**. Visible light is just a narrow band of a vast family of waves called the **electromagnetic spectrum**.\n\nLearning the spectrum explains color, why sunscreen matters, how Wi-Fi works, how night-vision cameras see heat, and why astronomers build separate telescopes for radio, infrared, and X-rays. It’s the user manual for light in all its forms."
              }
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                "markdown": "**All these waves are the same thing -- electromagnetic waves -- differing only in wavelength and frequency.** They all travel at the speed of light. Lined up from longest wavelength (lowest frequency, lowest energy) to shortest (highest frequency, highest energy):\n\n**Radio -> Microwave -> Infrared -> Visible -> Ultraviolet -> X-ray -> Gamma ray**\n\n- **Radio** -- longest waves; carry broadcast, Wi-Fi, and signals from spacecraft.\n- **Microwave** -- heat water in food; also carry phone and radar signals.\n- **Infrared (IR)** -- felt as **heat**; TV remotes and night-vision cameras use it.\n- **Visible light** -- the thin slice your eyes detect, from **red** (longest visible wavelength) through orange, yellow, green, blue, to **violet** (shortest).\n- **Ultraviolet (UV)** -- just past violet; gives sunburns. The ozone layer blocks most.\n- **X-rays** -- short, energetic; pass through soft tissue but not bone, so we image skeletons.\n- **Gamma rays** -- shortest, most energetic; from nuclear reactions and the most violent events in space.\n\n**Shorter wavelength means higher frequency and more energy.** That’s why gamma rays and X-rays are dangerous (high energy) while radio waves pass through you harmlessly.\n\n**Color is your eye reading visible wavelengths.** An object looks red because it **reflects red wavelengths** and absorbs the rest; a leaf looks green because it reflects green. A black object absorbs nearly all visible light; a white one reflects nearly all. Mixing the colors of light back together gives white -- which is why sunlight, containing them all, looks white.\n\n**The invisible universe.** Earth’s atmosphere blocks most UV, X-rays, and gamma rays -- good for life, hard for astronomy. So we launch space telescopes that see in **infrared** (peering through dust to newborn stars), **X-ray** (catching gas swirling into black holes), and more. Each band of the spectrum reveals things the others can’t. Combine them and you get the full picture of the cosmos."
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                      "Energy rises with frequency (and falls with wavelength).",
                      "Radio waves have very long wavelengths (low frequency, low energy).",
                      "X-rays have very short wavelengths (high frequency, high energy)."
                    ],
                    "answer": "X-rays carry far more energy -- which is exactly why X-rays can be hazardous and radio waves are harmless to pass through."
                  },
                  {
                    "title": "Why a leaf looks green",
                    "problem": "White sunlight (all colors) falls on a green leaf. Why does the leaf look green?",
                    "steps": [
                      "White light contains every visible color.",
                      "The leaf absorbs most colors but reflects green.",
                      "Only the reflected green reaches your eyes."
                    ],
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"answer": "The leaf reflects green wavelengths and absorbs the others, so green is what you see. Color is the light an object reflects."
},
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"title": "Seeing through dust",
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"steps": [
"Visible light is scattered and blocked by dust.",
"Longer-wavelength infrared passes through dust more easily.",
"An infrared telescope can detect the warm glow of the hidden stars."
],
"answer": "An infrared telescope. Its longer waves slip through the dust, revealing stars being born -- which visible-light telescopes can’t see."
}
]
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"Radio, microwave, infrared, visible, ultraviolet, X-ray, and gamma rays are all electromagnetic waves at different wavelengths.",
"Shorter wavelength means higher frequency and more energy (gamma/X-ray energetic; radio gentle).",
"Visible light is a thin slice, red (long) through violet (short).",
"Color is the visible light an object reflects; white light contains all colors.",
"Different telescopes (infrared, X-ray, radio) reveal parts of the universe visible light can’t show."
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"the only kind of light there is",
"one small band of the electromagnetic spectrum",
"a type of sound",
"faster than radio waves"
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in both directions."
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              "Infrared",
              "Radio",
              "Gamma rays"
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            "answer": 1,
            "difficulty": 2,
            "explanation": "Infrared radiation is felt as heat and used by remotes and night-vision cameras."
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            "scope": "CONCEPT_CHECK",
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            "prompt": "Shorter-wavelength electromagnetic waves carry more energy.",
            "answer": true,
            "difficulty": 2,
            "explanation": "Shorter wavelength means higher frequency and more energy -- which is why X-rays and gamma
rays are energetic and hazardous."
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            "options": [
              "Radio",
              "Infrared",
              "Visible light",
              "X-rays"
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            "answer": [],
            "difficulty": 3,
            "hint": "Radio is the longest; X-rays are very short.",
            "explanation": "Longest to shortest: radio -> infrared -> visible -> X-rays, matching increasing frequency
and energy."
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              "reflects red wavelengths and absorbs the rest",
              "is hot",
              "absorbs only red light"
            ],
            "answer": 1,
            "difficulty": 3,
            "hint": "Color is the light that bounces back to you.",
            "explanation": "The apple reflects red wavelengths to your eyes and absorbs the others, so it appears
red."
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              "Visible light",
              "Gamma rays"
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            "answer": 3,
            "difficulty": 2,
            "hint": "Shortest wavelength wins.",
            "explanation": "Gamma rays have the shortest wavelength and highest energy, making them the most
penetrating and hazardous."
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that block visible light.",
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            "explanation": "Infrared’s longer wavelengths penetrate dust, revealing newborn stars hidden from

visible-light telescopes."
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                "no light"
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              "hint": "Think of what sunlight contains.",
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                "sub": "On the airless Moon, the sunlit ground bakes hotter than boiling water while the shadow beside it freezes colder than Antarctica. Temperature is stranger -- and simpler -- than it seems."
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                "markdown": "Cooking, weather, climate, engines, refrigerators, spacecraft design -- all are about **heat**. But two everyday words get mixed up constantly: **temperature** and **heat** are not the same thing.\n\nGet the difference straight and suddenly you understand why a tiny spark can be hotter than a warm bath yet carry far less energy, why metal feels colder than wood at the same temperature, and what scientists mean by ’absolute zero,’ the coldest anything can ever be."
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standstill -- and you reach **absolute zero**, about **-273 degC**, the bottom of the temperature scale. Nothing can be colder, because particles can't move less than 'barely at all.' Scientists use the **Kelvin (K)** scale, which starts at absolute zero (0 K = -273 degC)."
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                    "steps": [
                      "Temperature is the average particle speed: the spark wins easily (1000 vs 40 degC).",
                      "Thermal energy is the total, which also depends on the number of particles.",
                      "The bath has vastly more particles than the tiny spark."
                    ],
                    "answer": "The spark is hotter (higher temperature), but the bath holds far more total thermal energy -- so the bath could burn you while the spark can't."
                  },
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                    "title": "Which way does heat flow?",
                    "problem": "You wrap your warm hands around a cold glass of soda. In which direction does thermal energy flow, and why do your hands feel cold?",
                    "steps": [
                      "Heat always flows from hotter to colder.",
                      "Your hands are warmer than the glass.",
                      "So thermal energy flows out of your hands into the glass."
                    ],
                    "answer": "Heat flows from your hands into the cold glass. Your hands feel cold because they're losing thermal energy -- not because cold is entering them."
                  },
                  {
                    "title": "Converting to Kelvin",
                    "problem": "Absolute zero is about -273 degC. What is a comfortable room temperature of 27 degC on the Kelvin scale?",
                    "steps": [
                      "Kelvin starts at absolute zero: 0 K = -273 degC.",
                      "Add 273 to the Celsius value.",
                      "$27 + 273 = 300$."
                    ],
                    "answer": "300 K. The Kelvin scale just shifts Celsius so that zero is the coldest possible temperature."
                  }
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              }
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              "content": {
                "markdown": "Physicists have spent over a century chasing absolute zero -- and they can get astonishingly close, but never quite arrive. In special labs, atoms have been cooled to **billionths of a degree** above 0 K, colder than anywhere in nature, including deep space (which sits at about 2.7 K, warmed by the Big Bang's afterglow).\n\nWhy bother? Near absolute zero, matter does bizarre things: some metals become **superconductors** that carry electricity with zero resistance, and gases form a strange new state called a **Bose-Einstein condensate** where atoms blur into a single quantum blob.\n\nReaching exactly 0 K turns out to be impossible -- a deep law of thermodynamics forbids it. You can get forever closer, like an asymptote you never touch. The coldest cold is a destination no one can quite reach."
              }
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```
          "All matter is made of constantly moving particles.",
          "Temperature = average particle motion; thermal energy = total energy of all the particles.",
          "A spark is hotter than a bath, but the bath holds far more total thermal energy.",
          "Heat is thermal energy flowing from hot to cold; ’cold’ is just less thermal energy.",
          "Absolute zero (~= -273  degC, 0 K) is the coldest possible temperature."
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        "the average motion of an object’s particles",
        "how much an object weighs",
        "how big an object is"
      ],
      "answer": 1,
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      "explanation": "Temperature reflects the average speed of the particles; faster average motion means a higher temperature."
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      "difficulty": 2,
      "explanation": "Heat always flows from hot to cold on its own. Cold is simply the absence of thermal energy."
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      "options": [
        "The spark is colder",
        "The spark has very few particles, so little total thermal energy",
        "Water can’t burn you",
        "Sparks aren’t real heat"
      ],
      "answer": 1,
      "difficulty": 3,
      "explanation": "Despite its high temperature, the spark holds little total thermal energy because it has so few particles."
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      "answer": {
        "value": 298,
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      "difficulty": 2,
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      "explanation": "$25 + 273 = 298$ K."
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      "prompt": "You hold an ice cube and your hand feels cold. What is actually happening?",
      "options": [
        "Cold flows from the ice into your hand",
        "Thermal energy flows from your hand into the ice",
        "Your hand makes cold",
        "Nothing flows"
      ],
      "answer": 1,
      "difficulty": 3,
      "hint": "Heat flows hot to cold.",
      "explanation": "Thermal energy leaves your warmer hand and flows into the ice; that loss is what feels
```

```
cold."
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energy.",
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              "explanation": "Same temperature means the same average motion, but a larger object has more particles and
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              "options": [
                "move faster",
                "move slower",
                "disappear",
                "get heavier"
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              "answer": 1,
              "difficulty": 2,
              "hint": "Lower temperature, less motion.",
              "explanation": "Cooling slows the particles; cooling toward absolute zero slows them to a near
standstill."
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                "water freezes",
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                "metal melts",
                "the Sun forms"
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              "hint": "It’s the coldest cold.",
              "explanation": "At absolute zero (~= -273  degC, 0 K) particle motion is at its theoretical minimum --
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evenly from one heater? How does the Sun’s heat reach Earth across empty space? And how does a thermos keep cocoa hot
for hours?\n\nEvery answer is one of three ways heat moves: **conduction, convection, and radiation**. Knowing them
explains cooking, home insulation, weather, ocean currents, and how engineers stop a spacecraft from cooking on its Sun
side and freezing on its shadow side."
              }
            },
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                "markdown": "Heat (thermal energy) travels from hot to cold by three methods:\n\n**1. Conduction -- heat
 through direct contact.** Fast-jiggling particles bump into their slower neighbors, passing energy along without the
material itself moving. Touch a hot pan and heat conducts into your hand. **Metals are great conductors** (their
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particles pass energy easily), which is why a metal handle heats up fast and metal feels cold to the touch -- it’s rapidly conducting heat *away* from your hand. **Wood, plastic, air, and foam are insulators** -- poor conductors that slow heat down.\n\n**2. Convection -- heat carried by a moving fluid.** In liquids and gases, heated material expands, becomes less dense, and **rises**; cooler material sinks to take its place, setting up a circulating **convection current**. This is how a pot of water heats throughout, how a room warms from one radiator, and how ocean currents and winds carry heat around the whole planet. (You met this exact engine driving the weather!)\n\n**3. Radiation -- heat as electromagnetic waves.** Hot objects glow with infrared (and, if hot enough, visible) light, carrying energy as waves that need **no material at all**. This is the only way heat crosses the **vacuum of space**, which is how the Sun’s energy reaches Earth. A campfire warms your face by radiation; so does a glowing toaster.\n\n**Beating all three: the thermos.** A vacuum flask keeps a drink hot (or cold) by fighting every path: a **vacuum** layer between two walls stops conduction and convection (no particles to carry heat), and **shiny silvered walls reflect radiation** back in. The same triple defense protects spacecraft and astronauts from wild temperature swings -- multilayer insulation and reflective surfaces are a thermos scaled up for orbit."
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"steps": [
"Spoon handle: heat moves through solid metal by direct contact -- conduction.",
"Sun across space: no medium, so energy travels as waves -- radiation.",
"Rising hot air circulating: a moving fluid carries heat -- convection."
],
"answer": "(a) conduction, (b) radiation, (c) convection -- one example of each of the three methods."
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"title": "Why metal feels colder than wood",
"problem": "A metal bench and a wooden bench have sat outside all night at the same temperature. Why does the metal feel colder when you touch it?",
"steps": [
"Both are at the same temperature, so neither is truly colder.",
"Metal is a great conductor; wood is an insulator.",
"Metal pulls heat from your hand quickly; wood barely does."
],
"answer": "They’re equally cold, but the metal conducts heat away from your hand far faster, so it *feels* colder. ’Feeling cold’ is really ’losing heat quickly.’"
},
{
"title": "How the thermos wins",
"problem": "A thermos keeps cocoa hot for hours. Explain how it blocks all three kinds of heat transfer.",
"steps": [
"A vacuum between its walls has no particles, blocking conduction and convection.",
"Shiny silvered surfaces reflect infrared radiation back toward the drink.",
"With every path blocked, heat escapes only very slowly."
],
"answer": "Vacuum stops conduction and convection; the reflective coating stops radiation. Blocking all three keeps the cocoa hot -- the same trick that thermally protects spacecraft."
}
]
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 reach the crew.\n\nThe Space Shuttle wore over 20,000 individual heat-resistant tiles, each one an insulator so good you could hold its edge moments after its center glowed red-hot. Mastering conduction, convection, and radiation is, quite literally, a matter of life and death."
              }
            },
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                  "Conduction: heat through direct contact (metals conduct well; wood/air/foam insulate).",
                  "Convection: heat carried by a rising-and-sinking fluid (currents in water, air, oceans).",
                  "Radiation: heat as electromagnetic waves -- the only kind that crosses empty space (the Sun).",
                  "’Feeling cold’ from metal is really fast conduction pulling heat from your hand.",
                  "A thermos (and spacecraft insulation) blocks all three: vacuum stops conduction/convection, shiny walls reflect radiation."
                ],
                "formulas": [
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                    "label": "Three paths of heat",
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                  }
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                "convection",
                "conduction",
                "radiation",
                "insulation"
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              "answer": 1,
              "difficulty": 2,
              "explanation": "Conduction transfers heat by particle-to-particle contact, which metals do very well."
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              "scope": "CONCEPT_CHECK",
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              "prompt": "How does the Sun’s heat reach Earth across empty space?",
              "options": [
                "Conduction",
                "Convection",
                "Radiation",
                "It doesn’t"
              ],
              "answer": 2,
              "difficulty": 2,
              "explanation": "Only radiation (electromagnetic waves) can cross a vacuum -- no medium is required."
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              "kind": "TRUE_FALSE",
              "prompt": "Warm air rising and cool air sinking to circulate heat through a room is convection.",
              "answer": true,
              "difficulty": 2,
              "explanation": "Convection is heat carried by a moving fluid as warm, less-dense material rises and cool material sinks."
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              "options": {
                "left": [
                  "Sun warming your face",
                  "Hot handle of a metal pot",
                  "Boiling water churning in a pot",
                  "Foam cup keeping coffee warm"
                ],
                "right": [
                  "Radiation",
                  "Conduction",
                  "Convection",
                  "Insulation (blocks transfer)"
                ]
              },
```

```
              "answer": [
                "Radiation",
                "Conduction",
                "Convection",
                "Insulation (blocks transfer)"
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              "difficulty": 3,
              "explanation": "The Sun radiates, the metal handle conducts, boiling water convects, and a foam cup
insulates by blocking transfer."
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              "prompt": "On a cold morning, a metal railing feels colder than a wooden one at the same temperature
because metal...",
              "options": [
                "is actually colder",
                "conducts heat away from your hand faster",
                "radiates cold",
                "is heavier"
              ],
              "answer": 1,
              "difficulty": 3,
              "hint": "Both are the same temperature.",
              "explanation": "Metal conducts heat out of your hand quickly, so it feels colder even though both are
equally cold."
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              "prompt": "A vacuum is an excellent insulator against conduction and convection because it has almost no
particles to carry heat.",
              "answer": true,
              "difficulty": 3,
              "hint": "What do conduction and convection both need?",
              "explanation": "Both conduction and convection need particles; a vacuum has almost none, so it blocks them
 -- the secret of a thermos."
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              "options": [
                "Copper",
                "Aluminum",
                "Foam (with trapped air)",
                "Iron"
              ],
              "answer": 2,
              "difficulty": 2,
              "hint": "Metals conduct; trapped air does not.",
              "explanation": "Foam traps air, a poor conductor, making it a good insulator; the metals are all good
conductors."
            },
            {
              "scope": "PRACTICE",
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              "prompt": "A shiny, silvered surface on a thermos helps by...",
              "options": [
                "conducting heat faster",
                "reflecting heat radiation back toward the drink",
                "increasing convection",
                "adding weight"
              ],
              "answer": 1,
              "difficulty": 3,
              "hint": "Shiny surfaces reflect infrared.",
              "explanation": "The reflective coating bounces infrared radiation back, cutting heat loss by the one path
a vacuum can’t stop."
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        }
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          "markdown": "**The state of matter depends on how its particles are arranged and how fast they move.**\n\n- **Solid** -- particles packed tightly in a fixed pattern, only vibrating in place. Fixed shape and volume (ice, rock, metal).\n- **Liquid** -- particles still close but free to slide past each other. Fixed volume, but takes the shape of its container (water, oil).\n- **Gas** -- particles far apart, zooming freely. No fixed shape or volume; it fills whatever space it’s in (steam, air).\n- **Plasma** -- a super-heated gas whose atoms have been torn apart into charged pieces. It glows and responds to magnetism. Rare on Earth (lightning, neon signs) but the **most common state in the universe**: stars, including the Sun, are giant balls of plasma.\n\n**Adding energy moves you ’up’ (solid -> liquid -> gas -> plasma); removing it moves you ’down.’** The named changes:\n\n- **Melting** (solid -> liquid) and **freezing** (liquid -> solid).\n- **Boiling/evaporation** (liquid -> gas) and **condensation** (gas -> liquid).\n- **Sublimation** -- solid straight to gas, skipping liquid (dry ice, and ice on a freezing-cold dry day).\n\n**The surprising part: during a change of state, the temperature holds steady.** When ice is melting, all the added energy goes into *breaking the particles loose* from their fixed pattern, not into speeding them up -- so an ice-water mix stays at 0  degC until the last ice melts. This ’hidden’ energy is called **latent heat**.\n\n**This is why steam burns worse than boiling water.** Steam carries all the extra energy that was poured in to turn water into gas. When it touches your skin and condenses back to liquid, it dumps that whole load of latent heat into you -- far more than the same mass of 100  degC water. The same principle, run in reverse, is why **sweating cools you**: evaporating sweat carries energy away from your skin as it turns to gas."
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              "problem": "Name the change of state: (a) a puddle disappearing on a sunny day, (b) dew forming on grass overnight, (c) dry ice ’smoking’ without melting.",
              "steps": [
                "Puddle to invisible vapor: liquid -> gas.",
                "Water vapor in air becoming liquid droplets: gas -> liquid.",
                "Solid carbon dioxide turning straight to gas: solid -> gas."
              ],
              "answer": "(a) evaporation, (b) condensation, (c) sublimation. Adding energy drives (a) and (c); removing it drives (b)."
            },
            {
              "title": "The melting plateau",
              "problem": "You heat a beaker of ice-water steadily. While ice is still melting, the temperature won’t rise above 0  degC. Where is the energy going?",
              "steps": [
                "Added energy can speed particles up (temperature) or break their arrangement (state change).",
                "During melting, it goes into breaking ice’s rigid structure.",
                "Only once all the ice has melted does added energy start raising the temperature."
              ],
              "answer": "Into latent heat -- breaking the solid structure apart. The temperature stays at 0  degC until the last ice melts, then climbs again."
            },
            {
              "title": "Why steam is dangerous",
              "problem": "Both are at 100  degC, yet steam burns far worse than boiling water. Why?",
              "steps": [
                "Turning water to steam required pouring in latent heat.",
                "Steam carries that extra energy.",
                "Landing on skin, it condenses and releases all that latent heat into you."
              ],
              "answer": "Steam holds a large amount of latent heat from boiling. Condensing on your skin, it releases that energy as well as its 100  degC heat, causing a worse burn."
            }
          ]
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            }
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              "intro": "Five problems to phase through."
            }
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            "content": {
              "markdown": "On Earth, solids, liquids, and gases feel like the whole story -- plasma seems exotic,
glimpsed only in lightning, neon signs, and flames. But zoom out, and the picture flips: **over 99% of the ordinary
(visible) matter in the universe is plasma.**\n\nEvery star, including our Sun, is an enormous sphere of plasma -- atoms
 so hot their electrons have been stripped away. The glowing gas of nebulae is plasma. The solar wind streaming past
Earth is plasma, and when it slams into our atmosphere near the poles it lights up the **auroras**.\n\nScientists are
even trying to bottle star-stuff on Earth: **fusion reactors** heat hydrogen plasma to over 100 million  degC -- hotter
than the Sun’s core -- held in place by powerful magnets, chasing the dream of clean energy from the same reaction that
powers the stars."
            }
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                "States of matter depend on particle arrangement and motion: solid, liquid, gas, plasma.",
                "Adding energy goes up the ladder (melting, boiling); removing it goes down (freezing,
condensation).",
                "Sublimation skips liquid (dry ice: solid -> gas).",
                "During a state change the temperature holds steady -- energy goes into latent heat, not speeding
particles up.",
                "Plasma is rare on Earth but the most common state in the universe (stars, including the Sun)."
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              "formulas": [
                {
                  "label": "The state ladder",
                  "tex": "\\text{solid} \\leftrightarrow \\text{liquid} \\leftrightarrow \\text{gas} \\leftrightarrow
\\text{plasma}"
                }
              ]
            }
          }
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            "prompt": "In which state are particles packed in a fixed pattern, only vibrating in place?",
            "options": [
              "Solid",
              "Liquid",
              "Gas",
              "Plasma"
            ],
            "answer": 0,
            "difficulty": 1,
            "explanation": "Solids have particles locked in a fixed arrangement, giving a fixed shape and volume."
          },
          {
            "scope": "CONCEPT_CHECK",
            "kind": "MCQ",
            "prompt": "A solid turning directly into a gas (like dry ice) is called...",
            "options": [
              "melting",
              "condensation",
              "sublimation",
              "freezing"
            ],
            "answer": 2,
            "difficulty": 2,
            "explanation": "Sublimation is the change from solid straight to gas, skipping the liquid state."
          },
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  "prompt": "While ice is melting, the temperature of the ice-water mixture keeps rising steadily.",
  "answer": false,
  "difficulty": 3,
  "explanation": "It holds at 0 degC while melting -- the energy goes into latent heat (breaking the structure), not raising the temperature."
},
{
  "scope": "PRACTICE",
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  "prompt": "Order these states from the LEAST particle energy to the MOST.",
  "options": [
    "Solid",
    "Liquid",
    "Gas",
    "Plasma"
  ],
  "answer": [],
  "difficulty": 2,
  "hint": "Adding energy moves you up the ladder.",
  "explanation": "Least to most energy: solid -> liquid -> gas -> plasma, the order in which adding energy changes the state."
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{
  "scope": "PRACTICE",
  "kind": "MCQ",
  "prompt": "Steam at 100 degC burns worse than water at 100 degC because steam...",
  "options": [
    "is hotter",
    "carries extra latent heat it releases when it condenses on you",
    "moves faster",
    "is acidic"
  ],
  "answer": 1,
  "difficulty": 3,
  "hint": "Think about the energy poured in to boil it.",
  "explanation": "Steam holds the latent heat of vaporization; condensing on skin it dumps that extra energy, worsening the burn."
},
{
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  "kind": "MCQ",
  "prompt": "Most of the ordinary matter in the universe is in which state?",
  "options": [
    "Solid",
    "Liquid",
    "Gas",
    "Plasma"
  ],
  "answer": 3,
  "difficulty": 2,
  "hint": "Think about what stars are made of.",
  "explanation": "Stars are plasma, and they make up over 99% of the visible matter, so plasma is the most common state."
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  "prompt": "Sweating cools you because evaporating sweat carries energy away from your skin as it turns to gas.",
  "answer": true,
  "difficulty": 3,
  "hint": "Evaporation absorbs latent heat.",
  "explanation": "Turning liquid sweat into vapor takes energy (latent heat), drawn from your skin, which cools you."
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    "Fixed shape and volume",
    "Fixed volume, takes its container’s shape",
    "No fixed shape or volume; fills any space",
    "A charged, glowing state"
  ],
  "answer": 2,
  "difficulty": 2,
  "hint": "Gas particles zoom freely.",
  "explanation": "A gas has neither fixed shape nor volume -- its fast, far-apart particles fill whatever

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space they’re in."
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          "slug": "energy-heat-efficiency",
          "title": "Energy, Heat, and Efficiency",
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                "sub": "Energy is never lost -- yet every engine, bulb, and body pays a ’heat tax’ that can never be
fully refunded. This one-way leak is among the deepest laws in physics."
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obsess over **efficiency**? Why does a phone get warm? Why can’t we build a machine that runs forever?\n\nThe answer is
that while energy’s *total* is conserved, in every real transformation some always escapes as **low-grade heat** that
spreads out and becomes hard to use again. Understanding this ’heat tax’ is the key to engines, power plants, climate,
and why **perpetual motion is impossible**."
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            },
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              "content": {
                "markdown": "**Two laws of energy, working together:**\n\n**Law 1 -- energy is conserved.** The total
amount never changes; it only transforms (you saw this with kinetic, potential, chemical, and the rest).\n\n**Law 2 --
energy spreads out, and some always becomes waste heat.** In every real transformation, a portion of the energy turns
into low-grade **thermal energy** that disperses into the surroundings. It’s still ’there’ -- total energy is conserved
-- but it’s spread too thin to do useful work. Heat is the universe’s tax collector, and the tax always gets
paid.\n\n**Efficiency measures how much of the input energy did the useful job:**\n\n$$\\text{efficiency} =
\\frac{\\text{useful energy out}}{\\text{total energy in}} \\times 100\\%$$\n\n- An old incandescent bulb turns only
about **10%** of its electrical energy into light -- the other **90%** is wasted heat (they’re warm to the touch!). LED
bulbs reach **80-90%**, which is why they’ve taken over.\n- A car engine is roughly **25-30%** efficient; most of the
fuel’s energy roars away as heat through the exhaust and radiator.\n- A human body is about **25%** efficient at turning
 food energy into motion -- the rest keeps you warm (handy in winter!).\n\n**Efficiency is always less than 100%.** No
real machine can turn *all* its input into useful output, because some always escapes as heat. This is exactly why a
**perpetual motion machine is impossible**: it would have to recycle energy with no losses forever, and the heat tax
forbids it. Every ’free energy’ machine ever proposed quietly breaks this law.\n\n**The deep idea -- entropy.** Energy
naturally flows from concentrated and useful (the chemical energy in fuel) toward spread-out and useless (warmth
dispersed into the air). This one-way drift toward disorder is called **entropy**, and its steady increase even points
the **arrow of time**: you can unscramble nothing for free. Improving efficiency means fighting this drift -- capturing
more useful energy before the tax takes its cut."
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                    "problem": "A motor takes in 500 J of electrical energy and delivers 350 J of useful mechanical
energy. What is its efficiency?",
                    "steps": [
                      "Use efficiency = useful out / total in x 100%.",
                      "$= \\dfrac{350}{500} \\times 100\\%$.",
                      "$= 70\\%$."
                    ],
                    "answer": "70% efficient. The other 150 J (30%) escaped as waste heat and sound -- the unavoidable
tax."
                  },
                  {
                    "title": "Where the bulb’s energy goes",
                    "problem": "An old incandescent bulb is about 10% efficient at making light. If it draws 60 J of
electrical energy each second, how much becomes light, and what happens to the rest?",
                    "steps": [
                      "Useful light = 10% of 60 J.",
                      "$0.10 \\times 60 = 6$ J of light per second.",
                      "The remaining 54 J becomes heat."
                    ],
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                    "answer": "Only about 6 J/s becomes light; 54 J/s is wasted as heat -- which is why old bulbs are
hot and LEDs (80-90% efficient) replaced them."
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                  {
                    "title": "Why perpetual motion fails",
                    "problem": "An inventor claims a wheel that, once spun, will turn forever and even power a light.
Using the laws of energy, what’s wrong?",
                    "steps": [
                      "Friction and air resistance always convert some motion to waste heat.",
                      "That heat spreads out and can’t be fully recovered.",
                      "So the wheel loses usable energy every cycle and must slow down."
                    ],
                    "answer": "It violates the heat tax (Law 2). Energy is conserved, but losses to heat mean the wheel
can’t sustain itself -- let alone power anything extra. Perpetual motion is impossible."
                  }
                ]
              }
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smoke and clamor of the Industrial Revolution. Engineers building steam engines desperately wanted to know: how much
work can we squeeze from a given amount of burning coal? Is there a limit?\n\nIn 1824 a young French engineer, **Sadi
Carnot**, proved there is. No engine, however cleverly built, can convert all its heat into work -- there’s a hard
ceiling set by the temperatures it runs between. From his analysis grew the **second law of thermodynamics** and the
concept of **entropy**.\n\nThe lesson echoes far beyond engines. Entropy’s relentless increase is why heat won’t flow
uphill on its own, why we age, and why time runs in one direction. A practical question about coal and steam led to one
of the most profound truths in all of science."
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                  "Energy is conserved (total never changes), but every real transformation leaks some as waste heat.",
                  "Efficiency = useful energy out / total energy in x 100%, always less than 100%.",
                  "Examples: old bulbs ~10%, car engines ~25-30%, LEDs ~80-90%.",
                  "Perpetual motion machines are impossible because the ’heat tax’ can never be avoided.",
                  "Entropy -- the spreading of energy toward disorder -- even sets the arrow of time."
                ],
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                    "label": "Efficiency",
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                  }
                ]
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              "answer": true,
              "difficulty": 2,
              "explanation": "Every real transformation loses some energy to low-grade heat that spreads out -- the
unavoidable ’heat tax’."
            },
            {
              "scope": "CONCEPT_CHECK",
              "kind": "MCQ",
              "prompt": "Efficiency is the ratio of...",
              "options": [
                "total energy in to useful energy out",
```

"useful energy out to total energy in",
"heat to light",
"force to distance"
],
"answer": 1,
"difficulty": 2,
"explanation": "Efficiency = useful energy out / total energy in, often written as a percentage."
},
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"options": [
"Magnets are too weak",
"Some energy is always lost to heat, so it can’t sustain itself",
"It would spin too fast",
"Gravity stops it"
],
"answer": 1,
"difficulty": 3,
"explanation": "Unavoidable losses to waste heat mean usable energy drains away each cycle -- it can’t run forever."
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"answer": {
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"tolerance": 0
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"hint": "useful / total x 100.",
"explanation": "$\\dfrac{150}{200} \\times 100 = 75\\%$."
},
{
"scope": "PRACTICE",
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"prompt": "A bulb draws 50 J of electrical energy and emits 5 J as light. What is its efficiency, as a percent?",
"answer": {
"value": 10,
"tolerance": 0
},
"difficulty": 3,
"hint": "Light is the useful output.",
"explanation": "$\\dfrac{5}{50} \\times 100 = 10\\%$ -- typical of an old incandescent bulb; the other 90% is heat."
},
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"options": [
"makes colder light",
"wastes far less energy as heat",
"uses no energy",
"reflects heat away"
],
"answer": 1,
"difficulty": 3,
"hint": "Higher efficiency means less waste heat.",
"explanation": "LEDs convert most of their energy to light (high efficiency), so little is wasted as heat -- they stay cool."
},
{
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"prompt": "When fuel burns in a car engine, most of its energy ends up as useful motion.",
"answer": false,
"difficulty": 3,
"hint": "Car engines are only ~25-30% efficient.",
"explanation": "Only about a quarter to a third becomes motion; most of the fuel’s energy leaves as waste heat through the exhaust and radiator."
},
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                "entropy",
                "momentum",
                "power"
              ],
              "answer": 1,
              "difficulty": 4,
              "hint": "It also points the arrow of time.",
              "explanation": "Entropy measures the spreading toward disorder; its steady increase even sets the
direction of time."
            }
          ]
        }
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      "description": "The physics of motion. Position and displacement, velocity and acceleration, the kinematic
equations, and free fall & projectiles -- taught with the calculus definitions surfaced.",
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          "difficulty": 2,
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                "scene": "earth",
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                "sub": "Before we can talk about how fast or how soon, we need a precise language for where an object is
 and how far it has travelled."
              }
            },
            {
              "kind": "CONTEXT",
              "title": "Why this matters",
              "content": {
                "markdown": "Every GPS fix, every spacecraft trajectory, every self-driving car begins with the same
question: *where is it, and how has that changed?*\n\nIn everyday speech we use \"distance\" loosely. Physics needs two
distinct ideas -- **distance** and **displacement** -- because they answer different questions and, as you’ll see, they
are often **not** the same number."
              }
            },
            {
              "kind": "CONCEPT",
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              "content": {
                "markdown": "**Position** ($x$) is an object’s location measured from a chosen reference point (the
origin) along a chosen axis. Position needs a sign: $+x$ to one side of the origin, $-x$ to the
other.\n\n**Displacement** ($\\Delta x$) is the *change* in position -- how far and in which direction the object ended
up relative to where it started:\n\n$$\\Delta x = x_f - x_i$$\n\nDisplacement is a **vector**: it has both size and
direction (a sign). \n\n**Distance** is the total length of the path actually travelled. Distance is a **scalar**: it
has size only, is never negative, and ignores direction.\n\nThe key consequence:\n\n- If you walk $5\\,\\text{m}$ east,
then $2\\,\\text{m}$ west, your **distance** is $7\\,\\text{m}$ but your **displacement** is $+3\\,\\text{m}$ (east).\n-
 Distance is always **greater than or equal to** the magnitude of displacement: $\\text{distance} \\ge |\\Delta x|$.
They are equal only when motion never reverses direction."
              }
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                    "problem": "Maya starts at $x_i = 0$, walks to $x = +5\\,\\text{m}$, then back to $x_f =
+3\\,\\text{m}$. Find her displacement and the distance she walked.",
                    "steps": [
                      "Displacement uses only start and end: $\\Delta x = x_f - x_i = 3 - 0 = +3\\,\\text{m}$.",
```

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                      "Distance is the whole path: $5\\,\\text{m}$ out $+ 2\\,\\text{m}$ back $= 7\\,\\text{m}$."
                    ],
                    "answer": "Displacement $= +3\\,\\text{m}$; distance $= 7\\,\\text{m}$."
                  },
                  {
                    "title": "One full lap",
                    "problem": "A runner completes exactly one lap of a $400\\,\\text{m}$ track, finishing where they began. Displacement? Distance?",
                    "steps": [
                      "End position equals start position, so $\\Delta x = 0$.",
                      "The path length is one full lap $= 400\\,\\text{m}$."
                    ],
                    "answer": "Displacement $= 0$; distance $= 400\\,\\text{m}$. A big distance can have zero displacement."
                  },
                  {
                    "title": "Choosing a sign",
                    "problem": "Taking east as positive, a car moves from $x_i=+8\\,\\text{m}$ to $x_f=+2\\,\\text{m}$. Which way did it move, and what is its displacement?",
                    "steps": [
                      "$\\Delta x = 2 - 8 = -6\\,\\text{m}$.",
                      "A negative result means the displacement points in the $-x$ direction (west)."
                    ],
                    "answer": "It moved $6\\,\\text{m}$ west; $\\Delta x = -6\\,\\text{m}$."
                  }
                ]
              }
            },
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                    "title": "Distance vs Displacement | Khan-style intro"
                  },
                  {
                    "youtubeId": "9Q4XAB1JjQc",
                    "title": "Position, Distance & Displacement"
                  }
                ]
              }
            },
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              "title": "Quick check",
              "content": {
                "intro": "Vectors vs scalars -- answer carefully."
              }
            },
            {
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              "title": "Practice problems",
              "content": {
                "intro": "Take the positive direction as stated in each problem."
              }
            },
            {
              "kind": "DEEPER_DIVE",
              "title": "Deeper dive: vectors in more than one dimension",
              "content": {
                "markdown": "On a line, direction is just a sign. In two or three dimensions, displacement becomes a true vector $\\vec{\\Delta r} = \\langle \\Delta x, \\Delta y, \\Delta z\\rangle$, and its magnitude comes from the Pythagorean theorem: $|\\vec{\\Delta r}| = \\sqrt{\\Delta x^2 + \\Delta y^2}$. Distance is still the length of the actual (possibly curved) path -- found with calculus as the integral of speed over time. We’ll meet that idea again."
              }
            },
            {
              "kind": "SUMMARY",
              "title": "Summary",
              "content": {
                "takeaways": [
                  "Position $x$ is location from an origin, with a sign for direction.",
                  "Displacement $\\Delta x = x_f - x_i$ is a vector (size + direction).",
                  "Distance is total path length: a non-negative scalar.",
                  "$\\text{distance} \\ge |\\Delta x|$, equal only if motion never reverses."
                ],
                "formulas": [
                  {
                    "label": "Displacement",
                    "tex": "\\Delta x = x_f - x_i"
```

```
          }
        ]
      }
    }
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      "prompt": "Displacement is a vector -- it has a direction.",
      "answer": true,
      "difficulty": 1,
      "explanation": "Yes. Displacement carries a sign (direction); distance does not."
    },
    {
      "scope": "CONCEPT_CHECK",
      "kind": "MCQ",
      "prompt": "You walk $4\\,\\text{m}$ east, then $4\\,\\text{m}$ west. Your displacement is:",
      "options": [
        "8 m",
        "0 m",
        "4 m east",
        "4 m west"
      ],
      "answer": 1,
      "explanation": "You end where you started, so $\\Delta x = 0$ (distance is 8 m)."
    },
    {
      "scope": "CONCEPT_CHECK",
      "kind": "MCQ",
      "prompt": "Compared with the magnitude of displacement, distance is:",
      "options": [
        "always greater than or equal",
        "always equal",
        "always smaller",
        "sometimes negative"
      ],
      "answer": 0,
      "explanation": "Distance $\\ge |\\Delta x|$ always; they’re equal only if direction never reverses."
    },
    {
      "scope": "CONCEPT_CHECK",
      "kind": "MATCHING",
      "prompt": "Match each symbol to what it represents.",
      "options": {
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          "$x$",
          "$\\Delta x$",
          "$x_i$",
          "$x_f$"
        ],
        "right": [
          "Position",
          "Displacement",
          "Initial position",
          "Final position"
        ]
      },
      "answer": [
        "Position",
        "Displacement",
        "Initial position",
        "Final position"
      ],
      "explanation": "$x$ is position, $\\Delta x = x_f - x_i$ is displacement, and $x_i$/$x_f$ are the start
and end positions."
    },
    {
      "scope": "CONCEPT_CHECK",
      "kind": "FILL_BLANK",
      "prompt": "Distance is a ___ (size only), while displacement is a ___ (size and direction).",
      "answer": [
        [
          "scalar"
        ],
        [
          "vector"
        ]
      ],
      "explanation": "Distance is a scalar; displacement is a vector."
    },
    {
```

```
        "scope": "CONCEPT_CHECK",
        "kind": "ORDER",
        "prompt": "Order the steps to find an object’s displacement.",
        "options": [
          "Choose an origin and a positive direction",
          "Record the initial position $x_i$",
          "Record the final position $x_f$",
          "Compute $\\Delta x = x_f - x_i$"
        ],
        "answer": [],
        "explanation": "Set up a reference, read off the start and end positions, then subtract."
      },
      {
        "scope": "CONCEPT_CHECK",
        "kind": "GRAPH",
        "prompt": "Sketch the position-time graph of an object moving at a constant $2\\,\\text{m/s}$ from the
origin -- a straight line through the origin with slope 2. Drag the points to trace it.",
        "answer": {
          "expr": "2*x",
          "domain": [
            0,
            5
          ],
          "variable": "x",
          "mode": "draw",
          "samples": 7
        },
        "difficulty": 3,
        "hint": "At $t = 0$ the position is 0; at $t = 5$ it is 10.",
        "explanation": "Constant velocity gives a straight line $x = 2t$ -- slope 2, through the origin."
      },
      {
        "scope": "PRACTICE",
        "kind": "NUMERIC",
        "prompt": "An object moves from $x_i = 2\\,\\text{m}$ to $x_f = 9\\,\\text{m}$. Displacement in metres?",
        "answer": {
          "value": 7,
          "tolerance": 0
        },
        "hint": "$\\Delta x = x_f - x_i$.",
        "explanation": "$9 - 2 = 7\\,\\text{m}$."
      },
      {
        "scope": "PRACTICE",
        "kind": "NUMERIC",
        "prompt": "Taking right as positive: you walk $6\\,\\text{m}$ right then $10\\,\\text{m}$ left.
Displacement in metres?",
        "answer": {
          "value": -4,
          "tolerance": 0
        },
        "difficulty": 3,
        "hint": "Add with signs: $+6 - 10$.",
        "explanation": "$+6 - 10 = -4\\,\\text{m}$ (4 m to the left)."
      },
      {
        "scope": "PRACTICE",
        "kind": "SYMBOLIC",
        "prompt": "Write displacement $\\Delta x$ as a formula in terms of the final position $x_f$ and the
initial position $x_i$. (Type it like ‘x_f - x_i‘.)",
        "answer": {
          "expr": "x_f - x_i",
          "vars": [
            "x_f",
            "x_i"
          ]
        },
        "difficulty": 2,
        "hint": "Displacement is the *change* in position.",
        "explanation": "$\\Delta x = x_f - x_i$ -- final minus initial position. (Any equivalent form, like $-x_i
+ x_f$, is accepted.)"
      },
      {
        "scope": "PRACTICE",
        "kind": "PROOF",
        "prompt": "Build the argument: if an object returns to where it started, its displacement is zero.",
        "options": [
          "Let the start position be $x_i$ and the end position be $x_f$.",
          "Returning to the start means $x_f = x_i$.",
          "Displacement is defined as $\\Delta x = x_f - x_i$.",
          "Substitute $x_f = x_i$ to get $\\Delta x = x_i - x_i$.",
          "Therefore $\\Delta x = 0$."
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              ],
              "answer": [],
              "difficulty": 4,
              "hint": "Start by naming the positions, end with the conclusion.",
              "explanation": "Naming the positions, using the return condition $x_f=x_i$, and substituting into $\\Delta
 x = x_f - x_i$ gives $\\Delta x = 0$."
            },
            {
              "scope": "PRACTICE",
              "kind": "NUMERIC",
              "prompt": "For that same trip (6 m right, then 10 m left), what total distance did you travel, in
metres?",
              "answer": {
                "value": 16,
                "tolerance": 0
              },
              "hint": "Distance ignores direction.",
              "explanation": "$6 + 10 = 16\\,\\text{m}$."
            },
            {
              "scope": "PRACTICE",
              "kind": "MCQ",
              "prompt": "A car drives $3\\,\\text{km}$ north then $3\\,\\text{km}$ south. Its displacement is:",
              "options": [
                "6 km",
                "0 km",
                "3 km north",
                "3 km south"
              ],
              "answer": 1,
              "hint": "Where does it end up?",
              "explanation": "It returns to the start, so displacement is 0 (distance is 6 km)."
            },
            {
              "scope": "PRACTICE",
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              "prompt": "It is possible to travel zero distance but have a non-zero displacement.",
              "answer": false,
              "hint": "Can you change position without moving along a path?",
              "explanation": "No. Any change in position requires travelling some path, so distance $\\ge |\\Delta x|$."
            }
          ]
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          "title": "Velocity & Speed",
          "tagline": "How fast -- and in which direction",
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          "xpReward": 150,
          "difficulty": 3,
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                "scene": "earth",
                "headline": "Rates of Change",
                "sub": "Speed tells you how fast. Velocity tells you how fast and which way. The difference is the whole
 game in physics."
              }
            },
            {
              "kind": "CONTEXT",
              "title": "Why this matters",
              "content": {
                "markdown": "A weather report saying \"wind 30 km/h\" is almost useless to a pilot -- they need the
*direction* too. That’s the difference between **speed** (a scalar) and **velocity** (a vector). Getting this
distinction right is the foundation for everything from projectile motion to orbital mechanics."
              }
            },
            {
              "kind": "CONCEPT",
              "title": "The big idea",
              "content": {
                "markdown": "**Average velocity** is displacement per unit time:\n\n$$\\bar v = \\frac{\\Delta
x}{\\Delta t}$$\n\nIt is a vector -- its sign is the sign of the displacement. Units are metres per second,
$\\text{m/s}$.\n\n**Average speed** is total distance per unit time:\n\n$$\\text{average speed} =
\\frac{\\text{distance}}{\\Delta t}$$\n\nIt is a scalar and never negative.\n\nThe **instantaneous velocity** is the
velocity at a single instant -- what a speedometer (plus a direction) reads. It is the limit of average velocity over a
vanishingly small time interval, i.e. the derivative of position with respect to time:\n\n$$v = \\lim_{\\Delta t \\to
0}\\frac{\\Delta x}{\\Delta t} = \\frac{dx}{dt}$$\n\nThat calculus definition is the bridge between this algebra-based
course and the full university treatment. **Instantaneous speed** is just the magnitude $|v|$.\n\n**Unit conversion**
```

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matters constantly. To convert $\\text{km/h}$ to $\\text{m/s}$, multiply by $\\tfrac{1000}{3600} = \\tfrac{1}{3.6}$. So
$72\\,\\text{km/h} = 72/3.6 = 20\\,\\text{m/s}$."
              }
            },
            {
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              "title": "Try it: motion grapher",
              "content": {
                "simId": "motion-graphs",
                "intro": "Set an initial velocity and zero acceleration to see constant-velocity motion: a straight,
sloped position-time line and a flat velocity line. The slope of $x(t)$ is the velocity."
              }
            },
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                    "title": "Straight-line sprint",
                    "problem": "A runner covers $100\\,\\text{m}$ in a straight line in $5.0\\,\\text{s}$. Average
velocity?",
                    "steps": [
                      "Motion is straight and one-way, so $\\Delta x = 100\\,\\text{m}$.",
                      "$\\bar v = \\Delta x/\\Delta t = 100/5.0 = 20\\,\\text{m/s}$."
                    ],
                    "answer": "$\\bar v = 20\\,\\text{m/s}$ in the direction of running."
                  },
                  {
                    "title": "Velocity vs speed on a round trip",
                    "problem": "From the lesson before: a runner does one $400\\,\\text{m}$ lap in $50\\,\\text{s}$,
finishing at the start. Average speed? Average velocity?",
                    "steps": [
                      "Average speed $= \\text{distance}/\\Delta t = 400/50 = 8\\,\\text{m/s}$.",
                      "Displacement is $0$, so $\\bar v = 0/50 = 0\\,\\text{m/s}$."
                    ],
                    "answer": "Average speed $= 8\\,\\text{m/s}$; average velocity $= 0$."
                  },
                  {
                    "title": "Converting units",
                    "problem": "A car’s speed is $90\\,\\text{km/h}$. Express it in $\\text{m/s}$.",
                    "steps": [
                      "Divide by $3.6$: $90 / 3.6 = 25$."
                    ],
                    "answer": "$25\\,\\text{m/s}$."
                  }
                ]
              }
            },
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                    "title": "Speed and Velocity"
                  }
                ]
              }
            },
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              "title": "Quick check",
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                "intro": "Mind the difference between speed and velocity."
              }
            },
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              "content": {
                "intro": "Use $1\\,\\text{m/s} = 3.6\\,\\text{km/h}$ where needed."
              }
            },
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              "content": {
                "markdown": "On a **position-time graph**, the slope of the line at a point *is* the instantaneous
velocity, $v = dx/dt$. A steeper line means faster motion; a downward slope means negative velocity; a flat line means
at rest. This geometric reading of a derivative is one of the most useful skills in all of physics -- and it works in
```

reverse too: the **area under a velocity-time graph** equals displacement, which is the integral $\\Delta x = \\int v\\,dt$."
              }
            },
            {
              "kind": "SUMMARY",
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              "content": {
                "takeaways": [
                  "Average velocity $\\bar v = \\Delta x/\\Delta t$ (vector).",
                  "Average speed = distance/time (scalar, never negative).",
                  "Instantaneous velocity is the derivative $v = dx/dt$ -- the slope of $x(t)$.",
                  "Convert km/h to m/s by dividing by 3.6."
                ],
                "formulas": [
                  {
                    "label": "Average velocity",
                    "tex": "\\bar v = \\dfrac{\\Delta x}{\\Delta t}"
                  },
                  {
                    "label": "Instantaneous velocity",
                    "tex": "v = \\dfrac{dx}{dt}"
                  }
                ]
              }
            }
          ],
          "questions": [
            {
              "scope": "CONCEPT_CHECK",
              "kind": "MCQ",
              "prompt": "Average velocity is defined as:",
              "options": [
                "distance / time",
                "$\\Delta x / \\Delta t$",
                "$\\Delta t / \\Delta x$",
                "speed x time"
              ],
              "answer": 1,
              "explanation": "Average velocity is displacement over time, $\\Delta x/\\Delta t$."
            },
            {
              "scope": "CONCEPT_CHECK",
              "kind": "TRUE_FALSE",
              "prompt": "Speed is the magnitude of velocity.",
              "answer": true,
              "explanation": "Correct -- speed is $|v|$, dropping the direction."
            },
            {
              "scope": "CONCEPT_CHECK",
              "kind": "MCQ",
              "prompt": "$72\\,\\text{km/h}$ equals how many $\\text{m/s}$?",
              "options": [
                "20",
                "72",
                "7.2",
                "200"
              ],
              "answer": 0,
              "explanation": "$72 / 3.6 = 20\\,\\text{m/s}$."
            },
            {
              "scope": "PRACTICE",
              "kind": "NUMERIC",
              "prompt": "A car travels $150\\,\\text{m}$ in a straight line in $6.0\\,\\text{s}$. Average speed in m/s?",
              "answer": {
                "value": 25,
                "tolerance": 0
              },
              "hint": "distance / time.",
              "explanation": "$150/6.0 = 25\\,\\text{m/s}$."
            },
            {
              "scope": "PRACTICE",
              "kind": "NUMERIC",
              "prompt": "An object’s displacement is $-40\\,\\text{m}$ over $8.0\\,\\text{s}$. Average velocity in m/s?",
              "answer": {
                "value": -5,
                "tolerance": 0
              },

```
              "hint": "Keep the sign: $\\Delta x/\\Delta t$.",
              "explanation": "$-40/8.0 = -5\\,\\text{m/s}$."
            },
            {
              "scope": "PRACTICE",
              "kind": "MCQ",
              "prompt": "Instantaneous velocity is the:",
              "options": [
                "derivative $dx/dt$",
                "derivative $dv/dt$",
                "product $x\\cdot t$",
                "ratio $\\Delta t/\\Delta x$"
              ],
              "answer": 0,
              "hint": "Rate of change of position.",
              "explanation": "$v = dx/dt$."
            },
            {
              "scope": "PRACTICE",
              "kind": "NUMERIC",
              "prompt": "Convert $36\\,\\text{km/h}$ to $\\text{m/s}$.",
              "answer": {
                "value": 10,
                "tolerance": 0
              },
              "hint": "Divide by 3.6.",
              "explanation": "$36/3.6 = 10\\,\\text{m/s}$."
            },
            {
              "scope": "PRACTICE",
              "kind": "TRUE_FALSE",
              "prompt": "If your average velocity over a trip is exactly zero, you ended where you started.",
              "answer": true,
              "hint": "Zero average velocity means zero displacement.",
              "explanation": "$\\bar v = 0 \\Rightarrow \\Delta x = 0$, so start and end coincide."
            },
            {
              "scope": "PRACTICE",
              "kind": "GRAPH",
              "prompt": "On a position-time graph, an object starts at the origin and moves at a constant 
$2\\,\\text{m/s}$. Enter its position $x$ as a function of time $t$.",
              "answer": {
                "expr": "2*t",
                "domain": [
                  0,
                  5
                ],
                "variable": "t"
              },
              "difficulty": 3,
              "hint": "Position = velocity x time when starting at the origin.",
              "explanation": "$x(t) = 2t$ -- a straight line through the origin with slope equal to the velocity."
            }
          ]
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          "title": "Acceleration",
          "tagline": "How quickly velocity itself changes",
          "estMinutes": 14,
          "xpReward": 150,
          "sections": [
            {
              "kind": "HERO",
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                "scene": "earth",
                "headline": "Changing Velocity",
                "sub": "A sports car’s bragging right -- 0 to 100 km/h in 3 seconds -- is a statement about 
acceleration: how fast the velocity changes."
              }
            },
            {
              "kind": "CONTEXT",
              "title": "Why this matters",
              "content": {
                "markdown": "Whenever an object speeds up, slows down, or changes direction, it is **accelerating**. 
Acceleration is what forces produce (next course: Newton’s laws), and it is the quantity that makes motion *interesting*
 -- constant velocity is easy; changing velocity is where the physics lives."
              }
            },
            {
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```

```
              "title": "The big idea",
              "content": {
                "markdown": "**Average acceleration** is the change in velocity per unit time:\n\n$$\\bar a =
\\frac{\\Delta v}{\\Delta t} = \\frac{v_f - v_i}{\\Delta t}$$\n\nUnits are metres per second per second,
$\\text{m/s}^2$.\n\nThe **instantaneous acceleration** is the derivative of velocity -- and therefore the second
derivative of position:\n\n$$a = \\frac{dv}{dt} = \\frac{d^2x}{dt^2}$$\n\n**Sign and direction.** Acceleration is a
vector. A common trap: *negative acceleration does not always mean \"slowing down.\"* What matters is the acceleration’s
 direction **relative to the velocity**:\n\n- If $a$ points the **same way** as $v$, the object speeds up.\n- If $a$
points **opposite** to $v$, the object slows down (this is what we usually call *deceleration*).\n\nSo a ball moving in
the $-x$ direction ($v<0$) with $a<0$ is actually speeding up."
              }
            },
            {
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              "title": "Try it: motion grapher",
              "content": {
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                "intro": "Give it an initial velocity and a non-zero acceleration. Watch the velocity line tilt (its
slope is $a$) and the position curve bend into a parabola. Try opposite signs of $v_0$ and $a$ to see slowing down."
              }
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                    "problem": "A car goes from rest to $30\\,\\text{m/s}$ in $6.0\\,\\text{s}$. Average acceleration?",
                    "steps": [
                      "$\\Delta v = 30 - 0 = 30\\,\\text{m/s}$.",
                      "$\\bar a = \\Delta v/\\Delta t = 30/6.0 = 5.0\\,\\text{m/s}^2$."
                    ],
                    "answer": "$\\bar a = 5.0\\,\\text{m/s}^2$."
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acceleration.",
                    "steps": [
                      "$\\Delta v = 0 - 20 = -20\\,\\text{m/s}$.",
                      "$\\bar a = -20/4.0 = -5.0\\,\\text{m/s}^2$."
                    ],
                    "answer": "$\\bar a = -5.0\\,\\text{m/s}^2$ -- negative because it opposes the motion."
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                    "problem": "A puck moves in the $-x$ direction at $v_i = -2\\,\\text{m/s}$ and reaches $v_f =
-8\\,\\text{m/s}$ in $3\\,\\text{s}$. Is it speeding up or slowing down?",
                    "steps": [
                      "$\\bar a = (-8 - (-2))/3 = -6/3 = -2\\,\\text{m/s}^2$.",
                      "$a$ and $v$ are both negative -- same direction -- so the speed $|v|$ increases."
                    ],
                    "answer": "Speeding up, with $\\bar a = -2\\,\\text{m/s}^2$."
                  }
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```

```
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\\xrightarrow{\\;d/dt\\;} v(t) \\xrightarrow{\\;d/dt\\;} a(t)$$\n\nDifferentiate position to get velocity, differentiate
 again to get acceleration. Run it backwards with integration: $v = \\int a\\,dt$ and $x = \\int v\\,dt$. When $a$ is
**constant**, doing those two integrals produces exactly the kinematic equations you’ll meet in the next lesson."
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              "Instantaneous acceleration $a = dv/dt = d^2x/dt^2$.",
              "Acceleration is a vector; compare its direction to the velocity.",
              "Same direction => speeding up; opposite => slowing down."
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              },
              {
                "label": "Instantaneous acceleration",
                "tex": "a = \\dfrac{dv}{dt} = \\dfrac{d^2x}{dt^2}"
              }
            ]
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            "$\\text{m/s}^2$",
            "m",
            "s"
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$\\text{m/s}^2$?",
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          "explanation": "Exactly -- $a = dv/dt$."
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          "prompt": "A rocket goes $0 \\to 100\\,\\text{m/s}$ in $20\\,\\text{s}$. Acceleration in
$\\text{m/s}^2$?",
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          "hint": "$\\Delta v/\\Delta t$.",
          "explanation": "$100/20 = 5\\,\\text{m/s}^2$."
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        {
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$\\text{m/s}^2$ (with sign)?",
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                "tolerance": 0
              },
              "hint": "$\\Delta v$ is negative.",
              "explanation": "$(0-30)/6 = -5\\,\\text{m/s}^2$."
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                "$dx/dt$",
                "$\\int v\\,dt$",
                "$x/t$"
              ],
              "answer": 0,
              "hint": "Differentiate position twice.",
              "explanation": "Acceleration is the second derivative of position."
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              "prompt": "Negative acceleration always means an object is slowing down.",
              "answer": false,
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              "explanation": "If velocity is also negative, a negative acceleration speeds the object up."
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              "prompt": "A cyclist accelerates at $2.5\\,\\text{m/s}^2$ from $5$ to $15\\,\\text{m/s}$. How many seconds
 does it take?",
              "answer": {
                "value": 4,
                "tolerance": 0
              },
              "hint": "$t = \\Delta v / a$.",
              "explanation": "$\\Delta v = 10$, $t = 10/2.5 = 4\\,\\text{s}$."
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-- position, velocity, and time."
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              "content": {
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 steady push. For these, you don’t need calculus on every problem -- four **kinematic equations** package the answers.
Master choosing the right one and you can solve the vast majority of intro-mechanics problems."
              }
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$v$, and displacement $\\Delta x$ over time $t$:\n\n$$v = v_0 + at$$\n$$\\Delta x = v_0 t + \\tfrac{1}{2}a t^2$$\n$$v^2
= v_0^2 + 2a\\,\\Delta x$$\n$$\\Delta x = \\tfrac{1}{2}(v_0 + v)\\,t$$\n\n**How to choose.** Each equation is missing
exactly one of the five quantities $\\{v_0, v, a, t, \\Delta x\\}$. Identify what you know and what you want, then pick
the equation that doesn’t involve the variable you neither know nor need:\n\n- No $\\Delta x$? Use $v = v_0 + at$.\n- No
 $v$? Use $\\Delta x = v_0 t + \\tfrac12 a t^2$.\n- No $t$? Use $v^2 = v_0^2 + 2a\\,\\Delta x$.\n- No $a$? Use $\\Delta
x = \\tfrac12(v_0+v)t$ (displacement = average velocity x time).\n\n**Where they come from (calculus).** Starting from
constant $a$: integrate once, $v = \\int a\\,dt = v_0 + at$. Integrate again, $\\Delta x = \\int v\\,dt = v_0 t +
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\\tfrac12 a t^2$. The other two follow by eliminating $t$ algebraically. That’s the whole derivation."
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\\tfrac12 a t^2$ and the velocity line is $v = v_0 + at$. Read values off the live readout and check them against the
equations."
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for $t = 5.0\\,\\text{s}$. Find $v$ and $\\Delta x$.",
                    "steps": [
                      "$v = v_0 + at = 10 + 2.0\\times 5.0 = 20\\,\\text{m/s}$.",
                      "$\\Delta x = v_0 t + \\tfrac12 a t^2 = 10(5) + \\tfrac12(2)(5^2) = 50 + 25 = 75\\,\\text{m}$."
                    ],
                    "answer": "$v = 20\\,\\text{m/s}$, $\\Delta x = 75\\,\\text{m}$."
                  },
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                    "title": "No time given",
                    "problem": "Starting from rest, a sled accelerates at $a = 3.0\\,\\text{m/s}^2$ over $\\Delta x =
24\\,\\text{m}$. Final speed?",
                    "steps": [
                      "No $t$, so use $v^2 = v_0^2 + 2a\\,\\Delta x$.",
                      "$v^2 = 0 + 2(3.0)(24) = 144$, so $v = \\sqrt{144} = 12\\,\\text{m/s}$."
                    ],
                    "answer": "$v = 12\\,\\text{m/s}$."
                  },
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                    "problem": "A car braking from $20\\,\\text{m/s}$ stops in $40\\,\\text{m}$. Find $a$, then the
stopping time.",
                    "steps": [
                      "No $t$: $v^2 = v_0^2 + 2a\\,\\Delta x \\Rightarrow 0 = 20^2 + 2a(40)$.",
                      "$400 = -80a \\Rightarrow a = -5.0\\,\\text{m/s}^2$.",
                      "Now use $v = v_0 + at$: $0 = 20 + (-5)t \\Rightarrow t = 4.0\\,\\text{s}$."
                    ],
                    "answer": "$a = -5.0\\,\\text{m/s}^2$; stopping time $= 4.0\\,\\text{s}$."
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              }
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                  "markdown": "These four equations are valid **only** for constant $a$. If acceleration varies with time,
 you must go back to calculus: $v(t) = v_0 + \\int_0^t a(t’)\\,dt’$ and $x(t) = x_0 + \\int_0^t v(t’)\\,dt’$. Air
resistance, springs (where $a$ depends on position), and rockets (changing mass) all break the constant-$a$ assumption
-- and lead to the richer methods of later mechanics courses."
                }
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                    "Each omits one variable -- choose the one missing your unknown/unneeded quantity.",
                    "They come from integrating constant $a$ twice.",
                    "$\\Delta x = \\tfrac12(v_0+v)t$ says displacement = average velocity x time."
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                      "tex": "v = v_0 + at"
                    },
                    {
                      "label": "Position-time",
                      "tex": "\\Delta x = v_0 t + \\tfrac{1}{2}a t^2"
                    },
                    {
                      "label": "Time-free",
                      "tex": "v^2 = v_0^2 + 2a\\,\\Delta x"
                    },
                    {
                      "label": "Average-velocity",
                      "tex": "\\Delta x = \\tfrac{1}{2}(v_0+v)t"
                    }
                  ]
                }
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                "options": [
                  "$v = v_0 + at$",
                  "$v^2 = v_0^2 + 2a\\,\\Delta x$",
                  "$\\Delta x = v_0 t + \\tfrac12 a t^2$",
                  "$\\Delta x = \\tfrac12(v_0+v)t$"
                ],
                "answer": 1,
                "explanation": "$v^2 = v_0^2 + 2a\\,\\Delta x$ contains no $t$."
              },
              {
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                "prompt": "The kinematic equations assume the acceleration is constant.",
                "answer": true,
                "explanation": "Yes -- they are derived for constant $a$."
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                  "tolerance": 0
                },
                "explanation": "$v = 0 + 4\\times 3 = 12\\,\\text{m/s}$."
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              {
                "scope": "PRACTICE",
                "kind": "NUMERIC",
                "prompt": "From rest, $a = 2.0\\,\\text{m/s}^2$ for $t = 5.0\\,\\text{s}$. Displacement in metres?",
                "answer": {
                  "value": 25,
                  "tolerance": 0
                },
                "hint": "$\\Delta x = \\tfrac12 a t^2$ (since $v_0=0$).",
                "explanation": "$\\tfrac12(2)(25) = 25\\,\\text{m}$."
              },
              {
                "scope": "PRACTICE",
                "kind": "NUMERIC",
```

```
              "prompt": "$v_0 = 8\\,\\text{m/s}$, $a = 0$, $t = 6\\,\\text{s}$. Displacement in metres?",
              "answer": {
                "value": 48,
                "tolerance": 0
              },
              "hint": "No acceleration => constant velocity.",
              "explanation": "$\\Delta x = v_0 t = 8\\times 6 = 48\\,\\text{m}$."
            },
            {
              "scope": "PRACTICE",
              "kind": "NUMERIC",
              "prompt": "From rest, $a = 3.0\\,\\text{m/s}^2$ over $\\Delta x = 24\\,\\text{m}$. Final speed in m/s?",
              "answer": {
                "value": 12,
                "tolerance": 0
              },
              "hint": "$v^2 = 2a\\,\\Delta x$.",
              "explanation": "$v^2 = 2(3)(24) = 144 \\Rightarrow v = 12\\,\\text{m/s}$."
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              "kind": "NUMERIC",
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$\\text{m/s}^2$ (with sign)?",
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                "tolerance": 0
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              "hint": "$0 = v_0^2 + 2a\\Delta x$.",
              "explanation": "$0 = 400 + 80a \\Rightarrow a = -5\\,\\text{m/s}^2$."
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                "final",
                "initial",
                "instantaneous"
              ],
              "answer": 0,
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              "explanation": "It’s displacement = average velocity x time."
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                "sub": "Drop a ball or fire it sideways -- gravity treats both the same. Projectile motion is just two
kinematics problems running at once."
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application of kinematics. The deep insight -- first nailed by Galileo -- is that **horizontal and vertical motion are
independent**: gravity only affects the vertical part."
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                "markdown": "Near Earth’s surface, gravity gives every object the same downward acceleration, the
**acceleration due to gravity**:\n\n$$g \\approx 9.8\\,\\text{m/s}^2 \\;(\\text{downward})$$\n\n**Free fall** is motion
under gravity alone (no air resistance). Taking *up* as positive, the acceleration is $a = -g$, and the kinematic
equations apply directly. For an object **dropped from rest** ($v_0 = 0$):\n\n$$v = gt, \\qquad h = \\tfrac{1}{2}g t^2,
\\qquad t = \\sqrt{\\tfrac{2h}{g}}$$\n\nCrucially, in a vacuum a hammer and a feather fall **together** -- mass does not
 affect the rate of fall.\n\n**Projectile motion** is two-dimensional motion under gravity. Split it into independent
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axes:\n\n- **Horizontal ($x$):** no acceleration, so velocity is constant: $x = v_{0x}\\,t$.\n- **Vertical ($y$):**
acceleration $-g$, so ordinary free-fall kinematics: $y = v_{0y}t - \\tfrac12 g t^2$.\n\nThe two share only **one**
thing: the **time** $t$. That’s the link you use to solve them. At the **top** of a trajectory the vertical velocity is
momentarily zero (the horizontal velocity never changes)."
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9.8\\,\\text{m/s}^2$) the times match $t = \\sqrt{2h/g}$ -- check it against the readout."
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9.8\\,\\text{m/s}^2$, find its speed and how far it falls.",
                    "steps": [
                      "$v = gt = 9.8\\times 2.0 = 19.6\\,\\text{m/s}$.",
                      "$h = \\tfrac12 g t^2 = \\tfrac12 (9.8)(2.0^2) = 19.6\\,\\text{m}$."
                    ],
                    "answer": "Speed $\\approx 19.6\\,\\text{m/s}$; distance $\\approx 19.6\\,\\text{m}$."
                  },
                  {
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                    "problem": "A ball is thrown upward at $v_0 = 14.7\\,\\text{m/s}$. How long until it reaches its
highest point? ($g = 9.8$)",
                    "steps": [
                      "At the top, $v = 0$. Use $v = v_0 - g t$.",
                      "$0 = 14.7 - 9.8\\,t \\Rightarrow t = 14.7/9.8 = 1.5\\,\\text{s}$."
                    ],
                    "answer": "$t = 1.5\\,\\text{s}$."
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10\\,\\text{m/s}^2$, how long until it lands?",
                    "steps": [
                      "Vertical motion sets the time; it starts with $v_{0y}=0$, so $h = \\tfrac12 g t^2$.",
                      "$45 = \\tfrac12 (10) t^2 = 5t^2 \\Rightarrow t^2 = 9 \\Rightarrow t = 3\\,\\text{s}$.",
                      "(Horizontal speed doesn’t affect the fall time.)"
                    ],
                    "answer": "$t = 3\\,\\text{s}$."
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                    "title": "Projectile Motion"
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                    "title": "Free Fall & the Feather/Hammer drop"
                  }
                ]
              }
            },
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              }
            },
            {
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              "content": {
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          "content": {
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(ignoring air) is $R = \\dfrac{v_0^2 \\sin(2\\theta)}{g}$, which is maximised at $\\theta = 45^\\circ$. Real projectiles
 feel **air resistance**, which lowers the optimal angle and shortens the range -- one reason a long jumper or a golf
ball doesn’t quite follow the ideal parabola. Modelling that drag requires the calculus-based methods of later
mechanics."
          }
        },
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              "Near Earth, gravity gives $a = g \\approx 9.8\\,\\text{m/s}^2$ downward, independent of mass.",
              "Free fall is kinematics with $a = -g$ (up positive).",
              "Projectile motion = independent horizontal (constant $v$) and vertical (free fall) motions.",
              "Horizontal and vertical share only the time $t$; at the peak $v_y = 0$."
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                "label": "Projectile (horizontal)",
                "tex": "x = v_{0x}\\,t"
              },
              {
                "label": "Range (ground level)",
                "tex": "R = \\dfrac{v_0^2 \\sin 2\\theta}{g}"
              }
            ]
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9.8\\,\\text{m/s}^2$), in m/s?",
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          "explanation": "$v = gt = 9.8\\times 3.0 = 29.4\\,\\text{m/s}$."
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          "explanation": "Gravity acts only vertically; the horizontal motion is unaffected."
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            "$98\\,\\text{m/s}^2$",
            "$0$"
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          "answer": 0,
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```

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              "$9.8\\,\\text{m/s}$",
              "v_0"
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            "hint": "It’s momentarily neither rising nor falling.",
            "explanation": "Vertical velocity is zero at the peak (then it reverses)."
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            "hint": "$45 = \\tfrac12 g t^2$.",
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                "markdown": "For two thousand years people believed motion needed a constant cause -- that a moving
object naturally slows and stops. Galileo and then Newton overturned this: objects slow down because of **friction**,
not because stopping is their natural state. Remove the friction and motion persists forever. This single idea launched
modern physics."
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motion at **constant velocity** (same speed, same straight-line direction), **unless acted on by a net external
force.**\n\nThe tendency of an object to resist changes to its motion is called **inertia**, and the measure of inertia
is the object’s **mass**. More mass => more inertia => harder to start, stop, or turn.\n\nThe crucial phrase is *net*
force. Forces are vectors; what matters is their sum:\n\n$$\\vec F_{net} = \\sum \\vec F$$\n\nWhen $\\vec F_{net} = 0$,
the object is in **equilibrium**: it stays at rest or keeps moving at constant velocity. A book on a table is in
equilibrium -- gravity pulls it down, the table’s normal force pushes up with equal strength, and they cancel.\n\nSo
constant velocity (including zero) needs **no** net force. Only a *change* in velocity -- acceleration -- requires one."
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                    "steps": [
                      "It isn’t accelerating (it’s at rest), so by the first law the net force must be zero.",
                      "Gravity (down) is exactly balanced by the table’s normal force (up)."
                    ],
                    "answer": "Zero net force -- the forces are balanced (equilibrium)."
                  },
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                    "problem": "A car brakes suddenly and the passengers lurch forward. Why?",
                    "steps": [
                      "Before braking, passengers move forward at the car’s speed.",
                      "By inertia, they tend to keep moving forward when the car decelerates.",
                      "The seatbelt supplies the backward force needed to change their motion."
                    ],
                    "answer": "Inertia keeps the passengers moving; the seatbelt provides the force to stop them."
                  },
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                    "title": "Puck on ice",
                    "problem": "A hockey puck glides across nearly frictionless ice. What keeps it moving?",
                    "steps": [
                      "Nothing needs to keep it moving -- that’s the first law.",
                      "With negligible friction and no net force, it continues at constant velocity."
                    ],
                    "answer": "No force is needed; it coasts at constant velocity until friction or a wall acts."
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                ]
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              "Objects keep their velocity (including rest) unless a net force acts.",
              "Inertia is resistance to changes in motion; mass measures it.",
              "What matters is the net (vector sum) force.",
              "Zero net force => equilibrium (rest or constant velocity)."
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          "explanation": "Constant velocity means no acceleration, so the net force is zero."
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            "shape",
            "colour"
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            "upward",
            "forward"
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          "answer": 0,
          "hint": "Is it accelerating?",
          "explanation": "It’s in equilibrium -- gravity and normal force cancel."
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        {
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              "prompt": "On perfectly frictionless ice, a sliding puck would keep moving at constant velocity indefinitely.",
              "answer": true,
              "hint": "First law with no net force.",
              "explanation": "With no net force, motion continues unchanged."
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                "a tennis ball",
                "a feather"
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              "answer": 0,
              "hint": "More mass, more inertia.",
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              "hint": "Recall the puck on ice.",
              "explanation": "Constant velocity needs zero net force; force is only needed to change velocity."
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                "friction",
                "gravity",
                "the engine"
              ],
              "answer": 0,
              "hint": "They tend to keep moving.",
              "explanation": "Inertia keeps them moving forward until a force (seatbelt) stops them."
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                "markdown": "Newton’s second law is the workhorse of classical physics. It predicts how rockets launch, how cars stop, how planets orbit, and how a tossed ball arcs. Give it the net force and the mass, and it hands you the acceleration -- and from kinematics, the entire future of the motion."
              }
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              }
```

```
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same force -- read the live $a = F/m$ value and confirm."
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                    "It points in the direction of the force."
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                    "$F_{net} = m a = 1500 \\times 2.0 = 3000\\,\\text{N}$."
                  ],
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                },
                {
                  "title": "Net of two forces",
                  "problem": "A $2.0\\,\\text{kg}$ box is pushed right with $10\\,\\text{N}$ while friction pulls left
 with $4\\,\\text{N}$. Find its acceleration.",
                  "steps": [
                    "Net force: $10 - 4 = 6\\,\\text{N}$ to the right.",
                    "$a = F_{net}/m = 6/2.0 = 3.0\\,\\text{m/s}^2$ to the right."
                  ],
                  "answer": "$a = 3.0\\,\\text{m/s}^2$ to the right."
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 force equals the rate of change of momentum,\n\n$$\\vec F_{net} = \\frac{d\\vec p}{dt}.$$\n\nWhen mass is constant this
 becomes $\\vec F = m\\dfrac{d\\vec v}{dt} = m\\vec a$ -- our familiar form. But the momentum version is more general:
it correctly handles systems whose **mass changes**, like a rocket burning fuel, and it survives into relativity where
$m\\vec a$ does not."
```

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                "Acceleration is in the direction of the net force.",
                "$1\\,\\text{N} = 1\\,\\text{kg*m/s}^2$.",
                "Start with a free-body diagram to find the net force."
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                {
                  "label": "Momentum form",
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              "$F = mv$",
              "$a = Fm$",
              "$F = m/a$"
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$\\text{m/s}^2$?",
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            "explanation": "By definition of the newton."
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              "tolerance": 0
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            "hint": "$a = F/m$.",
            "explanation": "$12/3 = 4\\,\\text{m/s}^2$."
          },
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            "prompt": "A $5.0\\,\\text{kg}$ object accelerates at $2.0\\,\\text{m/s}^2$. Net force in newtons?",
            "answer": {
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              "tolerance": 0
            },
            "hint": "$F = ma$.",
            "explanation": "$5 \\times 2 = 10\\,\\text{N}$."
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            "prompt": "A net force of $50\\,\\text{N}$ produces $a = 10\\,\\text{m/s}^2$. Find the mass in kg.",
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            "hint": "$m = F/a$.",
            "explanation": "$50/10 = 5\\,\\text{kg}$."
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              "is unchanged",
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            "answer": 0,
            "hint": "$a \\propto F$.",
            "explanation": "Acceleration is proportional to net force."
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            "scope": "PRACTICE",
            "kind": "NUMERIC",
            "prompt": "A $2.0\\,\\text{kg}$ box feels $10\\,\\text{N}$ right and $4\\,\\text{N}$ left. Acceleration in $\\text{m/s}^2$ (right positive)?",
            "answer": {
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              "tolerance": 0
            },
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            "explanation": "Net $= 6\\,\\text{N}$, $a = 6/2 = 3\\,\\text{m/s}^2$."
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```

```
                "steps": [
                  "By the third law, the reaction is equal and opposite.",
                  "The wall pushes back on you with exactly $50\\,\\text{N}$."
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                "answer": "$50\\,\\text{N}$, directed back toward you."
              },
              {
                "title": "Why don’t they cancel?",
                "problem": "If the wall pushes back on you just as hard, why isn’t the net effect zero?",
                "steps": [
                  "Your push acts on the *wall*; the wall’s push acts on *you*.",
                  "They act on different objects, so they never appear in the same net-force sum.",
                  "Whether you accelerate depends only on the forces acting on *you*."
                ],
                "answer": "The pair acts on two different bodies, so it doesn’t cancel for either one."
              },
              {
                "title": "How walking works",
                "problem": "What actually pushes you forward when you walk?",
                "steps": [
                  "Your foot pushes backward on the ground (action).",
                  "The ground pushes forward on you (reaction).",
                  "That forward reaction force is what accelerates you."
                ],
                "answer": "The ground’s forward reaction force on your foot."
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 cart, how can they ever move?\" The resolution: those two forces act on **different objects** (one on the cart, one on
the horse), so they don’t cancel. To see if the *cart* accelerates, add up only the forces on the cart: the horse’s
forward pull versus the cart’s friction. The horse, meanwhile, accelerates because the **ground** pushes its hooves
forward harder than the cart pulls it back. Same law, careful bookkeeping."
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              "The two forces act on different objects, so they don’t cancel.",
              "Walking, rockets, and recoil are all third-law effects.",
              "To find an object’s acceleration, sum only the forces on that object."
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              }
            ]
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      "nothing",
      "only heavy objects"
    ],
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    "explanation": "Each force of the pair acts on a different body."
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      "has no effect",
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    ],
    "answer": 0,
    "explanation": "The reaction force pushes the rocket up."
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    ],
    "answer": 0,
    "hint": "Equal and opposite.",
    "explanation": "The reaction is exactly $50\\,\\text{N}$."
  },
  {
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      "the ground’s reaction to your backward push",
      "your muscles acting directly on you",
      "gravity",
      "air pressure"
    ],
    "answer": 0,
    "hint": "Third law at your feet.",
    "explanation": "You push the ground back; it pushes you forward."
  },
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    },
    "hint": "Equal magnitudes.",
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            "explanation": "The reaction force is also $30\\,\\text{N}$."
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            "answer": false,
            "hint": "Those forces act on different objects.",
            "explanation": "The horse accelerates the cart via the ground’s forward push on the horse; the
action-reaction pair doesn’t cancel for either body."
          }
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        "title": "Friction",
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              "headline": "Resisting Motion",
              "sub": "Friction is why you can walk and why nothing slides forever -- a force born where surfaces
meet."
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            "kind": "CONTEXT",
            "title": "Why this matters",
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              "markdown": "Friction is the hidden force in nearly every mechanics problem. It lets tyres grip, brakes
work, and knots hold -- yet it also wastes energy as heat and wears parts down. Knowing when it helps, when it hurts,
and how to compute it is essential."
            }
          },
          {
            "kind": "CONCEPT",
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              "markdown": "**Friction** is a force that opposes (or tends to oppose) relative sliding between two
surfaces in contact. It always acts **parallel to the surface**, opposite to the motion or attempted motion.\n\nFriction
 comes in two flavours:\n\n- **Static friction** acts when surfaces are *not yet* sliding. It adjusts itself to match
whatever you apply, up to a maximum:\n\n$$f_s \\le \\mu_s N$$\n\n- **Kinetic friction** acts when surfaces *are*
sliding, with a roughly constant value:\n\n$$f_k = \\mu_k N$$\n\nHere $N$ is the **normal force** (the perpendicular
push between the surfaces) and $\\mu$ (mu) is the dimensionless **coefficient of friction** -- a property of the two
materials. Usually $\\mu_s > \\mu_k$, which is why it takes a bigger push to *start* something sliding than to keep it
sliding.\n\nKey facts: friction is proportional to the **normal force**, not to the contact area or speed (to a good
approximation), and it converts mechanical energy into **heat**."
            }
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block slides -- it gives way exactly when $\\tan\\theta > \\mu$."
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                  "problem": "A crate slides across the floor. The normal force is $40\\,\\text{N}$ and $\\mu_k =
0.25$. Find the kinetic friction force.",
                  "steps": [
                    "$f_k = \\mu_k N = 0.25 \\times 40 = 10\\,\\text{N}$.",
                    "It opposes the sliding direction."
                  ],
                  "answer": "$f_k = 10\\,\\text{N}$."
                },
                {
                  "title": "Will it move?",
                  "problem": "A box on the floor has a maximum static friction of $40\\,\\text{N}$. You push
horizontally with $30\\,\\text{N}$. Does it move, and what is the friction force?",
                  "steps": [
```

```
                      "Your push ($30\\,\\text{N}$) is below the maximum static friction ($40\\,\\text{N}$).",
                      "Static friction matches your push to keep it still."
                    ],
                    "answer": "It stays put; static friction is $30\\,\\text{N}$ (not 40)."
                  },
                  {
                    "title": "Friction on level ground",
                    "problem": "A $10\\,\\text{kg}$ box sits on level ground, $\\mu_k = 0.20$, $g =
9.8\\,\\text{m/s}^2$. Find the kinetic friction once it’s sliding.",
                    "steps": [
                      "On level ground the normal force equals the weight: $N = mg = 10 \\times 9.8 = 98\\,\\text{N}$.",
                      "$f_k = \\mu_k N = 0.20 \\times 98 = 19.6\\,\\text{N}$."
                    ],
                    "answer": "$f_k = 19.6\\,\\text{N}$."
                  }
                ]
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                    "title": "Friction (static and kinetic)"
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                "intro": "Static vs kinetic."
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              "content": {
                "intro": "Use $f = \\mu N$, and remember static friction only rises to its maximum."
              }
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              "kind": "DEEPER_DIVE",
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              "content": {
                "markdown": "It seems strange that a brick slides with the same friction lying flat or on its end,
despite very different contact areas. The reason: real surfaces touch only at tiny high spots (\"asperities\").
Spreading the same weight over a larger area lowers the pressure but increases the number of contact points -- the two
effects roughly cancel, so total friction depends on the **normal force**, not the apparent area. This is captured in
the simple law $f = \\mu N$, an empirical approximation that works remarkably well."
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                  "Friction opposes sliding and acts parallel to the surface.",
                  "Static: $f_s \\le \\mu_s N$ (adjusts up to a max). Kinetic: $f_k = \\mu_k N$.",
                  "$N$ is the normal force; usually $\\mu_s > \\mu_k$.",
                  "Friction depends on normal force (not area or speed) and makes heat."
                ],
                "formulas": [
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                    "tex": "f_s \\le \\mu_s N"
                  },
                  {
                    "label": "Kinetic friction",
                    "tex": "f_k = \\mu_k N"
                  }
                ]
              }
            }
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```
              "prompt": "The kinetic friction force is given by:",
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                "$\\mu_k N$",
                "$\\mu_k m$",
                "$N/\\mu_k$",
                "$\\mu_k v$"
              ],
              "answer": 0,
              "explanation": "$f_k = \\mu_k N$."
            },
            {
              "scope": "CONCEPT_CHECK",
              "kind": "TRUE_FALSE",
              "prompt": "Static friction can take any value from zero up to a maximum.",
              "answer": true,
              "explanation": "It adjusts to match the applied force until it reaches $\\mu_s N$."
            },
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              "options": [
                "opposite to (attempted) motion",
                "in the direction of motion",
                "straight down",
                "straight up"
              ],
              "answer": 0,
              "explanation": "Friction opposes relative sliding."
            },
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              "scope": "PRACTICE",
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              "prompt": "Normal force $40\\,\\text{N}$, $\\mu_k = 0.25$. Kinetic friction in newtons?",
              "answer": {
                "value": 10,
                "tolerance": 0
              },
              "hint": "$f_k = \\mu_k N$.",
              "explanation": "$0.25 \\times 40 = 10\\,\\text{N}$."
            },
            {
              "scope": "PRACTICE",
              "kind": "NUMERIC",
              "prompt": "A box has $N = 100\\,\\text{N}$ and $\\mu_s = 0.40$. Maximum static friction in newtons?",
              "answer": {
                "value": 40,
                "tolerance": 0
              },
              "hint": "$f_{s,\\max} = \\mu_s N$.",
              "explanation": "$0.40 \\times 100 = 40\\,\\text{N}$."
            },
            {
              "scope": "PRACTICE",
              "kind": "TRUE_FALSE",
              "prompt": "If you push a box with $30\\,\\text{N}$ and its maximum static friction is $40\\,\\text{N}$,
the box stays still.",
              "answer": true,
              "hint": "Compare push to the maximum.",
              "explanation": "Static friction matches the $30\\,\\text{N}$ push, so it doesn’t move."
            },
            {
              "scope": "PRACTICE",
              "kind": "MCQ",
              "prompt": "Friction usually converts mechanical energy into:",
              "options": [
                "heat",
                "light",
                "electricity",
                "sound only"
              ],
              "answer": 0,
              "hint": "Rub your hands together.",
              "explanation": "Friction dissipates energy mainly as heat."
            },
            {
              "scope": "PRACTICE",
              "kind": "NUMERIC",
              "prompt": "A $10\\,\\text{kg}$ box on level ground ($g=9.8$), $\\mu_k = 0.20$. Kinetic friction in
newtons? (Use $N = mg$.)",
              "answer": {
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```

```
              "tolerance": 0.3
            },
            "hint": "$N = mg$, then $f_k = \\mu_k N$.",
            "explanation": "$N = 98\\,\\text{N}$, $f_k = 0.20\\times 98 = 19.6\\,\\text{N}$."
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              "sub": "A ramp is the ultimate test of dynamics: gravity, the normal force, and friction all at once --
solved by choosing clever axes."
            }
          },
          {
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            "title": "Why this matters",
            "content": {
              "markdown": "Ramps, hills, slides, wheelchair access, and mountain roads are all inclined planes. The
technique you learn here -- tilting your coordinate axes and splitting gravity into components -- is the single most
important problem-solving skill in introductory mechanics."
            }
          },
          {
            "kind": "CONCEPT",
            "title": "The big idea",
            "content": {
              "markdown": "On an incline tilted at angle $\\theta$, gravity still points straight down -- but it’s
convenient to **tilt the axes** so one points *along* the ramp and the other *perpendicular* to it. Then split the
weight $mg$ into two components:\n\n- **Along the ramp (down-slope):** $\\;mg\\sin\\theta$ -- this is what tends to
accelerate the object down.\n- **Perpendicular to the ramp:** $\\;mg\\cos\\theta$ -- this is balanced by the **normal
force**, so\n\n$$N = mg\\cos\\theta.$$\n\n**Frictionless incline.** The only unbalanced force is $mg\\sin\\theta$
down-slope, so by $F=ma$:\n\n$$a = g\\sin\\theta.$$\n\nNotice the mass cancels -- every object slides down a
frictionless ramp with the same acceleration, just as in free fall (the $\\theta = 90^\\circ$ case gives $a =
g$).\n\n**With friction.** Kinetic friction $f_k = \\mu N = \\mu mg\\cos\\theta$ acts up-slope. The net down-slope force
 is $mg\\sin\\theta - \\mu mg\\cos\\theta$, giving\n\n$$a = g(\\sin\\theta - \\mu\\cos\\theta).$$\n\nThe block only
begins to slide when the down-slope pull exceeds maximum static friction -- that is, when $\\tan\\theta > \\mu$."
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              "simId": "incline-plane",
              "intro": "Dial in the angle and friction, watch the force components and the normal force, then release
the block to see $a = g(\\sin\\theta - \\mu\\cos\\theta)$ play out."
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                  "title": "Frictionless ramp",
                  "problem": "A block slides down a frictionless ramp inclined at $30^\\circ$. Find its acceleration
($g = 9.8\\,\\text{m/s}^2$).",
                  "steps": [
                    "With no friction, $a = g\\sin\\theta$.",
                    "$a = 9.8 \\times \\sin 30^\\circ = 9.8 \\times 0.5 = 4.9\\,\\text{m/s}^2$."
                  ],
                  "answer": "$a = 4.9\\,\\text{m/s}^2$ down the ramp (independent of mass)."
                },
                {
                  "title": "Normal force",
                  "problem": "A $4.0\\,\\text{kg}$ block rests on a $30^\\circ$ incline. Find the normal force ($g =
9.8$).",
                  "steps": [
                    "$N = mg\\cos\\theta = 4.0 \\times 9.8 \\times \\cos 30^\\circ$.",
                    "$= 39.2 \\times 0.866 \\approx 33.9\\,\\text{N}$."
                  ],
                  "answer": "$N \\approx 33.9\\,\\text{N}$ (less than the full weight of $39.2\\,\\text{N}$)."
                },
```

```
                  {
                    "title": "Ramp with friction",
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($g = 9.8$; $\\sin 37^\\circ \\approx 0.60$, $\\cos 37^\\circ \\approx 0.80$).",
                    "steps": [
                      "$a = g(\\sin\\theta - \\mu\\cos\\theta)$.",
                      "$= 9.8(0.60 - 0.20\\times 0.80) = 9.8(0.60 - 0.16) = 9.8\\times 0.44 \\approx
4.3\\,\\text{m/s}^2$."
                    ],
                    "answer": "$a \\approx 4.3\\,\\text{m/s}^2$ down the slope."
                  }
                ]
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                    "title": "Inclined plane problems"
                  }
                ]
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              }
            },
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              "title": "Practice problems",
              "content": {
                "intro": "Use $N = mg\\cos\\theta$ and $a = g(\\sin\\theta - \\mu\\cos\\theta)$."
              }
            },
            {
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              "content": {
                "markdown": "You *could* keep the usual horizontal/vertical axes, but then both the normal force and the
 acceleration would have two components each, and you’d solve a messy pair of simultaneous equations. By rotating the
axes to align with the ramp, the acceleration has only **one** non-zero component (along the slope) and the
perpendicular direction is in equilibrium ($N = mg\\cos\\theta$). Choosing smart coordinates to make a problem simple is
 a theme that runs all the way up to Lagrangian mechanics."
              }
            },
            {
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              "title": "Summary",
              "content": {
                "takeaways": [
                  "Tilt axes along/perpendicular to the ramp and split gravity.",
                  "Down-slope component $mg\\sin\\theta$; normal force $N = mg\\cos\\theta$.",
                  "Frictionless: $a = g\\sin\\theta$ (mass-independent).",
                  "With friction: $a = g(\\sin\\theta - \\mu\\cos\\theta)$; slides when $\\tan\\theta > \\mu$."
                ],
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                    "label": "Normal force",
                    "tex": "N = mg\\cos\\theta"
                  },
                  {
                    "label": "Frictionless accel.",
                    "tex": "a = g\\sin\\theta"
                  },
                  {
                    "label": "With friction",
                    "tex": "a = g(\\sin\\theta - \\mu\\cos\\theta)"
                  }
                ]
              }
            }
          ],
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              "kind": "MCQ",
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        "prompt": "The component of gravity along a frictionless incline (angle $\\theta$) is:",
        "options": [
          "$mg\\sin\\theta$",
          "$mg\\cos\\theta$",
          "$mg\\tan\\theta$",
          "$mg$"
        ],
        "answer": 0,
        "explanation": "The down-slope component is $mg\\sin\\theta$."
      },
      {
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        "kind": "MCQ",
        "prompt": "The normal force on an incline is:",
        "options": [
          "$mg\\cos\\theta$",
          "$mg\\sin\\theta$",
          "$mg$",
          "zero"
        ],
        "answer": 0,
        "explanation": "$N = mg\\cos\\theta$."
      },
      {
        "scope": "CONCEPT_CHECK",
        "kind": "TRUE_FALSE",
        "prompt": "A block starts to slide on an incline when $\\tan\\theta > \\mu$.",
        "answer": true,
        "explanation": "That’s when the down-slope pull exceeds maximum static friction."
      },
      {
        "scope": "PRACTICE",
        "kind": "NUMERIC",
        "prompt": "Frictionless incline at $30^\\circ$, $g = 9.8$. Acceleration down the slope in
$\\text{m/s}^2$?",
        "answer": {
          "value": 4.9,
          "tolerance": 0.1
        },
        "hint": "$a = g\\sin\\theta$.",
        "explanation": "$9.8 \\times 0.5 = 4.9\\,\\text{m/s}^2$."
      },
      {
        "scope": "PRACTICE",
        "kind": "NUMERIC",
        "prompt": "A $4.0\\,\\text{kg}$ block on a $30^\\circ$ incline ($g = 9.8$). Normal force in newtons?",
        "answer": {
          "value": 33.9,
          "tolerance": 0.6
        },
        "hint": "$N = mg\\cos\\theta$.",
        "explanation": "$4\\times 9.8\\times\\cos30^\\circ \\approx 33.9\\,\\text{N}$."
      },
      {
        "scope": "PRACTICE",
        "kind": "MCQ",
        "prompt": "As an incline gets steeper, the gravity component along the slope:",
        "options": [
          "increases",
          "decreases",
          "stays the same",
          "becomes zero"
        ],
        "answer": 0,
        "hint": "$\\sin\\theta$ grows with $\\theta$.",
        "explanation": "$mg\\sin\\theta$ increases as $\\theta$ increases."
      },
      {
        "scope": "PRACTICE",
        "kind": "TRUE_FALSE",
        "prompt": "On a frictionless incline, a heavier block accelerates faster than a lighter one.",
        "answer": false,
        "hint": "Look at $a = g\\sin\\theta$.",
        "explanation": "Mass cancels -- all objects share $a = g\\sin\\theta$."
      },
      {
        "scope": "PRACTICE",
        "kind": "NUMERIC",
        "prompt": "Incline $37^\\circ$, $\\mu_k = 0.20$, $g = 9.8$ ($\\sin37^\\circ\\approx0.60$,
$\\cos37^\\circ\\approx0.80$). Acceleration down-slope in $\\text{m/s}^2$?",
        "answer": {
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              "tolerance": 0.3
            },
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            "explanation": "$9.8(0.60 - 0.16) \\approx 4.3\\,\\text{m/s}^2$."
          }
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    "name": "The Outer Solar System",
    "subtitle": "Classical Mechanics * Work, Energy & Power",
    "description": "The currency of physics: energy. Work as force through a distance, kinetic and potential energy, the conservation of mechanical energy, and power -- the rate energy is transferred.",
    "gradeRange": "10-12",
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        "estMinutes": 14,
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              "headline": "Doing Work",
              "sub": "In physics, work has a precise meaning: a force only does work when it moves something along its own direction."
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            "title": "Why this matters",
            "content": {
              "markdown": "\"Work\" in physics is not effort or tiredness -- holding a heavy box perfectly still is exhausting but does **zero** work, because nothing moves. Defining work precisely is the gateway to the single most powerful idea in physics: **energy**. Work is exactly the mechanism by which energy is transferred from one object to another."
            }
          },
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            "title": "The big idea",
            "content": {
              "markdown": "**Work** is done when a force acts on an object and moves it through a displacement. For a constant force,\n\n$$W = F\\,d\\cos\\theta$$\n\nwhere $F$ is the force magnitude, $d$ is the displacement, and $\\theta$ is the angle between the force and the displacement.\n\n- If the force is **along** the motion ($\\theta = 0$): $W = Fd$ (maximum, positive work).\n- If the force is **perpendicular** to the motion ($\\theta = 90^\\circ$): $\\cos 90^\\circ = 0$, so $W = 0$. (Carrying a box horizontally does no work *against gravity* -- gravity points down, motion is sideways.)\n- If the force **opposes** the motion ($\\theta = 180^\\circ$): $W = -Fd$ (negative work, like friction removing energy).\n\nThe SI unit of work is the **joule**: $1\\ \\text{J} = 1\\ \\text{N}\\cdot\\text{m}$. Work is a **scalar** -- it has no direction, only a sign.\n\nThe deep meaning: **work is energy transferred.** Positive work adds energy to an object; negative work takes it away."
            }
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                  "problem": "You push a crate $3.0\\,\\text{m}$ with a horizontal force of $40\\,\\text{N}$. How much work do you do?",
                  "steps": [
                    "Force is along the motion, so $\\theta = 0$ and $\\cos\\theta = 1$.",
                    "$W = Fd = 40 \\times 3.0 = 120\\,\\text{J}$."
                  ],
                  "answer": "$W = 120\\,\\text{J}$."
                },
                {
                  "title": "Carrying horizontally",
                  "problem": "You carry a $50\\,\\text{N}$ bag horizontally for $10\\,\\text{m}$ at constant height. How much work does gravity do?",
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```
                "steps": [
                  "Gravity points down; the motion is horizontal.",
                  "$\\theta = 90^\\circ$, so $\\cos\\theta = 0$.",
                  "$W = 50 \\times 10 \\times 0 = 0$."
                ],
                "answer": "Zero -- gravity does no work on horizontal motion."
              },
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                "title": "Work at an angle",
                "problem": "A rope pulls a sled with $60\\,\\text{N}$ at $60^\\circ$ above horizontal, moving it
$5.0\\,\\text{m}$ along the ground. Find the work done by the rope. ($\\cos 60^\\circ = 0.5$.)",
                "steps": [
                  "$W = Fd\\cos\\theta = 60 \\times 5.0 \\times 0.5$.",
                  "$W = 150\\,\\text{J}$."
                ],
                "answer": "$W = 150\\,\\text{J}$."
              }
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 \\cdot \\vec d$. The cosine is what the dot product extracts: the part of the force that lies along the
displacement.\n\nWhen the force **varies** along the path, we slice the path into tiny pieces and add up the work on
each, which in the limit becomes an integral:\n\n$$W = \\int \\vec F \\cdot d\\vec r.$$\n\nThis is how you compute the
work done by a stretching spring, whose force grows with distance -- a calculation you’ll meet again in the next
lessons."
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              "Perpendicular force does zero work; opposing force does negative work.",
              "The unit of work is the joule, $1\\,\\text{J} = 1\\,\\text{N*m}$.",
              "Work is a scalar and represents energy transferred."
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              "explanation": "$40 \\times 3 = 120\\,\\text{J}$."
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                "value": 150,
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              "hint": "$W = Fd\\cos\\theta$.",
              "explanation": "$60 \\times 5 \\times 0.5 = 150\\,\\text{J}$."
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              "markdown": "Why does a car need four times the braking distance when it doubles its speed? Kinetic energy holds the answer: it grows with the **square** of speed. The work-energy theorem turns messy force-and-acceleration problems into clean energy bookkeeping."
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          {
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            }
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            }
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                    "$= \\tfrac12 (2.0)(9.0) = 9.0\\,\\text{J}$."
                  ],
                  "answer": "$K = 9.0\\,\\text{J}$."
                },
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                  "title": "Double the speed",
                  "problem": "A car’s speed doubles. By what factor does its kinetic energy change?",
                  "steps": [
                    "$K \\propto v^2$.",
                    "Doubling $v$ multiplies $v^2$ by 4."
```

```
                  ],
                  "answer": "Kinetic energy becomes 4 times larger."
                },
                {
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rest. Find its final speed.",
                  "steps": [
                    "$W_{net} = \\Delta K = \\tfrac12 m v_f^2 - 0$.",
                    "$50 = \\tfrac12 (4.0) v_f^2 = 2 v_f^2$, so $v_f^2 = 25$.",
                    "$v_f = 5.0\\,\\text{m/s}$."
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                  "answer": "$v_f = 5.0\\,\\text{m/s}$."
                }
              ]
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 Start with $W_{net} = \\int F\\,dx = \\int ma\\,dx$. Using $a = \\dfrac{dv}{dt}$ and $dx = v\\,dt$, a change of
variables gives $\\int m v\\,dv$, which integrates to $\\tfrac12 m v^2$. The $\\tfrac12$ and the square both fall
straight out of the calculus. This is why kinetic energy *has* to be $\\tfrac12 m v^2$ and nothing else."
            }
          },
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                "$K$ grows with the square of speed: double $v$ => quadruple $K$.",
                "Work-energy theorem: $W_{net} = \\Delta K$.",
                "Net positive work speeds up; net negative work slows down."
              ],
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                  "tex": "K = \\tfrac12 m v^2"
                },
                {
                  "label": "Work-energy theorem",
                  "tex": "W_{net} = \\Delta K = \\tfrac12 m v_f^2 - \\tfrac12 m v_i^2"
                }
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        "$\\tfrac12 mv^2$",
        "$mgh$",
        "$mv$",
        "$Fd$"
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      "explanation": "$\\tfrac12(2)(9) = 9\\,\\text{J}$."
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      "explanation": "$\\tfrac12(1)(16) = 8\\,\\text{J}$."
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        "tolerance": 0
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      "hint": "$W = \\tfrac12 mv^2$.",
      "explanation": "$50 = 2v^2 \\Rightarrow v = 5\\,\\text{m/s}$."
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      "kind": "MCQ",
      "prompt": "A car triples its speed. Its kinetic energy becomes:",
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        "3x larger",
        "6x larger",
        "unchanged"
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      "answer": 0,
      "hint": "$K \\propto v^2$.",
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    },
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              "hint": "$\\Delta K = \\tfrac12 m(v_f^2 - v_i^2)$.",
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                "markdown": "A drawn bow, a raised hammer, water behind a dam -- all hold energy purely because of their
 *position* or *configuration*. This stored energy is **potential energy**, and pairing it with kinetic energy lets us
solve motion problems by simple accounting, without ever computing a force."
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Two common kinds:\n\n**Gravitational potential energy** -- energy of height:\n\n$$U_g = mgh$$\n\nwhere $h$ is the height
 above a chosen reference level. Lift a mass higher and you store more energy; let it fall and that energy is released
as kinetic energy. Only *changes* in height matter, so you may put the zero wherever it’s convenient.\n\n**Elastic
potential energy** -- energy stored in a stretched or compressed spring:\n\n$$U_s = \\tfrac12 k x^2$$\n\nwhere $k$ is
the spring’s **stiffness** (spring constant) and $x$ is how far it’s stretched or compressed from its natural length.
(This comes from the spring force $F = kx$, **Hooke’s law**.)\n\nLike all energy, potential energy is measured in
**joules**. It is *potential* because it can be converted into motion at any time."
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shrink as the yellow KE bar grows -- position energy becoming motion energy."
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$1.5\\,\\text{m}$? ($g = 9.8$.)",
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                      "$U_g = mgh = 2.0 \\times 9.8 \\times 1.5$.",
                      "$= 29.4\\,\\text{J}$."
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energy.",
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                      "$= \\tfrac12 (200)(0.01) = 1.0\\,\\text{J}$."
                    ],
                    "answer": "$U_s = 1.0\\,\\text{J}$."
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                    "problem": "A spring is stretched twice as far. How does its stored energy change?",
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```
                "Doubling $x$ multiplies $x^2$ by 4."
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              "answer": "It becomes 4 times larger."
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slope** of the potential energy curve:\n\n$$F = -\\frac{dU}{dx}.$$\n\nFor a spring, $U = \\tfrac12 kx^2$, and its slope
is $kx$, giving the restoring force $F = -kx$ -- Hooke’s law falls right out. Objects always tend to roll \"downhill\"
on a potential-energy graph, toward lower energy. This picture of energy *landscapes* runs all the way up to chemistry
and quantum mechanics."
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            "Gravitational: $U_g = mgh$ (only changes in height matter).",
            "Elastic (spring): $U_s = \\tfrac12 kx^2$, with stiffness $k$.",
            "Spring energy grows with the square of the stretch.",
            "Potential energy can convert into kinetic energy."
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          "$\\tfrac12 kx^2$",
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      },
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          "$\\tfrac12 kx$",
          "$kx^2$"
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        "answer": 0,
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joules?",
        "answer": {
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        "explanation": "$\\tfrac12(200)(0.01) = 1.0\\,\\text{J}$."
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        "kind": "NUMERIC",
        "prompt": "A $5.0\\,\\text{kg}$ mass is raised $4.0\\,\\text{m}$ ($g = 9.8$). PE gained in joules?",
        "answer": {
          "value": 196,
          "tolerance": 1
        },
        "hint": "$mgh$.",
        "explanation": "$5 \\times 9.8 \\times 4 = 196\\,\\text{J}$."
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          "is unchanged"
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        "explanation": "$2^2 = 4$ times."
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0$.",
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        "explanation": "The reference level is arbitrary; only $\\Delta U$ has physical meaning."
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stays exactly the same."
              }
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swinging pendulum, a falling apple -- all obey it. It lets you find a final speed from a starting height in a single
line, sidestepping forces and time entirely."
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            },
            {
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              "content": {
                "markdown": "Define the **total mechanical energy** as kinetic plus potential:\n\n$$E = K +
U.$$\n\n**Conservation of mechanical energy:** when only gravity and springs act (no friction or air resistance), the
total mechanical energy stays **constant**:\n\n$$K_i + U_i = K_f + U_f.$$\n\nEnergy is not created or destroyed -- it
merely **converts** between forms. As a ball falls, $U_g = mgh$ shrinks while $K = \\tfrac12 mv^2$ grows by exactly the
same amount, keeping the sum fixed.\n\nA classic result: drop an object from height $h$ (from rest). All its potential
energy becomes kinetic energy at the bottom:\n\n$$mgh = \\tfrac12 m v^2 \\;\\Rightarrow\\; v = \\sqrt{2gh}.$$\n\nThe
mass cancels -- every object reaches the same speed from the same height (without friction).\n\n**With friction**,
mechanical energy is *not* conserved on its own: friction converts some into **heat**. But counting heat too, the
*total* energy of everything is still conserved -- that’s the universal law."
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moves -- that’s conservation of energy in action."
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($g = 9.8$, no air resistance).",
                    "steps": [
                      "All PE becomes KE: $mgh = \\tfrac12 mv^2$.",
                      "$v = \\sqrt{2gh} = \\sqrt{2 \\times 9.8 \\times 5.0} = \\sqrt{98}$.",
                      "$v \\approx 9.9\\,\\text{m/s}$."
                    ],
                    "answer": "$v \\approx 9.9\\,\\text{m/s}$ (mass-independent)."
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                    "problem": "A pendulum is released from a height $0.20\\,\\text{m}$ above its lowest point. How fast
 is it moving at the bottom? ($g = 9.8$.)",
                    "steps": [
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                      "$= \\sqrt{3.92} \\approx 2.0\\,\\text{m/s}$."
                    ],
                    "answer": "$v \\approx 2.0\\,\\text{m/s}$."
                  },
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                "That energy became heat in the surfaces."
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            "Without friction, $K_i + U_i = K_f + U_f$ -- energy is conserved.",
            "Energy converts between kinetic and potential, never vanishing.",
            "Falling from height $h$: $v = \\sqrt{2gh}$ (mass cancels).",
            "Friction converts mechanical energy to heat, but total energy is always conserved."
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 = \\frac{W}{t} = \\frac{\\Delta E}{t}.$$\n\nThe SI unit is the **watt**: $1\\ \\text{W} = 1\\ \\text{J/s}$. A 60-watt
bulb converts 60 joules of electrical energy every second.\n\nThere’s a second, very useful form. Since $W = Fd$,
dividing by time gives $P = F\\dfrac{d}{t} = Fv$ -- power equals force times speed:\n\n$$P = F v.$$\n\nThis is why a car
 needs far more engine power to maintain high speed (the drag force is large *and* you’re moving fast), and why climbing
 stairs quickly takes more power than climbing them slowly, even though the work (your weight times the height) is
identical.\n\nA handy energy unit comes from power: the **kilowatt-hour** is the energy of one kilowatt running for one
hour -- that’s what your utility actually charges for."
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                    "$= 150\\,\\text{W}$."
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                    "$P = W/t = 2940/10 = 294\\,\\text{W}$."
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$100\\text{-}150\\,\\text{W}$ for long periods (about a fifth of a horsepower), though a sprint cyclist can hit over
$1000\\,\\text{W}$ for a few seconds. Your body ’burns’ food energy at roughly $100\\,\\text{W}$ just resting -- the
same as an old incandescent bulb."
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                  "Equivalently, $P = Fv$ (force times speed).",
                  "Same work in less time means more power.",
                  "Energy can be billed in kilowatt-hours."
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"explanation": "$60 \\times 9.8 \\times 5 / 10 = 294\\,\\text{W}$."
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**impulse** -- stretching out the time over which momentum changes to soften the force. Momentum is also the conserved
quantity that lets us analyse collisions and rocket flight where energy methods alone fall short."
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has direction), measured in $\\text{kg*m/s}$. A heavy slow object and a light fast one can carry the same
momentum.\n\n**Impulse** is what changes momentum. From Newton’s second law in its original form, $\\vec F =
\\dfrac{d\\vec p}{dt}$, a force acting over a time interval delivers an **impulse**:\n\n$$\\vec J = \\vec F\\,\\Delta t
= \\Delta \\vec p.$$\n\nThis is the **impulse-momentum theorem**: the impulse on an object equals its change in
momentum. Impulse is measured in $\\text{N*s}$, which is the same as $\\text{kg*m/s}$.\n\nThe practical punchline: to
produce a given change in momentum (say, stopping a moving car), a **longer** contact time means a **smaller** force. An
 airbag increases $\\Delta t$, so the force on you, $F = \\Delta p / \\Delta t$, drops. Same $\\Delta p$, gentler ride."
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 things. Momentum is a **vector** and is conserved in *every* collision, even when kinetic energy is lost to heat.
Kinetic energy is a **scalar** and is only conserved in *elastic* collisions. Having both lets you solve for two
unknowns (two final velocities) in an elastic collision. Momentum also handles direction -- head-on versus glancing --
which energy alone can’t. The two laws are partners, not rivals."
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              "Impulse $\\vec J = \\vec F\\Delta t = \\Delta\\vec p$ (impulse-momentum theorem).",
              "Impulse is measured in $\\text{N*s} = \\text{kg*m/s}$.",
              "For a fixed $\\Delta p$, longer contact time means smaller force -- the airbag principle."
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              "explanation": "$50 \\times 4 = 200$."
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              "kind": "NUMERIC",
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              "answer": {
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momentum stays constant**:\n\n$$\\vec p_{i,\\text{total}} = \\vec p_{f,\\text{total}}.$$\n\nFor two objects interacting
(a collision, an explosion, a push-off):\n\n$$m_1 \\vec v_{1i} + m_2 \\vec v_{2i} = m_1 \\vec v_{1f} + m_2 \\vec
v_{2f}.$$\n\nWhy is it true? During the interaction, object 1 pushes on object 2 and (Newton’s third law) object 2
pushes back equally and oppositely. These internal forces give equal-and-opposite impulses, so whatever momentum one
object gains, the other loses. The **total** can’t change.\n\nConsequences you can see everywhere:\n\n- **Recoil.** Fire
 a bullet forward and the gun must move backward so the total stays the same (it started at zero).\n- **Push-off.** Two
skaters at rest shove apart; they fly in opposite directions with equal and opposite momenta.\n- **Sticking together.**
Two carts that couple move off with the combined momentum they had before.\n\nMomentum is a vector, so **directions
matter** -- choose a positive direction and give opposing velocities opposite signs."
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and they stick together. Find their common speed.",
                "steps": [
                  "Total momentum before: $2.0\\times3.0 + 1.0\\times0 = 6.0\\,\\text{kg*m/s}$.",
                  "After, combined mass is $3.0\\,\\text{kg}$ at speed $v$.",
                  "$3.0\\,v = 6.0 \\Rightarrow v = 2.0\\,\\text{m/s}$."
                ],
                "answer": "$v = 2.0\\,\\text{m/s}$."
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                "title": "Recoil of a skater",
                "problem": "A $50\\,\\text{kg}$ skater (at rest) throws a $2.0\\,\\text{kg}$ ball at
$5.0\\,\\text{m/s}$. Find the skater’s recoil speed.",
                "steps": [
                  "Total momentum starts at zero.",
                  "Ball: $2.0\\times5.0 = 10\\,\\text{kg*m/s}$ forward.",
                  "Skater must carry $10\\,\\text{kg*m/s}$ backward: $50\\,v = 10$."
                ],
                "answer": "$v = 0.20\\,\\text{m/s}$ backward."
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                  "The system started with zero momentum.",
                  "The bullet now carries forward momentum.",
                  "The rifle must carry equal backward momentum to keep the total zero."
                ],
                "answer": "Backward -- that’s recoil, required by momentum conservation."
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Each second it hurls hot exhaust out the back, and to keep total momentum constant, the rocket gains forward momentum.
This leads to the **Tsiolkovsky rocket equation**, $\\Delta v = v_e \\ln\\dfrac{m_0}{m_f}$, which says your change in
speed depends on the exhaust speed $v_e$ and the *ratio* of starting to ending mass. It’s why rockets are mostly fuel --
 and why getting to orbit is so hard."
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                "$m_1 v_{1i} + m_2 v_{2i} = m_1 v_{1f} + m_2 v_{2f}$.",
                "It follows from Newton’s third law (equal, opposite internal forces).",
                "Recoil, push-offs, and rockets are all momentum conservation.",
                "Momentum is a vector -- track directions with signs."
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              "speeds up"
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they stick. Combined speed in m/s?",
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momentum after, in $\\text{kg*m/s}$?",
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            "explanation": "It stays $12\\,\\text{kg*m/s}$."
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            "hint": "$p/m_{tot}$.",
            "explanation": "$12/4 = 3\\,\\text{m/s}$."
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            "hint": "$6/3$.",
            "explanation": "$6.0/3.0 = 2.0\\,\\text{m/s}$."
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              "undefined"
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            "hint": "What was it before?",
            "explanation": "It started at zero and stays zero."
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            "answer": {
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              "tolerance": 0.01
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            "hint": "$50v = 2\\times5$.",
            "explanation": "$10/50 = 0.20\\,\\text{m/s}$."
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            }
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                    "steps": [
                      "$v_f = \\dfrac{2.0\\times4.0 + 2.0\\times0}{2.0+2.0} = \\dfrac{8.0}{4.0}$.",
                      "$= 2.0\\,\\text{m/s}$."
                    ],
                    "answer": "$v_f = 2.0\\,\\text{m/s}$."
                  },
                  {
                    "title": "Energy lost",
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                    "steps": [
                      "Before: $K_i = \\tfrac12(2.0)(4.0)^2 = 16\\,\\text{J}$.",
                      "After: $K_f = \\tfrac12(4.0)(2.0)^2 = 8\\,\\text{J}$.",
                      "Lost: $16 - 8 = 8\\,\\text{J}$."
                    ],
                    "answer": "$8\\,\\text{J}$ lost to heat and deformation."
                  },
                  {
                    "title": "Car pileup",
                    "problem": "A $1000\\,\\text{kg}$ car at $10\\,\\text{m/s}$ rear-ends an identical parked car; they lock together. Find the wreck’s speed.",
                    "steps": [
                      "$v_f = \\dfrac{1000\\times10}{2000} = \\dfrac{10000}{2000}$.",
                      "$= 5.0\\,\\text{m/s}$."
                    ],
                    "answer": "$v_f = 5.0\\,\\text{m/s}$."
                  }
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              }
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              "content": {
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 of wood -- it embeds (perfectly inelastic), and the block-plus-bullet swings upward. The trick uses **both** big ideas
in sequence: *momentum conservation* during the fast embedding gives the block’s launch speed, then *energy
conservation* during the slow swing turns that speed into a measured height. Combining them recovers the bullet’s
original speed from nothing but a length measurement. It’s a beautiful reminder that momentum and energy are conserved
in **different phases** of the same event."
              }
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                  "Perfectly inelastic: objects stick and move together.",
                  "$v_f = (m_1v_1 + m_2v_2)/(m_1+m_2)$ from momentum conservation.",
                  "Momentum is conserved; kinetic energy is not.",
                  "Lost kinetic energy becomes heat, sound, and deformation.",
                  "Sticking gives the maximum possible kinetic-energy loss."
                ],
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                    "label": "Stick-together speed",
                    "tex": "v_f = \\dfrac{m_1 v_1 + m_2 v_2}{m_1 + m_2}"
                  }
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                "bounce apart elastically",
                "pass through each other",
                "speed up"
              ],
              "answer": 0,
              "explanation": "They couple and move as one."
            },
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stick. Final speed in m/s?",
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Combined speed in m/s?",
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              "explanation": "$5.0\\,\\text{m/s}$."
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stick. Combined speed in m/s?",
          "answer": {
            "value": 2,
            "tolerance": 0
          },
          "hint": "$12/6$.",
          "explanation": "$2.0\\,\\text{m/s}$."
        },
        {
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          "kind": "NUMERIC",
          "prompt": "Kinetic energy of a $2.0\\,\\text{kg}$ object at $4.0\\,\\text{m/s}$, in joules? (its energy
before sticking)",
          "answer": {
            "value": 16,
            "tolerance": 0
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          "hint": "$\\tfrac12 mv^2$.",
          "explanation": "$\\tfrac12(2)(16) = 16\\,\\text{J}$."
        },
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          "scope": "PRACTICE",
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          "prompt": "In an inelastic collision, the ’lost’ kinetic energy mostly becomes:",
          "options": [
            "heat and deformation",
            "more momentum",
            "more speed",
            "light"
          ],
          "answer": 0,
          "hint": "Where does crushing energy go?",
          "explanation": "It turns into heat, sound, and permanent deformation."
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stick. Combined speed in m/s?",
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          "explanation": "$1.5\\,\\text{m/s}$."
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 balls or atoms."
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            "markdown": "Truly elastic collisions are rare in everyday life but exact at the atomic scale -- gas
molecules, billiard balls, and Newton’s cradles come close. Because **both** momentum and kinetic energy are conserved,
elastic collisions have clean, predictable outcomes."
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along with momentum:\n\n$$m_1 v_{1i} + m_2 v_{2i} = m_1 v_{1f} + m_2 v_{2f}, \\qquad \\tfrac12 m_1 v_{1i}^2 + \\tfrac12
m_2 v_{2i}^2 = \\tfrac12 m_1 v_{1f}^2 + \\tfrac12 m_2 v_{2f}^2.$$\n\nTwo conservation laws mean you can solve for
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**both** final velocities.\n\nThe most important special case -- **equal masses**, one initially at rest: they simply
**exchange velocities.** The moving ball stops dead, and the struck ball moves off with the original speed. This is
exactly what a **Newton’s cradle** shows, and why a head-on cue ball stops while the object ball rolls away.\n\nGeneral
1-D formulas (useful, not worth memorising):\n\n$$v_{1f} = \\frac{m_1 - m_2}{m_1 + m_2}\\,v_{1i} + \\frac{2 m_2}{m_1 +
m_2}\\,v_{2i}, \\qquad v_{2f} = \\frac{m_2 - m_1}{m_1 + m_2}\\,v_{2i} + \\frac{2 m_1}{m_1 + m_2}\\,v_{1i}.$$\n\nA light
ball hitting a heavy one bounces nearly straight back; a heavy ball hitting a light one barely slows and sends the light
 one off at nearly twice its speed."
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Then try unequal masses and see how the bounce changes."
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rest. Find both final velocities.",
                    "steps": [
                      "Equal masses, one at rest -> they exchange velocities.",
                      "Mover stops; struck ball takes the $3.0\\,\\text{m/s}$."
                    ],
                    "answer": "First ball: $0$; second ball: $3.0\\,\\text{m/s}$."
                  },
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                    "title": "Newton’s cradle",
                    "problem": "Why does lifting one ball of a Newton’s cradle launch exactly one ball off the far
end?",
                    "steps": [
                      "The balls have equal mass and collide elastically.",
                      "Momentum and kinetic energy must both be conserved.",
                      "Only ’one ball out at the same speed’ satisfies both."
                    ],
                    "answer": "Conserving both momentum and energy forces one ball out at the incoming speed."
                  },
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                    "title": "Light hits heavy",
                    "problem": "In an elastic collision, a light ball strikes a much heavier ball at rest. Roughly what
happens to the light ball?",
                    "steps": [
                      "With $m_1 \\ll m_2$, the factor $(m_1-m_2)/(m_1+m_2) \\approx -1$.",
                      "So $v_{1f} \\approx -v_{1i}$."
                    ],
                    "answer": "It bounces back at nearly its original speed."
                  }
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 Whatever speed the objects approach each other at, they separate at the *same* speed:\n\n$$v_{1i} - v_{2i} = -(v_{1f} -
 v_{2f}).$$\n\nThis single equation, paired with momentum conservation, replaces the messy quadratic energy equation and
 gives both final velocities with only linear algebra. It also makes the equal-mass exchange obvious. Physicists use
this ’closing speed = separating speed’ rule constantly."
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                  "Equal masses (one at rest) exchange velocities.",
                  "A Newton’s cradle is the classic demonstration.",
                  "Light-on-heavy bounces back; heavy-on-light barely slows.",
                  "Shortcut: relative (closing) speed reverses on separation."
                ],
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v_{2f}^2"
                  },
                  {
                    "label": "Relative speed",
                    "tex": "v_{1i} - v_{2i} = -(v_{1f} - v_{2f})"
                  }
                ]
              }
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              "explanation": "That is the definition of elastic."
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                "stick together",
                "both stop",
                "both reverse"
              ],
              "answer": 0,
              "explanation": "Equal masses swap velocities."
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struck ball afterward, in m/s?",
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              },
              "explanation": "It takes the full $3.0\\,\\text{m/s}$."
            },
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              "kind": "NUMERIC",
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initially-moving ball afterward, in m/s?",
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              "explanation": "The mover stops; speed 0."
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              "inelastic sticking",
              "friction",
              "gravity only"
            ],
            "answer": 0,
            "hint": "One in, one out.",
            "explanation": "It’s a chain of elastic collisions."
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            "prompt": "Elastic collisions lose kinetic energy to heat.",
            "answer": false,
            "hint": "What does ’elastic’ mean?",
            "explanation": "Elastic collisions conserve kinetic energy."
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            "answer": {
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              "tolerance": 0
            },
            "hint": "Exchange rule.",
            "explanation": "A stops; B moves at $6.0\\,\\text{m/s}$."
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              "perfectly inelastic",
              "all collisions",
              "none"
            ],
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            "hint": "Only one keeps all the KE.",
            "explanation": "Only elastic collisions conserve kinetic energy."
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at the CM** and all external forces acted there. That’s why we could treat objects as points in earlier lessons.\n-**With no external force, the CM moves at constant velocity** -- even while the parts collide, explode, or spin. In a collision, the individual carts change motion, but their *combined* center of mass sails on undisturbed. This is just conservation of momentum seen from another angle: $\\vec p_{total} = M\\vec v_{cm}$.\n\nThe center of mass turns a complicated, wobbling system into a single point that obeys $\\vec F_{ext} = M\\vec a_{cm}$."
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                      "$= 1.0\\,\\text{m}$."
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                      "$= 2.0\\,\\text{m}$."
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              }
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                "$x_{cm} = (m_1x_1 + m_2x_2)/(m_1+m_2)$.",
                "It lies closer to the heavier mass; midway for equal masses.",
                "A system moves as if all mass were at the CM.",
                "With no external force, the CM moves at constant velocity."
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in m?",
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            "explanation": "$8.0/4.0 = 2.0\\,\\text{m}$."
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              "explanation": "The CM sits nearer the heavier object."
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motion has its own kinematics that exactly mirrors linear motion -- learn the dictionary between them and every equation
 you already know carries over."
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turn $= 2\\pi$ rad):\n\n- **Angular position** $\\theta$ -- the angle turned through.\n- **Angular velocity** $\\omega =
 \\dfrac{d\\theta}{dt}$ -- how fast it spins (rad/s).\n- **Angular acceleration** $\\alpha = \\dfrac{d\\omega}{dt}$ --
how fast the spin rate changes (rad/s^2).\n\nThese map one-to-one onto linear motion: $x\\to\\theta$, $v\\to\\omega$,
$a\\to\\alpha$. So the constant-acceleration equations look identical:\n\n$$\\omega = \\omega_0 + \\alpha t, \\qquad
\\theta = \\omega_0 t + \\tfrac12 \\alpha t^2, \\qquad \\omega^2 = \\omega_0^2 + 2\\alpha\\theta.$$\n\n**Linking spin to
 straight-line motion.** A point at radius $r$ from the axis traces an arc, and:\n\n$$s = r\\theta, \\qquad v =
```

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r\\omega, \\qquad a_{t} = r\\alpha.$$\n\nThe farther a point is from the axis, the faster it moves -- which is why the
rim of a wheel races while the hub barely moves."
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does a point on the rim move?",
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Find its angular acceleration.",
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                  "$= 2\\,\\text{rad/s}^2$."
                ],
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              },
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                "problem": "How many radians are in one complete revolution?",
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                  "$2\\pi \\approx 6.28$."
                ],
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conversion factor -- and what makes $v = r\\omega$ and $a_t = r\\alpha$ clean. Radians also make calculus tidy:
$\\dfrac{d}{dx}\\sin x = \\cos x$ only when $x$ is in radians. They’re the SI ’unit’ for angle precisely because they
simplify everything."
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                "These mirror $x$, $v$, $a$ -- same constant-acceleration equations.",
                "$s = r\\theta$, $v = r\\omega$, $a_t = r\\alpha$ link spin to linear motion.",
                "Points farther from the axis move faster.",
                "One revolution $= 2\\pi$ radians."
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              "m"
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            "answer": true,
            "explanation": "$v = r\\omega$ grows with $r$."
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            "explanation": "$10/5 = 2\\,\\text{rad/s}^2$."
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acceleration in m/s^2?",
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 (rad/s)",
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```
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              "explanation": "$2\\pi$ radians per revolution."
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              "explanation": "$3/0.25 = 12\\,\\text{rad/s}$."
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effect."
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stubborn bolt? Both are torque: the same force does more turning when applied farther from the pivot. Torque is to
rotation what force is to straight-line motion."
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it’s applied, and *at what angle*:\n\n$$\\tau = r F \\sin\\theta$$\n\nwhere $r$ is the distance from the pivot to where
the force acts, $F$ is the force, and $\\theta$ is the angle between them. The quantity $r\\sin\\theta$ is the **lever
arm** -- the perpendicular distance from the pivot to the force’s line of action. Torque is measured in
**newton-metres** ($\\text{N*m}$).\n\n- Maximum torque when the force is **perpendicular** ($\\theta = 90^\\circ$):
$\\tau = rF$.\n- **Zero** torque when the force points straight at the pivot ($\\theta = 0$) -- no twist at all.\n- A
**longer lever arm** (bigger $r$) means more torque for the same force.\n\nTorque has a sense: by convention,
counter-clockwise is **positive**, clockwise is **negative**.\n\n**Rotational equilibrium.** An object’s rotation
doesn’t change when the **net torque is zero**: the clockwise torques balance the counter-clockwise ones. A seesaw
balances when $m_1 g\\, d_1 = m_2 g\\, d_2$ -- a heavy child close to the pivot can be balanced by a lighter child
farther out. This is the **law of the lever**."
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larger torque and balances exactly when $m_1 d_1 = m_2 d_2$."
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                "steps": [
                  "Perpendicular force, so $\\sin\\theta = 1$.",
                  "$\\tau = rF = 0.30 \\times 20 = 6.0\\,\\text{N*m}$."
                ],
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                  "$30 \\times 2.0 = 20 \\times d_2$, so $d_2 = 60/20$.",
                  "$= 3.0\\,\\text{m}$."
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                "answer": "$3.0\\,\\text{m}$ from the pivot."
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                "steps": [
                  "The force points at the pivot, $\\theta = 0$.",
                  "$\\sin 0 = 0$."
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              "The lever arm $r\\sin\\theta$ is the perpendicular distance to the force’s line.",
              "Force toward the pivot gives zero torque; perpendicular gives the most.",
              "A longer lever arm means more torque.",
              "Rotational equilibrium: net torque is zero ($m_1d_1 = m_2d_2$ on a lever)."
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            "$\\tfrac12 I\\omega^2$",
            "$ma$"
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                "sub": "Mass resists acceleration; moment of inertia resists angular acceleration -- and depends on how that mass is spread out."
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                "markdown": "Why does a figure skater spin faster when they pull their arms in? Why is a solid disk easier to spin than a ring of the same mass? The answer is **moment of inertia** -- rotational mass -- which depends not just on how much mass there is, but on *where* it sits relative to the axis."
              }
            },
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              }
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                      "$I = mr^2 = 2.0 \\times (3.0)^2$.",
                      "$= 2.0 \\times 9.0 = 18\\,\\text{kg*m}^2$."
                    ],
                    "answer": "$I = 18\\,\\text{kg*m}^2$."
                  },
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              "$= 3.0\\,\\text{rad/s}^2$."
            ],
            "answer": "$\\alpha = 3.0\\,\\text{rad/s}^2$."
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            "problem": "A solid disk and a hoop have the same mass and radius. Which has the larger moment of
inertia?",
            "steps": [
              "Disk: $\\tfrac12 MR^2$; hoop: $MR^2$.",
              "The hoop’s mass is all at the rim, far from the axis."
            ],
            "answer": "The hoop -- twice the disk’s moment of inertia."
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gives a shortcut: if you know $I_{cm}$ about an axis through the center of mass, then about any parallel axis a distance
 $d$ away,\n\n$$I = I_{cm} + Md^2.$$\n\nThis is why a rod is much harder to spin about its end than about its middle --
you’ve added $Md^2$. It also explains why holding a heavy object far from your body strains you: you’ve increased the
effective moment of inertia your muscles must control."
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          "Mass farther from the axis contributes much more ($r^2$).",
          "A hoop has larger $I$ than a disk of equal mass and radius.",
          "Newton’s law for rotation: $\\tau_{net} = I\\alpha$.",
          "Parallel-axis theorem: $I = I_{cm} + Md^2$."
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            "tex": "\\tau_{net} = I\\alpha"
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              "$\\tau = mr$"
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$\\text{kg*m}^2$?",
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            "hint": "$\\alpha = \\tau/I$.",
            "explanation": "$12/4 = 3\\,\\text{rad/s}^2$."
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            "kind": "NUMERIC",
            "prompt": "A body has $I = 5.0\\,\\text{kg*m}^2$ and $\\alpha = 2.0\\,\\text{rad/s}^2$. Net torque in
N*m?",
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              "tolerance": 0
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            "hint": "$\\tau = I\\alpha$.",
            "explanation": "$5\\times2 = 10\\,\\text{N*m}$."
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              "they are equal",
              "depends on color"
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            "explanation": "The hoop has the larger $I = MR^2$."
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MR^2$, in $\\text{kg*m}^2$?",
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spin."
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classic race -- which object rolls downhill fastest -- is decided entirely by how rotational energy is shared.
Rotational kinetic energy completes our energy toolkit."
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mv^2$:\n\n$$K_{rot} = \\tfrac12 I \\omega^2.$$\n\nA **rolling** object (like a wheel or ball) has *both* kinds at once
-- its center moves *and* it spins:\n\n$$K_{total} = \\tfrac12 m v^2 + \\tfrac12 I \\omega^2.$$\n\nFor rolling **without
 slipping**, the two are linked by $v = r\\omega$.\n\nThis explains the famous downhill race. When objects roll down a
ramp, gravitational PE splits between translation and rotation. An object with a **larger** moment of inertia (for its
mass and radius) puts more energy into spin and less into forward speed -- so it rolls **slower**. The ranking from
fastest to slowest:\n\n$$\\text{sphere} \\;>\\; \\text{disk} \\;>\\; \\text{hoop}.$$\n\nA hoop (all mass at the rim,
largest $I$) always loses; a solid sphere always wins. Remarkably, the result is **independent of mass and radius** --
only the *shape* (the $I$-to-$mr^2$ ratio) matters."
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that kinetic energy would split between forward motion and spin -- the bigger the moment of inertia, the more goes into
spinning."
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Find its rotational kinetic energy.",
                  "steps": [
                    "$K_{rot} = \\tfrac12 I\\omega^2 = \\tfrac12(2.0)(10)^2$.",
                    "$= \\tfrac12(2.0)(100) = 100\\,\\text{J}$."
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                  "answer": "$K_{rot} = 100\\,\\text{J}$."
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mR^2$ (a disk), how much rotational kinetic energy does it have?",
                  "steps": [
```

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                      "For a disk, $K_{rot} = \\tfrac12 I\\omega^2 = \\tfrac12(\\tfrac12 mR^2)(v/R)^2 = \\tfrac14
mv^2$.",
                      "Translational is $\\tfrac12 mv^2 = 4\\,\\text{J}$, so $mv^2 = 8\\,\\text{J}$.",
                      "$K_{rot} = \\tfrac14 mv^2 = 2\\,\\text{J}$."
                    ],
                    "answer": "$2\\,\\text{J}$ of rotational kinetic energy."
                  },
                  {
                    "title": "The downhill race",
                    "problem": "A solid sphere, a disk, and a hoop roll from the same height. Which reaches the bottom
first?",
                    "steps": [
                      "Smaller $I$ (for its $mr^2$) leaves more energy for speed.",
                      "Sphere $\\tfrac25$, disk $\\tfrac12$, hoop $1$ (in units of $mR^2$)."
                    ],
                    "answer": "The solid sphere -- smallest moment of inertia, fastest roll."
                  }
                ]
              }
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 spun very fast stores enormous energy in a small package. Modern flywheel energy storage uses carbon-fibre rotors
spinning in vacuum at tens of thousands of rpm, magnetically levitated to kill friction. They charge and discharge in
seconds, survive millions of cycles, and stabilise power grids -- a mechanical ’battery’ running on pure rotational
kinetic energy."
              }
            },
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                  "A rolling object has both $\\tfrac12 mv^2$ and $\\tfrac12 I\\omega^2$.",
                  "Rolling without slipping links them via $v = r\\omega$.",
                  "Downhill race: sphere > disk > hoop (smaller $I$ wins).",
                  "The race result is independent of mass and radius."
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                  {
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                  }
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            "$I\\omega$",
            "$\\tfrac12 I\\omega$"
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        },
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KE in joules?",
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            "tolerance": 0
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          "answer": true,
          "explanation": "It moves forward and spins."
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        {
          "scope": "PRACTICE",
          "kind": "NUMERIC",
          "prompt": "A body with $I = 4.0\\,\\text{kg*m}^2$ spins at $\\omega = 3.0\\,\\text{rad/s}$. Rotational KE
in joules?",
          "answer": {
            "value": 18,
            "tolerance": 0
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          "hint": "$\\tfrac12 I\\omega^2$.",
          "explanation": "$\\tfrac12(4)(9) = 18\\,\\text{J}$."
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            "disk",
            "hoop",
            "they tie"
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          "hint": "Smallest moment of inertia.",
          "explanation": "The sphere puts the least energy into spin."
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            "they tie"
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          "explanation": "The hoop has all its mass at the rim, largest $I$."
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              "explanation": "That’s the rolling condition."
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              "content": {
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              }
            },
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              "content": {
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                    "steps": [
                      "Conserve $L$: $I_i\\omega_i = I_f\\omega_f$.",
                      "$6 \\times 2 = 2 \\times \\omega_f$, so $\\omega_f = 12/2$.",
                      "$= 6\\,\\text{rad/s}$."
                    ],
                    "answer": "$\\omega_f = 6\\,\\text{rad/s}$ -- three times faster."
                  },
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          "steps": [
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            "With $L$ conserved, $\\omega$ must rise to compensate."
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**symmetry** (Noether’s theorem): the laws of physics are the same in every **direction** -- *rotational symmetry* of
space. Because nature has no preferred orientation, angular momentum is conserved. This is why it holds everywhere, from
 the electron’s quantised spin to the rotation of entire galaxies, and why a gyroscope can keep a spacecraft pointed at
a distant star for years."
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        "Net torque changes $L$; with no external torque, $L$ is conserved.",
        "$I_i\\omega_i = I_f\\omega_f$: shrink $I$ and you spin faster.",
        "Skaters, divers, and pulsars all use this.",
        "It’s a vector along the spin axis -- the basis of gyroscopes."
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                "$mr$"
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              "explanation": "That is conservation of angular momentum."
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              "hint": "$L = I\\omega$.",
              "explanation": "$4\\times5 = 20$."
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                "faster",
                "slower",
                "at the same rate",
                "backward"
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              "answer": 0,
              "hint": "Smaller $I$, conserved $L$.",
              "explanation": "$\\omega$ rises to keep $L$ constant."
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              "scope": "PRACTICE",
              "kind": "NUMERIC",
              "prompt": "Conserving $L$: $I_i = 8$, $\\omega_i = 3$; if $I_f = 4$, find $\\omega_f$ (rad/s).",
              "answer": {
                "value": 6,
                "tolerance": 0
              },
              "hint": "$I_i\\omega_i = I_f\\omega_f$.",
              "explanation": "$24 = 4\\omega_f \\Rightarrow 6\\,\\text{rad/s}$."
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                "decreases",
                "stays the same",
                "stops"
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              "answer": 0,
              "hint": "$I$ shrinks, $L$ conserved.",
              "explanation": "Shrinking $I$ forces a huge increase in $\\omega$."
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              "prompt": "A spinning gyroscope resists changes to its axis direction because angular momentum is a
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vector.",
              "answer": true,
              "hint": "Changing direction needs a torque.",
              "explanation": "Its angular momentum vector wants to stay fixed."
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                "sub": "An object moving in a circle at steady speed is still accelerating -- because its direction never stops changing."
              }
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              "kind": "CONTEXT",
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              "content": {
                "markdown": "Earth races around the Sun at about $30\\,\\text{km/s}$, yet it neither speeds up nor flies away. Its **speed** is constant but its **velocity** is not -- velocity is a vector, and its direction turns continuously. A changing velocity means acceleration, and acceleration needs a force. Understanding that force is the key to every orbit."
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                "markdown": "In **uniform circular motion** an object moves in a circle of radius $r$ at constant speed $v$. The velocity is always **tangent** to the circle, so its direction changes constantly -- that is an acceleration even though the speed is fixed.\n\nThis **centripetal** (’center-seeking’) acceleration points toward the center:\n\n$$a_c = \\frac{v^2}{r}.$$\n\nBy Newton’s second law it requires a net inward **centripetal force**:\n\n$$F_c = m a_c = \\frac{m v^2}{r}.$$\n\nThe time for one full loop is the **period** $T$, and since the object covers a circumference $2\\pi r$ in that time,\n\n$$v = \\frac{2\\pi r}{T}.$$\n\nThe centripetal force is not a new kind of force -- it is whatever real force (gravity, tension, friction) happens to point toward the center."
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                    "steps": [
                      "Use $a_c = v^2/r$.",
                      "$a_c = 10^2 / 20 = 100/20$."
                    ],
                    "answer": "$a_c = 5\\,\\text{m/s}^2$ toward the center."
                  },
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                    "title": "Centripetal force",
                    "problem": "A $2\\,\\text{kg}$ mass in that same turn needs what inward force?",
                    "steps": [
                      "$F_c = m a_c$.",
                      "$F_c = 2 \\times 5$."
```

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                    ],
                    "answer": "$F_c = 10\\,\\text{N}$."
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                    "title": "Speed from the period",
                    "problem": "A point at radius $r = 1\\,\\text{m}$ completes a loop in $T = \\pi\\,\\text{s}$. Find
$v$.",
                    "steps": [
                      "$v = 2\\pi r / T$.",
                      "$v = 2\\pi(1)/\\pi = 2$."
                    ],
                    "answer": "$v = 2\\,\\text{m/s}$."
                  }
                ]
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            },
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              "content": {
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              }
            },
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              }
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                "markdown": "Over a tiny time $\\Delta t$ the velocity vector turns by a small angle but keeps the same
length. The **change** in velocity, $\\Delta \\vec v = \\vec v_f - \\vec v_i$, points toward the center of the circle --
 and $\\vec a = \\Delta \\vec v / \\Delta t$ inherits that direction. Taking the limit $\\Delta t \\to 0$ gives a vector
 that always points exactly at the center, with magnitude $v^2/r$. This is the geometric origin of centripetal
acceleration, and the same limiting argument is the derivative in disguise."
              }
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                  "Constant speed in a circle still means changing velocity -> acceleration.",
                  "Centripetal acceleration $a_c = v^2/r$ points to the center.",
                  "Centripetal force $F_c = mv^2/r$ is whatever real force points inward.",
                  "Period and speed are linked by $v = 2\\pi r/T$.",
                  "There is no outward ’centrifugal’ force in the inertial frame."
                ],
                "formulas": [
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                    "tex": "a_c = \\dfrac{v^2}{r}"
                  },
                  {
                    "label": "Centripetal force",
                    "tex": "F_c = \\dfrac{m v^2}{r}"
                  }
                ]
              }
            }
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              "answer": true,
              "explanation": "Its direction (hence velocity) changes, so it accelerates."
            },
            {
              "scope": "CONCEPT_CHECK",
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              "prompt": "The centripetal acceleration points:",
              "options": [
                "tangent to the circle",
                "toward the center",
                "away from the center",
                "straight down"
```

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          ],
          "answer": 1,
          "explanation": "Centripetal means center-seeking -- it points inward."
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          "answer": {
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          "explanation": "$a_c = 10^2/20 = 5\\,\\text{m/s}^2$."
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          "prompt": "Write centripetal acceleration $a_c$ in terms of speed $v$ and radius $r$. (Type like ‘v^2/r‘.)",
          "answer": {
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            "vars": [
              "v",
              "r"
            ]
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          "difficulty": 3,
          "hint": "Speed squared over radius.",
          "explanation": "$a_c = \\dfrac{v^2}{r}$."
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          "kind": "NUMERIC",
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          "answer": {
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            "tolerance": 0
          },
          "hint": "$F_c = mv^2/r$.",
          "explanation": "$F_c = 2 \\times 16 / 2 = 16\\,\\text{N}$."
        },
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          "prompt": "At fixed radius $r = 1\\,\\text{m}$, centripetal acceleration as a function of speed is $a = v^2$. Graph it (enter $a$ as a function of $v$, use $x$ for $v$).",
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            "domain": [
              0,
              5
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            "variable": "x"
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          "explanation": "$a = v^2$ -- a parabola: doubling the speed quadruples the acceleration."
        },
        {
          "scope": "PRACTICE",
          "kind": "MCQ",
          "prompt": "If the speed doubles at the same radius, the centripetal acceleration becomes:",
          "options": [
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            "halved",
            "four times larger",
            "unchanged"
          ],
          "answer": 2,
          "hint": "$a_c \\propto v^2$.",
          "explanation": "Acceleration scales with $v^2$, so it quadruples."
        },
        {
          "scope": "PRACTICE",
          "kind": "NUMERIC",
          "prompt": "A point at radius $2\\,\\text{m}$ completes a loop every $\\pi\\,\\text{s}$. Find its speed in m/s ($v = 2\\pi r/T$).",
          "answer": {
            "value": 4,
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              "hint": "$v = 2\\pi(2)/\\pi$.",
              "explanation": "$v = 4\\pi/\\pi = 4\\,\\text{m/s}$."
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                "sub": "The pull that drops an apple is the very same pull that bends the Moon’s path. Newton’s great
unification."
              }
            },
            {
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              "title": "From the apple to the Moon",
              "content": {
                "markdown": "Newton’s insight was audacious: the force pulling an apple to the ground is the *same*
force holding the Moon in orbit, just weaker with distance. Every pair of masses in the universe attracts, by one law,
with no exceptions yet found. That single equation explains tides, orbits, and the structure of galaxies."
              }
            },
            {
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              "content": {
                "markdown": "**Newton’s law of universal gravitation:** every two point masses attract along the line
joining them with a force\n\n$$F = G\\,\\frac{m_1 m_2}{r^2},$$\n\nwhere $m_1, m_2$ are the masses, $r$ is the distance
between their centers, and $G = 6.67\\times10^{-11}\\ \\text{N*m}^2/\\text{kg}^2$ is the gravitational constant.\n\nKey
features:\n\n- The force is **mutual and equal** on both bodies (Newton’s third law): the Earth pulls you down exactly
as hard as you pull it up.\n- It obeys an **inverse-square law**: triple the distance and the force drops to
$\\tfrac{1}{9}$.\n- Near Earth’s surface this reduces to $F = mg$ with $g = GM_\\oplus/R_\\oplus^2 \\approx
9.8\\,\\text{m/s}^2$."
              }
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              "content": {
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                "intro": "Gravity gives every mass the same acceleration near Earth’s surface. Drop objects and watch
them fall together -- mass cancels out."
              }
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                    "steps": [
                      "$F \\propto 1/r^2$.",
                      "Replace $r$ with $3r$: factor $1/3^2 = 1/9$."
                    ],
                    "answer": "The force becomes one-ninth as strong."
                  },
                  {
                    "title": "Surface gravity",
                    "problem": "Why do all objects fall at the same $g$ regardless of mass?",
                    "steps": [
                      "$F = GMm/R^2$ on a mass $m$.",
                      "Acceleration $a = F/m = GM/R^2$.",
                      "The $m$ cancels."
                    ],
                    "answer": "Acceleration is independent of the falling mass."
                  },
                  {
                    "title": "Reading the constant",
                    "problem": "In $F = Gm_1m_2/r^2$, what role does $G$ play?",
                    "steps": [
                      "It sets the absolute strength of gravity.",
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                    "It’s the same everywhere in the universe."
                  ],
                  "answer": "$G$ is the universal gravitational constant."
                }
              ]
            }
          },
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            }
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                "It is attractive, mutual, and equal on both bodies.",
                "It follows an inverse-square law with distance.",
                "$G$ is the universal gravitational constant.",
                "Surface gravity $g = GM/R^2$ is the same for all falling masses."
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              "$\\propto 1/r$",
              "$\\propto 1/r^2$",
              "independent of $r$"
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            "answer": 2,
            "explanation": "It is an inverse-square law: $F \\propto 1/r^2$."
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            "answer": true,
            "explanation": "Newton’s third law -- the gravitational forces are an equal, opposite pair."
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            "scope": "CONCEPT_CHECK",
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            "prompt": "If the distance between two masses triples, the force becomes:",
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              "9x larger",
              "1/3 as strong",
              "1/9 as strong"
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              ],
              "answer": 3,
              "explanation": "$1/3^2 = 1/9$."
            },
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$r$. (Type like ‘k/r^2’.)",
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                "vars": [
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                  "r"
                ]
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              "difficulty": 3,
              "hint": "Inverse square of $r$.",
              "explanation": "$F = \\dfrac{k}{r^2}$ where $k = Gm_1m_2$."
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              "scope": "PRACTICE",
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as a function of $r$, use $x$ for $r$).",
              "answer": {
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                "domain": [
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                  4
                ],
                "variable": "x"
              },
              "difficulty": 4,
              "hint": "Inverse-square curve.",
              "explanation": "$F = 1/r^2$ -- a steep curve that falls off quickly with distance."
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              "explanation": "Acceleration $g = GM/R^2$ is independent of the falling mass."
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              "scope": "PRACTICE",
              "kind": "MCQ",
              "prompt": "The constant $G$ in $F = Gm_1m_2/r^2$:",
              "options": [
                "changes from planet to planet",
                "is the same throughout the universe",
                "depends on the masses",
                "depends on the distance"
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              "answer": 1,
              "hint": "It’s the *universal* gravitational constant.",
              "explanation": "$G$ is a universal constant."
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                "Gravity on a mass $m$ near Earth is $F = GMm/R^2$.",
                "By Newton’s second law, $a = F/m$.",
                "Substitute: $a = GMm/(R^2 m)$.",
                "The $m$ cancels: $a = GM/R^2 = g$."
              ],
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              "difficulty": 5,
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              "explanation": "Dividing the gravitational force by the mass cancels $m$, leaving $g = GM/R^2$."
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              }
            },
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              "content": {
                "markdown": "Newton imagined a cannon firing horizontally ever faster. Slow shots fall to Earth; fast enough, and the ground curves away as quickly as the ball falls -- it never lands. That is an **orbit**: the same gravity that drops an apple, supplying exactly the centripetal force a circular path needs."
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            },
            {
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              "content": {
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              }
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                      "The satellite mass $m$ cancels.",
                      "$v = \\sqrt{GM/r}$."
                    ],
                    "answer": "Orbital speed depends only on $M$ and $r$."
                  },
                  {
                    "title": "Kepler scaling",
                    "problem": "Planet B orbits 4x farther than planet A. How many times longer is B’s year? (Use $T^2 \\propto r^3$.)",
                    "steps": [
                      "$T \\propto r^{3/2}$.",
                      "$4^{3/2} = (\\sqrt4)^3 = 2^3 = 8$."
                    ],
                    "answer": "B’s year is 8 times longer."
                  },
                  {
                    "title": "Equal areas",
                    "problem": "Where in its elliptical orbit does a comet move fastest?",
                    "steps": [
                      "Equal areas in equal times.",
                      "Near the Sun the radius is short, so it must sweep faster."
                    ],
                    "answer": "At its closest approach (perihelion)."
                  }
                ]
              }
            },
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                "intro": "Gravity supplies the centripetal force."
              }
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              "content": {
                "intro": "Use $v = \\sqrt{GM/r}$ and $T^2 \\propto r^3$."
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          }
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          "content": {
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$\\dfrac{2\\pi r}{T} = \\sqrt{\\dfrac{GM}{r}}$. Square both sides: $\\dfrac{4\\pi^2 r^2}{T^2} = \\dfrac{GM}{r}$.
Rearrange to isolate $T^2$: $T^2 = \\dfrac{4\\pi^2}{GM} r^3$. The proportionality constant $\\dfrac{4\\pi^2}{GM}$ is the
 same for every body orbiting that mass -- which is exactly why a single Kepler curve fits all the planets at once."
          }
        },
        {
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              "Gravity provides the centripetal force: $v = \\sqrt{GM/r}$.",
              "Orbital speed is independent of the orbiting mass; farther = slower.",
              "Kepler III: $T^2 \\propto r^3$.",
              "Orbits are ellipses; planets sweep equal areas in equal times."
            ],
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                "tex": "v = \\sqrt{\\dfrac{GM}{r}}"
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              {
                "label": "Kepler’s third law",
                "tex": "T^2 \\propto r^3"
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            "gravity",
            "the engine",
            "magnetism"
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            "$T^3 \\propto r^2$",
            "$T \\propto 1/r$"
          ],
          "answer": 1,
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              "k",
              "r"
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              "difficulty": 4,
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              "explanation": "$v = \\sqrt{\\dfrac{GM}{r}} = \\sqrt{\\dfrac{k}{r}}$."
            },
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              "prompt": "Planet B orbits 4 times farther than planet A. Using $T^2 \\propto r^3$, how many times longer
is B’s orbital period?",
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                "tolerance": 0
              },
              "hint": "$4^{3/2} = (\\sqrt 4)^3$.",
              "explanation": "$4^{3/2} = 2^3 = 8$ times longer."
            },
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function of $r$, use $x$ for $r$).",
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                "domain": [
                  0,
                  4
                ],
                "variable": "x"
              },
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              "explanation": "$T = r^{1.5}$ -- period rises faster than linearly, but slower than $r^2$."
            },
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                "farthest from the Sun",
                "closest to the Sun",
                "at constant speed throughout",
                "halfway along the orbit"
              ],
              "answer": 1,
              "hint": "Equal areas in equal times.",
              "explanation": "Near the Sun it sweeps the same area with a shorter radius, so it must move faster."
            },
            {
              "scope": "PRACTICE",
              "kind": "PROOF",
              "prompt": "Arrange the derivation of orbital speed for a circular orbit.",
              "options": [
                "Gravity equals the centripetal force: $\\dfrac{GMm}{r^2} = \\dfrac{mv^2}{r}$.",
                "Cancel $m$ from both sides.",
                "Multiply both sides by $r$: $\\dfrac{GM}{r} = v^2$.",
                "Take the square root: $v = \\sqrt{GM/r}$."
              ],
              "answer": [],
              "difficulty": 5,
              "hint": "Set the two forces equal, then solve for $v$.",
              "explanation": "Equating gravity to the centripetal requirement and solving gives $v = \\sqrt{GM/r}$."
            }
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      "name": "Binary Star Systems",
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              "content": {
                "markdown": "Binary stars trace each other in a steady, repeating dance -- the same periodic rhythm as a mass bouncing on a spring or a child on a swing. When the force pulling a system back to equilibrium grows in proportion to how far it’s displaced, you get **simple harmonic motion (SHM)**, the template for all oscillation."
              }
            },
            {
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              "content": {
                "markdown": "**Simple harmonic motion** occurs when the restoring force is proportional to displacement and points back toward equilibrium -- **Hooke’s law**:\n\n$$F = -kx,$$\n\nwhere $k$ is the stiffness and the minus sign means ’always back toward center’.\n\nThe motion in time is a sinusoid:\n\n$$x(t) = A\\cos(\\omega t),$$\n\nwith **amplitude** $A$ (the maximum displacement) and **angular frequency** $\\omega$. For a mass $m$ on a spring,\n\n$$\\omega = \\sqrt{\\frac{k}{m}}, \\qquad T = 2\\pi\\sqrt{\\frac{m}{k}}.$$\n\nThe **period** $T$ is the time for one full cycle; **frequency** $f = 1/T$ is cycles per second (hertz). Notice the period does **not** depend on amplitude -- a wider swing simply moves faster, taking the same time. Energy sloshes between elastic potential ($\\tfrac12 kx^2$) and kinetic ($\\tfrac12 mv^2$), summing to a constant."
              }
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                      "$\\sqrt{4/1} = 2$."
                    ],
                    "answer": "$\\omega = 2\\,\\text{rad/s}$."
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                    "title": "Frequency",
                    "problem": "An oscillator has period $T = 0.5\\,\\text{s}$. Find its frequency.",
                    "steps": [
                      "$f = 1/T$.",
                      "$1/0.5 = 2$."
                    ],
                    "answer": "$f = 2\\,\\text{Hz}$."
                  },
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                    "title": "Amplitude independence",
                    "problem": "Does a bigger push change a spring’s period?",
                    "steps": [
                      "$T = 2\\pi\\sqrt{m/k}$ has no $A$.",
                      "Period is fixed by $m$ and $k$."
                    ],
                    "answer": "No -- period is independent of amplitude."
                  }
                ]
              }
            },
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            {
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```
              "content": {
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            },
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              "content": {
                "markdown": "Spin a point steadily around a circle and watch its shadow on a wall: the shadow moves in exact simple harmonic motion. That’s why $x(t) = A\\cos(\\omega t)$ uses the same $\\omega$ as rotation. It also explains the deep link to calculus: differentiating $x(t) = A\\cos\\omega t$ gives velocity $-A\\omega\\sin\\omega t$ and acceleration $-A\\omega^2\\cos\\omega t = -\\omega^2 x$ -- acceleration proportional to displacement, the defining equation of SHM."
              }
            },
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                  "SHM arises from a restoring force $F = -kx$.",
                  "Motion is sinusoidal: $x(t) = A\\cos(\\omega t)$.",
                  "$\\omega = \\sqrt{k/m}$ and $T = 2\\pi\\sqrt{m/k}$.",
                  "Frequency $f = 1/T$, in hertz.",
                  "Period is independent of amplitude; energy trades between elastic and kinetic."
                ],
                "formulas": [
                  {
                    "label": "Hooke’s law",
                    "tex": "F = -kx"
                  },
                  {
                    "label": "Angular frequency",
                    "tex": "\\omega = \\sqrt{\\dfrac{k}{m}}"
                  },
                  {
                    "label": "Period",
                    "tex": "T = 2\\pi\\sqrt{\\dfrac{m}{k}}"
                  }
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                "mass only"
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              "answer": 1,
              "explanation": "$F = -kx$ -- proportional to displacement."
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            },
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              "prompt": "An oscillator has period $T = 0.5\\,\\text{s}$. Find its frequency in Hz.",
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              "explanation": "$f = 1/0.5 = 2$."
            },
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              "prompt": "Write the angular frequency $\\omega$ of a spring in terms of stiffness $k$ and mass $m$. (Type like ‘sqrt(k/m)‘.)",
```

```
              "answer": {
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                "vars": [
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                  "m"
                ]
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              "explanation": "$\\omega = \\sqrt{k/m}$."
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              "answer": false,
              "hint": "Does $T = 2\\pi\\sqrt{m/k}$ contain amplitude?",
              "explanation": "Period depends only on $m$ and $k$, not amplitude."
            },
            {
              "scope": "PRACTICE",
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function of $t$, use $x$ for $t$).",
              "answer": {
                "expr": "cos(x)",
                "domain": [
                  -6.28,
                  6.28
                ],
                "variable": "x"
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              "difficulty": 4,
              "hint": "Starts at the maximum.",
              "explanation": "$x(t) = \\cos t$ -- starts at $+1$ and oscillates."
            },
            {
              "scope": "PRACTICE",
              "kind": "NUMERIC",
              "prompt": "A $4\\,\\text{kg}$ mass on a spring with $k = 16\\,\\text{N/m}$. Find $\\omega$ in rad/s.",
              "answer": {
                "value": 2,
                "tolerance": 0
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              "hint": "$\\sqrt{16/4}$.",
              "explanation": "$\\sqrt{4} = 2$."
            },
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                "metres",
                "hertz",
                "newtons"
              ],
              "answer": 2,
              "hint": "Cycles per second.",
              "explanation": "Frequency is in hertz (Hz)."
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                "sub": "A wave carries energy across space while the medium itself just oscillates in place."
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            },
            {
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```
              "content": {
                "markdown": "The light reaching us from a binary star crossed light-years of space as a **wave**. So does the sound of a voice, the ripple on a pond, and the signal in a fibre. A wave is a self-propagating oscillation -- and one simple equation links its speed, frequency, and wavelength."
              }
            },
            {
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              "content": {
                "markdown": "A **wave** is a disturbance that carries energy through a medium (or, for light, through space) while the medium itself only oscillates locally.\n\nKey quantities:\n\n- **Wavelength** $\\lambda$: distance between successive crests.\n- **Frequency** $f$: oscillations per second (Hz).\n- **Amplitude**: the oscillation’s size (loudness for sound, brightness for light).\n- **Speed** $v$: how fast a crest travels.\n\nThe master relation:\n\n$$v = f\\lambda.$$\n\nFor a fixed speed, higher frequency means shorter wavelength. **Transverse** waves oscillate perpendicular to travel (light, a plucked string); **longitudinal** waves oscillate along the travel direction (sound, as compressions of air). Sound travels at about $340\\,\\text{m/s}$ in air; light at $3\\times10^8\\,\\text{m/s}$."
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                    "steps": [
                      "$v = f\\lambda$.",
                      "$5 \\times 2 = 10$."
                    ],
                    "answer": "$v = 10\\,\\text{m/s}$."
                  },
                  {
                    "title": "Find wavelength",
                    "problem": "Sound at $340\\,\\text{m/s}$ and $f = 170\\,\\text{Hz}$. Find $\\lambda$.",
                    "steps": [
                      "$\\lambda = v/f$.",
                      "$340/170 = 2$."
                    ],
                    "answer": "$\\lambda = 2\\,\\text{m}$."
                  },
                  {
                    "title": "Type of wave",
                    "problem": "Is sound transverse or longitudinal?",
                    "steps": [
                      "Air compresses along the travel direction.",
                      "Oscillation is parallel to motion."
                    ],
                    "answer": "Longitudinal."
                  }
                ]
              }
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              }
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                "intro": "Rearrange $v = f\\lambda$ as needed."
              }
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              }
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                "A wave carries energy while the medium oscillates in place.",
                "Wavelength lambda, frequency f, amplitude, and speed v describe it.",
                "The wave relation: $v = f\\lambda$.",
                "Transverse waves oscillate across travel; longitudinal along it.",
                "Sound is longitudinal (~340 m/s); light is transverse (~3x10^8 m/s)."
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                  "tex": "v = f\\lambda"
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              "value": 10,
              "tolerance": 0
            },
            "explanation": "$v = f\\lambda = 10\\,\\text{m/s}$."
          },
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            "prompt": "The wave relation is:",
            "options": [
              "$v = f/\\lambda$",
              "$v = f\\lambda$",
              "$v = \\lambda/f$",
              "$v = f + \\lambda$"
            ],
            "answer": 1,
            "explanation": "$v = f\\lambda$."
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            "scope": "CONCEPT_CHECK",
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            "prompt": "Sound waves in air are:",
            "options": [
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              "stationary",
              "electromagnetic"
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            "answer": 1,
            "explanation": "Air compresses along the direction of travel -- longitudinal."
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            "answer": {
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            "hint": "$\\lambda = v/f$.",
            "explanation": "$340/170 = 2\\,\\text{m}$."
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            "prompt": "Solve the wave relation for wavelength $\\lambda$ in terms of speed $v$ and frequency $f$.
(Type like ‘v/f’.)",
            "answer": {
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              "vars": [
                "v",
                "f"
              ]
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            "difficulty": 3,
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            "explanation": "$\\lambda = v/f$."
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              "explanation": "Since $v = f\\lambda$ is constant, $f$ up means $\\lambda$ down."
            },
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              "scope": "PRACTICE",
              "kind": "NUMERIC",
              "prompt": "A wave travels at $12\\,\\text{m/s}$ with wavelength $3\\,\\text{m}$. Find its frequency in
Hz.",
              "answer": {
                "value": 4,
                "tolerance": 0
              },
              "hint": "$f = v/\\lambda$.",
              "explanation": "$12/3 = 4\\,\\text{Hz}$."
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                "greenhouse effect",
                "Bernoulli effect"
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              "answer": 0,
              "hint": "Motion changes observed frequency.",
              "explanation": "Relative motion shifts the observed frequency -- the Doppler effect."
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          "estMinutes": 15,
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                "sub": "A force like gravity but vastly stronger -- and it comes in two signs, so it can push as well as
 pull."
              }
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              "title": "Glowing clouds of charge",
              "content": {
                "markdown": "A nebula glows because its gas is **ionized** -- atoms stripped into charged particles.
What governs them is the **electric force**, described by Coulomb’s law. It looks almost identical to Newton’s gravity,
but it is enormously stronger and, uniquely, can both attract and repel."
              }
            },
            {
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              "title": "The big idea",
              "content": {
                "markdown": "**Electric charge** comes in two kinds, positive and negative, measured in coulombs (C).
The rule: **like charges repel, opposite charges attract.**\n\n**Coulomb’s law** gives the force between two point
```

```
charges:\n\n$$F = k\\,\\frac{q_1 q_2}{r^2},$$\n\nwhere $r$ is their separation and $k \\approx 9\\times10^{9}\\
\\text{N*m}^2/\\text{C}^2$.\n\nCompare with gravity, $F = G\\dfrac{m_1 m_2}{r^2}$: both are **inverse-square** laws. The
 differences:\n\n- Charge has two signs, so the force can be attractive *or* repulsive; gravity only attracts.\n- The
electric force is far stronger -- about $10^{36}$ times stronger between two protons.\n\nTriple the distance and the
force drops to $\\tfrac19$, exactly as with gravity."
              }
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                    "steps": [
                      "$F \\propto 1/r^2$.",
                      "$1/3^2 = 1/9$."
                    ],
                    "answer": "It drops to one-ninth."
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                    "title": "Sign of the force",
                    "problem": "Two negative charges -- attract or repel?",
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                      "Like charges.",
                      "Like repels."
                    ],
                    "answer": "They repel."
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                      "Both inverse-square."
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              }
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              "content": {
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any charge is the **vector sum** of the separate Coulomb forces from every other charge, each computed as if the others
weren’t there. This simple additivity -- shared with gravity -- is what lets us handle complex arrangements, and it
underlies the electric field of any charged object, however intricate."
              }
            },
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                  "Charge is positive or negative; like repels, opposite attracts.",
                  "Coulomb’s law: $F = kq_1q_2/r^2$.",
                  "It is an inverse-square law, like gravity.",
                  "The electric force can attract or repel and is far stronger than gravity.",
                  "Forces from many charges add by superposition (vector sum)."
                ],
                "formulas": [
                  {
                    "label": "Coulomb’s law",
                    "tex": "F = k\\dfrac{q_1 q_2}{r^2}"
                  }
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        "repel",
        "do nothing",
        "merge"
      ],
      "answer": 1,
      "explanation": "Like charges repel."
    },
    {
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      "kind": "MCQ",
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      "options": [
        "$\\propto r$",
        "$\\propto 1/r$",
        "$\\propto 1/r^2$",
        "independent of $r$"
      ],
      "answer": 2,
      "explanation": "Inverse-square, like gravity."
    },
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      "prompt": "Unlike gravity, the electric force can repel as well as attract.",
      "answer": true,
      "explanation": "Charge has two signs, so the force can do both."
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      "prompt": "If the distance between two charges triples, the force becomes:",
      "options": [
        "3x larger",
        "9x larger",
        "1/3",
        "1/9"
      ],
      "answer": 3,
      "hint": "$1/3^2$.",
      "explanation": "Inverse-square: $1/9$."
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    {
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      "prompt": "With $k q_1 q_2 = c$ constant, write the Coulomb force $F$ as a function of distance $r$. (Type
 like ‘c/r^2‘.)",
      "answer": {
        "expr": "c/r^2",
        "vars": [
          "c",
          "r"
        ]
      },
      "difficulty": 3,
      "hint": "Inverse square of $r$.",
      "explanation": "$F = \\dfrac{c}{r^2}$."
    },
    {
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$r$, use $x$ for $r$).",
      "answer": {
        "expr": "1/x^2",
        "domain": [
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          4
        ],
        "variable": "x"
      },
      "difficulty": 4,
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            "hint": "Inverse-square curve.",
            "explanation": "$F = 1/r^2$ falls off steeply with distance."
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            "prompt": "Coulomb’s law and Newton’s gravitation share the same inverse-square form.",
            "answer": true,
            "hint": "Compare $q_1q_2/r^2$ and $m_1m_2/r^2$.",
            "explanation": "Both are inverse-square in the separation."
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            "prompt": "The total force from several charges on one charge is found by:",
            "options": [
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              "adding the forces as vectors",
              "averaging them",
              "ignoring distance"
            ],
            "answer": 1,
            "hint": "Superposition.",
            "explanation": "Forces add as vectors -- the superposition principle."
          }
        ]
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        "difficulty": 4,
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              "sub": "A charge reshapes the space around it into a field -- and any other charge feels that field, not
 the distant source directly."
            }
          },
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            "kind": "CONTEXT",
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            "content": {
              "markdown": "How does one charge ’know’ another is there across empty space? Physics answers with the
**field**: a charge fills the surrounding space with an electric field, and any charge placed there feels a force from
the local field. The field is the real, physical go-between."
            }
          },
          {
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            "content": {
              "markdown": "The **electric field** $E$ at a point is the force per unit positive charge placed
there:\n\n$$E = \\frac{F}{q}, \\qquad \\text{so} \\qquad F = qE.$$\n\nIts units are newtons per coulomb (N/C). A single
point charge $Q$ produces a field\n\n$$E = k\\,\\frac{Q}{r^2},$$\n\nagain inverse-square. The field points **away** from
 positive charges and **toward** negative ones -- the direction a small positive test charge would be pushed.\n\nField
lines picture it: denser lines mean a stronger field. Between two parallel charged plates the field is **uniform**
(constant strength and direction), the configuration inside capacitors and the setup for accelerating particles in a
straight line."
            }
          },
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                  "problem": "A charge $q = 2\\,\\text{C}$ sits in a field $E = 5\\,\\text{N/C}$. Find the force.",
                  "steps": [
                    "$F = qE$.",
                    "$2 \\times 5$."
                  ],
                  "answer": "$F = 10\\,\\text{N}$."
                },
                {
                  "title": "Field from force",
```

```
                    "problem": "A $3\\,\\text{C}$ charge feels $12\\,\\text{N}$. Find $E$.",
                    "steps": [
                      "$E = F/q$.",
                      "$12/3$."
                    ],
                    "answer": "$E = 4\\,\\text{N/C}$."
                  },
                  {
                    "title": "Direction",
                    "problem": "Which way does the field point near a positive charge?",
                    "steps": [
                      "Away from positive.",
                      "Toward negative."
                    ],
                    "answer": "Radially outward."
                  }
                ]
              }
            },
            {
              "kind": "CONCEPT_CHECK",
              "title": "Quick check",
              "content": {
                "intro": "$E = F/q$; field points away from positive."
              }
            },
            {
              "kind": "PRACTICE",
              "title": "Practice problems",
              "content": {
                "intro": "Relate field, force, and charge with $F = qE$."
              }
            },
            {
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              "title": "Deeper dive: energy and potential",
              "content": {
                "markdown": "Just as height in a gravity field stores potential energy, position in an electric field stores **electric potential energy**. The energy per unit charge is the **potential** $V$, measured in **volts**. A difference in potential -- a **voltage** -- is what drives charges through a circuit, exactly as a height difference drives water downhill. Field and potential are two views of the same thing: the field points ’downhill’ in potential, from high $V$ to low."
              }
            },
            {
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                "takeaways": [
                  "The electric field is force per unit charge: $E = F/q$.",
                  "A charge in a field feels $F = qE$.",
                  "A point charge makes a field $E = kQ/r^2$ (inverse-square).",
                  "Field points away from positive charges, toward negative.",
                  "Potential (volts) is the energy per unit charge; voltage drives current."
                ],
                "formulas": [
                  {
                    "label": "Field",
                    "tex": "E = \\dfrac{F}{q}"
                  },
                  {
                    "label": "Point-charge field",
                    "tex": "E = k\\dfrac{Q}{r^2}"
                  }
                ]
              }
            }
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              "answer": {
                "value": 10,
                "tolerance": 0
              },
              "explanation": "$F = qE = 10\\,\\text{N}$."
            },
            {
              "scope": "CONCEPT_CHECK",
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      "kind": "NUMERIC",
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        "value": 4,
        "tolerance": 0
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      "explanation": "$E = F/q = 4$."
    },
    {
      "scope": "CONCEPT_CHECK",
      "kind": "MCQ",
      "prompt": "The electric field near a positive charge points:",
      "options": [
        "toward the charge",
        "away from the charge",
        "in circles",
        "there is none"
      ],
      "answer": 1,
      "explanation": "Away from positive charges."
    },
    {
      "scope": "PRACTICE",
      "kind": "SYMBOLIC",
      "prompt": "Write the force $F$ on a charge $q$ in a field $E$. (Type like ‘q*E‘.)",
      "answer": {
        "expr": "q*E",
        "vars": [
          "q",
          "E"
        ]
      },
      "difficulty": 2,
      "hint": "Field times charge.",
      "explanation": "$F = qE$."
    },
    {
      "scope": "PRACTICE",
      "kind": "GRAPH",
      "prompt": "Taking $kQ = 1$, a point charge’s field is $E = 1/r^2$. Graph it (enter $E$ as a function of
$r$, use $x$ for $r$).",
      "answer": {
        "expr": "1/x^2",
        "domain": [
          0.5,
          4
        ],
        "variable": "x"
      },
      "difficulty": 4,
      "hint": "Inverse-square again.",
      "explanation": "$E = 1/r^2$ -- the same shape as the Coulomb force."
    },
    {
      "scope": "PRACTICE",
      "kind": "NUMERIC",
      "prompt": "A $0.5\\,\\text{C}$ charge sits in a $20\\,\\text{N/C}$ field. Find the force in newtons.",
      "answer": {
        "value": 10,
        "tolerance": 0
      },
      "hint": "$F = qE$.",
      "explanation": "$0.5 \\times 20 = 10$."
    },
    {
      "scope": "PRACTICE",
      "kind": "MCQ",
      "prompt": "Electric potential is measured in:",
      "options": [
        "newtons",
        "volts",
        "amps",
        "ohms"
      ],
      "answer": 1,
      "hint": "Energy per unit charge.",
      "explanation": "Potential is measured in volts."
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    {
      "scope": "PRACTICE",
      "kind": "TRUE_FALSE",
      "prompt": "Between two parallel charged plates the field is approximately uniform.",
```

```
              "answer": true,
              "hint": "Think of a capacitor.",
              "explanation": "Parallel plates produce a uniform field."
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          "title": "Circuits & Ohm’s Law",
          "tagline": "Voltage, current, resistance",
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                "scene": "sun",
                "headline": "Charge in Motion",
                "sub": "Give charges a push and a path, and they flow -- a current, governed by one beautifully simple
law."
              }
            },
            {
              "kind": "CONTEXT",
              "title": "From spark to circuit",
              "content": {
                "markdown": "Static charge is dramatic but fleeting. Channel charge into a steady **flow** through wires
 and you get a **circuit** -- the basis of every device you own. Three quantities -- voltage, current, resistance -- and
 one law connect them."
              }
            },
            {
              "kind": "CONCEPT",
              "title": "The big idea",
              "content": {
                "markdown": "Three quantities describe a simple circuit:\n\n- **Voltage** $V$ (volts): the energy per
charge driving the flow -- the ’push’.\n- **Current** $I$ (amperes): the rate of charge flow -- how much passes per
second.\n- **Resistance** $R$ (ohms): how strongly the path opposes the flow.\n\n**Ohm’s law** links them:\n\n$$V =
I\\,R.$$\n\nSo $I = \\dfrac{V}{R}$ -- more voltage drives more current; more resistance chokes it. The water analogy:
voltage is the pump pressure, current is the flow rate, resistance is a narrow pipe.\n\n**Electrical power** delivered
is\n\n$$P = VI = I^2R,$$\n\nin watts -- the rate energy is converted to light, heat, or motion."
              }
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                    "problem": "A $12\\,\\text{V}$ battery drives a $4\\,\\Omega$ resistor. Find the current.",
                    "steps": [
                      "$I = V/R$.",
                      "$12/4 = 3$."
                    ],
                    "answer": "$I = 3\\,\\text{A}$."
                  },
                  {
                    "title": "Find voltage",
                    "problem": "A current of $2\\,\\text{A}$ flows through $5\\,\\Omega$. Find the voltage.",
                    "steps": [
                      "$V = IR$.",
                      "$2 \\times 5 = 10$."
                    ],
                    "answer": "$V = 10\\,\\text{V}$."
                  },
                  {
                    "title": "Power",
                    "problem": "Find the power for $V = 12\\,\\text{V}$, $I = 3\\,\\text{A}$.",
                    "steps": [
                      "$P = VI$.",
                      "$12 \\times 3 = 36$."
                    ],
                    "answer": "$P = 36\\,\\text{W}$."
                  }
                ]
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            },
            {
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```
            "content": {
              "intro": "Ohm’s law: $V = IR$."
            }
          },
          {
            "kind": "PRACTICE",
            "title": "Practice problems",
            "content": {
              "intro": "Rearrange $V = IR$; compute power with $P = VI$."
            }
          },
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            "kind": "DEEPER_DIVE",
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            "content": {
              "markdown": "Resistors in **series** (one after another) add: $R = R_1 + R_2$ -- the current has more to
 fight through. In **parallel** (side by side) they combine as $\\dfrac1R = \\dfrac1{R_1} + \\dfrac1{R_2}$, giving
*less* total resistance because the current has more paths. These two rules, plus Ohm’s law, let you analyse any network
 -- and they’re the electrical cousins of how springs combine and how pipes share flow."
            }
          },
          {
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            "title": "Summary",
            "content": {
              "takeaways": [
                "Voltage (V) pushes; current (A) flows; resistance (Omega) opposes.",
                "Ohm’s law: $V = IR$, so $I = V/R$.",
                "More voltage -> more current; more resistance -> less current.",
                "Power $P = VI = I^2R$, in watts.",
                "Series resistances add; parallel resistances combine reciprocally."
              ],
              "formulas": [
                {
                  "label": "Ohm’s law",
                  "tex": "V = IR"
                },
                {
                  "label": "Power",
                  "tex": "P = VI"
                }
              ]
            }
          }
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            "answer": {
              "value": 3,
              "tolerance": 0
            },
            "explanation": "$I = V/R = 12/4 = 3$."
          },
          {
            "scope": "CONCEPT_CHECK",
            "kind": "NUMERIC",
            "prompt": "A current of $2\\,\\text{A}$ flows through $5\\,\\Omega$. Find the voltage in volts.",
            "answer": {
              "value": 10,
              "tolerance": 0
            },
            "explanation": "$V = IR = 10$."
          },
          {
            "scope": "CONCEPT_CHECK",
            "kind": "MCQ",
            "prompt": "Ohm’s law states:",
            "options": [
              "$V = I/R$",
              "$V = IR$",
              "$V = R/I$",
              "$V = I + R$"
            ],
            "answer": 1,
            "explanation": "$V = IR$."
          },
          {
            "scope": "PRACTICE",
            "kind": "SYMBOLIC",
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            "prompt": "Solve Ohm’s law for current $I$ in terms of $V$ and $R$. (Type like ‘V/R‘.)",
            "answer": {
              "expr": "V/R",
              "vars": [
                "V",
                "R"
              ]
            },
            "difficulty": 3,
            "hint": "Divide voltage by resistance.",
            "explanation": "$I = V/R$."
          },
          {
            "scope": "PRACTICE",
            "kind": "NUMERIC",
            "prompt": "Find the power for $V = 12\\,\\text{V}$ and $I = 3\\,\\text{A}$, in watts.",
            "answer": {
              "value": 36,
              "tolerance": 0
            },
            "hint": "$P = VI$.",
            "explanation": "$12 \\times 3 = 36$."
          },
          {
            "scope": "PRACTICE",
            "kind": "GRAPH",
            "prompt": "For a fixed resistance $R = 2\\,\\Omega$, voltage vs current is $V = 2I$. Graph it (enter $V$
as a function of $I$, use $x$ for $I$).",
            "answer": {
              "expr": "2*x",
              "domain": [
                0,
                5
              ],
              "variable": "x"
            },
            "difficulty": 3,
            "hint": "Linear: slope is the resistance.",
            "explanation": "$V = 2I$ -- a straight line whose slope is the resistance."
          },
          {
            "scope": "PRACTICE",
            "kind": "NUMERIC",
            "prompt": "Two $3\\,\\Omega$ resistors in series. Find the total resistance in ohms.",
            "answer": {
              "value": 6,
              "tolerance": 0
            },
            "hint": "Series resistances add.",
            "explanation": "$3 + 3 = 6\\,\\Omega$."
          },
          {
            "scope": "PRACTICE",
            "kind": "TRUE_FALSE",
            "prompt": "Adding resistors in parallel lowers the total resistance.",
            "answer": true,
            "hint": "More paths for current.",
            "explanation": "Parallel paths reduce total resistance."
          }
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photoelectric effect, and the quantum nature of light and matter -- the physics that explodes a star.",
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                "headline": "The Speed Limit of the Universe",
                "sub": "Hold the speed of light constant for everyone, and space and time themselves must bend to comply."
              }
            },
            {
              "kind": "CONTEXT",
              "title": "When stars explode",
              "content": {
                "markdown": "A supernova flings matter outward at a sizeable fraction of light speed, and converts mass into staggering energy. To describe such speeds, Newton’s mechanics fails and Einstein’s **special relativity** takes over -- built on two deceptively simple postulates with shocking consequences."
              }
            },
            {
              "kind": "CONCEPT",
              "title": "The big idea",
              "content": {
                "markdown": "Special relativity rests on two postulates:\n\n1. The laws of physics are the same in every inertial (non-accelerating) frame.\n2. The speed of light $c \\approx 3\\times10^8\\,\\text{m/s}$ is the same for **all** observers, regardless of their motion.\n\nThe second is the radical one. To keep $c$ constant, space and time must adjust:\n\n- **Time dilation:** a moving clock runs slow, by the factor $\\gamma = \\dfrac{1}{\\sqrt{1 - v^2/c^2}}$.\n- **Length contraction:** a moving object is shortened along its motion.\n- Nothing with mass can reach $c$ -- $\\gamma$ blows up as $v \\to c$.\n\nAnd mass itself is a form of energy, the most famous equation in physics:\n\n$$E = mc^2.$$\n\nA tiny mass holds enormous energy because $c^2$ is huge -- the engine of stars and supernovae alike."
              }
            },
            {
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              "title": "Worked examples",
              "content": {
                "examples": [
                  {
                    "title": "Gamma at low speed",
                    "problem": "Why is relativity invisible in daily life?",
                    "steps": [
                      "Everyday $v \\ll c$.",
                      "Then $v^2/c^2 \\approx 0$ and $\\gamma \\approx 1$."
                    ],
                    "answer": "Effects are negligible at ordinary speeds."
                  },
                  {
                    "title": "Rest energy",
                    "problem": "What does $E = mc^2$ say about a mass at rest?",
                    "steps": [
                      "Even at rest, mass has energy.",
                      "$E = mc^2$."
                    ],
                    "answer": "Mass is concentrated energy."
                  },
                  {
                    "title": "The cosmic limit",
                    "problem": "Can a spaceship reach the speed of light?",
                    "steps": [
                      "$\\gamma \\to \\infty$ as $v \\to c$.",
                      "Infinite energy needed."
                    ],
                    "answer": "No -- c is unreachable for matter."
                  }
                ]
              }
            },
            {
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              "title": "Quick check",
              "content": {
                "intro": "Light speed is constant for everyone; mass is energy."
              }
            },
            {
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              "title": "Practice problems",
              "content": {
                "intro": "Reason about the postulates and $E = mc^2$."
              }
```

```
            },
            {
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              "content": {
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              }
            },
            {
              "kind": "SUMMARY",
              "title": "Summary",
              "content": {
                "takeaways": [
                  "Postulate: light speed c is the same for all observers.",
                  "Moving clocks run slow (time dilation); moving lengths contract.",
                  "$\\gamma = 1/\\sqrt{1 - v^2/c^2}$ grows without bound as $v \\to c$.",
                  "No massive object can reach c.",
                  "Mass is energy: $E = mc^2$."
                ],
                "formulas": [
                  {
                    "label": "Mass-energy",
                    "tex": "E = mc^2"
                  },
                  {
                    "label": "Lorentz factor",
                    "tex": "\\gamma = \\dfrac{1}{\\sqrt{1 - v^2/c^2}}"
                  }
                ]
              }
            }
          ],
          "questions": [
            {
              "scope": "CONCEPT_CHECK",
              "kind": "MCQ",
              "prompt": "According to relativity, the speed of light is:",
              "options": [
                "faster for a moving observer",
                "the same for all observers",
                "slower in space",
                "infinite"
              ],
              "answer": 1,
              "explanation": "The constancy of $c$ is the key postulate."
            },
            {
              "scope": "CONCEPT_CHECK",
              "kind": "TRUE_FALSE",
              "prompt": "A fast-moving clock runs slow compared with a stationary one.",
              "answer": true,
              "explanation": "Time dilation -- moving clocks tick slower."
            },
            {
              "scope": "CONCEPT_CHECK",
              "kind": "MCQ",
              "prompt": "The equation $E = mc^2$ says that:",
              "options": [
                "energy and mass are unrelated",
                "mass is a form of energy",
                "light has no energy",
                "c depends on mass"
              ],
              "answer": 1,
              "explanation": "Mass and energy are equivalent."
            },
            {
              "scope": "PRACTICE",
              "kind": "SYMBOLIC",
              "prompt": "Write the rest energy $E$ of a mass $m$ (speed of light $c$). (Type like ‘m*c^2‘.)",
              "answer": {
                "expr": "m*c^2",
                "vars": [
                  "m",
                  "c"
                ]
              },
              "difficulty": 3,
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```
            "hint": "Mass times c squared.",
            "explanation": "$E = mc^2$."
          },
          {
            "scope": "PRACTICE",
            "kind": "TRUE_FALSE",
            "prompt": "A massive object can be accelerated to exactly the speed of light.",
            "answer": false,
            "hint": "What happens to $\\gamma$ as $v \\to c$?",
            "explanation": "It would require infinite energy -- impossible for matter."
          },
          {
            "scope": "PRACTICE",
            "kind": "MCQ",
            "prompt": "Relativistic effects are negligible in daily life because:",
            "options": [
              "c is small",
              "our speeds are tiny compared with c",
              "clocks are accurate",
              "mass is large"
            ],
            "answer": 1,
            "hint": "Compare $v$ to $c$.",
            "explanation": "At everyday speeds $v \\ll c$, so $\\gamma \\approx 1$."
          },
          {
            "scope": "PRACTICE",
            "kind": "TRUE_FALSE",
            "prompt": "Two events simultaneous for one observer may not be simultaneous for another.",
            "answer": true,
            "hint": "Simultaneity is relative.",
            "explanation": "Relativity makes simultaneity frame-dependent."
          },
          {
            "scope": "PRACTICE",
            "kind": "MCQ",
            "prompt": "The huge energy in $E = mc^2$ comes from:",
            "options": [
              "m being large",
              "$c^2$ being enormous",
              "addition of energies",
              "gravity"
            ],
            "answer": 1,
            "hint": "c ~= 3x10^8 m/s.",
            "explanation": "$c^2$ is an enormous multiplier."
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 of atoms. Those lines, and the **photoelectric effect** that won Einstein his Nobel Prize, revealed that light is
**quantized**: it comes in packets called **photons**, each carrying a fixed chunk of energy."
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depends only on its **frequency**:\n\n$$E = h f,$$\n\nwhere $h = 6.63\\times10^{-34}\\,\\text{J*s}$ is **Planck’s
constant**. Higher frequency (bluer light) means more energetic photons.\n\nThe **photoelectric effect** is the
clincher: shining light on a metal ejects electrons -- but only if the light’s frequency is high enough, no matter how
```

```
bright a low-frequency beam is. A smooth-wave picture can’t explain that; photons can: one photon, one electron, and it needs enough energy to free it.\n\nThis is **wave-particle duality**: light is both wave and particle, and matter (electrons, atoms) shows the same dual nature. It is the foundation of quantum mechanics."
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                    "problem": "Blue light or red light per photon?",
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                      "Blue has higher frequency.",
                      "$E = hf$ is larger."
                    ],
                    "answer": "Blue -- higher frequency, more energy."
                  },
                  {
                    "title": "Photoelectric threshold",
                    "problem": "Why doesn’t dim blue light fail but bright red light does?",
                    "steps": [
                      "Each photon’s energy is set by frequency.",
                      "Red photons individually too weak."
                    ],
                    "answer": "Energy per photon, not brightness, frees electrons."
                  }
                ]
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              }
            },
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                  "Photon energy: $E = hf$ (Planck’s constant $h$).",
                  "Higher frequency means more energetic photons.",
                  "The photoelectric effect shows energy depends on frequency, not brightness.",
                  "Wave-particle duality underlies quantum mechanics."
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              "$E = hf$",
              "$E = \\tfrac12 mv^2$",
              "$E = IR$"
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            "explanation": "$E = hf$ -- energy proportional to frequency."
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              "depends on brightness"
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              "vars": [
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                "f"
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            "explanation": "$E = hf$."
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              "variable": "x"
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            "explanation": "Below the threshold frequency, no single photon has enough energy, regardless of
brightness."
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            "explanation": "The particle (photon) picture explains the frequency threshold."
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              "time and temperature",
              "voltage and current"
            ],
            "answer": 1,
            "hint": "$\\Delta x \\Delta p \\gtrsim \\hbar/2$.",
            "explanation": "Position and momentum cannot both be known exactly."
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speed limit."
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reach $99.99\\%$ of light speed. At such speeds Newton’s $p = mv$ and $KE = \\tfrac12 mv^2$ quietly fail. Relativity
supplies the corrected laws -- and unifies energy and momentum into one elegant relation."
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v^2/c^2}$:\n\n- **Relativistic momentum:** $p = \\gamma m v$. As $v \\to c$, $\\gamma \\to \\infty$, so $p$ grows
without limit -- no finite push reaches $c$.\n- **Total energy:** $E = \\gamma m c^2$. At rest ($\\gamma = 1$) this is
$E = mc^2$; the extra over the rest energy is the **kinetic energy**, $KE = (\\gamma - 1)mc^2$.\n- **The master
relation** links them for any speed:\n\n$$E^2 = (pc)^2 + (mc^2)^2.$$\n\nFor a massless particle like a photon ($m = 0$),
 this collapses to $E = pc$ -- light carries momentum despite having no mass, which is how a **solar sail** is pushed by
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                    "$p = \\gamma m v \\approx m v$."
                  ],
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            "Set $m = 0$.",
            "$E^2 = (pc)^2$."
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          "answer": "$E = pc$ -- massless light still has momentum."
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            "Subtract rest energy $mc^2$."
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          "answer": "$KE = (\\gamma-1)mc^2$."
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        "Total energy: $E = \\gamma m c^2$; kinetic energy $(\\gamma-1)mc^2$.",
        "Master relation: $E^2 = (pc)^2 + (mc^2)^2$.",
        "Massless particles: $E = pc$ (light carries momentum).",
        "Rest mass is the frame-invariant $\\sqrt{E^2-(pc)^2}/c^2$."
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      "stays constant",
      "becomes negative"
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              "$(\\gamma-1)mc^2$",
              "$\\gamma mc^2$",
              "$mc^2$"
            ],
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                "c"
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            "explanation": "$E = pc$ for a massless particle."
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            "explanation": "$\\gamma \\approx 1$, recovering Newtonian momentum."
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              "charge",
              "frequency"
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            "difficulty": 4,
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                      "$E = \\Delta m c^2$ converts the lost mass."
                    ],
                    "answer": "The mass defect becomes energy."
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                    "problem": "Why can’t a star fuse iron for energy?",
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                      "Binding energy per nucleon peaks at iron.",
                      "Fusing beyond iron absorbs energy."
                    ],
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                  },
                  {
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                    "problem": "Why is a small $\\Delta m$ enough to power a star?",
                    "steps": [
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                      "$c^2$ is about $9\\times10^{16}$."
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            "The mass defect becomes energy: $E = \\Delta m c^2$.",
            "The Sun fuses hydrogen to helium, losing ~0.7% of the mass as light.",
            "Binding energy per nucleon peaks at iron -- fusion past iron yields no energy.",
            "Supernovae and neutron-star mergers forge the elements heavier than iron."
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          "spins faster",
          "is electrically neutral"
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          "helium"
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        "explanation": "Iron sits at the peak -- the most tightly bound nucleus."
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          "vars": [
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            "c"
          ]
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        "hint": "Einstein’s relation.",
        "explanation": "$E = m c^2$ (here $m$ is the mass defect $\\Delta m$)."
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        "answer": 2,
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              "explanation": "An iron core produces no fusion energy, so it collapses and rebounds as a supernova."
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ordinary stellar cores.",
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              "difficulty": 4,
              "hint": "Fusing past iron needs energy input.",
              "explanation": "Such extreme events supply the neutrons and energy to build the heaviest elements."
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probability."
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further isn’t heat -- it’s a purely **quantum** effect: degeneracy pressure, born from the wave nature of matter and the
 rule that identical particles can’t share a state. To understand it, we start with the strangest idea in physics:
matter is made of waves."
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Broglie**:\n\n$$\\lambda = \\frac{h}{p},$$\n\nwhere $p = mv$ is momentum and $h$ is Planck’s constant. Big objects have
absurdly tiny wavelengths (hence we never notice), but electrons’ wavelengths are atom-sized -- and they diffract like
waves.\n\nA quantum system is described by a **wavefunction** $\\psi(x)$. Born’s rule: $|\\psi(x)|^2$ is the
**probability density** of finding the particle at $x$. Quantum mechanics predicts **probabilities**, not
certainties.\n\nConsequences:\n\n- The **double-slit experiment**: single electrons build up an interference pattern --
each one passes through ’both slits’ as a wave.\n- **Heisenberg uncertainty:** $\\Delta x\\,\\Delta p \\gtrsim
\\dfrac{\\hbar}{2}$ -- position and momentum can’t both be sharp.\n\nThis is not ignorance; it’s how nature is built at
the smallest scales."
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              "$\\lambda = h/p$.",
              "Inversely."
            ],
            "answer": "Faster (more momentum) -> shorter wavelength."
          },
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            "problem": "Why don’t baseballs diffract?",
            "steps": [
              "Large $p$ -> tiny $\\lambda$.",
              "Wavelength far smaller than any slit."
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            "answer": "Their wavelength is unmeasurably small."
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              "Probability density."
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quantum state. Pack them tightly and they’re forced into ever-higher momentum states, creating **degeneracy pressure**
-- a resistance to compression that exists even at absolute zero, purely from quantum statistics. This pressure holds up
 white dwarfs (electron degeneracy) and neutron stars (neutron degeneracy), until gravity finally wins and a black hole
forms."
      }
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          "Every particle has a wavelength: $\\lambda = h/p$ (de Broglie).",
          "The wavefunction psi encodes the state; $|\\psi|^2$ is probability density.",
          "Quantum mechanics predicts probabilities, not certainties.",
          "Uncertainty: $\\Delta x\\,\\Delta p \\gtrsim \\hbar/2$.",
          "Pauli exclusion gives degeneracy pressure, holding up neutron stars."
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        "$\\lambda = h/p$",
        "$\\lambda = p/h$",
        "$\\lambda = h + p$"
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        "momentum",
        "wavelength"
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          "p"
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      "explanation": "$\\lambda = h/p$."
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      "explanation": "Big $p$ gives a negligibly small $\\lambda$."
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        "position and momentum",
        "energy and charge",
        "spin and mass"
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      "answer": 1,
      "hint": "$\\Delta x\\Delta p \\gtrsim \\hbar/2$.",
      "explanation": "Position and momentum trade off."
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        "Pauli exclusion principle",
        "strong magnetic field",
        "fusion reactions"
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      "explanation": "Pauli exclusion forces neutrons into high-momentum states."
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              "prompt": "Quantum mechanics generally predicts probabilities rather than certain outcomes.",
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              "hint": "Born’s rule.",
              "explanation": "Outcomes are probabilistic, governed by $|\\psi|^2$."
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              "content": {
                "markdown": "Each element emits light at its own exact set of colors -- a spectral ’barcode’ that lets
us read the composition of stars across the galaxy. Those discrete lines are direct evidence that electron energies in
atoms are **quantized**: allowed only at specific levels."
              }
            },
            {
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              "content": {
                "markdown": "When a wave is **confined**, only certain wavelengths fit -- like a guitar string, which
sounds only its fundamental and harmonics. A confined electron is the same: only specific **standing-wave** states are
allowed, so its energy is **quantized** into discrete levels $E_1, E_2, E_3, \\dots$\n\nKey consequences:\n\n- An
electron can only have an allowed energy, never one in between.\n- It jumps between levels by absorbing or emitting a
**photon** whose energy exactly matches the gap:\n\n$$E_{\\text{photon}} = E_{\\text{high}} - E_{\\text{low}} =
hf.$$\n\n- Because the gaps are fixed, the emitted colors are sharp and characteristic -- the spectral lines.\n\nThere
is a lowest level, the **ground state**, with nonzero energy: the uncertainty principle forbids an electron from sitting
 perfectly still at the bottom. Quantization, born from confinement plus the wave nature of matter, is why atoms are
stable and why each element glows its own color."
              }
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                    "problem": "An electron drops between levels differing by energy $\\Delta E$. What is emitted?",
                    "steps": [
                      "A photon of that energy.",
                      "$E = hf = \\Delta E$."
                    ],
                    "answer": "A photon with $hf = \\Delta E$."
                  },
                  {
                    "title": "Why lines are sharp",
                    "problem": "Why are atomic spectra discrete lines, not a smooth rainbow?",
                    "steps": [
                      "Energy gaps are fixed.",
                      "Only those photon energies appear."
                    ],
                    "answer": "Fixed gaps -> fixed colors."
                  },
                  {
                    "title": "Ground state",
                    "problem": "Can a confined electron have zero energy?",
                    "steps": [
                      "Uncertainty forbids being perfectly at rest.",
                      "Lowest state has $E > 0$."
                    ],
                    "answer": "No -- the ground state energy is nonzero."
                  }
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            },
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$F = ma$ for quantum systems: given the forces (a potential), it determines the allowed wavefunctions and their
energies. For a confined particle it yields exactly a discrete ladder of $\\psi$’s and $E$’s; for a free particle, a
continuum. Solving it for the hydrogen atom reproduces the observed spectrum to extraordinary precision -- one of the
great triumphs of physics."
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                  "Confining a wave allows only certain standing-wave states.",
                  "So a bound electron’s energy is quantized into discrete levels.",
                  "Jumps emit/absorb a photon with $hf = \\Delta E$.",
                  "Fixed gaps produce sharp spectral lines -- each element’s fingerprint.",
                  "The Schr?dinger equation determines the allowed states and energies."
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                    "tex": "hf = E_{\\text{high}} - E_{\\text{low}}"
                  }
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                "always zero",
                "unlimited"
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              "answer": 1,
              "explanation": "Confinement quantizes the energy."
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              "prompt": "An electron emits a photon when it drops to a lower energy level.",
              "answer": true,
              "explanation": "The energy difference leaves as a photon."
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              "prompt": "Sharp spectral lines occur because energy gaps are:",
              "options": [
                "random",
                "fixed",
                "zero",
                "continuous"
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              "answer": 1,
              "explanation": "Fixed gaps -> fixed photon energies -> sharp lines."
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            {
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                  "f"
                ]
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              "difficulty": 3,
              "hint": "Same as any photon.",
              "explanation": "$E = hf = \\Delta E$."
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              "prompt": "A confined electron can sit perfectly still with exactly zero energy.",
              "answer": false,
              "hint": "Uncertainty principle.",
              "explanation": "The ground state has nonzero energy."
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                "guitar string’s harmonics",
                "flowing river",
                "straight line"
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              "answer": 1,
              "hint": "Only certain notes fit.",
              "explanation": "A confined wave allows only specific standing modes."
            },
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                "Schr?dinger equation",
                "Ohm’s law",
                "ideal gas law",
                "Coulomb law"
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              "answer": 0,
              "hint": "It governs the wavefunction.",
              "explanation": "Schr?dinger’s equation determines quantum dynamics."
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              "explanation": "Distinct energy levels give each element its own spectrum."
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through space."
              }
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              "content": {
                "markdown": "Inside a neutron star, matter is crushed to nuclear density, and only quantum rules
describe it. At the heart of those rules is the **wavefunction** -- the object that replaced the classical idea of a
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particle with a definite position and velocity."
              }
            },
            {
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isn’t directly observable, but its square gives the **probability** of finding the particle at each location:\n\n$$P(x)
\\propto |\\psi(x)|^2.$$\n\nKey consequences:\n\n- Before measurement, the particle is in a **superposition** -- a blend
 of possible positions (and states). It has no single value.\n- **Measurement** forces a definite outcome at random,
weighted by $|\\psi|^2$ -- the wavefunction ’collapses.’\n- Total probability must equal 1: the particle is
*somewhere*.\n- The shape of $\\psi$ over time follows the **Schr?dinger equation**, the quantum law of motion.\n\nThis
is why quantum mechanics is **probabilistic**: it predicts the odds of outcomes, not certainties. Einstein resisted it
-- ’God does not play dice’ -- but experiment has sided with the dice every time."
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                      "$\\psi$ is a probability amplitude.",
                      "$|\\psi|^2$ is a probability density."
                    ],
                    "answer": "The likelihood of finding the particle there."
                  },
                  {
                    "title": "Superposition",
                    "problem": "Where is an electron before measurement?",
                    "steps": [
                      "It has no single position.",
                      "It is a weighted blend of possibilities."
                    ],
                    "answer": "In a superposition of locations."
                  },
                  {
                    "title": "Normalization",
                    "problem": "Why must the total probability be 1?",
                    "steps": [
                      "The particle exists somewhere.",
                      "Summing $|\\psi|^2$ over all space gives 1."
                    ],
                    "answer": "Probabilities must total 100%."
                  }
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                "intro": "$|\\psi|^2$ is probability; measurement collapses superposition."
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                "intro": "Reason about wavefunctions and probability."
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state of the other, no matter how far apart -- **entanglement**, what Einstein called ’spooky action at a distance.’ It
transmits no usable signal faster than light, but it is real and now powers quantum computing and quantum cryptography.
The wavefunction, once a bookkeeping device, has become engineering."
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                  "A particle is described by a wavefunction $\\psi$.",
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        "$|\\psi|^2$ gives the probability of finding it at each point.",
        "Before measurement a particle is in a superposition of possibilities.",
        "Measurement collapses $\\psi$ to a random outcome weighted by $|\\psi|^2$.",
        "Quantum mechanics predicts probabilities, not certainties."
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      "the probability of finding the particle there",
      "the particle’s charge",
      "the particle’s mass"
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    "difficulty": 3,
    "explanation": "$|\\psi|^2$ is the probability density."
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    "answer": true,
    "difficulty": 3,
    "explanation": "Superposition: a blend of possibilities until measured."
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      "exact outcomes every time",
      "the probabilities of outcomes",
      "nothing measurable",
      "only classical results"
    ],
    "answer": 1,
    "difficulty": 2,
    "explanation": "It is fundamentally probabilistic."
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    "prompt": "When a measurement is made, the wavefunction:",
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      "disappears forever",
      "collapses to a definite outcome",
      "doubles",
      "becomes negative"
    ],
    "answer": 1,
    "difficulty": 3,
    "hint": "One result is selected.",
    "explanation": "Measurement collapses $\\psi$ to a single outcome, weighted by $|\\psi|^2$."
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    "hint": "The particle exists.",
    "explanation": "Normalization: integrating $|\\psi|^2$ over all space gives 1."
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      "Schr?dinger equation",
```

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          "Maxwell equation",
          "Einstein equation"
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        "explanation": "The Schr?dinger equation governs the wavefunction’s evolution."
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          "sub": "A neutron star resists collapse not with heat or fuel, but with a quantum law about how particles may be packed."
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          "markdown": "A neutron star has burned out -- no fusion fights gravity. Yet it doesn’t collapse to a point. The reason is purely quantum: **degeneracy pressure**, a consequence of the **Pauli exclusion principle**. And the same quantum strangeness lets the Sun fuse at all, through **tunnelling**."
        }
      },
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              "steps": [
                "Classically they can’t overcome repulsion.",
                "Their wave nature lets a few tunnel through."
              ],
              "answer": "Quantum tunnelling enables fusion."
            },
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              "title": "No two alike",
              "problem": "What does Pauli exclusion forbid?",
              "steps": [
                "Identical fermions can’t share a state.",
                "Packing forces higher-energy states."
              ],
              "answer": "Two identical fermions in the same quantum state."
            },
            {
              "title": "Holding up a dead star",
              "problem": "What supports a neutron star against gravity?",
              "steps": [
                "No fusion remains.",
                "Neutron degeneracy pressure resists compression."
              ],
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            "answer": "Quantum degeneracy pressure."
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      "content": {
        "markdown": "At nineteen, on a ship to England, **Subrahmanyan Chandrasekhar** calculated that electron degeneracy pressure can support a white dwarf only up to about **1.4 solar masses**. Beyond it, gravity wins. The establishment mocked him for years; he was right, and won the Nobel Prize. Above his limit, stars collapse to neutron stars -- held by neutron degeneracy -- and above a further limit, to black holes, where no known pressure can stop the fall."
      }
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          "Quantum tunnelling lets particles pass barriers they classically couldn’t -- enabling stellar fusion.",
          "The Pauli exclusion principle forbids identical fermions sharing a quantum state.",
          "Degeneracy pressure (needing no heat) supports white dwarfs and neutron stars.",
          "The Chandrasekhar limit (~1.4 solar masses) caps a white dwarf’s mass.",
          "Beyond these limits, collapse continues to a neutron star or black hole."
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        "gain mass",
        "stop existing"
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      "difficulty": 3,
      "explanation": "Its wave nature gives a probability of being found beyond the barrier."
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      "prompt": "Two identical electrons cannot occupy the same quantum state.",
      "answer": true,
      "difficulty": 3,
      "explanation": "The Pauli exclusion principle forbids it."
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        "neutron degeneracy pressure",
        "magnetism",

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              "radiation pressure"
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            "difficulty": 3,
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              "fuse despite their electric repulsion",
              "escape the Sun",
              "become photons"
            ],
            "answer": 1,
            "difficulty": 3,
            "hint": "Classically they couldn’t get close enough.",
            "explanation": "A few protons tunnel through the repulsion barrier, igniting fusion."
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          {
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            "prompt": "Degeneracy pressure requires the star to stay hot.",
            "answer": false,
            "difficulty": 3,
            "hint": "It comes from the exclusion principle, not heat.",
            "explanation": "Degeneracy pressure is purely quantum and persists even in a cold, burnt-out star."
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            "prompt": "Above the Chandrasekhar limit (~1.4 solar masses), a white dwarf:",
            "options": [
              "becomes stable forever",
              "collapses further (to a neutron star or black hole)",
              "turns into the Sun",
              "loses all gravity"
            ],
            "answer": 1,
            "difficulty": 4,
            "hint": "Electron degeneracy pressure is no longer enough.",
            "explanation": "Gravity overwhelms degeneracy pressure, driving collapse."
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              "sub": "Einstein’s deepest idea: gravity isn’t a force pulling through space -- it’s the shape of space
and time themselves."
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                      "Both feel identical locally.",
                      "Equivalence principle."
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                    "answer": "Gravity ~= acceleration."
                  },
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                    "title": "Light and gravity",
                    "problem": "Does gravity affect light?",
                    "steps": [
                      "Light follows spacetime geodesics.",
                      "Curved near mass."
                    ],
                    "answer": "Yes -- light bends (lensing)."
                  },
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                    "title": "Time",
                    "problem": "Where do clocks run slower?",
                    "steps": [
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                      "Lower clocks tick slow."
                    ],
                    "answer": "Deeper in a gravity well."
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                ]
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              }
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              }
            },
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                  "Mass-energy curves spacetime; free objects follow geodesics.",
                  "Gravity is geometry, not a force.",
                  "Predictions: light bending, gravitational time dilation, orbit precession.",
                  "The Einstein field equations relate curvature to energy."
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              "explanation": "That is exactly the equivalence principle."
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                "bends",
                "stops",
                "is unaffected"
              ],
              "answer": 1,
              "explanation": "Light follows curved geodesics -- gravitational lensing."
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              "answer": true,
              "hint": "Gravitational time dilation.",
              "explanation": "GPS satellites must correct for this effect."
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              "prompt": "Objects in free fall follow paths called:",
              "options": [
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                "geodesics",
                "asymptotes",
                "secants"
              ],
              "answer": 1,
              "hint": "Straightest paths in curved spacetime.",
              "explanation": "Free-fall paths are geodesics."
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not.",
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              "hint": "A famous early success.",
              "explanation": "GR precisely accounts for Mercury’s perihelion precession."
```

```
            },
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                "sound waves",
                "gravitational waves",
                "radio waves",
                "light waves"
              ],
              "answer": 1,
              "hint": "Predicted by Einstein.",
              "explanation": "Gravitational waves are propagating spacetime curvature."
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                "color"
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              "hint": "The right-hand side $T_{\\mu\\nu}$.",
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horizon."
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steeply that a region forms from which **nothing escapes** -- a **black hole**. It is general relativity taken to its
logical extreme, and a real object we have now photographed."
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small enough radius and the escape velocity reaches the speed of light $c$ -- beyond that boundary, nothing, not even
light, can escape. That is a **black hole**.\n\nThe boundary is the **event horizon**, at the **Schwarzschild
radius**:\n\n$$r_s = \\frac{2GM}{c^2}.$$\n\nIt is proportional to mass: double the mass, double the horizon. For the
Sun, $r_s \\approx 3\\,\\text{km}$; for Earth, about $9\\,\\text{mm}$.\n\n- Cross the horizon and all paths lead inward
to the **singularity** -- the future itself points down.\n- The horizon isn’t a surface of matter; it’s a boundary in
spacetime.\n- **Hawking radiation:** quantum effects let black holes slowly evaporate, the deepest known link between
gravity, quantum mechanics, and thermodynamics."
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                      "$r_s = 2GM/c^2 \\propto M$.",
                      "Doubles."
```

```
                ],
                "answer": "The Schwarzschild radius doubles."
              },
              {
                "title": "What is the horizon?",
                "problem": "Is the event horizon a physical wall?",
                "steps": [
                  "No matter there.",
                  "A boundary in spacetime."
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                "problem": "Why does no light escape?",
                "steps": [
                  "Escape velocity exceeds $c$.",
                  "Nothing can go faster than light."
                ],
                "answer": "Light itself can’t climb out."
              }
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              "The event horizon is the boundary of no return.",
              "Schwarzschild radius $r_s = 2GM/c^2$, proportional to mass.",
              "Inside, all paths lead to the singularity.",
              "Hawking radiation makes black holes slowly evaporate."
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            "the boundary of no return",
            "its center",
            "made of light"
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          "answer": 1,
          "explanation": "It’s the boundary beyond which nothing escapes."
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    "$M$",
    "$1/M$",
    "$\\sqrt{M}$"
  ],
  "answer": 1,
  "explanation": "$r_s = 2GM/c^2 \\propto M$."
},
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  "answer": true,
  "explanation": "Escape velocity there exceeds $c$."
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  "answer": {
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    "vars": [
      "G",
      "M",
      "c"
    ]
  },
  "difficulty": 4,
  "hint": "Two G M over c squared.",
  "explanation": "$r_s = \\dfrac{2GM}{c^2}$."
},
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    "3x larger",
    "9x larger",
    "1/3"
  ],
  "answer": 1,
  "hint": "$r_s \\propto M$.",
  "explanation": "Linear in mass -- triples."
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  "scope": "PRACTICE",
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  "options": [
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    "slowly evaporate",
    "grow without limit",
    "emit no energy"
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  "hint": "A quantum effect at the horizon.",
  "explanation": "They radiate and slowly lose mass."
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    "Schwarzschild radius",
    "wavelength",
```

"half-life"
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"hint": "Inside the horizon.",
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"sub": "Einstein realized that a person in free fall feels no gravity at all -- the seed that grew into a new theory of gravity."
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"steps": [
"Both press you to the floor identically.",
"No local experiment distinguishes them."
],
"answer": "No -- that is the equivalence principle."
},
{
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"problem": "Are orbiting astronauts beyond Earth’s gravity?",
"steps": [
"Gravity is still strong in orbit.",
"They are in continuous free fall."
],
"answer": "No -- they float because they’re freely falling."
},
{
"title": "Equal fall rates",
"problem": "Why do all objects fall at the same rate in a vacuum?",
"steps": [
"Inertial mass equals gravitational mass.",
"The mass cancels in the motion."
],
"answer": "Because the two kinds of mass are equal."
}

```
        ]
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      "content": {
        "intro": "Gravity ~= acceleration; free fall ~= weightlessness."
      }
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          "Free fall feels like weightlessness; orbiting astronauts are in free fall.",
          "Inertial mass equals gravitational mass, so all objects fall at the same rate.",
          "Gravity is reinterpreted as motion through curved spacetime.",
          "Gravity bends light -- confirmed in the 1919 eclipse."
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        "in continuous free fall",
        "weightless by magnetism",
        "moving slowly"
      ],
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      "difficulty": 3,
      "explanation": "Gravity is strong there; they free-fall around the Earth, so they float."
    },
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        "inertial mass equals gravitational mass",
        "heavy things resist gravity",
        "gravity ignores mass entirely"
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              "explanation": "The equality of the two masses makes the fall rate independent of mass."
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              "hint": "Confirmed in 1919.",
              "explanation": "Starlight bending near the Sun confirmed the prediction."
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                "crushed flat",
                "nothing"
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              "difficulty": 3,
              "hint": "Equivalence principle.",
              "explanation": "The acceleration mimics Earth’s gravity exactly."
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                "friction",
                "a kind of light"
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              "answer": 1,
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              "hint": "Free fall erases gravity.",
              "explanation": "Gravity becomes geometry -- motion through curved spacetime."
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                "sub": "Time runs slower deeper in a gravitational field -- by a sliver on Earth, dramatically near a
black hole."
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violent rearrangement of mass should send **ripples** through spacetime itself. Both predictions are now confirmed, one
in your phone’s GPS, the other by detecting colliding black holes."
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 a gravitational well). A clock at sea level ticks slightly slower than one on a mountaintop. The effect is tiny on
Earth but real -- and near a black hole’s horizon, time nearly stops as seen from far away.\n\n**GPS proves it daily.**
Satellite clocks run faster (weaker gravity up high) and slower (their orbital speed, special relativity) -- net, they
gain about $38$ microseconds per day. Uncorrected, GPS would drift kilometers off within hours. Your phone’s navigation
is a working relativity experiment.\n\n**Gravitational waves.** When massive objects accelerate violently -- two black
holes spiralling together -- they radiate **ripples in spacetime** that stretch and squeeze space as they pass,
travelling at the speed of light. In 2015, **LIGO** detected such a wave from two merging black holes a billion
light-years away, stretching its 4 km arms by less than the width of a proton. Predicted by Einstein in 1916, confirmed
```

```
a century later -- and opening a brand-new way to observe the universe, by listening instead of looking."
              }
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                      "Sea level is deeper in the well."
                    ],
                    "answer": "The sea-level clock runs slightly slower."
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                      "Position errors grow to kilometers."
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                    "answer": "Navigation fails without the correction."
                  },
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                    "problem": "What are gravitational waves?",
                    "steps": [
                      "Accelerating masses ripple spacetime.",
                      "The ripples stretch and squeeze space."
                    ],
                    "answer": "Ripples in spacetime, travelling at light speed."
                  }
                ]
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telescopes worldwide swung to the same patch of sky to catch the light, gamma rays, and X-rays from the same event. For
the first time, the universe was observed in both gravitational waves and light at once -- **multi-messenger
astronomy**. The collision was seen forging gold and platinum, confirming where the heaviest elements come from. A new
sense has been added to astronomy."
              }
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                  "GPS satellites correct for relativity (~38 microseconds/day) to stay accurate.",
                  "Near a black hole horizon, time nearly stops as seen from afar.",
                  "Accelerating masses emit gravitational waves -- ripples in spacetime at light speed.",
                  "LIGO’s 2015 detection confirmed Einstein’s 1916 prediction and opened gravitational-wave astronomy."
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            "clocks run the same everywhere",
            "only in space"
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          "difficulty": 3,
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 atoms in its atmosphere soak up specific wavelengths. Reading those lines reveals composition, and their **Doppler
shifts** reveal motion -- rotation, orbital wobble from unseen planets, and the recession of distant galaxies. A single
spectrum thus encodes temperature, size, chemistry, and velocity: astrophysics is, in large part, the art of decoding
light."
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        "Wien’s law: hotter stars peak at bluer (shorter) wavelengths.",
        "Stefan-Boltzmann: $L = 4\\pi R^2\\sigma T^4$.",
        "Luminosity is hugely sensitive to temperature ($T^4$) and size ($R^2$).",
        "Apparent brightness also falls off with distance (inverse-square)."
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            "Mass decides the fate.",
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            "High-mass: supernova -> neutron star or black hole; heavy elements are forged in stars."
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age, and fate. It is the most important diagram in all of astrophysics."
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vertical axis against **surface temperature** on the horizontal (running hot-to-cool, left-to-right). Stars are not
scattered randomly -- they cluster into groups:\n\n- **The main sequence** -- a diagonal band from hot-bright (top-left)
 to cool-dim (bottom-right) where stars spend most of their lives fusing hydrogen. The Sun sits here, mid-band. A star’s
 **position on the main sequence is set by its mass**: massive stars are hot, bright, blue, and short-lived; small stars
 are cool, dim, red, and long-lived.\n- **Red giants** -- cool but luminous (top-right): huge, swollen, late-life
stars.\n- **White dwarfs** -- hot but faint (bottom-left): tiny, dense stellar corpses.\n\nA star **traces a path**
across the diagram as it ages: born onto the main sequence, swelling into a red giant when its core hydrogen runs low,
then ending as a white dwarf (or, if massive, exploding). Reading a star’s spot on the HR diagram tells you what it is
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age. The most massive stars burn out first, peeling off the top of the main sequence. The point where the main sequence
’turns off’ acts as a clock: the lower the turn-off, the older the cluster. This **main-sequence turn-off** method dates
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                  "Most stars lie on the main sequence (hydrogen fusion); position there is set by mass.",
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                  "Stars trace paths across the diagram as they age.",
                  "A cluster’s main-sequence turn-off reveals its age."
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                      "Parallax is the apparent shift from Earth’s orbit.",
                      "Closer objects shift more."
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                    "answer": "A larger parallax means a nearer star."
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                      "Brightness $\\propto 1/d^2$.",
                      "Double $d$ gives $1/4$."
                    ],
                    "answer": "One quarter as bright."
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                  {
                    "title": "Reaching the farthest",
                    "problem": "How are the most distant galaxies’ distances found?",
                    "steps": [
                      "Their light is redshifted by expansion.",
                      "Hubble’s law converts redshift to distance."
                    ],
                    "answer": "By redshift and Hubble’s law."
                  }
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              "Standard candles have known luminosity; apparent brightness gives distance via the inverse-square law.",
              "Cepheids and Type Ia supernovae are the key standard candles.",
              "Redshift plus Hubble’s law gives the distances of the farthest galaxies.",
              "Each rung calibrates the next, mapping the universe to its edge."
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            "a star’s mass",
            "redshift"
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            "1/d^2",
            "d",
            "d^2"
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          "answer": 1,
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            "difficulty": 3,
            "explanation": "The inverse-square law: brightness  prop  1/d^2."
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              "vars": [
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                "d"
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            "hint": "Inverse square.",
            "explanation": "$b \\propto L/d^2$."
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              "1/6 as bright",
              "1/9 as bright",
              "3x as bright"
            ],
            "answer": 2,
            "difficulty": 3,
            "hint": "Square the factor.",
            "explanation": "Brightness  prop  1/d^2, so 3x distance gives 1/9 the brightness."
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              "temperature",
              "spin"
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            "answer": 1,
            "difficulty": 3,
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            "explanation": "Redshift plus Hubble’s law gives distances beyond the reach of candles."
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temperature difference."
            }
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glowing in X-rays. To understand such a system -- or a steam engine, or a star -- we need **thermodynamics**: the
physics of temperature, heat, and the statistical behavior of countless particles."
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**Heat** $Q$ is energy transferred *because of* a temperature difference, and it always flows from hot to cold.\n\nThe
**absolute (Kelvin)** scale starts at **absolute zero** ($0\\,\\text{K} = -273^\\circ\\text{C}$), where particle motion
is minimal.\n\nFor an **ideal gas**, pressure $P$, volume $V$, and temperature $T$ obey the **ideal gas law**:\n\n$$PV =
 nRT,$$\n\nwhere $n$ is the amount of gas and $R$ the gas constant. Consequences at fixed amount:\n\n- At constant $T$:
$P$ and $V$ trade off ($PV$ constant) -- squeeze a gas and pressure rises.\n- At constant $V$: heating raises pressure
($P \\propto T$).\n\n**Kinetic theory** explains why: pressure is just countless molecules drumming on the walls, and
raising $T$ makes them hit harder and more often."
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                    "Until equal temperature."
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                  "answer": "From hot to cold."
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                  "problem": "At constant T, you halve a gas’s volume. What happens to pressure?",
                  "steps": [
                    "$PV$ constant.",
                    "Half V -> double P."
                  ],
                  "answer": "Pressure doubles."
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                  "problem": "What is special about 0 K?",
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                    "Lowest possible temperature.",
                    "Minimal particle motion."
                  ],
                  "answer": "Particle motion is minimized."
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$10^{23}$ particles. Their speeds follow the **Maxwell-Boltzmann distribution**, and temperature is precisely a measure
of the spread of that distribution. This bridge -- from the random motion of individual particles to the smooth laws of
pressure and temperature -- is one of physics’ great unifications, connecting mechanics, probability, and thermodynamics
 into one framework."
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              "Heat flows from hot to cold.",
              "Kelvin starts at absolute zero ($-273 degC$).",
              "Ideal gas law: $PV = nRT$.",
              "At constant T, P and V trade off; statistical mechanics underlies it all."
            ],
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                "tex": "PV = nRT"
              }
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            "size of particles",
            "charge of particles"
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          "explanation": "Temperature reflects average particle kinetic energy."
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          "options": [
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            "hot to cold",
            "in both directions equally",
            "only upward"
          ],
          "answer": 1,
          "explanation": "Spontaneously from hot to cold."
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            "$P = IR$",
            "$E = mc^2$",
            "$F = ma$"
          ],
          "answer": 0,
          "explanation": "$PV = nRT$."
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          "answer": {
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            "vars": [
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              "R",
              "T",
              "V"
            ]
          },
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          "explanation": "$P = \\dfrac{nRT}{V}$."
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          "scope": "PRACTICE",
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              "explanation": "Boyle’s law behavior: $P \\propto 1/V$."
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              "answer": {
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                "tolerance": 1
              },
              "hint": "$0\\,K = -273 degC$.",
              "explanation": "$0\\,\\text{K} \\approx -273^\\circ\\text{C}$."
            },
            {
              "scope": "PRACTICE",
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              "prompt": "At fixed volume and amount (constant set to 1), pressure vs temperature is $P = T$. Graph it (enter $P$ as a function of $T$, use $x$ for $T$).",
              "answer": {
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                "domain": [
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                  5
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                "variable": "x"
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              "difficulty": 3,
              "hint": "$P \\propto T$ -- a line.",
              "explanation": "At constant $V$, $P$ is proportional to $T$."
            },
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                "molecules colliding with the walls",
                "magnetism",
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              "explanation": "Countless molecular impacts create pressure."
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              }
            },
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              "content": {
                "markdown": "Drop a glass and it shatters; you never see shards leap back together. Heat spreads out; it never spontaneously concentrates. This one-way character of nature is the **second law of thermodynamics**, and it is woven from a single quantity: **entropy**."
              }
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              "content": {
                "markdown": "**Entropy** $S$ measures the disorder of a system -- more precisely, the number of microscopic arrangements consistent with its overall state.\n\nThe **second law of thermodynamics:** the total entropy
```

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of an isolated system never decreases.\n\n$$\\Delta S_{\\text{total}} \\ge 0.$$\n\nConsequences:\n\n- Heat flows hot ->
cold because that spreads energy out, raising entropy.\n- No engine can convert heat **entirely** into work -- some is
always 'lost' to entropy. The best possible efficiency is the **Carnot limit**, set by the hot and cold temperatures.\n-
 Order can increase **locally** (a fridge, a growing organism) only by dumping *more* entropy into the
surroundings.\n\nEntropy gives time its **arrow**: the future is the direction of increasing entropy. It is the deep
reason perpetual-motion machines are impossible."
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                    "steps": [
                      "Spreading energy raises entropy.",
                      "Second law favors it."
                    ],
                    "answer": "It increases total entropy."
                  },
                  {
                    "title": "Engine limit",
                    "problem": "Can an engine turn 100% of heat into work?",
                    "steps": [
                      "Some heat must be exhausted.",
                      "Entropy forbids full conversion."
                    ],
                    "answer": "No -- never 100% efficient."
                  },
                  {
                    "title": "Local order",
                    "problem": "How can a fridge make things colder (more ordered)?",
                    "steps": [
                      "It exports entropy as waste heat.",
                      "Total entropy still rises."
                    ],
                    "answer": "By increasing entropy elsewhere."
                  }
                ]
              }
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              }
            },
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              }
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more uniformly, and usable energy dwindles. In the far future the universe approaches a state of maximum entropy -- the
**heat death** -- where everything is the same temperature and no work can be extracted. Galaxy clusters, with their
slowly cooling X-ray gas, are part of this grand, irreversible drift toward equilibrium."
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                  "Second law: total entropy of an isolated system never decreases.",
                  "Heat flows hot->cold because it raises entropy.",
                  "No engine is 100% efficient (Carnot limit).",
                  "Entropy defines the arrow of time."
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      "charge"
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    "answer": false,
    "explanation": "The second law: total entropy never decreases."
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      "built",
      "reversed"
    ],
    "answer": 1,
    "explanation": "Entropy forbids complete heat-to-work conversion."
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    "prompt": "Heat flows from hot to cold because it increases total entropy.",
    "answer": true,
    "hint": "Energy spreads out.",
    "explanation": "The spontaneous direction raises entropy."
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    "prompt": "A refrigerator can lower entropy locally by increasing it more in the surroundings.",
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    "hint": "Total still rises.",
    "explanation": "Local order is paid for by greater disorder elsewhere."
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    "hint": "Past vs future.",
    "explanation": "Increasing entropy points toward the future."
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      "Hooke’s law",
      "Snell’s law"
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lunch forever."
            }
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              "markdown": "The million-degree gas between galaxies, a car engine, your own metabolism -- all obey the
same handful of laws. Having met entropy and the second law, we now lay out the **full set**, the foundation of all
thermal physics."
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equilibrium with a third, they’re in equilibrium with each other. This is what makes **temperature** a meaningful,
measurable quantity -- it’s why a thermometer works.\n\n**First law (conservation of energy).** Energy is conserved; you
 can change a system’s internal energy by adding **heat** $Q$ or doing **work** $W$ on it:\n\n$$\\Delta U = Q -
W.$$\n\nYou can’t create energy from nothing -- *’you can’t win.’*\n\n**Second law (entropy).** The total entropy of an
isolated system never decreases; heat flows spontaneously hot -> cold; no process is perfectly efficient -- *’you can’t
break even.’* This law gives time its arrow.\n\n**Third law.** As temperature approaches **absolute zero**, entropy
approaches a minimum, and absolute zero itself can never be fully reached -- *’you can’t get out of the
game.’*\n\nTogether they bound what any machine, chemical reaction, or living thing can do. They are among the most
universal laws in science -- Einstein expected them to be the one part of physics never overturned."
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energy?",
                  "steps": [
                    "Energy is conserved.",
                    "$\\Delta U = Q - W$."
                  ],
                  "answer": "$\\Delta U = Q - W$."
                },
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                  "problem": "Which law guarantees temperature is well-defined?",
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                    "Equilibrium is transitive.",
                    "That’s the zeroth law."
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                  "answer": "The zeroth law."
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            "steps": [
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              "The third law forbids fully reaching it."
            ],
            "answer": "No -- absolute zero is unreachable."
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      }
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 behavior of countless particles. Entropy counts the number of microscopic arrangements that look the same large-scale;
disordered states have vastly more arrangements, so systems drift toward them simply by probability. Boltzmann’s
tombstone bears his equation, $S = k \\log W$, linking entropy $S$ to the number of microstates $W$ -- heat and
disorder, explained by counting."
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          "First law: energy is conserved, $\\Delta U = Q - W$ (’you can’t win’).",
          "Second law: total entropy never decreases (’you can’t break even’).",
          "Third law: absolute zero can never be fully reached.",
          "Statistical mechanics explains the laws by counting microscopic arrangements."
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        "conservation of energy",
        "temperature is relative",
        "absolute zero is reachable"
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      "difficulty": 2,
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              "first law",
              "second law",
              "third law"
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            "answer": true,
            "difficulty": 3,
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                "W"
              ]
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            "explanation": "$\\Delta U = Q - W$."
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              "second and third laws",
              "first and second laws",
              "zeroth and first laws",
              "third and first laws"
            ],
            "answer": 1,
            "difficulty": 3,
            "hint": "Energy, then entropy.",
            "explanation": "First law: can’t create energy (’win’); second law: can’t be perfectly efficient (’break
even’)."
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            "explanation": "More arrangements (microstates) means higher entropy; systems drift toward the most
probable, disordered states."
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into work."
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                "markdown": "Heat engines drove the Industrial Revolution and still generate most of our electricity.
The second law sets an unbreakable limit on them -- discovered by a young engineer studying steam, and as fundamental as
 any law in physics."
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**some** of it into useful work, and dumps the rest into a cold reservoir at $T_c$. It *must* dump some -- the second
law forbids turning heat entirely into work.\n\nThe **maximum possible efficiency**, achieved only by an ideal
(reversible) **Carnot** engine, depends only on the two temperatures (in kelvin):\n\n$$\\eta_{max} = 1 -
\\frac{T_c}{T_h}.$$\n\nLessons from this formula:\n\n- A bigger temperature gap (hotter $T_h$ or colder $T_c$) means
higher possible efficiency.\n- Perfect efficiency ($\\eta = 1$) would need $T_c = 0$ (absolute zero) -- impossible by
the third law.\n- Real engines, with friction and irreversibility, always fall **below** the Carnot limit.\n\nThis is
why power plants run boilers as hot as materials allow, and why no ’perfect’ engine can exist. Run an engine in reverse
and you get a **refrigerator** or **heat pump**, using work to move heat from cold to hot -- paying the entropy bill
with the work input."
              }
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efficiency?",
                    "steps": [
                      "Use $\\eta_{max} = 1 - T_c/T_h$.",
                      "$= 1 - 300/600$.",
                      "$= 1 - 0.5 = 0.5$."
                    ],
                    "answer": "50% -- and a real engine between these temperatures does worse."
                  },
                  {
                    "title": "Why dump heat?",
                    "problem": "Why can’t an engine turn all its input heat into work?",
                    "steps": [
                      "The second law forbids it.",
                      "Some heat must go to the cold reservoir."
                    ],
                    "answer": "Exhaust heat is unavoidable."
                  },
                  {
                    "title": "Boosting efficiency",
                    "problem": "How do you raise an engine’s efficiency ceiling?",
                    "steps": [
                      "Increase the temperature gap.",
                      "Raise $T_h$ or lower $T_c$."
                    ],
                    "answer": "Widen the gap between hot and cold."
                  }
                ]
              }
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              }
            },
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              }
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forgotten. Yet his idealized engine set the gold standard every real engine is measured against, and his reasoning
birthed the second law and the concept of entropy. Remarkably, the Carnot limit depends *only* on temperatures -- not on
 the working substance, design, or fuel. It’s a statement about the universe, not about engineering, which is why it has
 never been beaten."
              }
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```
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            "Maximum (Carnot) efficiency: $\\eta_{max} = 1 - T_c/T_h$ (temperatures in kelvin).",
            "A larger hot-cold temperature gap allows higher efficiency.",
            "Perfect efficiency would need $T_c = 0$ -- impossible; real engines do worse than Carnot.",
            "Run in reverse, an engine becomes a refrigerator or heat pump."
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          "run at absolute zero",
          "create energy"
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        "difficulty": 3,
        "explanation": "The second law requires dumping some heat; full conversion is impossible."
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efficiency.",
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          "the cold reservoir at absolute zero",
          "more fuel",
          "a bigger engine"
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        "answer": 1,
        "difficulty": 4,
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(Use 1 - Tc/Th.)",
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        "hint": "1 - 200/800.",
        "explanation": "$1 - 200/800 = 1 - 0.25 = 0.75$."
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        "prompt": "Write the Carnot efficiency in terms of cold and hot temperatures $T_c$ and $T_h$. (Type like
‘1 - T_c/T_h‘.)",
        "answer": {
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          "vars": [
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            "T_h"
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          ]
        },
        "difficulty": 4,
        "hint": "One minus the temperature ratio.",
        "explanation": "$\\eta_{max} = 1 - T_c/T_h$."
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          "refrigerator or heat pump",
          "battery",
          "turbine"
        ],
        "answer": 1,
        "difficulty": 3,
        "hint": "It pumps heat ’uphill’.",
        "explanation": "A refrigerator/heat pump uses work input to move heat from cold to hot."
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            "sub": "A tiny, dense core binds protons and neutrons -- and releasing a sliver of that binding powers stars and reactors."
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          "content": {
            "markdown": "Every atom in the observable universe has a **nucleus** -- protons and neutrons bound impossibly tightly. Understanding it explains where the elements come from, how the Sun shines, how we date ancient bones, and the energy that lights cities and, in another form, levels them."
          }
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        {
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            "markdown": "The **nucleus** contains **protons** (positive) and **neutrons** (neutral), bound by the **strong nuclear force**, which overpowers the electric repulsion between protons at tiny range. The number of protons (the **atomic number**) defines the element.\n\nUnstable nuclei undergo **radioactive decay**, emitting:\n\n- **Alpha** ($\\alpha$): a helium nucleus -- heavy, stopped by paper.\n- **Beta** ($\\beta$): an electron from a neutron turning into a proton -- stopped by aluminum.\n- **Gamma** ($\\gamma$): a high-energy photon -- needs thick lead.\n\nDecay is governed by a constant **half-life** -- the time for half a sample to decay -- regardless of the amount, which makes it a precise clock for **radiometric dating**.\n\nThe energy comes from $E = mc^2$: the products weigh slightly *less* than the original, and that missing mass (the **mass defect**) is released. **Fission** splits heavy nuclei; **fusion** joins light ones -- the Sun’s power source."
          }
        },
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                      "4 years: 20 g."
                    ],
                    "answer": "20 g (two half-lives)."
                  },
                  {
                    "title": "What defines the element",
                    "problem": "Which count sets the chemical element?",
                    "steps": [
                      "The atomic number.",
                      "= number of protons."
                    ],
                    "answer": "The proton count."
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                    "title": "Energy source",
                    "problem": "Where does nuclear energy come from?",
                    "steps": [
                      "Products are lighter.",
                      "$E = mc^2$ on the mass defect."
                    ],
                    "answer": "Converted mass defect."
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              "title": "Practice problems",
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              }
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                  "Atomic number (protons) defines the element.",
                  "Radioactivity: alpha, beta, gamma emissions.",
                  "Decay has a constant half-life -- a precise clock.",
                  "Nuclear energy comes from the mass defect, $E = mc^2$ (fission & fusion)."
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                    "tex": "E = mc^2"
                  }
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        "electrons in motion",
        "photons"
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      "explanation": "Atomic number = proton count."
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        "tolerance": 0
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      "explanation": "Two half-lives: $80\\to40\\to20$."
    },
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      "options": [
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        "beta",
        "gamma",
        "they’re equal"
      ],
      "answer": 2,
      "explanation": "Gamma rays need thick lead to stop."
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        "tolerance": 0
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      "hint": "Two half-lives.",
      "explanation": "$1000\\to500\\to250$."
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      "hint": "Two protons, two neutrons.",
      "explanation": "Alpha = a helium-4 nucleus."
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        "the mass defect via $E=mc^2$",
        "friction",
        "electric current"
      ],
      "answer": 1,
      "hint": "Products are lighter.",
      "explanation": "Lost mass becomes energy."
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      "prompt": "The Sun is powered by nuclear:",
      "options": [
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        "fusion",
        "decay",
        "combustion"
      ],
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      "hint": "Joining light nuclei.",
      "explanation": "Hydrogen fusion powers the Sun."
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              "explanation": "Half-life is constant, independent of amount."
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particles."
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              "content": {
                "markdown": "Keep dividing matter -- molecules, atoms, nuclei, protons -- and you eventually reach
particles with no known substructure: the **fundamental particles**. The **Standard Model** organizes them into a
remarkably compact scheme that has survived every experimental test for half a century."
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            {
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              "content": {
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(six types) -- combine in threes to form protons and neutrons. A proton is two up quarks and one down.\n- **Leptons**
(six types) -- include the **electron** and the elusive, nearly massless **neutrinos**.\n\nForces are carried by
**force-carrier particles** (bosons):\n\n- **Photon** -- electromagnetism.\n- **Gluons** -- the strong force binding
quarks.\n- **W and Z bosons** -- the weak force (radioactive beta decay).\n\nThe **Higgs boson**, confirmed in 2012, is
tied to the field that gives many particles their mass.\n\nWhat’s missing: the Standard Model does **not** include
**gravity**, and it doesn’t explain **dark matter** or **dark energy** -- which together outweigh ordinary matter ~20 to
 1. The deepest theory we have is still incomplete."
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                    "steps": [
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                      "Two up, one down."
                    ],
                    "answer": "Two up quarks and one down quark."
                  },
                  {
                    "title": "Force carrier",
                    "problem": "Which particle carries the electromagnetic force?",
                    "steps": [
                      "The photon.",
                      "Massless boson."
                    ],
                    "answer": "The photon."
                  },
                  {
                    "title": "The gap",
                    "problem": "Name a force the Standard Model omits.",
                    "steps": [
                      "It excludes gravity.",
                      "Plus dark matter/energy."
                    ],
                    "answer": "Gravity."
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              }
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              "content": {
                "intro": "Quarks + leptons; forces via bosons."
              }
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              "content": {
                "intro": "Identify particles and their roles."
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              "content": {
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 particle is an excitation -- a ’ripple’ -- of an underlying field filling all of space. QFT unites quantum mechanics
with special relativity and yields the most precisely tested predictions in science (the electron’s magnetic moment
matches theory to twelve digits). Extending it to include gravity -- a **quantum theory of gravity** -- remains the
great unfinished goal of fundamental physics."
              }
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                  "The Standard Model lists the fundamental particles.",
                  "Quarks build protons and neutrons; leptons include electrons and neutrinos.",
                  "Forces are carried by bosons: photon, gluons, W/Z.",
                  "The Higgs boson relates to particle mass.",
                  "It omits gravity, dark matter, and dark energy -- it’s incomplete."
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                  }
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                "photons",
                "neutrinos"
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              "explanation": "Three quarks each."
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                "photon",
                "W boson",
                "Higgs"
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              "answer": 1,
              "explanation": "The photon carries the electromagnetic force."
            },
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              "prompt": "The Standard Model includes a complete description of gravity.",
              "answer": false,
              "explanation": "Gravity is notably absent from the Standard Model."
            },
            {
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              "kind": "MCQ",
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            "prompt": "Which is a lepton?",
            "options": [
              "up quark",
              "electron",
              "gluon",
              "photon"
            ],
            "answer": 1,
            "hint": "Electrons and neutrinos.",
            "explanation": "The electron is a lepton."
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            "kind": "MCQ",
            "prompt": "The strong force between quarks is carried by:",
            "options": [
              "photons",
              "gluons",
              "neutrinos",
              "W bosons"
            ],
            "answer": 1,
            "hint": "It glues quarks together.",
            "explanation": "Gluons carry the strong force."
          },
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            "scope": "PRACTICE",
            "kind": "MCQ",
            "prompt": "The particle confirmed in 2012, tied to mass, is the:",
            "options": [
              "neutrino",
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              "muon",
              "gluon"
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            "answer": 1,
            "hint": "Discovered at the LHC.",
            "explanation": "The Higgs boson."
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            "answer": false,
            "hint": "Major gaps.",
            "explanation": "Neither is part of the Standard Model."
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              "classical mechanics",
              "optics"
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into datable clocks."
            }
          },
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              "title": "Timing the universe",
              "content": {
                "markdown": "Having met the nucleus and radioactivity, we now quantify it. The **half-life** turns
random nuclear decay into one of science’s most powerful tools, dating everything from prehistoric campfires to the age
of the Earth and the cosmos."
              }
            },
            {
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              "content": {
                "markdown": "A radioactive nucleus is unstable and will eventually **decay**, emitting radiation. You
can’t predict *when* a single nucleus decays -- it’s random -- but a huge collection decays at a precise statistical
rate.\n\nThe **half-life** $t_{1/2}$ is the time for **half** of the nuclei in a sample to decay. After each half-life,
half of what remains is left:\n\n$$N = N_0 \\left(\\tfrac{1}{2}\\right)^{n}, \\qquad n = \\frac{t}{t_{1/2}}.$$\n\nSo
after 1 half-life, $1/2$ remains; after 2, $1/4$; after 3, $1/8$; and so on. Half-lives range from fractions of a second
 to billions of years.\n\nThree decay types differ in penetrating power:\n\n- **Alpha** ($\\alpha$): a helium nucleus;
heavy, stopped by paper.\n- **Beta** ($\\beta$): a fast electron; stopped by aluminum.\n- **Gamma** ($\\gamma$):
high-energy photon; needs thick lead or concrete.\n\n**Radiometric dating** uses this clock. **Carbon-14** (half-life
~5,700 years) dates organic remains up to ~50,000 years; **uranium** isotopes (billions of years) date rocks and
meteorites -- that’s how we know the Earth is about 4.5 billion years old."
              }
            },
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                      "$n = 15/5 = 3$ half-lives.",
                      "Halve three times: $80 \\to 40 \\to 20 \\to 10$."
                    ],
                    "answer": "10 g remain."
                  },
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                      "$(1/2)^4$.",
                      "$= 1/16$."
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                    "answer": "One sixteenth."
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                      "Carbon-14’s half-life is only ~5,700 years.",
                      "After a billion years, none is left."
                    ],
                    "answer": "Long half-life isotopes suit old samples."
                  }
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              }
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            },
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**carbon-14**. When they die, intake stops and the carbon-14 decays with its ~5,700-year half-life. Measuring how much
remains gives the time since death. Carbon dating revolutionized archaeology -- dating the Dead Sea Scrolls, ?tzi the
Iceman, and cave art -- and earned Willard Libby the Nobel Prize. The randomness of single nuclei becomes, in bulk, an
exquisitely reliable clock."
              }
```

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          "Half-life is the time for half a sample to decay; after n half-lives, $(1/2)^n$ remains.",
          "Alpha (paper), beta (aluminum), gamma (lead) differ in penetrating power.",
          "Radiometric dating uses known half-lives as clocks.",
          "Carbon-14 dates organic remains; uranium dates rocks and the 4.5-billion-year-old Earth."
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        "a nucleus to double",
        "radiation to stop"
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      "options": [
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        "beta",
        "gamma",
        "all equal"
      ],
      "answer": 2,
      "difficulty": 3,
      "explanation": "Gamma rays are high-energy photons requiring dense shielding."
    },
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      "hint": "6 h = 3 half-lives; halve three times.",
      "explanation": "$240 \\to 120 \\to 60 \\to 30$ g after 3 half-lives."
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        "tolerance": 0.001
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    },
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                "very long half-lives (e.g., uranium)",
                "no half-life",
                "only carbon-14"
              ],
              "answer": 1,
              "difficulty": 3,
              "hint": "The clock must still be ’ticking’.",
              "explanation": "Long-half-life isotopes like uranium suit very old samples; carbon-14 would be long gone."
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four."
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Astonishingly, every push, pull, glow, and decay in the cosmos comes down to **four fundamental interactions**, each
with its own strength, range, and carrier particle."
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strongest; binds quarks into protons and neutrons and holds the nucleus together against electric repulsion. Very short
range (nuclear size). Carrier: **gluon**.\n- **Electromagnetism** -- acts between electric charges; governs light,
chemistry, electronics, and all of everyday solidity. Infinite range, but attraction and repulsion. Carrier:
**photon**.\n- **Weak force** -- responsible for certain radioactive decays and for the reactions that let stars fuse.
Extremely short range. Carriers: **W and Z bosons**.\n- **Gravity** -- by far the weakest, but always attractive and
infinite in range, so it dominates at astronomical scales: planets, stars, galaxies. Carrier (theorized): the
**graviton**, not yet detected.\n\nA paradox: gravity rules the universe at large yet is fantastically feeble -- a tiny
magnet lifts a paperclip against the pull of the entire Earth. It dominates only because it’s always attractive and
never cancels, while electric charges mostly balance out.\n\n**Unification** is a grand goal of physics.
Electromagnetism and the weak force have been merged into one **electroweak** force. Theories aim to unite the strong
force too, and ultimately gravity -- a ’theory of everything’ -- but gravity stubbornly resists, the great unsolved
problem of fundamental physics."
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                      "The strong force is stronger at short range."
                    ],
                    "answer": "The strong force."
                  },
                  {
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                    "problem": "How can gravity rule the cosmos while being the weakest force?",
                    "steps": [
                      "Gravity is always attractive and never cancels.",
                      "Charges mostly balance out at large scale."
```

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          ],
          "answer": "It accumulates over huge masses."
        },
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          "title": "Force carriers",
          "problem": "Which particle carries the electromagnetic force?",
          "steps": [
            "EM acts between charges.",
            "Its carrier is the photon."
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        "Strong force binds nuclei (gluons); electromagnetism acts on charge (photons).",
        "The weak force drives some radioactive decays (W and Z bosons).",
        "Gravity is weakest but infinite-range and always attractive, so it dominates the cosmos.",
        "Electromagnetism and the weak force are unified (electroweak); fuller unification is unsolved."
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      "the weak force",
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    "explanation": "The strong force binds nucleons, overcoming electric repulsion at short range."
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          "weak",
          "gravity"
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        "difficulty": 2,
        "explanation": "Gravity is by far the weakest, though infinite in range and always attractive."
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        "difficulty": 3,
        "explanation": "Photons mediate electromagnetism."
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            "Strong force",
            "Weak force",
            "Gravity (theorized)"
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            "Gluon",
            "W and Z bosons",
            "Graviton"
          ]
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        "answer": [
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          "Gluon",
          "W and Z bosons",
          "Graviton"
        ],
        "difficulty": 3,
        "explanation": "Photon (EM), gluon (strong), W/Z (weak), graviton (gravity, not yet detected)."
      },
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          "the fastest",
          "always attractive and never cancels",
          "carried by photons",
          "short-ranged"
        ],
        "answer": 1,
        "difficulty": 3,
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        "explanation": "Gravity always adds up over huge masses, while electric charges largely cancel."
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        "difficulty": 4,
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        "explanation": "The electroweak theory unified them, confirmed by the discovery of the W and Z bosons."
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growing."
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the farther one is, the faster it flees. The startling conclusion: the **universe is expanding**. Run that expansion
backward and everything began, packed together, in a Big Bang."
              }
            },
            {
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H_0\\,d,$$\n\nwhere $H_0$ is the **Hubble constant**. This is exactly what uniform **expansion of space** looks like
from any point -- there is no center; every observer sees everyone else receding.\n\nThe evidence is **cosmological
redshift**: light from distant galaxies is stretched to longer (redder) wavelengths as space expands while it travels.
More distant galaxies show greater redshift.\n\n- A common analogy: dots on an inflating balloon -- every dot moves away
 from every other, with no special center.\n- Reversing the expansion implies a hot, dense beginning ~13.8 billion years
 ago.\n- $1/H_0$ gives a rough estimate of the age of the universe."
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                    "steps": [
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                      "Twice the distance -> twice the speed."
                    ],
                    "answer": "B recedes twice as fast."
                  },
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                    "title": "Redshift",
                    "problem": "What does a galaxy’s redshift tell us?",
                    "steps": [
                      "Light stretched by expansion.",
                      "More redshift = farther/faster."
                    ],
                    "answer": "It is moving away; more shift, more distant."
                  },
                  {
                    "title": "No center",
                    "problem": "Where is the center of the expansion?",
                    "steps": [
                      "Every observer sees recession.",
                      "Like a balloon’s surface."
                    ],
                    "answer": "There is no center."
                  }
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              }
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 -- it’s **accelerating**. Some unknown component, dubbed **dark energy**, makes up about 68% of the universe and pushes
 it apart ever faster. Its nature is the single biggest mystery in cosmology. The expansion is governed by the
**Friedmann equations** (from general relativity), which connect the universe’s expansion rate to everything it contains
 -- matter, radiation, and dark energy."
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                  "Hubble’s law: $v = H_0 d$.",
                  "Cosmological redshift stretches light as space expands.",
                  "There is no center -- every observer sees the same recession.",
                  "Dark energy is accelerating the expansion."
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                    "tex": "v = H_0\\,d"
                  }
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                "inversely proportional to distance",
                "zero"
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              "explanation": "$v = H_0 d$."
            },
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                "redshifted",
                "unchanged",
                "invisible"
              ],
              "answer": 1,
              "explanation": "Expansion stretches it to longer (red) wavelengths."
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              "answer": false,
              "explanation": "Every observer sees the same recession -- no center."
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(Type like ‘H*d’.)",
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                "vars": [
                  "H",
                  "d"
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            ]
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          "hint": "Hubble’s law.",
          "explanation": "$v = H_0 d$."
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          "kind": "NUMERIC",
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          "answer": {
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            "tolerance": 0
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          "hint": "$v \\propto d$.",
          "explanation": "Three times the distance, three times the speed."
        },
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 use $x$ for $d$).",
          "answer": {
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            "domain": [
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              5
            ],
            "variable": "x"
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          "hint": "Linear through the origin.",
          "explanation": "$v = H_0 d$ is a straight line; its slope is $H_0$."
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          "prompt": "The accelerating expansion is attributed to:",
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          "hint": "~68% of the universe.",
          "explanation": "Dark energy drives the acceleration."
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 detect today."
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 three independent pillars of evidence. And it leaves us facing the universe’s deepest puzzle: most of what exists is
invisible ’dark’ stuff we cannot yet identify."
```

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                "Cooled to microwaves."
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            {
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              "problem": "What hints that galaxies hold extra unseen mass?",
              "steps": [
                "Stars orbit too fast at the edges.",
                "Visible matter can’t hold them."
              ],
              "answer": "Galactic rotation curves."
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                "About 5%.",
                "The rest is dark."
              ],
              "answer": "About 5%."
            }
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        "The universe began hot and dense ~13.8 billion years ago.",
        "Evidence: Hubble expansion, the CMB, and primordial element abundances.",
        "The CMB is the cooled afterglow of the early universe.",
        "~5% ordinary matter, ~27% dark matter, ~68% dark energy.",
        "About 95% of the universe is of unknown nature -- the frontier."
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      "sunlight scattered by dust"
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      "black holes",
      "antimatter"
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    "hint": "Extra unseen mass.",
    "explanation": "Stars orbit too fast for the visible mass -- dark matter."
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      "tolerance": 1
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    "hint": "The visible part.",
    "explanation": "About 5%."
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    "options": [
      "Hubble expansion",
      "the CMB",
      "primordial element abundances",
      "the photoelectric effect"
    ],
    "answer": 3,
```

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              "hint": "One belongs to quantum physics.",
              "explanation": "The photoelectric effect is unrelated to Big Bang evidence."
            },
            {
              "scope": "PRACTICE",
              "kind": "MCQ",
              "prompt": "Dark energy is associated with:",
              "options": [
                "holding galaxies together",
                "accelerating cosmic expansion",
                "nuclear fusion",
                "the strong force"
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              "answer": 1,
              "hint": "~68% of the universe.",
              "explanation": "Dark energy accelerates the expansion."
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              "answer": false,
              "hint": "It’s the expansion of space itself.",
              "explanation": "Space itself expands; it wasn’t an explosion into a pre-existing void."
            }
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          "xpReward": 190,
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                "sub": "In a sliver of its first second, the universe ballooned by a factor of trillions of trillions --
 and seeded every galaxy."
              }
            },
            {
              "kind": "CONTEXT",
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              "content": {
                "markdown": "The Big Bang theory is superbly supported, yet on its own it leaves puzzles: why is the
universe so uniform, and so precisely flat? **Cosmic inflation** -- a burst of exponential expansion in the first
instant -- was proposed to solve them, and made predictions later confirmed in the cosmic microwave background."
              }
            },
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              "content": {
                "markdown": "**Inflation** proposes that in the first tiny fraction of a second (around $10^{-35}$ s
after the start), the universe underwent a brief phase of **exponential expansion**, ballooning by a factor of at least
$10^{26}$ almost instantly -- space itself expanding faster than light (which is allowed; no information travels through
 it).\n\nThis elegantly solves two puzzles:\n\n- **The horizon problem.** Opposite sides of the sky have nearly
identical temperature, yet seem too far apart to have ever exchanged heat. Inflation says they *were* in contact before
being stretched far apart -- explaining the uniformity.\n- **The flatness problem.** The universe’s geometry is observed
 to be remarkably flat. Inflation stretches any initial curvature smooth, like inflating a balloon makes its surface
look flat up close.\n\nAnd a bonus: tiny **quantum fluctuations** in the early universe were stretched to cosmic scale
by inflation, becoming the slight density variations that later grew, under gravity, into **galaxies and galaxy
clusters**. The largest structures trace back to quantum jitters. These fluctuations are seen as faint temperature
ripples in the CMB, matching inflation’s predictions remarkably well."
              }
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                    "problem": "Why is the sky’s temperature so uniform in all directions?",
                    "steps": [
                      "Distant regions seem never to have been in contact.",
                      "Inflation says they were, before being stretched apart."
                    ],
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```
            "answer": "Inflation explains the uniformity."
          },
          {
            "title": "Flatness",
            "problem": "How does inflation make the universe appear flat?",
            "steps": [
              "Rapid expansion stretches any curvature.",
              "Locally, it looks flat."
            ],
            "answer": "Stretching smooths the geometry."
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          {
            "title": "Seeds of galaxies",
            "problem": "What became the seeds of galaxies?",
            "steps": [
              "Quantum fluctuations were stretched by inflation.",
              "They grew under gravity."
            ],
            "answer": "Stretched quantum fluctuations."
          }
        ]
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      "content": {
        "intro": "Inflation: brief exponential expansion solving horizon and flatness."
      }
    },
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      "title": "Practice problems",
      "content": {
        "intro": "Reason about inflation and the early universe."
      }
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      "content": {
        "markdown": "Special relativity caps how fast anything moves **through** space at $c$. But the expansion of space itself isn’t motion through space -- it’s the stretching of the distances between things, and that has no speed limit. During inflation, and even today for very distant galaxies, the space between us and them grows so fast that those galaxies recede faster than light and their future light can never reach us. The cosmic speed limit applies to objects, not to the stage they sit on."
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          "It solves the horizon problem (uniform temperature) and flatness problem (flat geometry).",
          "Space can expand faster than light because nothing travels through it.",
          "Stretched quantum fluctuations seeded galaxies and clusters.",
          "Inflation’s predicted ripples match the cosmic microwave background."
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        "cooling only",
        "no change"
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      "difficulty": 3,
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```

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            "explanation": "It resolves the horizon (uniformity) and flatness problems."
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              "leftover stars",
              "quantum fluctuations stretched by inflation",
              "black holes",
              "comets"
            ],
            "answer": 1,
            "difficulty": 3,
            "explanation": "Tiny quantum fluctuations, stretched to cosmic scale, grew into structure."
          },
          {
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            "prompt": "Space itself can expand faster than light, since nothing is moving through space.",
            "answer": true,
            "difficulty": 4,
            "hint": "The limit applies to motion through space.",
            "explanation": "Expansion stretches distances; it isn’t motion through space, so $c$ doesn’t cap it."
          },
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            "prompt": "The ’horizon problem’ that inflation solves is that distant regions of the sky:",
            "options": [
              "are different temperatures",
              "are the same temperature despite seeming never in contact",
              "are empty",
              "are too close"
            ],
            "answer": 1,
            "difficulty": 4,
            "hint": "Uniformity needs past contact.",
            "explanation": "Inflation lets now-distant regions have been in contact before being stretched apart."
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            "scope": "PRACTICE",
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              "cosmic microwave background",
              "ocean",
              "Moon"
            ],
            "answer": 1,
            "difficulty": 3,
            "hint": "The Big Bang’s afterglow.",
            "explanation": "The CMB’s faint fluctuations match inflation’s predictions."
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              "sub": "Will the cosmos freeze, rip, or collapse? Dark energy holds the deciding vote -- and the journey
ends here."
            }
          },
          {
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                "steps": [
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                  "Dark energy pushes space apart."
                ],
                "answer": "Gravity versus dark energy."
              },
              {
                "title": "The 1998 surprise",
                "problem": "What did supernova observations in 1998 reveal?",
                "steps": [
                  "Expansion was expected to slow.",
                  "Instead it is accelerating."
                ],
                "answer": "Accelerating expansion, driven by dark energy."
              },
              {
                "title": "The favored ending",
                "problem": "Given accelerating expansion, what is the likely fate?",
                "steps": [
                  "Expansion continues forever.",
                  "Stars die; entropy maxes out."
                ],
                "answer": "The Big Freeze (heat death)."
              }
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          }
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          "content": {
            "markdown": "Dark energy is the deepest mystery in physics. It behaves like an energy built into space itself, pushing it apart, and its measured value is bafflingly small compared with naive quantum predictions -- off by some 120 orders of magnitude, the worst estimate in the history of science. Is it Einstein’s ’cosmological constant,’ a new field, or a sign that gravity itself needs revising on cosmic scales? Whoever answers will rewrite physics. The
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journey through the cosmos ends not with all questions closed, but with the greatest one wide open -- an invitation to
the next explorer."
              }
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                  "Three scenarios: Big Crunch (collapse), Big Freeze (heat death), Big Rip (everything torn apart).",
                  "In 1998, expansion was found to be accelerating, revealing dark energy dominates (~68%).",
                  "The favored fate is the Big Freeze: a cold, dark, ever-expanding future.",
                  "Dark energy’s nature is the deepest open question in physics."
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                    "tex": "v = H d"
                  }
                ]
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                "the strong force",
                "sunlight",
                "magnetism"
              ],
              "answer": 0,
              "difficulty": 3,
              "explanation": "Gravity pulls together; dark energy pushes apart."
            },
            {
              "scope": "CONCEPT_CHECK",
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              "prompt": "Observations show the universe’s expansion is accelerating.",
              "answer": true,
              "difficulty": 3,
              "explanation": "The 1998 supernova measurements revealed accelerating expansion, driven by dark energy."
            },
            {
              "scope": "CONCEPT_CHECK",
              "kind": "MCQ",
              "prompt": "If the universe keeps expanding forever and cools toward maximum entropy, this fate is the:",
              "options": [
                "Big Crunch",
                "Big Freeze (heat death)",
                "Big Bang",
                "Big Rip"
              ],
              "answer": 1,
              "difficulty": 3,
              "explanation": "Endless expansion and cooling toward maximum entropy is the heat death."
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$d$. (Type like ‘H*d‘.)",
              "answer": {
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                "vars": [
                  "H",
                  "d"
                ]
              },
              "difficulty": 3,
              "hint": "Speed proportional to distance.",
              "explanation": "$v = H d$ -- galaxies recede faster the farther they are."
            },
            {
              "scope": "PRACTICE",
              "kind": "MCQ",
              "prompt": "A ’Big Rip’ would occur if dark energy:",
              "options": [
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                "disappeared",
                "strengthened over time, tearing apart galaxies and atoms",
                "became gravity",
                "stopped expansion"
              ],
              "answer": 1,
              "difficulty": 4,
              "hint": "Runaway acceleration.",
              "explanation": "Strengthening dark energy could accelerate expansion enough to rip everything apart."
            },
            {
              "scope": "PRACTICE",
              "kind": "MCQ",
              "prompt": "A ’Big Crunch’ (collapse) would require:",
              "options": [
                "dark energy to dominate",
                "gravity to dominate and reverse the expansion",
                "no matter",
                "faster light"
              ],
              "answer": 1,
              "difficulty": 3,
              "hint": "Gravity must win.",
              "explanation": "Only if gravity overcame expansion would the universe collapse back together -- now
considered unlikely."
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      "subtitle": "Where the numbers begin",
      "description": "A sunny green at the city’s edge. Count the ducks, name the shapes in the flower beds, and crack
the patterns on the garden path -- the very first steps into mathematics.",
      "gradeRange": "K-1",
      "scaleLabel": "1 Counting Lane",
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          "xpReward": 100,
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                "headline": "One, Two, Three...",
                "sub": "You’re standing at the gates of Kindergarten Park. Ducks paddle on the pond, swings sway in the
breeze -- and everywhere you look, there are things to count."
              }
            },
            {
              "kind": "CONTEXT",
              "title": "You’re standing at the park gates",
              "content": {
                "markdown": "Look around the park. Three ducks float on the pond. Five flowers grow by the bench. Two
children laugh on the swings.\n\nNumbers are how we answer the question **\"how many?\"** Counting is one of the very
first kinds of math a person ever learns -- and it is everywhere. We count birthday candles, cookies on a plate, steps
up to the slide, and friends at the table.\n\nToday you’ll learn to count carefully and correctly, all the way to ten.
Get this right, and every other piece of math you ever meet will stand on top of it."
              }
            },
            {
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              "content": {
                "markdown": "Counting sounds easy -- but there is a secret rule that makes it work. The rule is **one
number for one thing**. This is called *one-to-one counting*.\n\nWhen you count the ducks, you point at **one** duck and
 say **\"one.\"** You point at the **next** duck and say **\"two.\"** Then **\"three.\"** Each duck gets exactly one
number, and you never skip a duck or count the same duck twice.\n\nThe number words always go in the same
order:\n\n**one, two, three, four, five, six, seven, eight, nine, ten.**\n\nThere is a second secret, and it is a big
one. The **last number you say tells you how many there are altogether.** If you count three ducks -- \"one, two,
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three\" -- then there are **3 ducks**. That last word, *three*, is the answer to \"how many?\" Grown-ups call this idea *cardinality*, but you can just call it **the last word counts**.\n\nWe can also write each number with a special symbol called a **numeral**:\n\n- 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\n\nSo \"five flowers\" can be written as **5 flowers**. The numeral 5 and the word *five* mean the very same thing."
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              }
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                    "problem": "At the party stand there are these balloons: ????. How many balloons are there?",
                    "steps": [
                      "Point at the first balloon and say \"one.\"",
                      "Point at the next and say \"two,\" then \"three,\" then \"four.\"",
                      "The last number you said was *four*."
                    ],
                    "answer": "There are 4 balloons. The last word you count is the answer."
                  },
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                    "title": "Don’t count twice",
                    "problem": "Sam counts a row of 3 toy cars but accidentally points at the middle car two times. He says \"one, two, three, four.\" Did he get the right answer?",
                    "steps": [
                      "There are only 3 cars, but Sam said four numbers.",
                      "He broke the rule: one number for one thing.",
                      "Counting the middle car twice gave a number that is too big."
                    ],
                    "answer": "No -- there are only 3 cars. Each thing gets counted exactly once."
                  },
                  {
                    "title": "What comes next?",
                    "problem": "You are counting and you just said \"six.\" What number comes next?",
                    "steps": [
                      "The number words go in a fixed order.",
                      "After six comes seven."
                    ],
                    "answer": "Seven comes next: ...six, seven, eight..."
                  }
                ]
              }
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                    "title": "Counting 1-10 Song"
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                    "youtubeId": "bGetqbqDVaA",
                    "title": "Let’s Count to 10"
                  }
                ]
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            },
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              "title": "Quick check",
              "content": {
                "intro": "Three quick questions -- count carefully!"
              }
            },
            {
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              "title": "Practice problems",
              "content": {
                "intro": "Show what you know. Use a finger to point and count if it helps!"
              }
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            },
            {
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              "content": {
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**cannot** touch -- like claps, jumps, or knocks on a door.\n\nTry it: clap your hands and count each clap. *Clap* --
one. *Clap* -- two. *Clap* -- three. You just counted sounds! The same rule works: one number for each clap, and the
last number tells you how many.\n\nThis is a big deal, because soon you’ll count things like minutes, dollars, and miles
 -- none of which you can hold in your hand."
              }
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                  "Use one number for one thing -- never skip and never double-count.",
                  "The number words always go in the same order: one to ten.",
                  "The last number you say tells you how many there are altogether.",
                  "Each number has a numeral: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10."
                ],
                "formulas": []
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            }
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              "options": [
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                "4",
                "5",
                "6"
              ],
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              "explanation": "Count them one at a time -- one, two, three, four, five. There are 5 stars."
            },
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              "prompt": "When you count, the last number you say tells how many there are.",
              "answer": true,
              "explanation": "Yes! If you count to four, there are 4 things. The last word is the answer."
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              "options": [
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                "nine",
                "ten",
                "five"
              ],
              "answer": 1,
              "explanation": "The order is seven, eight, nine, ten. After eight comes nine."
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              },
              "hint": "Point and count: one, two, three.",
              "explanation": "There are 3 apples."
            },
            {
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              "kind": "MCQ",
              "prompt": "Which row has 6 dots?",
              "options": [
                "- - - -",
                "- - - - -",
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                "- - -"
              ],
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            "answer": 2,
            "hint": "Count the dots in each row.",
            "explanation": "The third row has six dots: 1, 2, 3, 4, 5, 6."
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            "answer": false,
            "hint": "Remember the one-to-one rule.",
            "explanation": "No -- each thing gets exactly one number, or your count will be too big."
          },
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            "kind": "NUMERIC",
            "prompt": "Clap three times, then two more. How many claps in all? Type the number.",
            "answer": {
              "value": 5,
              "tolerance": 0
            },
            "hint": "Keep counting past three: four, five.",
            "explanation": "Three claps and two more makes 5 claps."
          },
          {
            "scope": "PRACTICE",
            "kind": "MCQ",
            "prompt": "What number comes right after nine?",
            "options": [
              "eight",
              "ten",
              "eleven",
              "seven"
            ],
            "answer": 1,
            "hint": "You’re almost at the end of the count.",
            "explanation": "After nine comes ten -- the top of today’s count."
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              "sub": "The pond is a circle. The picnic blanket is a square. The flag on the hill is a triangle. Once
you spot shapes, you’ll see them everywhere."
            }
          },
          {
            "kind": "CONTEXT",
            "title": "You’re standing on the picnic lawn",
            "content": {
              "markdown": "Spread out your picnic blanket and look at its edges. Four straight sides, four corners --
that’s a **square**. Glance at the round pond: no corners at all, just one smooth curve -- a **circle**. Up on the hill,
 a flag makes a **triangle** with three sides.\n\nShapes are the building blocks of everything we draw and build.
Knowing their names and what makes each one special is the start of **geometry** -- the math of shapes and space. And it
 all begins right here in the park."
            }
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            "kind": "CONCEPT",
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              "markdown": "We tell shapes apart by counting two things: their **sides** (the straight edges) and their
 **corners** (where two sides meet, also called *vertices*).\n\n- **Circle** -- perfectly round. It has **0 sides** and
**0 corners**, just one smooth curve. Think of a wheel, a coin, or the Sun.\n- **Triangle** -- **3 sides** and **3
corners**. *Tri* means three! Think of a slice of pizza or a party hat.\n- **Square** -- **4 sides** and **4 corners**,
and all four sides are the **same length**. Think of a cracker or a window pane.\n- **Rectangle** -- **4 sides** and **4
 corners**, but the sides come in **two long and two short**. A door and a phone are rectangles.\n\nHere’s a clever
idea: a **square is a special rectangle** where every side happens to be equal. So all squares are rectangles, but not
all rectangles are squares!\n\nShapes can be **big or small, and turned any way**, and they are still the same shape. A
triangle balanced on its point is still a triangle. Turning a shape never changes what it is -- only its sides and
corners decide its name."
            }
          },
```

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              }
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                    "title": "Count the sides",
                    "problem": "A road sign has 3 straight sides and 3 corners. What shape is it?",
                    "steps": [
                      "Count the sides: 3.",
                      "A shape with three sides is a triangle (*tri* = three)."
                    ],
                    "answer": "It is a triangle."
                  },
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                    "title": "Square or rectangle?",
                    "problem": "A window has 4 corners. Two sides are long and two sides are short. Is it a square or a rectangle?",
                    "steps": [
                      "Both squares and rectangles have 4 sides and 4 corners.",
                      "A square needs all four sides equal.",
                      "This window has long and short sides, so they are not all equal."
                    ],
                    "answer": "It is a rectangle (a square would have all sides the same length)."
                  },
                  {
                    "title": "The turned triangle",
                    "problem": "Mia draws a triangle, then turns her paper upside down. Is it still a triangle?",
                    "steps": [
                      "Turning a shape does not add or remove sides.",
                      "It still has 3 sides and 3 corners."
                    ],
                    "answer": "Yes -- it is still a triangle, just pointing a different way."
                  }
                ]
              }
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                    "title": "The Shapes Song"
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                    "title": "Learn 2D Shapes"
                  }
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              }
            },
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              "title": "Quick check",
              "content": {
                "intro": "Count those sides and corners!"
              }
            },
            {
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              "title": "Practice problems",
              "content": {
                "intro": "Find the shapes hiding in the questions below."
              }
            },
            {
              "kind": "DEEPER_DIVE",
              "title": "Deeper dive: shapes that build other shapes",
              "content": {
                "markdown": "Shapes love to team up. Put two triangles together along their long sides and you can make a **square** or a **rectangle**! Put a triangle on top of a square and you’ve drawn a little **house**.\n\nBuilders and
```

artists use this trick all the time. Bridges are full of triangles because triangles are very strong and don’t wobble. Next time you see a climbing frame at the park, look for the triangles holding it steady."
            }
          },
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                "We name shapes by their number of sides and corners.",
                "Circle: 0 sides, 0 corners -- perfectly round.",
                "Triangle: 3 sides, 3 corners.",
                "Square: 4 equal sides and 4 corners; a rectangle has 2 long and 2 short sides.",
                "Turning or resizing a shape never changes what it is."
              ],
              "formulas": []
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              "3",
              "4",
              "5"
            ],
            "answer": 1,
            "explanation": "A triangle has 3 sides. *Tri* means three!"
          },
          {
            "scope": "CONCEPT_CHECK",
            "kind": "TRUE_FALSE",
            "prompt": "A circle has 4 corners.",
            "answer": false,
            "explanation": "A circle has no corners and no straight sides -- it is perfectly round."
          },
          {
            "scope": "CONCEPT_CHECK",
            "kind": "MCQ",
            "prompt": "Which shape has 4 sides that are all the same length?",
            "options": [
              "Triangle",
              "Circle",
              "Square",
              "Rectangle"
            ],
            "answer": 2,
            "explanation": "A square has 4 equal sides. A rectangle has long and short sides."
          },
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            "kind": "MCQ",
            "prompt": "A slice of pizza is shaped most like which shape?",
            "options": [
              "Circle",
              "Triangle",
              "Square",
              "Rectangle"
            ],
            "answer": 1,
            "hint": "Count the points and straight edges.",
            "explanation": "A pizza slice has three sides and a point -- a triangle."
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            "kind": "NUMERIC",
            "prompt": "How many corners does a square have? Type the number.",
            "answer": {
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              "tolerance": 0
            },
            "hint": "Look at each place two sides meet.",
            "explanation": "A square has 4 corners (vertices)."
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            "prompt": "If you turn a triangle upside down, it becomes a square.",

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          "answer": false,
          "hint": "Does turning change the number of sides?",
          "explanation": "Turning never changes a shape. An upside-down triangle is still a triangle."
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          "options": [
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            "Rectangle",
            "Circle",
            "Triangle"
          ],
          "answer": 1,
          "hint": "Are all four sides the same length?",
          "explanation": "Long and short sides means it’s a rectangle, not a square."
        },
        {
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          "kind": "MCQ",
          "prompt": "Which shape has no straight sides at all?",
          "options": [
            "Triangle",
            "Rectangle",
            "Circle",
            "Square"
          ],
          "answer": 2,
          "hint": "Which one is perfectly round?",
          "explanation": "A circle is one smooth curve with no straight sides."
        }
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      "estMinutes": 11,
      "xpReward": 100,
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            "headline": "What Comes Next?",
            "sub": "Red tulip, yellow tulip, red tulip, yellow tulip... The flower beds of the park are full of patterns waiting to be cracked."
          }
        },
        {
          "kind": "CONTEXT",
          "title": "You’re standing by the flower beds",
          "content": {
            "markdown": "Walk along the garden path. The gardener planted the tulips in a special order: **red, yellow, red, yellow, red, yellow**. Can you guess what color comes next?\n\nThat repeating order is called a **pattern**. Patterns are everywhere -- in the stripes on a bee, the days of the week, the tiles on a floor, and the beat of a song.\n\nFinding patterns is one of the most powerful things a mathematician does. When you spot a pattern, you can **predict what comes next** -- even far into the future. That’s a kind of magic you’ll use for the rest of your life in math."
          }
        },
        {
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          "title": "How patterns work",
          "content": {
            "markdown": "A **pattern** is something that **repeats in the same order again and again**. The part that repeats is called the **unit** -- the little chunk that keeps coming back.\n\nMathematicians give patterns nicknames using letters. Each new thing gets a letter:\n\n- **A B A B** -- like *red, yellow, red, yellow*. The unit is \"A B\" and it repeats.\n- **A A B A A B** -- like *clap, clap, stomp, clap, clap, stomp*. The unit is \"A A B.\"\n- **A B C A B C** -- like *red, yellow, blue, red, yellow, blue*. The unit is \"A B C.\"\n\nTo figure out **what comes next**, do three things:\n\n1. **Find the unit** -- the part that repeats.\n2. **Check where you are** in the unit.\n3. **Say the next piece** of the unit.\n\nFor *red, yellow, red, yellow, red, ___* the unit is \"red, yellow.\" We just said red, so next comes **yellow**.\n\nPatterns don’t have to be colors. They can be **shapes** (circle, square, circle, square), **sounds** (loud, soft, loud, soft), **movements** (jump, spin, jump, spin), or **numbers** (1, 2, 1, 2). The idea is always the same: find the repeating unit, and you can predict forever."
          }
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 unit -- find the repeat and you’ll know what comes next!"
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                  "problem": "????? ___ -- what color comes next?",
                  "steps": [
                    "The unit that repeats is red, blue.",
                    "We just said red, so we are halfway through the unit.",
                    "The next piece of the unit is blue."
                  ],
                  "answer": "Blue (?). The pattern is red, blue, repeating."
                },
                {
                  "title": "A longer unit",
                  "problem": "??????? ___ -- what comes next: clap or stomp?",
                  "steps": [
                    "The repeating unit is clap, clap, stomp (A A B).",
                    "We just did clap, clap... so we are two-thirds through the unit.",
                    "The next piece is the stomp."
                  ],
                  "answer": "A stomp (?). The unit clap-clap-stomp starts over."
                },
                {
                  "title": "Is it a pattern?",
                  "problem": "Leo lines up blocks: red, blue, green, yellow, orange. Is this a repeating pattern?",
                  "steps": [
                    "Look for a unit that repeats.",
                    "Every block is a different color -- nothing repeats.",
                    "Without a repeat, there is no pattern to predict."
                  ],
                  "answer": "No -- it’s just different colors in a row. A pattern must repeat."
                }
              ]
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                  "title": "Patterns for Kids"
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            }
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            "content": {
              "markdown": "Most park patterns repeat, but some patterns **grow**. Look at a staircase: 1 step, then 2
steps, then 3 steps, then 4. Each time, the pattern adds **one more**.\n\nGrowing patterns don’t just repeat -- they
change by the same amount each time. *1, 2, 3, 4...* grows by one. *2, 4, 6, 8...* grows by two. Spotting how much a
pattern grows by is a sneak peek at adding and even multiplying, which you’ll explore as you walk deeper into the City
of Numbers."
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          },
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```

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          "A pattern repeats in the same order again and again.",
          "The unit is the chunk that keeps repeating (like A B or A A B).",
          "To predict, find the unit, see where you are, and say the next piece.",
          "Patterns can be colors, shapes, sounds, movements, or numbers.",
          "If nothing repeats, it is not a pattern."
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        "formulas": []
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    }
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        "? square",
        "? circle",
        "nothing"
      ],
      "answer": 1,
      "explanation": "The unit is triangle, square. We just had a triangle, so a square comes next."
    },
    {
      "scope": "CONCEPT_CHECK",
      "kind": "TRUE_FALSE",
      "prompt": "A pattern must repeat in the same order.",
      "answer": true,
      "explanation": "Yes -- repeating in the same order is exactly what makes something a pattern."
    },
    {
      "scope": "CONCEPT_CHECK",
      "kind": "MCQ",
      "prompt": "What is the repeating unit in: red, red, blue, red, red, blue?",
      "options": [
        "red",
        "red, blue",
        "red, red, blue",
        "blue, blue"
      ],
      "answer": 2,
      "explanation": "The chunk that repeats is red, red, blue (an A A B pattern)."
    },
    {
      "scope": "PRACTICE",
      "kind": "MCQ",
      "prompt": "?????? ___ -- what comes next?",
      "options": [
        "? apple",
        "? banana",
        "? grapes",
        "nothing"
      ],
      "answer": 1,
      "hint": "The unit is apple, banana.",
      "explanation": "After apple comes banana -- the unit apple-banana repeats."
    },
    {
      "scope": "PRACTICE",
      "kind": "MCQ",
      "prompt": "Which of these IS a repeating pattern?",
      "options": [
        "1, 2, 3, 4, 5",
        "red, blue, green, yellow",
        "A, B, A, B, A, B",
        "cat, dog, fish"
      ],
      "answer": 2,
      "hint": "Look for a chunk that comes back.",
      "explanation": "A, B, A, B repeats the unit A B. The others don’t repeat."
    },
    {
      "scope": "PRACTICE",
      "kind": "TRUE_FALSE",
      "prompt": "In the pattern clap, clap, stomp, clap, clap, stomp -- the unit is clap, clap, stomp.",
      "answer": true,
      "hint": "What part keeps coming back?",
      "explanation": "Correct -- clap, clap, stomp is the repeating unit (A A B)."
```

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          },
          {
            "scope": "PRACTICE",
            "kind": "MCQ",
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            "options": [
              "? square",
              "? triangle",
              "? circle",
              "nothing"
            ],
            "answer": 1,
            "hint": "The unit is square, square, triangle.",
            "explanation": "We just had two squares, so the triangle finishes the unit."
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            "kind": "MCQ",
            "prompt": "A staircase goes 1 step, 2 steps, 3 steps, 4 steps. How does this pattern change each time?",
            "options": [
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              "It adds one more",
              "It stays the same",
              "It takes one away"
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            "answer": 1,
            "hint": "Compare each number to the one before.",
            "explanation": "It grows by adding one more step each time -- a growing pattern."
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        "estMinutes": 10,
        "xpReward": 100,
        "sections": [
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            "content": {
              "scene": "park",
              "headline": "More, Less, or the Same?",
              "sub": "Two friends compare their handfuls of acorns at the snack stand. Who has more? Comparing is its
own kind of counting magic."
            }
          },
          {
            "kind": "CONTEXT",
            "title": "You’re standing at the snack stand",
            "content": {
              "markdown": "Two squirrels-in-training, Pip and Bo, empty their pockets at the snack stand. Pip has 5
acorns. Bo has 3 acorns. Who has **more**?\n\nEvery day we compare amounts. Who has more stickers? Are there enough cups
 for everyone? Is this pile bigger than that one? Learning to say **more**, **less**, and **equal** turns counting into
comparing -- and comparing is the doorway to adding, subtracting, and one day even algebra."
            }
          },
          {
            "kind": "CONCEPT",
            "title": "Comparing amounts",
            "content": {
              "markdown": "There are three ways two amounts can compare:\n\n- **More** (greater) -- one group has a
bigger amount.\n- **Less** (fewer) -- one group has a smaller amount.\n- **Equal** (the same) -- both groups have
exactly the same amount.\n\nHow do we tell? Two ways:\n\n**1. Match them up.** Line up Pip’s acorns next to Bo’s, one
for one. If Pip still has some left over after every one of Bo’s is matched, then Pip has **more**. If they match
perfectly with none left over, they are **equal**.\n\n**2. Count and compare.** Count each group. The bigger number is
more. Pip has 5, Bo has 3. Since **5 is more than 3**, Pip has more -- and Bo has **less**.\n\nIn counting order -- 1,
2, 3, 4, 5, 6... -- numbers that come **later** are **bigger**. So 5 comes after 3, which means 5 is more than 3. The
number 8 is more than 6 because 8 comes later when you count.\n\nMathematicians even have tiny symbols for this, which
you’ll meet soon:\n\n- the same as: **=** (5 = 5)\n- more than: **>** (5 > 3)\n- less than: **<** (3 < 5)\n\nFor now,
the words *more*, *less*, and *equal* are your trusty tools."
            }
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            "content": {
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              "intro": "Drop acorns onto each side of the balance scale. The heavier side tips down -- that’s the side
 with more. Make the two sides equal to balance it!"
            }
          },
```

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              "title": "Comparing together",
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                    "title": "Who has more?",
                    "problem": "Ada has 6 crayons. Ben has 4 crayons. Who has more?",
                    "steps": [
                      "Count each: Ada has 6, Ben has 4.",
                      "In counting order, 6 comes after 4.",
                      "Later means bigger, so 6 is more than 4."
                    ],
                    "answer": "Ada has more crayons (6 is more than 4)."
                  },
                  {
                    "title": "Just enough?",
                    "problem": "There are 4 children and 4 cups. Is that more cups, fewer cups, or equal?",
                    "steps": [
                      "Match one cup to each child.",
                      "Every child gets a cup and no cups are left over.",
                      "Same amount on both sides."
                    ],
                    "answer": "Equal -- 4 cups for 4 children, exactly the same."
                  },
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                    "title": "Which is less?",
                    "problem": "A red box holds 7 marbles. A blue box holds 9 marbles. Which box has fewer marbles?",
                    "steps": [
                      "Compare 7 and 9.",
                      "7 comes before 9 when you count.",
                      "Earlier means smaller, so 7 is less than 9."
                    ],
                    "answer": "The red box has fewer (7 is less than 9)."
                  }
                ]
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              }
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              "content": {
                "markdown": "Here’s a trick mathematicians teach to remember the symbols **>** and **<**. Pretend the
symbol is a hungry alligator’s mouth -- and the alligator always wants to eat the **bigger** pile!\n\nSo the open mouth
points at the larger number: **5 > 3** (the mouth opens toward the 5), and **3 < 5** (the mouth opens toward the 5
again). The pointy end always points at the smaller number. You’ll use these alligator jaws all the way into algebra."
              }
            },
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                  "Amounts can be more, less, or equal.",
                  "Match one-to-one, or count and compare the numbers.",
                  "Numbers later in the count are bigger; earlier numbers are smaller.",
```

```
          "Equal means exactly the same amount on both sides.",
          "Later you’ll write these as =, >, and < symbols."
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    }
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      "options": [
        "4",
        "7",
        "They are equal"
      ],
      "answer": 1,
      "explanation": "7 comes after 4 when you count, so 7 is more."
    },
    {
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      "kind": "TRUE_FALSE",
      "prompt": "If two groups match one-to-one with none left over, they are equal.",
      "answer": true,
      "explanation": "Exactly -- a perfect match with nothing left over means the amounts are equal."
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      "kind": "MCQ",
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      "options": [
        "9",
        "6",
        "They are equal"
      ],
      "answer": 1,
      "explanation": "6 comes before 9 when counting, so 6 is less."
    },
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      "kind": "MCQ",
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      "options": [
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        "Tom",
        "Equal"
      ],
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      "hint": "Which number is later in the count, 8 or 5?",
      "explanation": "8 is more than 5, so Maya has more."
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      "kind": "MCQ",
      "prompt": "There are 5 plates and 5 sandwiches. How do they compare?",
      "options": [
        "More plates",
        "More sandwiches",
        "Equal"
      ],
      "answer": 2,
      "hint": "Match each plate to a sandwich.",
      "explanation": "5 and 5 are the same -- they are equal."
    },
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      "prompt": "The number 3 is more than the number 6.",
      "answer": false,
      "hint": "Which comes later when you count?",
      "explanation": "No -- 3 comes before 6, so 3 is less than 6."
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      "kind": "MCQ",
      "prompt": "A hungry alligator mouth eats the bigger number: 8 ___ 2. Which way does it open?",
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        "8 > 2",
        "8 < 2"
      ],
      "answer": 0,
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```
            "hint": "The mouth opens toward the bigger pile.",
            "explanation": "8 is bigger, so the mouth opens toward it: 8 > 2."
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            "answer": {
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            "hint": "Equal means the same amount.",
            "explanation": "Equal means the same, so Bo also needs 4 acorns."
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    "description": "Chalkboards and lunch counts: addition, subtraction, place value, and two-digit regrouping -- the foundations every later number skill is built on.",
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      "glow": "#C77F00"
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              "Both give 8."
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              "The amount is unchanged."
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          "The plus sign is $+$; the equals sign means \"the same amount as\".",
          "Order doesn’t matter: $a + b = b + a$.",
          "Adding zero leaves a number unchanged.",
          "On a number line, adding jumps to the right."
```

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      "explanation": "Start at 3, count on four: 4, 5, 6, 7."
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      },
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      "explanation": "$6 + 2 = 8$."
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      "kind": "NUMERIC",
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              "hint": "Count on three from 4.",
              "explanation": "$4 + 3 = 7$."
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              "prompt": "$4 + 6 = 6 + 4$.",
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away and comparing."
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left?** And when two reading groups have 9 and 6 books, **how many more** does the bigger group have?\n\nBoth questions
are answered by **subtraction** -- the partner of addition. Subtraction tells us what remains after taking some away,
and it tells us the gap, or *difference*, between two amounts."
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them. We write it with the **minus sign** $-$, and the answer is called the **difference**.\n\n$$8 - 3 = 5$$\n\nRead it
as *\"eight minus three equals five.\"*\n\nKey facts:\n\n- **Counting back.** To find $8 - 3$, start at 8 and count back
 three: 7, 6, 5. The number you land on is the difference.\n- **Order matters.** Unlike addition, $7 - 2$ is *not* the
same as $2 - 7$. Subtraction is **not** commutative -- you must keep the larger amount first (for now).\n- **Subtracting
 zero changes nothing:** $6 - 0 = 6$. And any number minus itself is zero: $6 - 6 = 0$.\n- **Subtraction undoes
addition.** Since $4 + 5 = 9$, it must be that $9 - 5 = 4$. They are *inverse* operations -- this is your best way to
**check** a subtraction.\n\nOn a number line, subtracting is **jumps to the left**: start at the first number and hop
backward by the second."
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                    "steps": [
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                      "Count back three: 7, 6, 5.",
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                "You land on 5."
              ],
              "answer": "$8 - 3 = 5$."
            },
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              "steps": [
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                "$9 - 6 = 3$."
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              "answer": "3 more books."
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                "Add the answer back: $4 + 5 = 9$.",
                "It matches the start, so the answer is right."
              ],
              "answer": "$4 + 5 = 9$ confirms $9 - 5 = 4$."
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 makes 8?\"** That hidden number is the answer -- it’s called the *missing addend*. Counting up from 3 to 8 (4, 5, 6, 7,
 8 -- that’s 5 steps) gives the same 5.\n\nThis \"think-addition\" strategy is how cashiers make change and how you’ll
later solve equations like $3 + x = 8$. Subtraction and addition are two views of the very same relationship."
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            "The minus sign is $-$; the answer is the difference.",
            "Order matters: $a - b \\ne b - a$ in general.",
            "Subtracting zero changes nothing; a number minus itself is 0.",
            "Subtraction undoes addition -- use it to check your work."
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            {
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    "explanation": "Any number minus itself is 0."
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    "hint": "Subtracting zero changes nothing.",
    "explanation": "$5 - 0 = 5$."
  },
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of ten** with a rubber band, then counts the loose ones left over. Four bundles and seven singles: **47
pencils**.\n\nThis bundling is the secret behind every number we write. Our number system is built on **tens**, and the
*position* of each digit tells you how many ones, tens, hundreds, and so on it stands for. This idea -- **place value**
-- is what lets ten little symbols (0-9) write any number at all."
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its right:\n\n- the **ones place** (right) -- worth 1 each\n- the **tens place** (left) -- worth 10 each\n\nTake the
number **34**:\n\n$$34 = \\underbrace{3}_{\\text{tens}}\\,\\underbrace{4}_{\\text{ones}} = 3 \\times 10 + 4 \\times 1 =
30 + 4.$$\n\nThe **3** is not just \"three\" -- because it sits in the tens place, it is worth **30**. The **4** in the
ones place is worth **4**. Writing a number as $30 + 4$ is called **expanded form**.\n\nSo the same digit means
different amounts depending on *where* it sits. In **34** the 3 means 30; in **43** the 3 means just 3. Place is
everything.\n\nThe big rule that ties it together: **ten ones make one ten.** Whenever you gather ten loose ones, you
bundle them into a single ten and move one place to the left -- exactly what the rubber band did to the pencils."
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                    "Tens are worth 10 each: $5 \\times 10 = 50$."
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                  "answer": "The 5 is worth 50."
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                    "7 ones $= 7$.",
                    "$60 + 7 = 67$."
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                  "answer": "67."
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                "So $48 = 40 + 8$."
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              "answer": "$48 = 40 + 8$."
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            "Two-digit numbers have a tens place and a ones place.",
            "$34 = 3 \\text{ tens} + 4 \\text{ ones} = 30 + 4$ (expanded form).",
            "Ten ones bundle into one ten -- the heart of base ten.",
            "The same digit means different amounts in different places."
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        "explanation": "$60 = 6$ tens."
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        "kind": "NUMERIC",
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        "answer": {
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        "hint": "$50 + 3$.",
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          "$200 + 8$",
          "$20 + 80$"
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        "explanation": "The 9 is in the tens place: 90."
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carry them over."
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  "content": {
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column by column, a method that scales up to any size of number."
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and add each column, starting from the **ones** on the right.\n\n**When a column doesn’t reach ten** (no regrouping),
just add each column:\n\n$$23 + 14: \\quad 3 + 4 = 7 \\ (\\text{ones}), \\quad 2 + 1 = 3 \\ (\\text{tens})
\\;\\Rightarrow\\; 37.$$\n\n**When the ones reach ten or more**, you must **regroup** (also called **carrying**). Ten
ones bundle into one ten, which you carry into the tens column:\n\n$$27 + 15: \\quad 7 + 5 = 12.$$\n\nThat’s 1 ten and 2
ones -- write the **2** in the ones place and **carry the 1** ten on top of the tens column. Then add the tens,
including the carry:\n\n$$2 + 1 + 1 = 4 \\;\\Rightarrow\\; 42.$$\n\nThe rule in one line: **add each column right to
left, and whenever a column hits 10 or more, carry the extra ten to the next column.** It’s the same \"ten ones make one
ten\" bundling from place value -- now put to work."
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          "Tens: $3 + 1\\,(\\text{carry}) = 4$.",
          "Combine: 4 tens and 5 ones."
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          "Tens: $2 + 1 + 1\\,(\\text{carry}) = 4$.",
          "Combine: 4 tens and 2 ones."
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                  "Add columns from the right (ones first).",
                  "If a column reaches 10 or more, write the ones digit and carry the ten.",
                  "Add the carry into the next column.",
                  "Carrying is just \"ten ones make one ten\" again."
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```

```
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              "hint": "$6+8=14$, carry 1.",
              "explanation": "Ones 14 (write 4, carry 1), tens $4+2+1=7$: 74."
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              "hint": "No carry needed.",
              "explanation": "Ones $0+5=5$, tens $5+3=8$: 85."
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              "prompt": "$19 + 19 = \\,$?",
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                "value": 38,
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              "hint": "$9+9=18$, carry 1.",
              "explanation": "Ones 18 (write 8, carry 1), tens $1+1+1=3$: 38."
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                "write 1, carry 3",
                "write 13, carry 0",
                "write 3, carry 0"
              ],
              "answer": 0,
              "hint": "$6 + 7 = 13 = $ 1 ten and 3 ones.",
              "explanation": "$6+7=13$: write the 3, carry the 1 ten."
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short, borrow a ten."
              }
            },
            {
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```
like addition, we subtract two-digit numbers **column by column** using place value -- but now the tricky part is what to do when the top ones digit is too small to subtract from."
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                  "Tens: $4 - 2 = 2$.",
                  "Combine: 2 tens and 4 ones."
                ],
                "answer": "$47 - 23 = 24$."
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                  "Now $12 - 8 = 4$; tens drop from 5 to 4.",
                  "Tens: $4 - 1 = 3$."
                ],
                "answer": "$52 - 18 = 34$."
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              {
                "title": "Check by adding",
                "problem": "Subtract $42 - 17$ and check.",
                "steps": [
                  "Borrow: $12 - 7 = 5$ (ones), $3 - 1 = 2$ (tens) -> 25.",
                  "Check: $25 + 17 = 42$.",
                  "It matches the start."
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                "answer": "$42 - 17 = 25$ (checked: $25 + 17 = 42$)."
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change 42 into \"3 tens and 12 ones,\" that’s still $30 + 12 = 42$; we only **renamed** it so the ones column has enough
 to subtract from.\n\nThis is exactly the reverse of carrying: carrying bundles ten ones *up* into a ten, while
borrowing breaks a ten *down* into ten ones. The same column method, with borrowing, works for hundreds and thousands --
 and checking by adding back always tells you if you got it right."
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                "If the top ones digit is smaller than the bottom, borrow a ten.",
                "Borrowing renames one ten as ten ones -- value is unchanged.",
                "Borrowing is carrying in reverse.",
                "Check by adding the answer back to the number subtracted."
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                }
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            "answer": {
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            "hint": "Borrow: $12 - 7$.",
            "explanation": "Borrow: $12-7=5$, $3-1=2$: 25."
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            "hint": "Borrow: $10 - 4$.",
            "explanation": "Borrow: $10-4=6$, $5-2=3$: 36."
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            "kind": "NUMERIC",
            "prompt": "$85 - 30 = \\,$?",
            "answer": {
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              "tolerance": 0
            },
            "hint": "No borrow needed.",
            "explanation": "Ones $5-0=5$, tens $8-3=5$: 55."
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          {
            "scope": "PRACTICE",
            "kind": "NUMERIC",
            "prompt": "Check by adding: if $53 - 19 = 34$, what is $34 + 19$?",
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            "hint": "Adding back undoes the subtraction.",
            "explanation": "$34 + 19 = 53$, confirming the difference."
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              "4 is more than 6",
              "they are equal"
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            "explanation": "Top ones digit 4 is smaller than 6, so borrow a ten."
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```

a few are left in the tin. To describe \"how much\" each person has, whole numbers aren't enough -- we need **fractions**.\n\nFractions are everywhere: half a cup of flour, a quarter past three, three-quarters of a tank of gas. Master them at the bakery and you've unlocked the rest of arithmetic."
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                      "You ate 3 parts -> numerator 3."
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                    "answer": "$\\tfrac{3}{8}$ of the pie."
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                    "problem": "A chocolate bar has 6 equal pieces. You take 5. What fraction is left?",
                    "steps": [
                      "6 equal parts in all.",
                      "5 taken means $6 - 5 = 1$ left.",
                      "One part out of six."
                    ],
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                    "answer": "Four quarters make one whole."
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shared equally among 4 children, each child gets $\\tfrac{3}{4}$ of a cookie. The fraction bar is literally a division
sign!\n\nThis is why $\\tfrac{4}{4} = 1$ (4 shared among 4 is 1 each) and why $\\tfrac{0}{5} = 0$. It also hints at
decimals: $\\tfrac{3}{4} = 3 \\div 4 = 0.75$, which you’ll explore in the last lesson on this street."
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                "Denominator (bottom) = equal parts in the whole; numerator (top) = parts taken.",
                "The parts must be equal in size.",
                "A unit fraction has 1 on top; $\\tfrac{b}{b} = 1$.",
                "A larger denominator means more, smaller pieces."
              ],
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                  "tex": "\\tfrac{b}{b} = 1"
                }
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              "factor"
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            "explanation": "The bottom number is the denominator."
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              "the parts you have",
              "the total amount",
              "nothing"
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            "answer": 1,
            "explanation": "The numerator counts the parts you have."
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            "explanation": "Yes -- the denominator is the number of equal parts."
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              "$\\tfrac{3}{8}$",
              "$\\tfrac{3}{5}$",
              "$\\tfrac{5}{8}$"
```

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              "hint": "Parts eaten over total parts.",
              "explanation": "3 of 8 parts is $\\tfrac{3}{8}$."
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              "hint": "$\\tfrac{4}{4} = 1$.",
              "explanation": "Four quarters make one whole."
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                "$\\tfrac{3}{4}$",
                "$\\tfrac{5}{6}$"
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              "answer": 1,
              "hint": "A unit fraction has 1 on top.",
              "explanation": "$\\tfrac{1}{5}$ has numerator 1."
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              "scope": "PRACTICE",
              "kind": "TRUE_FALSE",
              "prompt": "The fraction $\\tfrac{4}{4}$ equals one whole.",
              "answer": true,
              "hint": "Numerator equals denominator.",
              "explanation": "$\\tfrac{4}{4} = 1$."
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                "$\\tfrac{1}{5}$"
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the same pie into 4 slices and you take 2 -- that’s $\\tfrac24$. You got **exactly the same amount of pie**. These are
**equivalent fractions**: different numbers, same value. Recognising them is the key to comparing and adding fractions
later."
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            },
            {
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              }
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                      "$\\tfrac{6 \\div 2}{8 \\div 2} = \\tfrac{3}{4}$.",
                      "3 and 4 share no factor > 1."
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                "Multiply top and bottom by the same number to build an equivalent.",
                "Divide top and bottom by a common factor to simplify.",
                "Lowest terms: no common factor greater than 1 remains.",
                "Cross-multiplying tests whether two fractions are equal."
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                      "Smaller denominator -> bigger piece: 2 < 5."
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                      "$\\tfrac23 = \\tfrac{8}{12}$, $\\tfrac34 = \\tfrac{9}{12}$.",
                      "$9 > 8$."
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        "A common denominator is any common multiple of the bottoms.",
        "Benchmarking against $\\tfrac12$ gives quick estimates."
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matching the slice size."
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 Boxed together, that’s $\\tfrac34$ of a pie. But what if the slices are *different sizes* -- say $\\tfrac12$ and
$\\tfrac13$? Then we first have to make the pieces match."
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\\qquad \\frac56 - \\frac16 = \\frac{4}{6} = \\frac23.$$\n\nAlways **simplify** the result if you can ($\\tfrac46 =
\\tfrac23$).\n\n**Different denominators.** You can’t add fourths to thirds any more than you can add apples to oranges
-- first rewrite both with a **common denominator**:\n\n$$\\frac12 + \\frac13: \\quad \\text{use 6} \\;\\Rightarrow\\;
\\frac{3}{6} + \\frac{2}{6} = \\frac{5}{6}.$$\n\nThe recipe:\n\n1. Find a common denominator (a common multiple of the
bottoms).\n2. Rewrite each fraction with that denominator (build equivalents).\n3. Add or subtract the numerators; keep
the denominator.\n4. Simplify the answer.\n\nThe whole game is **making the slices the same size** before you combine
them."
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Same-size slices simply add up."
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                      "$\\tfrac46$ simplifies: divide by 2.",
                      "$\\tfrac46 = \\tfrac23$."
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\\frac{c}{d} = \\frac{ad + bc}{bd}.$$\n\nFor $\\tfrac12 + \\tfrac13$: $a d + b c = 1\\cdot3 + 2\\cdot1 = 5$ over $bd =
6$, giving $\\tfrac56$ -- the same answer. It always works, though using the **least** common denominator keeps the
numbers small and the simplifying easy. This same cross-pattern reappears constantly in algebra when you add algebraic
fractions."
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"Different denominators: rewrite with a common denominator first.",
"Build equivalents so the slice sizes match, then combine.",
"Always simplify the final answer.",
"The formula $\\tfrac{a}{b}+\\tfrac{c}{d}=\\tfrac{ad+bc}{bd}$ always works."
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"$\\tfrac16$",
"$\\tfrac24$"
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all three is one of the most useful skills in everyday math."
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beyond:\n\n$$0.5 = \\frac{5}{10} = \\frac12, \\qquad 0.25 = \\frac{25}{100} = \\frac14, \\qquad 0.75 = \\frac{75}{100} =
 \\frac34.$$\n\nThe first digit after the dot is **tenths**, the second is **hundredths**. To turn a fraction into a
decimal, **divide** the top by the bottom: $\\tfrac34 = 3 \\div 4 = 0.75$.\n\n**Percent** means **\"per hundred\"** --
the symbol $\\%$ literally stands for $/100$:\n\n$$50\\% = \\frac{50}{100} = 0.5, \\qquad 25\\% = \\frac{25}{100} =
0.25.$$\n\nSo the three forms convert easily:\n\n- **Fraction -> decimal:** divide top by bottom.\n- **Decimal ->
percent:** multiply by 100 (move the dot two places right): $0.5 \\to 50\\%$.\n- **Percent -> decimal:** divide by 100
(move the dot two places left): $75\\% \\to 0.75$.\n\nMemorising a few anchors makes this instant: $\\tfrac12 = 0.5 =
50\\%$, $\\tfrac14 = 0.25 = 25\\%$, $\\tfrac34 = 0.75 = 75\\%$, $\\tfrac{1}{10} = 0.1 = 10\\%$."
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$0.5$, and $50\\%$ all describe the same shaded pie."
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                  "$= \\tfrac14$."
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0.3333\\ldots$ -- the 3s never stop. We write this as $0.\\overline{3}$, with a bar over the repeating part. Fractions
whose denominators (in lowest terms) are built only from 2s and 5s give *terminating* decimals like $0.25$; all others
**repeat** forever in a pattern. Remarkably, every repeating decimal can be turned back into an exact fraction -- so
fractions and decimals describe exactly the same set of numbers, the *rationals*."
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              "Fraction -> decimal: divide top by bottom.",
              "Percent means per hundred: $50\\% = \\tfrac{50}{100} = 0.5$.",
              "Decimal -> percent: x100; percent -> decimal: /100.",
              "Anchors: $\\tfrac12=0.5=50\\%$, $\\tfrac14=0.25=25\\%$, $\\tfrac34=0.75=75\\%$."
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      "one"
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    "explanation": "Percent means per hundred."
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      "$\\tfrac25$",
      "$\\tfrac{1}{25}$"
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    "explanation": "$0.25 = \\tfrac{25}{100} = \\tfrac14$."
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              "explanation": "$0.5$ is the largest."
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                "sub": "Two apples for every orange, three parts water to one part juice -- a ratio compares amounts side by side."
              }
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              "content": {
                "markdown": "The fruit seller stacks 6 apples next to 4 oranges. A juice stand mixes 3 cups of water with 1 cup of concentrate. To describe these comparisons precisely, we use **ratios** -- the foundation of rates, proportions, and percents, and one of the most useful ideas in everyday math."
              }
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                "markdown": "A **ratio** compares two quantities. The ratio of 6 apples to 4 oranges can be written three ways:\n\n$$6 : 4 \\qquad \\frac{6}{4} \\qquad \\text{\"6 to 4\".}$$\n\nLike fractions, ratios can be **simplified** by dividing both parts by a common factor:\n\n$$6 : 4 = 3 : 2.$$\n\nTwo flavours of ratio:\n\n- **Part-to-part** compares one group to another: apples to oranges, $6 : 4 = 3 : 2$.\n- **Part-to-whole** compares one group to the total: apples to all fruit, $6 : 10 = 3 : 5$ (since $6 + 4 = 10$).\n\nPart-to-whole ratios are just fractions in disguise -- apples are $\\tfrac{6}{10} = \\tfrac35$ of the fruit.\n\nOrder matters: \"apples to oranges\" $6:4$ is **not** the same as \"oranges to apples\" $4:6$. Always keep the quantities in the order named."
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          "$= 3 : 2$."
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          "Red to total is $6 : 15$.",
          "Divide by 3: $2 : 5$."
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        "steps": [
          "Ratio is $2 : 1$.",
          "6 cups flour is $3\\times$ the flour.",
          "Sugar scales the same: $3 \\times 1 = 3$."
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  "content": {
    "markdown": "Ratios can compare **three or more** quantities at once. A fruit punch recipe might be $2 :
 3 : 5$ (orange : pineapple : soda). The parts still scale together: doubling gives $4 : 6 : 10$. And the **total
parts** ($2 + 3 + 5 = 10$) let you turn any total amount into the right share -- for 20 cups of punch, each ’part’ is 2
cups, so you’d use 4, 6, and 10 cups. This ’parts’ thinking is exactly how recipes, paint mixes, and concrete are scaled
 in the real world."
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      "Simplify ratios by dividing both parts by a common factor.",
      "Part-to-part compares groups; part-to-whole compares to the total.",
      "Part-to-whole ratios are fractions.",
      "Order matters: $a:b \\ne b:a$."
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      "nothing"
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      "$3:2$",
      "$2:3$",
      "$12:8$"
    ],
    "answer": 1,
    "explanation": "Divide both by 2: $3:2$."
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    "answer": true,
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      "$8:12$",
      "$2:3$",
      "$12:20$"
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    "answer": 0,
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    "explanation": "$12:8 = 3:2$."
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      "$2:5$",
      "$9:15$",
      "$2:3$"
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    "answer": 1,
    "hint": "Total is 15.",
    "explanation": "$6:15 = 2:5$."
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              "tolerance": 0
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            "hint": "6 is $3\\times$ the flour.",
            "explanation": "Sugar scales to $3\\times1 = 3$ cups."
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                    "$= \\$2$ per pound."
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                  "answer": "$2 per pound."
                },
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                    "$150 \\div 3 = 50$.",
                    "Units: miles per hour."
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            },
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                "First: $4 \\div 2 = 2$, so \\$2/L.",
                "Second: $9 \\div 3 = 3$, so \\$3/L.",
                "$2 < 3$."
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              "answer": "\\$4 for 2 litres is the better deal."
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**typing speed** is words per minute; **heart rate** is beats per minute; a **density** is grams per cubic centimetre;
even a **wage** is dollars per hour. Each squeezes a two-number relationship into a single, comparable figure.
Supermarkets are even required to print unit prices on shelf tags so shoppers can compare a giant box and a small one
fairly -- math protecting your wallet."
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            "A unit rate has denominator 1 -- the amount per single unit.",
            "Find a unit rate by dividing.",
            "Lower price per unit = better deal; more distance per hour = faster.",
            "Equivalent ratios scale together; the unit rate has 1 on the bottom."
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      "explanation": "$120/2 = 60$ mph."
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      "answer": true,
      "explanation": "Like miles and hours."
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      "answer": {
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      "hint": "Divide 6 by 3.",
      "explanation": "$6 \\div 3 = 2$, so \\$2 per pound."
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        "tolerance": 0
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      "hint": "Distance / time.",
      "explanation": "$150/3 = 50$ mph."
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      "prompt": "12 cookies shared among 4 children. Cookies per child?",
      "answer": {
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      "hint": "Divide.",
      "explanation": "$12/4 = 3$."
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        "\\$9 for 3 L",
        "they are equal",
        "cannot tell"
      ],
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      "explanation": "\\$2/L vs \\$3/L -- the first is cheaper."
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        "tolerance": 0
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      "explanation": "$20/4 = 5$ pages/min."
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missing piece."
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stay in the same ratio, you have a **proportion** -- and cross-multiplication finds any missing value. It’s the engine
behind scaling recipes, reading maps, mixing chemicals, and converting currencies."
              }
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              "content": {
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\\frac{c}{d}.$$\n\nThe powerful tool for proportions is **cross-multiplication**: multiply each numerator by the other
denominator. If the ratios are equal, the cross-products are equal:\n\n$$\\frac{a}{b} = \\frac{c}{d}
\\;\\Longleftrightarrow\\; a \\times d = b \\times c.$$\n\nThis lets you **solve for an unknown**. To solve
$\\dfrac{x}{4} = \\dfrac{6}{8}$:\n\n$$8x = 4 \\times 6 = 24 \\;\\Rightarrow\\; x = 3.$$\n\nThe recipe for any proportion
 problem:\n\n1. Write the two ratios, keeping units in the **same order** on top and bottom.\n2. Cross-multiply.\n3.
Solve the resulting simple equation.\n\nCross-multiplication is also the quickest **test** of whether two ratios are
equal: $\\tfrac23$ and $\\tfrac69$ give $2\\times9 = 18$ and $3\\times6 = 18$ -- equal, so they form a proportion."
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                      "Divide by 8: $x = 3$."
                    ],
                    "answer": "$x = 3$."
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                    "problem": "If 2 pencils cost 50 cents, what do 8 pencils cost (same rate)?",
                    "steps": [
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                      "Cross-multiply: $2x = 50 \\times 8 = 400$.",
                      "$x = 200$ cents."
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                      "They’re equal."
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to find the real distance for 3.5 cm you solve $\\dfrac{1}{50} = \\dfrac{3.5}{x}$. Architects’ blueprints, model trains,
 and even the enlargement on a photocopier all rely on keeping a ratio constant. Two shapes with all sides in the same
ratio are called **similar** -- the geometry version of a proportion, which you’ll meet over in the Architect’s
Quarter."
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            "Use it to solve for an unknown value.",
            "Keep units in the same order top and bottom.",
            "Cross-products also test whether two ratios are equal."
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              "markdown": "A sign reads ’25% off’ on an $80 coat. To know your saving, you find **25% of 80**.
Percents run sales, taxes, tips, and interest -- being fast and confident with ’percent of a number’ is one of the most
practical skills in all of math."
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\\frac{p}{100} \\times n.$$\n\nFor example, $25\\%$ of $80$:\n\n$$\\frac{25}{100} \\times 80 = 0.25 \\times 80 =
20.$$\n\nHandy mental anchors:\n\n- **10%** -- just move the decimal one place: $10\\%$ of $50$ is $5$.\n- **50%** --
half: $50\\%$ of $90$ is $45$.\n- **25%** -- a quarter: $25\\%$ of $80$ is $20$.\n- **1%** -- move the decimal two
places: $1\\%$ of $300$ is $3$.\n\nYou can build other percents from these: $30\\%$ is three lots of $10\\%$; $15\\%$ is
 $10\\% + 5\\%$ (and $5\\%$ is half of $10\\%$). This is how people compute tips in their heads."
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$25\\%$ makes ‘percent of a number’ click -- a quarter of 80 is 20."
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                      "$0.25 \\times 80 = 20$."
                    ],
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                      "$50 \\to 5.0$."
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                      "$5\\%$ is half of that $= 2$.",
                      "$15\\% = 4 + 2 = 6$."
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 $50\\%$ of $8$ -- and the second is obviously $4$! This works because both are $\\dfrac{p \\times q}{100}$. When one
number is an easy percent (like 50 or 25 or 10), flip the problem to make it trivial. It’s a favourite trick for fast
mental math and a neat reminder that multiplication doesn’t care about order."
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              "$10\\%$: move the decimal one place; $1\\%$: two places.",
              "$50\\%$ is half; $25\\%$ is a quarter.",
              "Build other percents from these anchors (e.g. $15\\% = 10\\% + 5\\%$).",
              "$p\\%$ of $q$ equals $q\\%$ of $p$."
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          "explanation": "$0.25 \\times 80 = 20$."
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            "hint": "$10\\%$ of 60.",
            "explanation": "$0.10 \\times 60 = \\$6$."
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            "explanation": "$100\\%$ is the whole amount."
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                      "$\\dfrac{10}{50}\\times100\\% = 20\\%$."
                    ],
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                  },
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                    "problem": "A $100 item is $30\\%$ off. Find the sale price.",
                    "steps": [
                      "You pay $100\\% - 30\\% = 70\\%$.",
                      "$0.70 \\times 100 = 70$."
                    ],
                    "answer": "$70."
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                  {
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                    "problem": "A quantity falls from 40 to 30. Find the percent decrease.",
                    "steps": [
                      "Change $= 40 - 30 = 10$.",
                      "$\\dfrac{10}{40}\\times100\\% = 25\\%$."
                    ],
                    "answer": "$25\\%$ decrease."
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 No! Start at $100; up 50% gives $150; down 50% of *150* removes $75, leaving **$75**. The catch is that each percent is
 taken from a *different* base. This is why a stock that drops 50% must then *gain 100%* just to recover, and why
’percent off’ and ’percent on’ never simply cancel. Always check what the percent is a percentage *of*."
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                  "Up is an increase; down is a decrease.",
                  "Always compare to the original (starting) value.",
                  "Discount: pay $(100 - p)\\%$ of the price; markup adds $p\\%$.",
                  "Successive percent changes use different bases and don’t cancel."
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      "explanation": "Change over original, times 100%."
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      "explanation": "You pay $75\\%$ of 80 = 60."
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      "explanation": "$0.70\\times100 = \\$70$."
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      "answer": {
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      "hint": "$\\tfrac{10}{40}$.",
      "explanation": "$25\\%$."
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        "tolerance": 0
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      "explanation": "$50 + 10 = \\$60$."
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              }
            },
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the same variable part. $3x$ and $5x$ are like terms, so $3x + 5x = 8x$. But $3x$ and $3x^2$ are **not** like terms, and
 neither are $3x$ and $3y$ -- their variable parts differ, so they can’t be merged. Combining like terms is the
workhorse move you’ll use to tidy every expression and equation from here on."
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              "An expression combines numbers, variables, and operations -- no equals sign.",
              "A coefficient multiplies a variable; a constant stands alone.",
              "Evaluate by substituting a value and using the order of operations.",
              "Combine like terms (same variable part) to simplify."
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          "prompt": "Which of these is an *expression* (not an equation)?",
          "options": [
            "$2x + 5$",
            "$2x + 5 = 9$",
            "$x = 3$",
            "$7 = 7$"
```

```
  ],
  "answer": 0,
  "explanation": "An expression has no equals sign; the others all do."
},
{
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  "kind": "NUMERIC",
  "prompt": "Evaluate $3x + 2$ when $x = 5$.",
  "answer": {
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    "tolerance": 0
  },
  "explanation": "$3(5) + 2 = 17$."
},
{
  "scope": "CONCEPT_CHECK",
  "kind": "MCQ",
  "prompt": "In $4y - 9$, the coefficient is:",
  "options": [
    "$y$",
    "$4$",
    "$-9$",
    "$5$"
  ],
  "answer": 1,
  "explanation": "The coefficient multiplies the variable: $4$."
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  "kind": "NUMERIC",
  "prompt": "Evaluate $5n - 3$ when $n = 4$.",
  "answer": {
    "value": 17,
    "tolerance": 0
  },
  "hint": "Multiply, then subtract.",
  "explanation": "$5(4) - 3 = 20 - 3 = 17$."
},
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  "scope": "PRACTICE",
  "kind": "SYMBOLIC",
  "prompt": "Write an expression for ’seven more than a number $n$’. (Use $n$.)",
  "answer": {
    "expr": "n + 7",
    "vars": [
      "n"
    ]
  },
  "difficulty": 2,
  "hint": "’More than’ means add.",
  "explanation": "Seven more than $n$ is $n + 7$ (equivalently $7 + n$)."
},
{
  "scope": "PRACTICE",
  "kind": "SYMBOLIC",
  "prompt": "Combine like terms: simplify $3x + 5x$. (Use $x$.)",
  "answer": {
    "expr": "8*x",
    "vars": [
      "x"
    ]
  },
  "difficulty": 3,
  "hint": "Add the coefficients.",
  "explanation": "$3x + 5x = 8x$."
},
{
  "scope": "PRACTICE",
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  "prompt": "$3x$ and $3y$ are like terms and can be combined into $6xy$.",
  "answer": false,
  "hint": "Do they have the same variable part?",
  "explanation": "Different variables -- they are not like terms and cannot be combined."
},
{
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  "kind": "NUMERIC",
  "prompt": "A ticket costs \\$2 per person. Evaluate the cost $2p$ for a group of $p = 6$, in dollars.",
  "answer": {
    "value": 12,
    "tolerance": 0
  },
```

```
            "hint": "$2p$ with $p = 6$.",
            "explanation": "$2(6) = 12$, so \\$12."
          }
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              "sub": "An equation is a balance scale. Whatever you do to one side, do to the other, and the scale
stays level -- until the unknown stands alone."
            }
          },
          {
            "kind": "CONTEXT",
            "title": "The fountain puzzle",
            "content": {
              "markdown": "Carved on the plaza fountain: \"A number plus 7 equals 12. What is the number?\" You could
guess -- but algebra gives a method that never fails, even when the numbers get ugly. The trick is to think of the
equals sign as the pivot of a **balance scale**."
            }
          },
          {
            "kind": "CONCEPT",
            "title": "The big idea",
            "content": {
              "markdown": "An **equation** says two expressions are equal: $x + 7 = 12$. To **solve** it is to find
the value of the variable that makes the statement true.\n\nThe golden rule: **whatever you do to one side, do to the
other.** This keeps the scale balanced. To isolate the variable, undo what’s done to it using the **inverse
operation**:\n\n- Addition is undone by subtraction (and vice versa).\n- Multiplication is undone by division (and vice
versa).\n\nTo solve $x + 7 = 12$, subtract $7$ from **both** sides:\n\n$$x + 7 - 7 = 12 - 7 \\;\\Rightarrow\\; x =
5.$$\n\n**Check** by substituting back: $5 + 7 = 12$. [x] Always check -- it catches mistakes instantly."
            }
          },
          {
            "kind": "SIMULATION",
            "title": "Try it: the balance scale",
            "content": {
              "simId": "compare-balance",
              "intro": "Add or remove the same amount from both pans and watch the scale stay level. That is exactly
what solving an equation does."
            }
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                  "problem": "Solve $x + 7 = 12$.",
                  "steps": [
                    "Subtract 7 from both sides.",
                    "$x = 12 - 7$."
                  ],
                  "answer": "$x = 5$."
                },
                {
                  "title": "Undo multiplication",
                  "problem": "Solve $4x = 20$.",
                  "steps": [
                    "Divide both sides by 4.",
                    "$x = 20 \\div 4$."
                  ],
                  "answer": "$x = 5$."
                },
                {
                  "title": "Undo subtraction",
                  "problem": "Solve $x - 3 = 8$.",
                  "steps": [
                    "Add 3 to both sides.",
                    "$x = 8 + 3$."
                  ],
```

```
                    "answer": "$x = 11$."
                  }
                ]
              }
            },
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              "kind": "CONCEPT_CHECK",
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              }
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              }
            },
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              "content": {
                "markdown": "Equality is a statement that two quantities are *the same number*. If $a = b$, then adding
the same amount $c$ to each keeps them the same: $a + c = b + c$. The same holds for subtracting, multiplying, or
dividing by a nonzero number. These are the **properties of equality**, and they are the rigorous reason the
balance-scale picture always works -- not just a trick, but a theorem."
              }
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                  "An equation states two expressions are equal.",
                  "Solve by isolating the variable.",
                  "Use inverse operations: + undoes -, x undoes /.",
                  "Do the same thing to both sides to stay balanced.",
                  "Always check by substituting your answer back in."
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                    "tex": "a = b \\Rightarrow a + c = b + c"
                  }
                ]
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              "options": [
                "add 7 to both sides",
                "subtract 7 from both sides",
                "multiply both sides by 7",
                "divide both sides by 12"
              ],
              "answer": 1,
              "explanation": "Undo the $+7$ with $-7$ on both sides."
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              "prompt": "Solve $x + 7 = 12$. What is $x$?",
              "answer": {
                "value": 5,
                "tolerance": 0
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              "explanation": "$x = 12 - 7 = 5$."
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              "kind": "NUMERIC",
              "prompt": "Solve $4x = 20$. What is $x$?",
              "answer": {
                "value": 5,
                "tolerance": 0
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              "explanation": "Divide both sides by 4: $x = 5$."
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            "kind": "NUMERIC",
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            "answer": {
              "value": 11,
              "tolerance": 0
            },
            "hint": "Add 3 to both sides.",
            "explanation": "$x = 8 + 3 = 11$."
          },
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            "scope": "PRACTICE",
            "kind": "NUMERIC",
            "prompt": "Solve $\\dfrac{x}{2} = 9$. What is $x$?",
            "answer": {
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              "tolerance": 0
            },
            "hint": "Multiply both sides by 2.",
            "explanation": "$x = 9 \\times 2 = 18$."
          },
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            "options": [
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              "Divide both sides by 4.",
              "Simplify: $x = 5$.",
              "Check: $4(5) = 20$. [x]"
            ],
            "answer": [],
            "difficulty": 3,
            "hint": "Begin with the equation; finish by checking.",
            "explanation": "Divide both sides by 4 to isolate $x$, then verify the solution."
          },
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            "scope": "PRACTICE",
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            "prompt": "If $a = b$, then $a + 5 = b + 5$.",
            "answer": true,
            "hint": "Adding the same to equal things keeps them equal.",
            "explanation": "This is the addition property of equality."
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            "scope": "PRACTICE",
            "kind": "NUMERIC",
            "prompt": "Solve $x + 15 = 9$. What is $x$?",
            "answer": {
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              "tolerance": 0
            },
            "hint": "Subtract 15 from both sides.",
            "explanation": "$x = 9 - 15 = -6$."
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        "tagline": "Undo in reverse order",
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              "headline": "Peel It Back",
              "sub": "Two operations stand between you and the answer. Undo them in reverse -- like taking off shoes
before socks."
            }
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            "kind": "CONTEXT",
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            "content": {
              "markdown": "A taxi charges a \\$3 flat fee plus \\$2 per mile, and your ride cost \\$11. How many
miles? That’s the equation $2m + 3 = 11$. There are two operations wrapped around $m$ -- a multiply and an add -- so we
```

```
undo them one at a time, in the right order."
            }
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**reverse order** of operations:\n\n1. **First undo addition/subtraction** (the constant $b$).\n2. **Then undo
multiplication/division** (the coefficient $a$).\n\nFor $2m + 3 = 11$:\n\n$$2m + 3 - 3 = 11 - 3 \\;\\Rightarrow\\; 2m =
8 \\;\\Rightarrow\\; \\frac{2m}{2} = \\frac{8}{2} \\;\\Rightarrow\\; m = 4.$$\n\nWhy reverse order? When you *evaluate*
$2m + 3$ you multiply first, then add. To *undo*, you reverse the sequence: subtract first, then divide. Think of
unwrapping a gift -- last layer on goes first layer off."
            }
          },
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                  "problem": "Solve $2m + 3 = 11$.",
                  "steps": [
                    "Subtract 3: $2m = 8$.",
                    "Divide by 2: $m = 4$.",
                    "Check: $2(4) + 3 = 11$. [x]"
                  ],
                  "answer": "$m = 4$."
                },
                {
                  "title": "Add, then divide",
                  "problem": "Solve $3x - 5 = 16$.",
                  "steps": [
                    "Add 5: $3x = 21$.",
                    "Divide by 3: $x = 7$."
                  ],
                  "answer": "$x = 7$."
                },
                {
                  "title": "A negative coefficient",
                  "problem": "Solve $-2x + 1 = 9$.",
                  "steps": [
                    "Subtract 1: $-2x = 8$.",
                    "Divide by $-2$: $x = -4$."
                  ],
                  "answer": "$x = -4$."
                }
              ]
            }
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              "intro": "Constant first, coefficient second."
            }
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            "content": {
              "intro": "Two moves each: undo the add/subtract, then the multiply/divide."
            }
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            "content": {
              "markdown": "The same two-step logic rearranges **formulas**. To solve $y = mx + b$ for $x$, treat $y$,
$m$, $b$ as known: subtract $b$ from both sides to get $y - b = mx$, then divide by $m$ to get $x = \\dfrac{y - b}{m}$.
This is exactly how scientists rearrange equations to isolate whatever quantity they need -- the skill scales all the
way to physics and calculus."
            }
          },
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            "content": {
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                "A two-step equation looks like $ax + b = c$.",
                "Undo the constant first (add/subtract).",
```

"Undo the coefficient second (multiply/divide).",
"It’s the order of operations run in reverse.",
"Check your solution by substituting back."
],
"formulas": [
{
"label": "Solve a two-step",
"tex": "ax + b = c \\Rightarrow x = \\dfrac{c - b}{a}"
}
]
}
}
],
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"subtract 3 from both sides",
"add 3 to both sides",
"subtract 2x"
],
"answer": 1,
"explanation": "Undo the constant first: subtract 3."
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{
"scope": "CONCEPT_CHECK",
"kind": "NUMERIC",
"prompt": "Solve $2x + 3 = 11$. What is x?",
"answer": {
"value": 4,
"tolerance": 0
},
"explanation": "$2x = 8$, so $x = 4$."
},
{
"scope": "CONCEPT_CHECK",
"kind": "NUMERIC",
"prompt": "Solve $3x - 5 = 16$. What is x?",
"answer": {
"value": 7,
"tolerance": 0
},
"explanation": "$3x = 21$, so $x = 7$."
},
{
"scope": "PRACTICE",
"kind": "NUMERIC",
"prompt": "Solve $5x + 2 = 27$. What is x?",
"answer": {
"value": 5,
"tolerance": 0
},
"hint": "Subtract 2, then divide by 5.",
"explanation": "$5x = 25$, so $x = 5$."
},
{
"scope": "PRACTICE",
"kind": "NUMERIC",
"prompt": "A taxi charges \\$3 plus \\$2 per mile. If a ride costs \\$11, how many miles? Solve $2m + 3 = 11$.",
"answer": {
"value": 4,
"tolerance": 0
},
"hint": "Subtract the flat fee first.",
"explanation": "$2m = 8$, so $m = 4$ miles."
},
{
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"Add 5 to both sides: $3x = 21$.",
"Divide both sides by 3: $x = 7$.",
"Check: $3(7) - 5 = 16$. [x]"
],
"answer": [],
"difficulty": 4,

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              "hint": "Undo the subtraction before the multiplication.",
              "explanation": "Add 5, then divide by 3, then verify."
            },
            {
              "scope": "PRACTICE",
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              "prompt": "Solve $y = mx + b$ for $x$. Write $x$ in terms of $y$, $m$, and $b$. (Type like ‘(y - b)/m‘.)",
              "answer": {
                "expr": "(y - b)/m",
                "vars": [
                  "y",
                  "m",
                  "b"
                ]
              },
              "difficulty": 5,
              "hint": "Subtract $b$, then divide by $m$.",
              "explanation": "$x = \\dfrac{y - b}{m}$."
            },
            {
              "scope": "PRACTICE",
              "kind": "NUMERIC",
              "prompt": "Solve $-2x + 1 = 9$. What is $x$?",
              "answer": {
                "value": -4,
                "tolerance": 0
              },
              "hint": "Subtract 1, then divide by $-2$.",
              "explanation": "$-2x = 8$, so $x = -4$."
            }
          ]
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          "xpReward": 150,
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                "headline": "Three Ways to Say It",
                "sub": "A rule can live as a sentence, an expression, or a line on a graph. Fluency means moving freely
between all three."
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            },
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              "title": "The same idea, three languages",
              "content": {
                "markdown": "\"Each extra topping adds \\$1.50 to a \\$8 pizza.\" That sentence hides a rule. Written in
 symbols it’s $C = 1.5t + 8$. Drawn on a graph it’s a straight line. Real mathematical power comes from translating
between **words**, **symbols**, and **pictures** at will."
              }
            },
            {
              "kind": "CONCEPT",
              "title": "The big idea",
              "content": {
                "markdown": "**Words -> symbols.** Watch the keywords:\n\n- ’sum’, ’more than’, ’increased by’ -> $+$\n-
 ’difference’, ’less than’, ’decreased by’ -> $-$\n- ’product’, ’times’, ’of’ -> $\\times$\n- ’quotient’, ’per’,
’divided by’ -> $\\div$\n\n’Three less than twice a number $x$’ is $2x - 3$ (note the order: subtract from
$2x$).\n\n**Symbols -> graph.** A linear expression $mx + b$, drawn as $y = mx + b$, is a straight line:\n\n- $b$ is the
 **$y$-intercept** -- where the line crosses the vertical axis ($x = 0$).\n- $m$ is the **slope** -- how much $y$
changes for each step of $1$ in $x$.\n\nFor $y = 2x - 1$: start at $-1$ on the $y$-axis, then rise $2$ for every $1$
across. A positive slope rises left-to-right; a negative slope falls."
              }
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                "intro": "Plot values and see how adding or scaling shifts and stretches them along the line -- the
one-dimensional shadow of a graph."
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                  "’Three less than’ subtracts 3 from it."
                ],
                "answer": "$2x - 3$."
              },
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                "title": "Read a line",
                "problem": "For $y = 2x - 1$, give the slope and $y$-intercept.",
                "steps": [
                  "Compare to $y = mx + b$.",
                  "$m = 2$, $b = -1$."
                ],
                "answer": "Slope $2$, intercept $-1$."
              },
              {
                "title": "Build the cost rule",
                "problem": "A \\$8 pizza plus \\$1.50 per topping $t$. Write the cost $C$.",
                "steps": [
                  "Per-topping cost: $1.5t$.",
                  "Add the base \\$8."
                ],
                "answer": "$C = 1.5t + 8$."
              }
            ]
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        },
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          "content": {
            "intro": "Translate and read carefully -- order matters."
          }
        },
        {
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          "title": "Practice problems",
          "content": {
            "intro": "Move between words, symbols, and the graph."
          }
        },
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          "title": "Deeper dive: slope as a rate",
          "content": {
            "markdown": "Slope is really a **rate of change** -- the same idea as a unit rate or a speed. In $C =
1.5t + 8$, the slope $1.5$ is ’dollars per topping’. In $d = 60t$, the slope $60$ is ’miles per hour’. Later, in
calculus, the slope of a curve at a single point becomes the **derivative** -- but it is the very same idea you’re using
 now: how fast one quantity changes as another moves."
          }
        },
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              "Translate keywords into $+$, $-$, $\\times$, $\\div$.",
              "’Less than’ and ’subtracted from’ reverse the order.",
              "$y = mx + b$ is a line: $m$ is slope, $b$ is the $y$-intercept.",
              "Slope is the change in $y$ per step of 1 in $x$.",
              "Slope is a rate of change -- the seed of the derivative."
            ],
            "formulas": [
              {
                "label": "Slope-intercept line",
                "tex": "y = mx + b"
              }
            ]
          }
        }
      ],
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          "kind": "SYMBOLIC",
          "prompt": "Write ’three less than twice a number $x$’. (Use $x$.)",
```

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      "answer": {
        "expr": "2*x - 3",
        "vars": [
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        ]
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      "difficulty": 3,
      "hint": "Twice the number is $2x$; subtract 3 from it.",
      "explanation": "’Three less than $2x$’ is $2x - 3$."
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      "kind": "MCQ",
      "prompt": "For $y = 2x - 1$, the $y$-intercept is:",
      "options": [
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        "$-1$",
        "$1$",
        "$0$"
      ],
      "answer": 1,
      "explanation": "In $y = mx + b$, the intercept is $b = -1$."
    },
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      "kind": "NUMERIC",
      "prompt": "For $y = 2x - 1$, what is the slope?",
      "answer": {
        "value": 2,
        "tolerance": 0
      },
      "explanation": "The slope is the coefficient of $x$: $2$."
    },
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      "kind": "SYMBOLIC",
      "prompt": "Write ’the sum of a number $n$ and 4, all doubled’. (Use $n$; type like ‘2(n + 4)‘.)",
      "answer": {
        "expr": "2*(n + 4)",
        "vars": [
          "n"
        ]
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      "difficulty": 4,
      "hint": "Double the whole sum.",
      "explanation": "$2(n + 4)$, which expands to $2n + 8$."
    },
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      "scope": "PRACTICE",
      "kind": "GRAPH",
      "prompt": "Graph the line $y = 2x - 1$. Enter $y$ as a function of $x$.",
      "answer": {
        "expr": "2*x - 1",
        "domain": [
          -5,
          5
        ],
        "variable": "x"
      },
      "difficulty": 3,
      "hint": "Slope 2, intercept -1.",
      "explanation": "$y = 2x - 1$ -- a line through $(0,-1)$ rising 2 for each step right."
    },
    {
      "scope": "PRACTICE",
      "kind": "GRAPH",
      "prompt": "A \\$8 pizza costs \\$1.50 per topping $x$. Graph the total cost. Enter cost as a function of $x$.",
      "answer": {
        "expr": "1.5*x + 8",
        "domain": [
          0,
          10
        ],
        "variable": "x"
      },
      "difficulty": 4,
      "hint": "Base plus per-topping rate.",
      "explanation": "$C = 1.5x + 8$ -- intercept 8 (the base price), slope 1.5 (per topping)."
    },
    {
      "scope": "PRACTICE",
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                "is horizontal",
                "is vertical"
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              "answer": 1,
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              "explanation": "Negative slope falls from left to right."
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              "prompt": "Simplify $2(n + 4)$ by expanding. (Use $n$.)",
              "answer": {
                "expr": "2*n + 8",
                "vars": [
                  "n"
                ]
              },
              "difficulty": 3,
              "hint": "Distribute the 2.",
              "explanation": "$2(n + 4) = 2n + 8$."
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      "name": "The Architect’s Quarter",
      "subtitle": "Lines, angles, and area",
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                "sub": "Every building begins with angles. Learn the rules that govern how lines cross and corners
turn."
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              "kind": "CONTEXT",
              "title": "At the drafting table",
              "content": {
                "markdown": "An architect’s blueprint is a web of lines meeting at precise angles. Get one angle wrong
and walls won’t close, roofs won’t seal. A handful of simple angle rules let you find every missing measure on a drawing
 without a protractor."
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                "markdown": "An **angle** measures the turn between two rays from a shared vertex, in **degrees**
($^\\circ$).\n\n- **Acute**: less than $90^\\circ$. **Right**: exactly $90^\\circ$. **Obtuse**: between $90^\\circ$ and
$180^\\circ$. **Straight**: $180^\\circ$.\n\nThe key relationships:\n\n- **Complementary** angles sum to $90^\\circ$.\n-
 **Supplementary** angles sum to $180^\\circ$.\n- Angles on a **straight line** sum to $180^\\circ$; angles **around a
point** sum to $360^\\circ$.\n- When two lines cross, the opposite (**vertical**) angles are **equal**.\n\nThese let you
 chase missing angles: if one angle on a line is $130^\\circ$, the angle beside it is $180^\\circ - 130^\\circ =
50^\\circ$."
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angle size you’ll use throughout geometry."
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                      "$90 - 32 = 58$."
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                    "answer": "$58^\\circ$."
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                      "$180 - 130 = 50$."
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                    "answer": "$50^\\circ$."
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                    "problem": "Three angles around a point are $90^\\circ$, $120^\\circ$, and $x$. Find $x$.",
                    "steps": [
                      "They sum to $360^\\circ$.",
                      "$x = 360 - 90 - 120 = 150$."
                    ],
                    "answer": "$x = 150^\\circ$."
                  }
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              "title": "Practice problems",
              "content": {
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              }
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              "content": {
                "markdown": "When a line (a **transversal**) crosses two **parallel** lines, it creates matched angle
pairs: **corresponding** angles are equal, **alternate interior** angles are equal, and **co-interior** angles are
supplementary. These facts are the engine of geometric proof -- they let you transfer an angle from one line to another
and conclude, for example, that the three angles of any triangle sum to exactly $180^\\circ$."
              }
            },
            {
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                  "Angles are measured in degrees; acute < 90 < obtuse < 180 = straight.",
                  "Complementary angles sum to 90 deg.",
                  "Supplementary angles sum to 180 deg.",
                  "Angles on a line sum to 180 deg; around a point, 360 deg.",
                  "Vertical (opposite) angles are equal."
                ],
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      "obtuse",
      "straight"
    ],
    "answer": 2,
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    "explanation": "$90 - 32 = 58$."
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    "hint": "They sum to 180 deg.",
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    "hint": "They sum to 360 deg.",
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    "hint": "Think of an X.",
    "explanation": "Vertical angles are always equal."
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      "vars": [
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      ]
    },
    "difficulty": 3,
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              "hint": "Supplements sum to 180.",
              "explanation": "The supplement of $a$ is $180 - a$."
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              "options": [
                "$a + b = 180^\\circ$ (angles on a straight line).",
                "$b + c = 180^\\circ$ (angles on a straight line).",
                "So $a + b = b + c$ (both equal $180^\\circ$).",
                "Subtract $b$ from both sides: $a = c$."
              ],
              "answer": [],
              "difficulty": 4,
              "hint": "Use two straight-line sums, then cancel the shared angle.",
              "explanation": "Both pairs sum to 180 deg, so $a + b = b + c$; cancelling $b$ gives $a = c$."
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                "markdown": "An architect pricing a plot needs two numbers: the **perimeter** (how much fencing or trim to buy) and the **area** (how much floor or paint to cover). They measure different things -- and confusing them is one of the most common -- and expensive -- mistakes."
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                    "steps": [
                      "Area $= l w = 5 \\times 3 = 15$.",
                      "Perimeter $= 2(5+3) = 16$."
                    ],
                    "answer": "Area $15\\,\\text{m}^2$, perimeter $16\\,\\text{m}$."
                  },
                  {
                    "title": "Triangle",
                    "problem": "A triangle has base $8$ and height $5$. Find its area.",
                    "steps": [
                      "$A = \\tfrac12 b h$.",
                      "$A = \\tfrac12 (8)(5) = 20$."
                    ],
                    "answer": "Area $= 20$ square units."
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                  "$A = s^2$.",
                  "Replace $s$ with $2s$: $(2s)^2 = 4s^2$."
                ],
                "answer": "The area becomes four times larger."
              }
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$180^\\circ$, and the two together tile a parallelogram of base $b$ and height $h$, whose area is $bh$. So one triangle
is $\\tfrac12 b h$. This ’double it, then halve’ trick -- turning an unfamiliar shape into a familiar one -- is a
recurring strategy in geometry and, later, in integration when we find the area under a curve."
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              "Rectangle: $P = 2(l+w)$, $A = lw$.",
              "Triangle: $A = \\tfrac12 bh$.",
              "Circle: $C = 2\\pi r$, $A = \\pi r^2$.",
              "Area uses square units; doubling a side quadruples a square’s area."
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              },
              {
                "label": "Triangle area",
                "tex": "A = \\tfrac12 b h"
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              {
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                "tex": "A = \\pi r^2"
              }
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            "tolerance": 0
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          "explanation": "$A = 5 \\times 3 = 15\\,\\text{m}^2$."
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          "explanation": "$P = 2(5+3) = 16\\,\\text{m}$."
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            "length units (m)",
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            "no units",
            "degrees"
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          "answer": 1,
          "explanation": "Area is a surface, so it uses square units."
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          "answer": {
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            "tolerance": 0
          },
          "hint": "$A = \\tfrac12 bh$.",
          "explanation": "$\\tfrac12(8)(5) = 20$."
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          "answer": {
            "expr": "s^2",
            "vars": [
              "s"
            ]
          },
          "difficulty": 2,
          "hint": "Side times side.",
          "explanation": "$A = s \\times s = s^2$."
        },
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          "scope": "PRACTICE",
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          "prompt": "Graph the area of a square as a function of its side length $s$. Enter area as a function of $s$ (use $x$ for $s$).",
          "answer": {
            "expr": "x^2",
            "domain": [
              0,
              6
            ],
            "variable": "x"
          },
          "difficulty": 3,
          "hint": "$A = s^2$.",
          "explanation": "$A = s^2$ -- a parabola: area grows with the square of the side."
        },
        {
          "scope": "PRACTICE",
          "kind": "MCQ",
          "prompt": "If a square’s side doubles, its area becomes:",
          "options": [
            "doubled",
            "tripled",
            "four times larger",
            "unchanged"
          ],
          "answer": 2,
          "hint": "$A = s^2$.",
          "explanation": "$(2s)^2 = 4s^2$, four times larger."
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            "tolerance": 1
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          "hint": "$A = \\pi r^2$.",
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              "headline": "a^2 + b^2 = c^2",
              "sub": "The most famous equation in geometry -- and the secret behind every square corner ever built."
            }
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            "content": {
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                  "steps": [
                    "$c^2 = 3^2 + 4^2 = 9 + 16 = 25$.",
                    "$c = \\sqrt{25}$."
                  ],
                  "answer": "$c = 5$."
                },
                {
                  "title": "Find a leg",
                  "problem": "Hypotenuse $13$, one leg $5$. Find the other leg.",
                  "steps": [
                    "$a^2 = 13^2 - 5^2 = 169 - 25 = 144$.",
                    "$a = \\sqrt{144}$."
                  ],
                  "answer": "$a = 12$."
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                  "problem": "Do sides $6, 8, 10$ form a right triangle?",
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                    "Check $6^2 + 8^2 = 36 + 64 = 100$.",
                    "$10^2 = 100$. They match."
                  ],
                  "answer": "Yes -- it’s a right triangle."
                }
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 and the straight-line distance between them is the hypotenuse of a right triangle with those legs. So $d =
\\sqrt{\\Delta x^2 + \\Delta y^2}$ -- the **distance formula** is just the Pythagorean theorem in coordinates. Extend it
 to three dimensions and you get $d = \\sqrt{\\Delta x^2 + \\Delta y^2 + \\Delta z^2}$, the length of a vector -- a tool
 you’ll use constantly in physics."
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                  "The hypotenuse $c$ is opposite the right angle and is longest.",
                  "$a^2 + b^2 = c^2$.",
                  "Hypotenuse: $c = \\sqrt{a^2 + b^2}$; leg: $a = \\sqrt{c^2 - b^2}$.",
                  "The distance formula is this theorem in coordinates."
                ],
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                    "label": "Pythagorean theorem",
                    "tex": "a^2 + b^2 = c^2"
                  },
                  {
                    "label": "Hypotenuse",
                    "tex": "c = \\sqrt{a^2 + b^2}"
                  }
                ]
              }
            }
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                "opposite the right angle",
                "always vertical",
                "one of the legs"
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              "answer": 1,
              "explanation": "The hypotenuse is opposite the right angle and is the longest side."
            },
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                "tolerance": 0
              },
              "explanation": "$c = \\sqrt{9+16} = 5$."
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          "explanation": "$\\sqrt{169 - 25} = \\sqrt{144} = 12$."
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            "vars": [
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              "b"
            ]
          },
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          "hint": "Square root of the sum of the squared legs.",
          "explanation": "$c = \\sqrt{a^2 + b^2}$."
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            "only if scaled",
            "cannot tell"
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          "hint": "Check $6^2 + 8^2$ vs $10^2$.",
          "explanation": "$36 + 64 = 100 = 10^2$, so yes."
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          "options": [
            "Apply the theorem: $a^2 + b^2 = c^2$.",
            "The hypotenuse squared is the sum of the squared legs.",
            "Take the positive square root of both sides.",
            "Conclude $c = \\sqrt{a^2 + b^2}$."
          ],
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            "tolerance": 0
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          "hint": "$d = \\sqrt{\\Delta x^2 + \\Delta y^2}$.",
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single answer -- there are three useful ones: the **mean**, the **median**, and the **mode**. Each tells a slightly
different story, and knowing which to trust is the heart of statistics."
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            "markdown": "Three **measures of center** summarize a data set:\n\n- **Mean** (average): add all values,
 divide by how many. $\\bar x = \\dfrac{\\sum x}{n}$.\n- **Median**: the middle value when the data is sorted (the
average of the two middle values if $n$ is even).\n- **Mode**: the value that appears most often (there may be none or
several).\n\nFor $2, 3, 3, 6, 11$:\n\n- Mean $= \\dfrac{2+3+3+6+11}{5} = \\dfrac{25}{5} = 5$.\n- Median $= 3$ (the
middle of five sorted values).\n- Mode $= 3$ (it appears twice).\n\nThe **mean** is pulled by extreme values
(**outliers**); the **median** resists them. A single huge salary drags the mean up while the median barely moves --
which is why incomes are usually reported as medians."
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                  "Middle of five is the 3rd."
                ],
                "answer": "Median $= 5$."
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              {
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                "steps": [
                  "A few huge incomes inflate the mean.",
                  "The median ignores their size."
                ],
                "answer": "The median resists outliers."
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is ideal -- it uses every value. For **skewed** data or data with outliers (incomes, house prices), the **median** is
more honest. The **mode** is the only center that works for **categorical** data -- you can’t average ’favourite
colour’, but you can name the most common one."
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                  "Median = middle value of sorted data.",
                  "Mode = most frequent value.",
                  "The mean is sensitive to outliers; the median resists them.",
                  "Use the mode for categorical data."
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                  }
                ]
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              "explanation": "$24/4 = 6$."
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              "explanation": "Sorted: $2,4,5,7,9$ -- middle is 5."
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              "explanation": "$60/3 = 20$."
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              "explanation": "Sorted: $1,1,3,4,5,9$; middle two are 3 and 4, mean $3.5$."
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              "mode",
              "range"
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            "explanation": "The mode (most common) is the only center for categorical data."
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            "explanation": "$10 \\times 4 = 40$."
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                    "$11 - 3 = 8$."
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                ],
                "answer": "Range $= 8$."
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                "steps": [
                  "Both have mean 5.",
                  "A has range 0; B has range 8."
                ],
                "answer": "Set B is far more spread out."
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                  "Range uses only the extremes.",
                  "IQR uses the middle half, ignoring outliers."
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                "answer": "IQR resists outliers."
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typical distance of a value from the mean. You compute it by averaging the squared distances from the mean (the
**variance**) and taking the square root. Squaring keeps everything positive and weights big deviations more heavily.
Standard deviation is the backbone of the bell curve, opinion-poll margins of error, and quality control in
engineering."
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              "Range = max - min (uses only extremes).",
              "Quartiles split sorted data into quarters; IQR = Q_3 - Q_1.",
              "IQR resists outliers; range does not.",
              "Center + spread together describe data far better than either alone."
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"explanation": "$11 - 3 = 8$."
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"sum / count"
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"hint": "Max minus min.",
"explanation": "$20 - 4 = 16$."
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"maximum",
"mode"
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"explanation": "The IQR uses the middle 50%, so outliers don’t affect it."
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"explanation": "No spread at all, so range $= 0$."
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"explanation": "$5 + 30 = 35$."
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"vars": [
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"m"
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              }
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                "markdown": "Every bet, insurance policy, and weather forecast rests on **probability** -- a precise number for how likely an event is. Get the probability right and you price risk correctly; get it wrong and you lose. The basics are simple counting, but they power a vast field."
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              "content": {
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                      "$3/6$."
                    ],
                    "answer": "$P = \\tfrac12 = 0.5$."
                  },
                  {
                    "title": "Complement",
                    "problem": "If $P(\\text{rain}) = 0.3$, find $P(\\text{no rain})$.",
                    "steps": [
                      "Use $1 - P$.",
                      "$1 - 0.3 = 0.7$."
                    ],
                    "answer": "$0.7$."
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                  {
                    "title": "Two coins",
                    "problem": "Find $P(\\text{two heads})$ tossing two fair coins.",
                    "steps": [
                      "Independent: multiply.",
                      "$\\tfrac12 \\times \\tfrac12$."
                    ],
                    "answer": "$\\tfrac14 = 0.25$."
                  }
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$P(\\text{heads}) = 0.5$, but ten tosses might give seven heads. The **law of large numbers** says that as the number of
 trials grows, the observed fraction of heads converges to $0.5$. This is why casinos and insurers -- who run millions
of trials -- can profit reliably from probabilities, even though any single outcome is uncertain."
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                  "Equally likely: favorable / total.",
                  "Complement: $P(\\text{not }A) = 1 - P(A)$.",
                  "Independent events multiply: $P(A\\text{ and }B) = P(A)P(B)$.",
                  "Probability describes the long run, not a single trial."
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              "explanation": "1 of 2 outcomes: $0.5$."
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              "explanation": "$3/6 = 0.5$."
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              "options": [
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                "0.5",
                "1",
                "1.5"
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              "answer": 3,
              "explanation": "Probabilities lie between 0 and 1."
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            "tolerance": 0.001
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          "explanation": "$0.5 \\times 0.5 = 0.25$."
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            "vars": [
              "P"
            ]
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          "hint": "Complement rule.",
          "explanation": "$P(\\text{not }A) = 1 - P(A)$."
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          "answer": {
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            "tolerance": 0.001
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          "hint": "Reds over total.",
          "explanation": "$2/5 = 0.4$."
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out a bolt; put in $3$, get out $7$. A **function** is the mathematical version -- a rule that assigns to each input
**exactly one** output. That predictability is what makes functions the language of science and engineering."
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          {
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1,$$\n\nread ’$f$ of $x$ equals $2x+1$’. Here $x$ is the **input**; $f(x)$ is the **output**.\n\n- To **evaluate**,
substitute: $f(3) = 2(3) + 1 = 7$.\n- The **domain** is the set of allowed inputs; the **range** is the set of outputs
produced.\n- The defining rule: **one input -> one output**. A relation that sends one input to two different outputs is
 *not* a function.\n\n**Vertical line test:** on a graph, if any vertical line crosses the curve more than once, it
fails -- that input would have two outputs, so it isn’t a function."
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                  "steps": [
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                    "$2(3) + 1 = 7$."
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                  "answer": "$f(3) = 7$."
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                    "$4 - 1 = 3$."
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and $g(x) = x + 1$, then $f(g(x)) = f(x+1) = 2(x+1) = 2x + 2$. This **composition**, written $(f \\circ g)(x)$, is how
complex processes are built from simple steps -- and reversing a function gives its **inverse**, the idea behind
logarithms and square roots."
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          "$f(x)$ notation: input $x$, output $f(x)$.",
          "Evaluate by substituting the input.",
          "Domain = allowed inputs; range = outputs produced.",
          "Vertical line test: one input can’t have two outputs."
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          }
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      "explanation": "$5 - 3(2) = 5 - 6 = -1$."
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        "is a function",
        "has positive slope"
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      "explanation": "If a vertical line hits the curve twice, one input has two outputs -- not a function."
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            "hint": "$9 + 6$.",
            "explanation": "$3^2 + 2(3) = 9 + 6 = 15$."
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            "explanation": "Domain = inputs; range = outputs."
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slope is its rate."
            }
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          {
            "kind": "CONTEXT",
            "title": "Steady machines",
            "content": {
              "markdown": "A conveyor that adds the same number of parts each minute, a tank that drains at a fixed
rate -- these are **linear** processes. Their defining feature is a **constant rate of change**, which shows up as a
straight-line graph. Linear functions are the simplest and most common model in all of engineering."
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line.\n\n- $m$ is the **slope** -- the change in output per unit change in input: $m = \\dfrac{\\Delta y}{\\Delta x} =
\\dfrac{\\text{rise}}{\\text{run}}$.\n- $b$ is the **$y$-intercept** -- the output when $x = 0$, where the line crosses
the vertical axis.\n\nSlope between two points $(x_1, y_1)$ and $(x_2, y_2)$:\n\n$$m = \\frac{y_2 - y_1}{x_2 -
x_1}.$$\n\nA positive slope rises, a negative slope falls, a zero slope is horizontal. The bigger $|m|$, the steeper the
 line."
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                    "$= \\dfrac{6}{2} = 3$."
                  ],
                  "answer": "Slope $m = 3$."
```

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                "title": "Read the line",
                "problem": "Give the slope and intercept of $f(x) = -2x + 5$.",
                "steps": [
                  "Compare to $mx + b$.",
                  "$m = -2$, $b = 5$."
                ],
                "answer": "Slope $-2$, intercept $5$."
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                "title": "Evaluate",
                "problem": "For $f(x) = 3x - 4$, find $f(2)$.",
                "steps": [
                  "$3(2) - 4$.",
                  "$6 - 4 = 2$."
                ],
                "answer": "$f(2) = 2$."
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              "Linear functions $f(x) = mx + b$ graph as straight lines.",
              "Slope $m$ = rise/run = change in output per unit input.",
              "$b$ is the $y$-intercept (output at $x = 0$).",
              "Slope between points: $(y_2-y_1)/(x_2-x_1)$.",
              "Parallel lines share a slope; perpendicular slopes are negative reciprocals."
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              },
              {
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                "tex": "f(x) = mx + b"
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      "falling"
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    "explanation": "$3(2) - 4 = 2$."
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      "domain": [
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      "variable": "x"
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      "$3$",
      "$-\\tfrac13$"
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    "answer": 2,
    "hint": "Parallel lines share a slope.",
    "explanation": "Same slope, $3$."
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        6
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      "variable": "x"
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    "difficulty": 3,
    "hint": "Slope ?, intercept 0.",
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cable bridge."
            }
          },
          {
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            "title": "Arcs in the skyline",
            "content": {
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all trace the same curve, a **parabola**. It’s the graph of a **quadratic function**, the next step up from linear, and
the first time a graph bends."
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 graph is a **parabola**.\n\n- If $a > 0$ the parabola opens **upward** (a valley); if $a < 0$ it opens **downward** (a
hill).\n- The turning point is the **vertex** -- the minimum or maximum.\n- The parabola is **symmetric** about the
vertical line through its vertex.\n- The simplest case $f(x) = x^2$ has its vertex at the origin and opens
upward.\n\nAdding a constant shifts it vertically: $x^2 - 4$ is $x^2$ slid down 4, crossing the $x$-axis at $x = \\pm 2$
 (its **roots**, where $f(x) = 0$)."
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                    "$9 - 4 = 5$."
                  ],
                  "answer": "$f(3) = 5$."
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                  "problem": "Where does $f(x) = x^2 - 9$ equal zero?",
                  "steps": [
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                    "$x^2 = 9$, so $x = \\pm 3$."
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                    "$x^2 + 2x + x + 2$."
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**vertex**. This **vertex form** makes the graph’s position obvious: $h$ shifts it horizontally, $k$ vertically. Getting
 there from $ax^2 + bx + c$ is called **completing the square** -- the same algebraic move that produces the quadratic
formula $x = \\dfrac{-b \\pm \\sqrt{b^2 - 4ac}}{2a}$ for the roots."
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                  "$a > 0$ opens up (min); $a < 0$ opens down (max).",
                  "The vertex is the turning point; the parabola is symmetric about it.",
                  "Roots are where $f(x) = 0$ (x-axis crossings).",
                  "Vertex form $a(x-h)^2 + k$ reveals the vertex $(h,k)$."
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                  {
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                    "tex": "x = \\dfrac{-b \\pm \\sqrt{b^2 - 4ac}}{2a}"
                  }
                ]
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              "explanation": "Quadratics graph as parabolas."
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              "kind": "MCQ",
              "prompt": "If $a < 0$ in $ax^2 + bx + c$, the parabola opens:",
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                "it doesn’t"
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```
          ],
          "answer": 1,
          "explanation": "Negative leading coefficient opens downward."
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            "vars": [
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          "hint": "Multiply each pair of terms.",
          "explanation": "$(x+1)(x+2) = x^2 + 3x + 2$."
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            "domain": [
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              3
            ],
            "variable": "x"
          },
          "difficulty": 4,
          "hint": "$x^2$ shifted down 4.",
          "explanation": "A valley parabola with vertex $(0,-4)$, crossing the axis at $x = \\pm 2$."
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          "kind": "MCQ",
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            "$x = 3$ and $x = -3$",
            "$x = 0$",
            "no real roots"
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          "answer": 1,
          "hint": "Solve $x^2 = 9$.",
          "explanation": "$x = \\pm 3$."
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          "explanation": "Vertex at the origin, opening upward."
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            "tolerance": 0
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          "hint": "$2(4) + 1$.",
          "explanation": "$2(2)^2 + 1 = 8 + 1 = 9$."
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unreachable."
              }
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              "content": {
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 up to its top, and trigonometry does the rest. The **sine, cosine, and tangent** ratios turn an angle into a length --
the surveyor’s, astronomer’s, and engineer’s most basic tool."
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**adjacent**, and the **hypotenuse**. The three trig ratios are:\n\n$$\\sin\\theta = \\frac{\\text{opp}}{\\text{hyp}},
\\qquad \\cos\\theta = \\frac{\\text{adj}}{\\text{hyp}}, \\qquad \\tan\\theta =
\\frac{\\text{opp}}{\\text{adj}}.$$\n\nRemember **SOH-CAH-TOA**. Since $\\tan\\theta = \\dfrac{\\text{opp}}{\\text{adj}}
 = \\dfrac{\\sin\\theta}{\\cos\\theta}$, the three are linked.\n\nKey reference values:\n\n- $\\sin 30^\\circ =
\\tfrac12$, $\\cos 60^\\circ = \\tfrac12$.\n- $\\sin 45^\\circ = \\cos 45^\\circ = \\tfrac{\\sqrt2}{2} \\approx
0.707$.\n- $\\tan 45^\\circ = 1$.\n\nGiven an angle and one side, these ratios find the others; given two sides, the
**inverse** functions ($\\sin^{-1}$, etc.) recover the angle."
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                    "steps": [
                      "$\\sin 30^\\circ = \\text{opp}/10$.",
                      "opp $= 10 \\times \\tfrac12$."
                    ],
                    "answer": "Opposite $= 5$."
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                    "problem": "If opp $= 3$ and adj $= 4$, find $\\tan\\theta$.",
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                      "$= 3/4$."
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                    "problem": "You know the adjacent side and want the hypotenuse. Which ratio?",
                    "steps": [
                      "Adjacent and hypotenuse appear together in cosine.",
                      "$\\cos\\theta = \\text{adj}/\\text{hyp}$."
                    ],
                    "answer": "Use cosine."
                  }
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triangle. The **law of sines**, $\\dfrac{a}{\\sin A} = \\dfrac{b}{\\sin B} = \\dfrac{c}{\\sin C}$, links each side to
its opposite angle. The **law of cosines**, $c^2 = a^2 + b^2 - 2ab\\cos C$, is the Pythagorean theorem with a correction
 term for the angle -- it reduces to $a^2 + b^2 = c^2$ exactly when $C = 90^\\circ$ (since $\\cos 90^\\circ = 0$)."
        }
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            "Label sides as opposite, adjacent, hypotenuse relative to the angle.",
            "SOH-CAH-TOA: sin = opp/hyp, cos = adj/hyp, tan = opp/adj.",
            "$\\tan\\theta = \\sin\\theta/\\cos\\theta$.",
            "Know the 30-45-60 reference values.",
            "Inverse trig functions recover an angle from two sides."
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            {
              "label": "Tangent",
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            }
          ]
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          "opp/hyp",
          "opp/adj",
          "hyp/opp"
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        "explanation": "SOH: sine = opposite / hypotenuse."
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        "explanation": "$10 \\times 0.5 = 5$."
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```
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            "tangent",
            "Pythagoras only"
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          "explanation": "$\\tan 45^\\circ = 1$."
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            "vars": [
              "x"
            ]
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          "difficulty": 4,
          "hint": "Tangent is a ratio of the other two.",
          "explanation": "$\\tan x = \\dfrac{\\sin x}{\\cos x}$."
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60^\\circ = 0.5$).",
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            "tolerance": 0
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          "hint": "$\\cos = $ adj/hyp.",
          "explanation": "$20 \\times 0.5 = 10$."
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          "prompt": "$\\sin 45^\\circ$ equals $\\cos 45^\\circ$.",
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          "explanation": "Both equal $\\tfrac{\\sqrt2}{2} \\approx 0.707$."
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        },
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that. The **unit circle** extends sine and cosine to *any* angle, and **radians** give angles a natural measure tied
directly to arc length, the unit calculus and physics demand."
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positive $x$-axis, the point on the circle is\n\n$$(\\cos\\theta, \\sin\\theta).$$\n\nSo **cosine is the $x$-coordinate
and sine is the $y$-coordinate** -- definitions that work for *any* angle, including those past $90^\\circ$ and negative
 ones.\n\n**Radians** measure an angle by the **arc length** it subtends on the unit circle. A full circle is $2\\pi$
radians $= 360^\\circ$, so:\n\n$$180^\\circ = \\pi \\text{ rad}, \\qquad 90^\\circ = \\tfrac{\\pi}{2}, \\qquad 1\\text{
rad} \\approx 57.3^\\circ.$$\n\nConvert with $\\text{rad} = \\text{deg} \\times \\dfrac{\\pi}{180}$. Key values: $\\cos
0 = 1$, $\\sin 0 = 0$, $\\sin\\tfrac{\\pi}{2} = 1$, $\\cos\\pi = -1$."
              }
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                      "$90\\pi/180 = \\pi/2$."
                    ],
                    "answer": "$\\tfrac{\\pi}{2}$ rad."
                  },
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                      "$(1, 0)$."
                    ],
                    "answer": "$(1, 0)$."
                  },
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                    "steps": [
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                      "There $y < 0$."
                    ],
                    "answer": "Negative."
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                ]
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            },
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              "title": "Deeper dive: why radians make calculus clean",
              "content": {
                "markdown": "Radians aren’t just convenient -- they’re *necessary* for calculus. Only in radians is
$\\lim_{x \\to 0} \\dfrac{\\sin x}{x} = 1$, which makes the derivative of $\\sin x$ exactly $\\cos x$. In degrees that
derivative picks up an ugly factor of $\\pi/180$. Because radians tie angle directly to arc length, every rotational
formula in physics -- angular velocity, torque, wave phase -- is written in radians."
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            },
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                    "Cosine is the x-coordinate; sine is the y-coordinate.",
                    "This extends sine and cosine to any angle.",
                    "A full turn is 2pi radians = 360 deg; pi rad = 180 deg.",
                    "Convert with rad = deg x pi/180."
                  ],
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                    {
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                      "tex": "(\\cos\\theta, \\sin\\theta)"
                    },
                    {
                      "label": "Degrees to radians",
                      "tex": "\\text{rad} = \\text{deg}\\times\\dfrac{\\pi}{180}"
                    }
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                  "$\\cos\\theta$",
                  "$\\tan\\theta$",
                  "$\\theta$"
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              },
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                  "$2\\pi$",
                  "$360$",
                  "$\\tfrac{\\pi}{2}$"
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                "answer": 1,
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                "explanation": "At 90 deg the $y$-coordinate is 1."
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type $\\pi \\approx 3.1416$).",
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                "hint": "$180^\\circ = \\pi$ rad.",
                "explanation": "$180^\\circ = \\pi \\approx 3.1416$ rad."
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                "kind": "NUMERIC",
                "prompt": "What is $\\cos\\pi$?",
                "answer": {
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                "hint": "Point at 180 deg is $(-1, 0)$.",
                "explanation": "$\\cos\\pi = -1$."
              },
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              "answer": true,
              "hint": "Below the x-axis, y is negative.",
              "explanation": "In the lower half of the circle the $y$-coordinate (sine) is negative."
            },
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              "answer": {
                "value": 1.571,
                "tolerance": 0.01
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              "hint": "$90 \\times \\pi/180$.",
              "explanation": "$\\pi/2 \\approx 1.571$ rad."
            },
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              "scope": "PRACTICE",
              "kind": "NUMERIC",
              "prompt": "What is $\\cos 0$?",
              "answer": {
                "value": 1,
                "tolerance": 0.001
              },
              "hint": "Point at angle 0 is $(1,0)$.",
              "explanation": "$\\cos 0 = 1$."
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current."
              }
            },
            {
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              "content": {
                "markdown": "Watch the water below the observatory bridge rise and fall in a smooth, repeating swell. 
Plot a point’s height over time and you get the **sine curve** -- nature’s signature for anything that repeats: 
heartbeats, radio, AC power, the seasons."
              }
            },
            {
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              "content": {
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the $x$-coordinate traces $y = \\cos\\theta$.\n\nBoth are **periodic** with period $2\\pi$ -- they repeat every full 
turn -- and oscillate between $-1$ and $1$.\n\n- **Amplitude**: the peak height. For $y = A\\sin x$ it is $|A|$.\n- 
**Period**: the length of one cycle. For $y = \\sin(Bx)$ it is $\\dfrac{2\\pi}{B}$ -- a larger $B$ squeezes the wave.\n-
 $\\sin x$ starts at $0$ and rises; $\\cos x$ starts at its peak $1$. Cosine is just sine shifted left by 
$\\tfrac{\\pi}{2}$.\n\nThese two controls -- amplitude and period -- let a sine wave model a pure tone, a tide, or an 
alternating current."
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                      "$A = 3$."
```

```
            ],
            "answer": "Amplitude $= 3$."
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              "$B = 2$, so $2\\pi/2$."
            ],
            "answer": "Period $= \\pi$."
          },
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            "problem": "What is $\\cos x$ at $x = 0$ versus $\\sin x$ at $x = 0$?",
            "steps": [
              "$\\cos 0 = 1$ (peak).",
              "$\\sin 0 = 0$ (zero)."
            ],
            "answer": "Cosine starts at 1; sine starts at 0."
          }
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      }
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      "content": {
        "intro": "Read amplitude and period; graph the waves."
      }
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period, and phase you can place a sine wave anywhere. The astonishing **Fourier theorem** then says *any* repeating
signal -- a square wave, a violin note, a heartbeat -- is a sum of pure sines and cosines of different frequencies. That
 decomposition powers MP3s, image compression, and nearly all of signal processing."
      }
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          "Period is 2pi; they repeat every full turn.",
          "Amplitude $|A|$ in $A\\sin x$ sets the peak height.",
          "Period $2\\pi/B$ in $\\sin(Bx)$ sets the cycle length.",
          "Cosine is sine shifted left by pi/2."
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            "tex": "y = A\\sin x \\Rightarrow \\text{amp} = |A|"
          },
          {
            "label": "Period",
            "tex": "y = \\sin(Bx) \\Rightarrow T = \\dfrac{2\\pi}{B}"
          }
        ]
      }
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        "tolerance": 0
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      "explanation": "Amplitude is $|A| = 3$."
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        "$1$",
        "$\\tfrac{\\pi}{2}$"
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      "answer": 1,
      "explanation": "Sine repeats every $2\\pi$."
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        "$1$",
        "$-1$",
        "undefined"
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      "answer": 1,
      "explanation": "Cosine starts at its peak, 1."
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      "hint": "Starts at 0, peaks at 1.",
      "explanation": "The standard sine wave: 0 at the origin, oscillating between -1 and 1."
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      "answer": {
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        "domain": [
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          6.28
        ],
        "variable": "x"
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      "difficulty": 3,
      "hint": "Starts at its peak, 1.",
      "explanation": "Cosine is sine shifted left by pi/2 -- it starts at 1."
    },
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        "domain": [
          -6.28,
          6.28
        ],
        "variable": "x"
      },
      "difficulty": 4,
      "hint": "Double the height.",
      "explanation": "Same wave, stretched vertically to peak at +/-2."
    },
    {
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        "$2\\pi$",
```

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          "$\\pi$",
          "$\\tfrac{\\pi}{2}$"
        ],
        "answer": 2,
        "hint": "Period $= 2\\pi/B$ with $B = 2$.",
        "explanation": "$2\\pi/2 = \\pi$."
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        "answer": true,
        "hint": "By pi/2.",
        "explanation": "$\\cos x = \\sin(x + \\tfrac{\\pi}{2})$ -- a horizontal shift."
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            "sub": "Multiply by the same factor again and again, and quantities explode -- or vanish --
astonishingly fast."
          }
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          "content": {
            "markdown": "A rumor that doubles its reach each hour, a savings account compounding interest, a
radioactive sample halving every few years -- all follow the same law. **Exponential** change multiplies by a constant
factor each step, in contrast to **linear** change, which adds a constant. The difference becomes staggering quickly."
          }
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            "markdown": "An **exponential function** has the variable in the **exponent**:\n\n$$f(x) =
a\\,b^{x},$$\n\nwhere $a$ is the starting value and $b$ is the **growth factor**.\n\n- If $b > 1$, it’s **growth** (e.g.
 $b = 2$ doubles each step).\n- If $0 < b < 1$, it’s **decay** (e.g. $b = \\tfrac12$ halves each step).\n\nKey contrasts
 with linear functions:\n\n- Linear *adds* a fixed amount; exponential *multiplies* by a fixed factor.\n- Exponential
growth has a constant **doubling time**; decay has a constant **half-life**.\n- The graph of $b^x$ is always positive,
rising steeply (growth) or falling toward zero (decay) but never reaching it.\n\nThe special base $e \\approx 2.718$
gives the smoothest, ’natural’ exponential $e^x$, central to all of calculus and continuous growth."
          }
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                "steps": [
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                  "$= 8$."
                ],
```

```
            "answer": "$f(3) = 8$."
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            "title": "Decay",
            "problem": "A sample halves each year, starting at $80\\,\\text{g}$. How much after 2 years?",
            "steps": [
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              "Year 2: $20$."
            ],
            "answer": "$20\\,\\text{g}$."
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            "title": "Growth vs linear",
            "problem": "Compare adding 3 each step vs tripling each step, from 1, after 3 steps.",
            "steps": [
              "Linear: $1,4,7,10$.",
              "Exponential: $1,3,9,27$."
            ],
            "answer": "Exponential reaches 27 vs 10."
          }
        ]
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      }
    },
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        "intro": "Evaluate and graph exponentials."
      }
    },
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      "content": {
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$\\left(1 + \\tfrac1n\\right)^n \\to e$ as $n \\to \\infty$. The function $e^x$ is the unique exponential whose **rate
of growth equals its own value** -- its derivative is itself. That self-replicating property makes $e^x$ the backbone of
 continuous growth, differential equations, and the link between exponentials and the trigonometric waves via Euler’s
formula."
      }
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          "Exponential functions $f(x) = a\\,b^x$ put the variable in the exponent.",
          "$b > 1$ is growth; $0 < b < 1$ is decay.",
          "Exponentials multiply by a fixed factor; linear functions add.",
          "Growth has a doubling time; decay has a half-life.",
          "The natural base $e \\approx 2.718$ gives $e^x$, central to calculus."
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          }
        ]
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      "explanation": "$2^3 = 8$."
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    "options": [
      "growth",
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      "linear change",
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    "answer": 1,
    "explanation": "A base below 1 shrinks each step -- decay."
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    "kind": "NUMERIC",
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    "answer": {
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      "tolerance": 0
    },
    "explanation": "$80 \\to 40 \\to 20$."
  },
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      "domain": [
        -3,
        4
      ],
      "variable": "x"
    },
    "difficulty": 4,
    "hint": "Doubles each step; always positive.",
    "explanation": "$2^x$ rises steeply and never touches zero."
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    "answer": {
      "expr": "(1/2)^x",
      "domain": [
        -3,
        4
      ],
      "variable": "x"
    },
    "difficulty": 4,
    "hint": "Halves each step.",
    "explanation": "$(1/2)^x$ falls toward zero as $x$ grows."
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    "kind": "NUMERIC",
    "prompt": "For $f(x) = 3^x$, find $f(2)$.",
    "answer": {
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      "tolerance": 0
    },
    "hint": "$3^2$.",
    "explanation": "$3^2 = 9$."
  },
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    "hint": "Multiplying beats adding in the long run.",
    "explanation": "For a base > 1, exponential growth outpaces any line eventually."
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asking a single question: *to what power must we raise the base?*"
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b^{x} = y.$$\n\nIn words: $\\log_b y$ is the power you raise $b$ to in order to get $y$. For example $\\log_2 8 = 3$
because $2^3 = 8$.\n\nCommon bases: $\\log_{10}$ (’log’, used for pH and decibels) and $\\log_e = \\ln$ (the ’natural
log’).\n\nThe **laws of logarithms** turn multiplication into addition:\n\n$$\\log(xy) = \\log x + \\log y, \\quad
\\log\\!\\frac{x}{y} = \\log x - \\log y, \\quad \\log(x^n) = n\\log x.$$\n\nThese laws are why logarithms historically
powered slide rules and hand computation, and why they appear whenever we solve for a variable stuck in an exponent."
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                      "$2^3 = 8$."
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\\log 50$, so $x = \\dfrac{\\log 50}{\\log 2} \\approx 5.64$. This ’take the log of both sides’ move solves for any
variable trapped in an exponent -- exactly what’s needed for half-life problems, compound interest, and the time
constants of every charging capacitor and cooling cup of coffee."
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              "$\\log_b y$ is the power giving $y$ from base $b$.",
              "Common logs: base 10 (’log’) and base $e$ (’ln’).",
              "Product -> sum: $\\log(xy) = \\log x + \\log y$; powers come down: $\\log(x^n) = n\\log x$.",
              "Logs solve for a variable in an exponent."
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              "hint": "$3 + 2$.",
              "explanation": "$\\log_2 32 = 5$."
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**approach**, not the value *at* $a$ -- the function need not even be defined there.\n\n- For a ’nice’ (continuous)
function, just substitute: $\\lim_{x\\to 2}(x^2) = 4$.\n- The famous case is $f(x) = \\dfrac{x^2 - 1}{x - 1}$. At $x =
1$ it’s $\\tfrac00$ -- undefined. But for every other $x$ it simplifies to $x + 1$, so as $x \\to 1$ the function
approaches $2$. The limit is $2$ even though $f(1)$ doesn’t exist.\n\nA limit exists only if the approach from the
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                      "$2(3) + 1 = 7$."
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                      "Cancel: $x + 1$.",
                      "As $x\\to1$: $2$."
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                      "Then there’s no single value."
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\\to 0$. That limit, $\\displaystyle\\lim_{h\\to 0}\\dfrac{f(x+h)-f(x)}{h}$, is the **derivative** -- the exact
instantaneous rate of change. Every idea in differential calculus rests on the limit you’ve just met."
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                  "It concerns the approach, not the value at $a$.",
                  "For continuous functions, substitute to evaluate.",
                  "Factor and cancel to resolve $0/0$ forms.",
                  "A limit exists only if left and right approaches agree."
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    "explanation": "It simplifies to $x + 1 \\to 2$."
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    "explanation": "$5 - 0 = 5$."
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      "$x = 0$"
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with a limit."
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giving the exact rate of change at a point. It is the central object of differential calculus."
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f(x)}{h}.$$\n\nIt measures the **instantaneous rate of change**: velocity is the derivative of position; acceleration is
 the derivative of velocity.\n\nThe workhorse is the **power rule**:\n\n$$\\frac{d}{dx}\\,x^{n} =
n\\,x^{\\,n-1}.$$\n\nSo $\\dfrac{d}{dx}x^2 = 2x$, $\\dfrac{d}{dx}x^3 = 3x^2$, and the derivative of a constant is $0$ (a
 flat line has no slope). The derivative of $x$ itself is $1$.\n\nReading a derivative: where $f’ > 0$ the function
rises; where $f’ < 0$ it falls; where $f’ = 0$ it has a flat spot -- a peak, valley, or plateau."
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                    "$f’(3) = 6$."
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                    "Slope is 0."
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                "Power rule: $\\frac{d}{dx}x^n = nx^{n-1}$.",
                "The derivative of a constant is 0.",
                "$f’ > 0$ rising, $f’ < 0$ falling, $f’ = 0$ flat."
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                }
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        "vars": [
          "x"
        ]
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$x$).",
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          5
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        "variable": "x"
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        "Divide by $h$: $2x + h$.",
        "Let $h\\to0$: the limit is $2x$."
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meet."
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 time, calculus gives **rules** that combine known derivatives. Master a handful and you can differentiate any
polynomial instantly."
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$\\dfrac{d}{dx}\\big(c\\,f(x)\\big) = c\\,f’(x)$. A constant factor rides along.\n- **Sum rule:** $\\dfrac{d}{dx}\\big(f
 + g\\big) = f’ + g’$. Differentiate term by term.\n- **Constant:** the derivative of any constant is $0$.\n\nTogether
these differentiate any polynomial. For $f(x) = 3x^2 + 5x - 4$:\n\n$$f’(x) = 3(2x) + 5(1) - 0 = 6x + 5.$$\n\nThe
**second derivative** $f’’(x)$ -- the derivative of the derivative -- measures how the rate itself changes (in motion,
that’s acceleration). For more complex functions there are the product, quotient, and chain rules, but the
constant-multiple and sum rules already unlock all of polynomial calculus."
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```

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 -- differentiate the outer function, then multiply by the derivative of the inside. For example
$\\dfrac{d}{dx}(x^2+1)^3 = 3(x^2+1)^2 \\cdot 2x$. The chain rule is the most-used rule in all of calculus, because
almost every real function is a composition of simpler ones."
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            "The derivative of a constant is 0.",
            "These rules differentiate any polynomial.",
            "The second derivative is the derivative of the derivative."
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curve."
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quantities -- the area under a curve, the distance from a speed graph, the total from a rate."
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Reversing the power rule:\n\n$$\\int x^{n}\\,dx = \\frac{x^{\\,n+1}}{n+1} + C \\quad (n \\ne -1).$$\n\nAdd one to the
power, divide by the new power. The $+C$ is the **constant of integration** -- any constant differentiates to zero, so
antiderivatives form a family. For example $\\int 2x\\,dx = x^2 + C$.\n\nThe **definite integral** $\\int_a^b f(x)\\,dx$
 is the **signed area** under $f$ between $a$ and $b$. The **Fundamental Theorem of Calculus** ties the two
together:\n\n$$\\int_a^b f(x)\\,dx = F(b) - F(a),$$\n\nwhere $F$ is any antiderivative. So areas -- once found by
painstaking limits of rectangles -- collapse to evaluating an antiderivative at two points."
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                      "$3^2 - 0^2 = 9$."
                    ],
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 extra area is a sliver of width $h$ and height about $f(x)$, so $A(x+h) - A(x) \\approx f(x)\\,h$. Divide by $h$ and
let $h\\to0$: $A’(x) = f(x)$. The area function *is* an antiderivative -- that is the Fundamental Theorem of Calculus,
```

```
the bridge uniting the two halves of the subject."
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              "Power rule for integrals: $\\int x^n dx = \\frac{x^{n+1}}{n+1} + C$.",
              "The $+C$ reflects the family of antiderivatives.",
              "A definite integral is the signed area under the curve.",
              "Fundamental Theorem: $\\int_a^b f\\,dx = F(b) - F(a)$."
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              {
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                "tex": "\\int_a^b f\\,dx = F(b) - F(a)"
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            "$F’ = f$",
            "$F’’ = f$",
            "$F = f’$"
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**vectors**: ordered lists of numbers. Linear algebra is the study of vectors and the transformations acting on them,
and it is the mathematical engine of computer graphics, machine learning, and modern physics."
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b_1+b_2)$.\n- **Scalar multiplication** stretches: $k(a, b) = (ka, kb)$.\n- The **magnitude** (length) comes from the
Pythagorean theorem:\n\n$$|\\vec v| = \\sqrt{a^2 + b^2}.$$\n\nThe **dot product** combines two vectors into a number:
$(a_1,b_1)\\cdot(a_2,b_2) = a_1a_2 + b_1b_2$. It measures how much they point the same way -- zero means
**perpendicular**. These few operations, scaled to many dimensions, are all of applied linear algebra."
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                      "$(4, 7)$."
                    ],
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                    "problem": "Compute $(1, 2)\\cdot(3, 4)$.",
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rule generalizes: add componentwise, and $|\\vec v| = \\sqrt{v_1^2 + \\cdots + v_n^2}$. A grayscale image is a vector
with one component per pixel; a document can be a vector of word counts. Machine learning lives in spaces of thousands
of dimensions, where the dot product still measures similarity exactly as it does for two arrows."
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                  "Add and scale componentwise.",
                  "Magnitude $|\\vec v| = \\sqrt{a^2 + b^2}$.",
                  "Dot product $a_1a_2 + b_1b_2$ measures alignment; 0 means perpendicular.",
                  "Everything generalizes to n dimensions."
```

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          "b"
        ]
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acting on vectors. A matrix is both a table of data and a **linear transformation**, and that dual nature makes it the
central object of the data sciences."
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is\n\n$$A = \\begin{pmatrix} a & b \\\\ c & d \\end{pmatrix}.$$\n\nOperations:\n\n- **Add** matrices of the same shape
entrywise.\n- **Scalar multiply** by multiplying every entry.\n- **Matrix x vector** transforms the vector:
$\\begin{pmatrix} a & b \\\\ c & d \\end{pmatrix}\\begin{pmatrix} x \\\\ y \\end{pmatrix} = \\begin{pmatrix} ax + by
\\\\ cx + dy \\end{pmatrix}$.\n\nMatrix multiplication chains transformations -- and is **not commutative**: $AB \\ne
BA$ in general, because doing a rotation then a stretch differs from the reverse. The **identity matrix**
$\\begin{pmatrix}1&0\\\\0&1\\end{pmatrix}$ leaves every vector unchanged."
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                    "$(2, 3)$."
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                  "problem": "Multiply $\\begin{pmatrix}1&2\\\\3&4\\end{pmatrix}$ by 2.",
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```

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            "$\\begin{pmatrix}2&4\\\\6&8\\end{pmatrix}$."
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        "Add and scalar-multiply entrywise.",
        "Matrix x vector transforms the vector.",
        "Matrix multiplication is not commutative: $AB \\ne BA$.",
        "The identity matrix leaves vectors unchanged."
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],
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why a zero determinant breaks everything: you can’t divide by zero, so a singular matrix has no inverse and no unique solution."
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              "$\\det = 0$: no unique solution (parallel/identical equations).",
              "The inverse A^{-1} exists only when $\\det \\ne 0$."
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              "explanation": "A zero determinant means no inverse."
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              "explanation": "The identity has determinant 1."
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example\n\n$$\\frac{dy}{dt} = k\\,y$$\n\nsays ’the rate of change of $y$ is proportional to $y$ itself’.\n\nA
**solution** is a function that satisfies the equation. Crucially, solutions are **families** of functions; a specific
**initial condition** (a known value at one time) pins down which one.\n\n- The **order** of a DE is the highest
derivative it contains. $\\dfrac{dy}{dt} = ky$ is first-order; $\\dfrac{d^2x}{dt^2} = -\\omega^2 x$ is second-order.\n-
To **check** a proposed solution, differentiate it and substitute. For $\\dfrac{dy}{dt} = 2y$, try $y = e^{2t}$: its
derivative is $2e^{2t} = 2y$. [x]\n\nThe leap from algebra is profound: the unknown isn’t a number, it’s an entire
**function**."
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gap’ becomes Newton’s law of cooling $\\dfrac{dT}{dt} = -k(T - T_{\\text{env}})$. ’Population grows but is capped by
resources’ becomes the logistic equation. Once written, the DE is solved (by hand or numerically) and its solution
**predicts the future** -- which is exactly how weather, epidemics, and spacecraft trajectories are forecast."
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                  "A solution is a function that satisfies it.",
                  "Solutions come in families; an initial condition picks one.",
                  "Order = the highest derivative present.",
                  "Check a solution by differentiating and substituting."
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              "explanation": "An initial condition pins down the constants."
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solution\n\n$$y(t) = y_0\\,e^{kt},$$\n\nwhere $y_0 = y(0)$ is the initial value.\n\n- If $k > 0$: **exponential growth**
 (population, interest).\n- If $k < 0$: **exponential decay** (radioactivity, cooling).\n\nWhy $e$? Because
$\\dfrac{d}{dt}e^{kt} = k\\,e^{kt}$ -- the exponential is the unique function (up to scale) whose derivative is
proportional to itself, exactly what the DE demands.\n\nThe **half-life** (decay) or **doubling time** (growth) is
constant and depends only on $|k|$: doubling time $= \\dfrac{\\ln 2}{k}$. This is why carbon dating works -- decay is
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                  ],
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              "$k > 0$ is growth; $k < 0$ is decay.",
              "$e^{kt}$ works because its derivative is proportional to itself.",
              "Doubling time / half-life depends only on $|k|$.",
              "Separation of variables derives the solution."
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            "$y = k/t$"
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              "variable": "x"
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                    "steps": [
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                      "$\\dfrac1n \\to 0$."
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sequence whose terms bunch arbitrarily close together (a **Cauchy sequence**) actually converges -- to a real number.
The rationals fail this: $3, 3.1, 3.14, 3.141, \\dots$ bunches up but its limit $\\pi$ isn’t rational. Completeness is
the axiom that plugs the ’holes’ in the number line and guarantees that limits, suprema, and integrals exist."
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              "It converges to $L$ if terms stay arbitrarily close to $L$.",
              "epsilon-N definition: for any epsilon>0, eventually $|a_n - L| < epsilon$.",
              "$1/n \\to 0$; $(-1)^n$ diverges.",
              "Completeness of the reals guarantees Cauchy sequences converge."
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            "diverges",
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          "answer": 2,
          "explanation": "It alternates and never settles."
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            "explanation": "Convergence must hold for every epsilon > 0, no matter how small."
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            "hint": "Divide top and bottom by n.",
            "explanation": "$\\dfrac{3}{1 + 1/n} \\to 3$."
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            "explanation": "Completeness of the reals guarantees it."
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theory, geometry, and the symmetries of physical law."
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for all $a, b \\in G$, $a * b \\in G$.\n2. **Associativity:** $(a * b) * c = a * (b * c)$.\n3. **Identity:** there is an
 $e$ with $e * a = a * e = a$ for all $a$.\n4. **Inverses:** every $a$ has an $a^{-1}$ with $a * a^{-1} =
e$.\n\nExamples:\n\n- The integers under **addition**: identity $0$, inverse of $5$ is $-5$. [x] A group.\n- The
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integers under **multiplication**: $0$ has no inverse, and most elements lack one. [ ] Not a group.\n- The rotations of a square (0 deg, 90 deg, 180 deg, 270 deg) under composition. [x] A finite group.\n\nIf the operation is also **commutative** ($a*b = b*a$), the group is **abelian**. From these spare axioms an enormous, rigorous theory unfolds."
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                      "Identity is 1.",
                      "But 2 has no integer inverse."
                    ],
                    "answer": "No -- inverses fail."
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                  "Closure, associativity, an identity, and inverses for all.",
                  "Integers under addition form a group; under multiplication they don’t.",
                  "An abelian group also has $a*b = b*a$.",
                  "Groups formalize symmetry -- central to modern physics."
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            "explanation": "Most integers lack a multiplicative inverse, so it’s not a group."
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              "gives zero",
              "is commutative"
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beyond any doubt, for infinitely many cases at once. Two techniques do most of the heavy lifting: **induction** (for
statements about all integers) and **contradiction** (assume the opposite and derive an absurdity)."
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steps:\n\n1. **Base case:** show $P(1)$ is true.\n2. **Inductive step:** assume $P(k)$ (the hypothesis) and prove
$P(k+1)$.\n\nThen $P(n)$ holds for all $n$ -- like dominoes: the first falls, and each knocks over the next.\n\n**Proof
by contradiction** proves a statement by assuming its **negation** and deriving an impossibility. The classic: to show
$\\sqrt2$ is irrational, assume $\\sqrt2 = a/b$ in lowest terms and reach a contradiction (both $a$ and $b$ turn out
even).\n\nOther tools include **direct proof** (chain implications from hypothesis to conclusion) and **proof by
counterexample** (one example disproves a universal claim). Mastery of these is what separates doing mathematics from
merely using it."
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                    "Derive an impossibility."
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                    "2 is prime and even."
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show that any consistent formal system rich enough for arithmetic contains true statements it cannot prove -- and cannot
 prove its own consistency. This doesn’t make mathematics shaky; it reveals that truth outruns provability. It reshaped
logic, foundations, and our understanding of what computation and reasoning can ever achieve."
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              "Induction: prove the base case, then the inductive step.",
              "Contradiction: assume the negation, derive an absurdity.",
              "One counterexample disproves a universal claim.",
              "G?del showed some truths are unprovable within a system."
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            "contradiction",
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            "a base case",
            "nothing"
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              "Assume it holds for $n = k$: $1 + \\cdots + k = \\dfrac{k(k+1)}{2}$.",
              "Add $(k+1)$ to both sides.",
              "Simplify the right side to $\\dfrac{(k+1)(k+2)}{2}$ -- the formula for $n = k+1$.",
              "By induction, the formula holds for all $n$."
            ],
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            "difficulty": 5,
            "hint": "Base case, hypothesis, add the next term, simplify, conclude.",
            "explanation": "Verify $n=1$, assume for $k$, add $(k+1)$, simplify to the $k+1$ form, then conclude for
all $n$."
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              "Assume $\\sqrt2 = \\dfrac{a}{b}$ in lowest terms.",
              "Then $2b^2 = a^2$, so $a^2$ is even, hence $a$ is even.",
              "Write $a = 2c$, giving $2b^2 = 4c^2$, so $b^2 = 2c^2$.",
              "Then $b^2$ is even, so $b$ is even too.",
              "But then $a$ and $b$ share the factor 2 -- contradicting ’lowest terms’."
            ],
            "answer": [],
            "difficulty": 5,
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            "explanation": "Assuming a lowest-terms fraction forces both numerator and denominator even --
impossible."
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              "1 is not prime",
              "primes are infinite"
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            "answer": 1,
            "hint": "Find an even prime.",
            "explanation": "2 is prime and even -- a counterexample."
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            "hint": "The domino effect.",
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